Orthopedics AIIMS 2017

Question 1:

A 5 year old boy fell on his outstretched hand. X-ray is shown below. Which of the following vessel is most commonly affected?

a) Radial artery

b) Brachial artery

c) Ulnar artery

d) Cubital vein

Question 2:

True regarding Galeazzi fracture-dislocation:

a) Radial collateral ligament tear with interosseous membrane tear with radial shaft fracture

b) Interosseous membrane tear with ulnar shaft fracture

c) Interosseous membrane tear with triangular fibrocartilage complex (TFCC) tear and radial shaft fracture

d) Interosseous membrane tear with triangular fibrocartilage complex (TFCC) tear and ulnar shaft fracture

Question 3:

A 25-year-old male presented to the emergency department with pain and swelling of the right knee. The radiograph showed the following finding. What is the best treatment for the given condition?

- **a)** Tension band wiring with K-wires
- **b)** Tension band wiring with cancellous screws
- **c)** Cylinder cast
- **d)** Patellectomy

Question 4:

Which of the following movements is being tested in the picture given below?

a) Internal rotation of hip

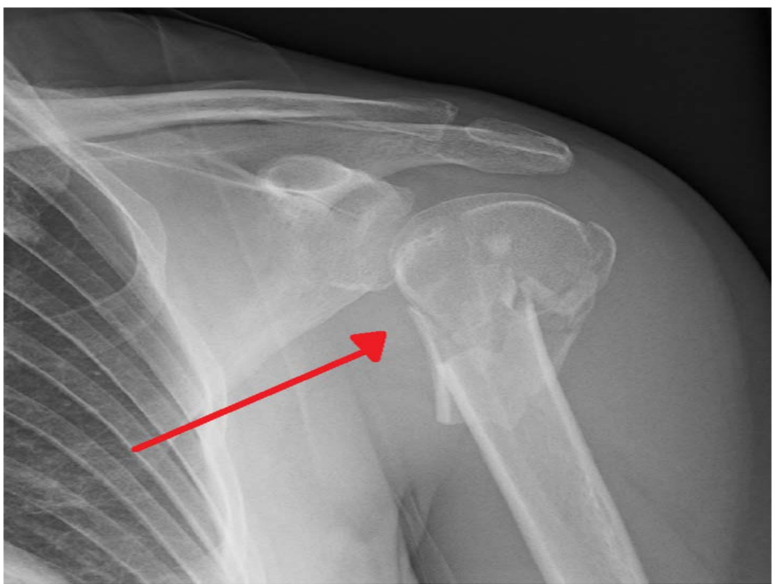
b) Abduction

c) Flexion

d) External rotation of hip

Question 5:

Which is the classification system used for the fracture shown in the x-ray given below?



a) Schatzker classification

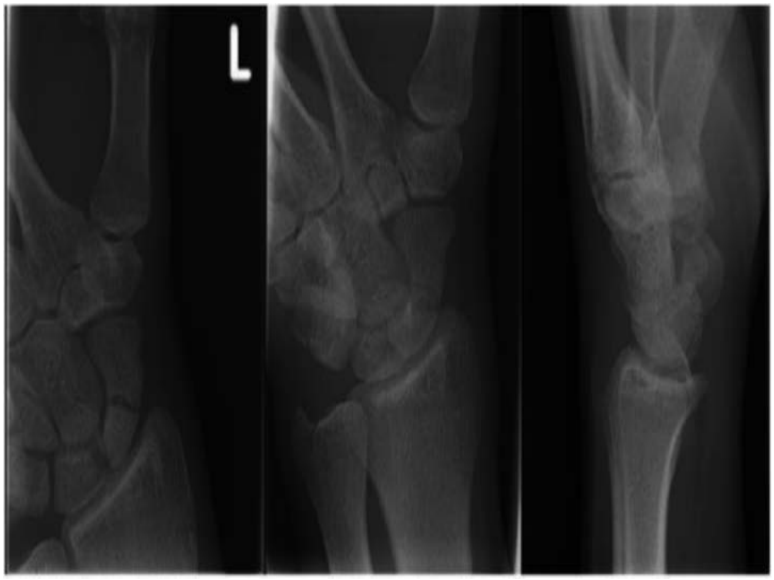
b) Ideberg classification

c) Neer classification

d) Gartland classification

Question 6:

An elderly male presented with a history of fall on outstretched hand following which he developed swelling of the wrist. His X-ray is shown below. What is your diagnosis?



a) Fracture scaphoid

b) Fracture scaphoid with perilunate dislocation

c) Fracture hamate

d) Fracture capitate

Question 7:

Gustillo anderson classification type IIIb is?

a) Open fracture, wound more than 1 cm , less than 10 cm in length without extensive soft tissue damage, flaps, avulsions

b) Open fracture, wound less than 1 cm in length without extensive soft tissue damage, flaps, avulsions

c) Open fracture, wound more than 10 cm in length with extensive soft tissue loss and periosteal stripping and bone damage

d) Open facture, wound more than 10 cm in length associated with an arterial injury

Question 8:

An overweight man (90 kg) presented with complaints of not being able to squat and rest pain of left hip for 6 months duration. He had a past history of steroid and creatine use. His X-ray revealed destruction of the articular surface of femoral head with the presence of subchondral cysts and positive crescent sign. His MRI is suggestive of marrow edema. What is your diagnosis?

- **a)** Tuberculosis of hip
- **b)** Fracture neck of femur
- **c)** Osteochondroma of femur
- **d)** Osteonecrosis of femoral head

Question 9:

All of the following are associated with Sprengel shoulder except?

- **a)** Diastematomyelia
- **b)** Klippelfeil syndrome
- **c)** Dextrocardia
- **d)** Congenital scoliosis

Question 10:

A man with a POP cast for forearm fracture is prescribed analgesics. The earliest way to check for compartment syndrome is to look for:

- **a)** Disappearance of pulse by displacing the cast
- **b)** Discolouration of fingers
- **c)** Decreased response to analgesics
- **d)** Odour and discharge

Question 11:

Which of the following is not true about the posterior cruciate ligament:

- **a)** Extra synovial
- **b)** Prevent the posterior displacement of tibia

c) Primary action is to prevent the internal rotation of knee joint

d) Attached to anterolateral aspect of medial condyle

Question 12:

A patient presented with lower limb weakness. The examiner places one hand under the patient's heel and the patient is then asked to raise his other leg against downward resistance. What is the test being performed?

a) Hoover’s Test

b) Mc Bride Test

c) Waddell’s Test

d) O’Donoghue’s Test

Question 13:

March fracture is seen in the ____ of second metatarsal bone.

a) Head

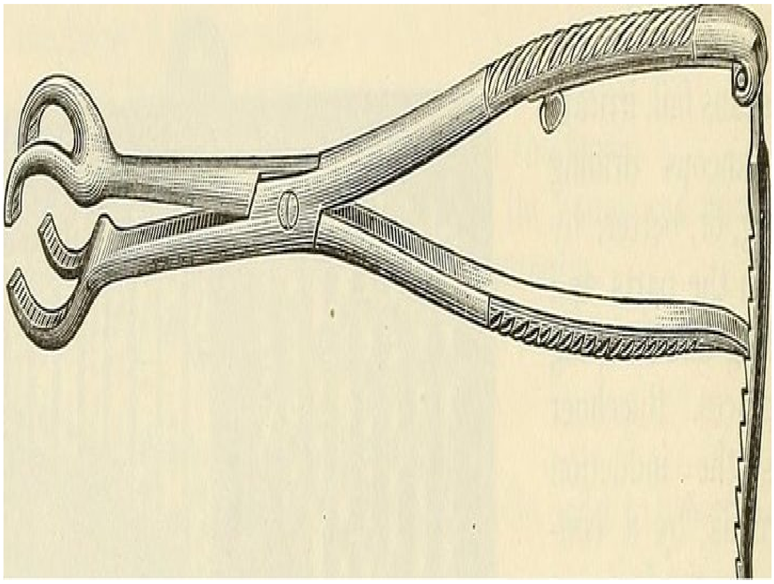
b) Neck

c) Shaft

d) Base

Question 14:

Identify the name of the instrument given below:



a) Bent Hohmann retractor

b) Bone holding forceps

c) Bone distractor

d) Bone curette

Question 15:

A patient came to the casualty after a RTA with fracture of femur. On 3rd day of admission, he developed sudden breathlessness. Which of the following is the most common cause?

a) Hypovolemic shock

b) Pulmonary hypertension

c) Post trauma MI

d) Fat embolism

Question 16:

Hand, knee gait in polio paralysis is due to weakness of the ________.

a) Popliteus

b) Quadriceps

c) Gastro-soleus

d) Gluteus maximus

Question 17:

The most common sequela of traumatic dislocation of the shoulder in young adults is_____.

a) Subscapular tendinitis

b) Rotator cuff injury

c) Recurrent dislocation of shoulder

d) Frozen shoulder

Question 18:

After a fracture, the growth of callus will be increased by which of the following?

a) Strict immobilization

b) Repeated tiny movements at the fracture site

c) Intermedullary nail

d) K wire fixation

Question 19:

Haglund deformity is seen around which of the following joints?

a) Knee

b) Ankle

c) Wrist

d) Elbow

Question 20:

An elderly patient slipped in the bathroom and sustained injury over the left hip. The radiograph is shown below. His attitude of the leg will be:

a) Shortened and abducted

b) Lengthened and internally rotated

c) Shortened and externally rotated

d) Flexed, adducted and internally rotated

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	d
5	c
6	a
7	c
8	d
9	c
10	c
11	c
12	a
13	b
14	b

15	d
16	b
17	c
18	b
19	b
20	c

Detailed Explanations

Solution to Question 1:

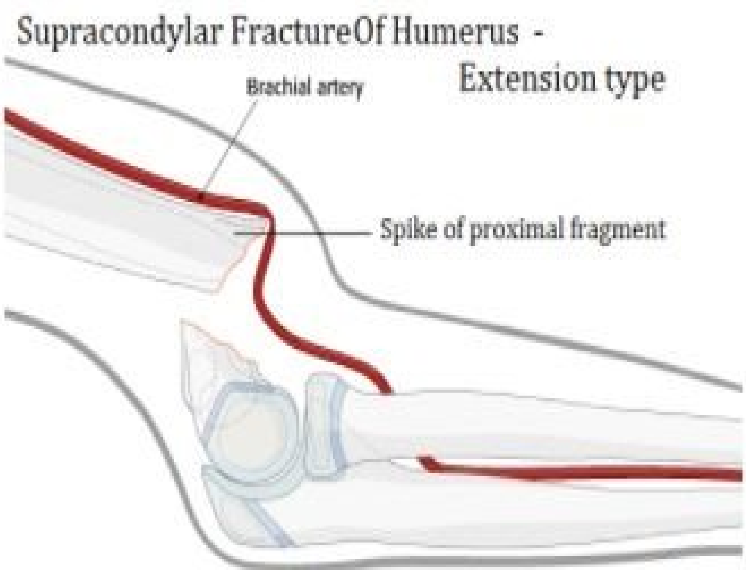
The given X-ray shows a supracondylar fracture of the humerus and injury to the brachial artery is the most common vascular complication.

The fracture most commonly caused by a fall on an outstretched hand in a child is the supracondylar fracture of the humerus. As the hand strikes the ground, the elbow is forced into hyperextension resulting in a fracture of the humerus above the condyles.

• The brachial artery is usually injured by the sharp edge of the proximal fragment.

• Sometimes flexor muscles of the forearm - flexor pollicis longus and flexor digitorum profundus (medial-half) may suffer ischemic damage leading to Volkmann's ischemia leading to compartment syndrome.

While injury to the brachial artery is the commonest vascular complication, however, malunion is the most common complication of supracondylar fracture of the humerus.



Other delayed complications:

• Malunion - Cubitus varus / Gunstock deformity

- Volkmann’s ischaemic contracture
- Myositis ossificans

Solution to Question 2:

Galeazzi fractures consist of a fracture of the radial shaft (lower one-third) with dislocation of the distal radioulnar joint (DRUJ).

Triangular Fibrocartilage Complex (TFCC)

The TFCC serves as the medial continuation of the distal articular surface of the radius as well as a static stabilizer of the DRUJ.

It consists of:

- Articular disc
- Dorsal radioulnar ligament - primary stabilizer
- Palmar radioulnar ligament - primary stabilizer
- The meniscus homologue
- The ulnar collateral ligament
- The sheath of the extensor carpi ulnaris

Other options:

Option A: Mentions radial collateral ligament but TFCC consists of the ulnar collateral ligament.

Options B and D: Mention ulnar shaft fracture when Galeazzi fracture is a radial shaft fracture.

Note: Galeazzi # are more common than Monteggia # and may be associated with ulnar nerve damage.

Solution to Question 3:

The radiograph shows a displaced patellar fracture. The best treatment for the given condition is tension band wiring with K- wires. In this method, two K-wires are used to transfix the reduced patellar fragments and a flexible wire is looped around them tightly.

Treatment options for patellar fracture are:

• Undisplaced patellar fractures with minimal joint incongruity and intact extensor retinaculum - cylinder cast

• Displaced patellar fractures - tension band wiring

• Severely comminuted fractures with difficulty in reducing fragments - patellectomy

Note: Tension band wiring with cancellous screw is done when compressive forces are also needed along with tensile forces to resist the displacement of fractures e.g., olecranon fractures.

Solution to Question 4:

External rotation of hip is being tested in the given image.

Rotatory movements of the hip joint can be measured with the knee flexed or extended. However, more precise measurements are obtained when the rotation is tested with the flexed knee. The procedure involves the following steps.

• The patient is made to lie supine.

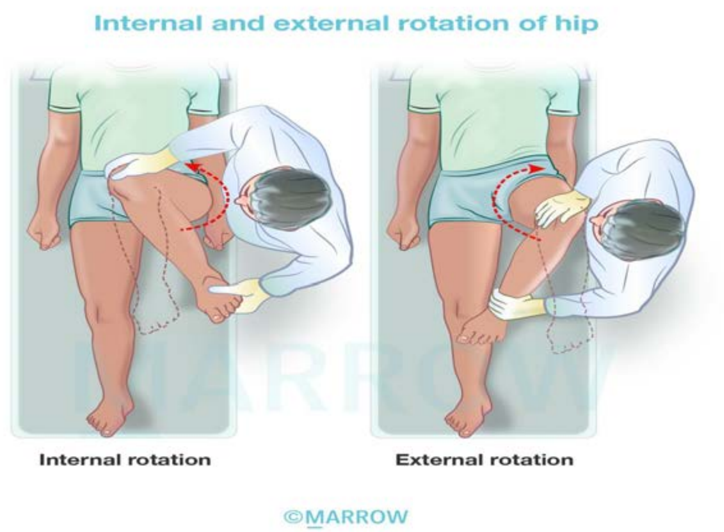
• The examiner holds the knee with one hand and ankle with the other hand.

• The knee and hip are flexed to 90 degrees.

• Rotation movement is produced at the hip joint by stabilising the knee and moving the leg as the lever.

• If the hip is rotated such that the leg moves medially, external rotation of the hip occurs.

• If the hip is rotated such that the leg moves laterally, internal rotation of the hip occurs.

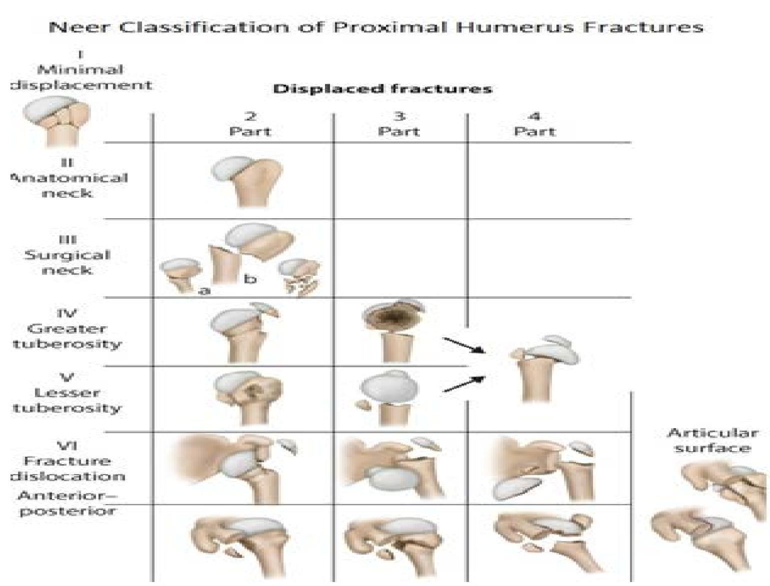


Solution to Question 5:

The given image is of the fracture of the proximal humerus. The most widely accepted classification is the Neer classification, which involves 4 major segments in injuries:

- Head of humerus
- Greater tuberosity
- Lesser tuberosity
- Shaft of humerus

Neer classification distinguishes between the number of displaced fragments, with displacement defined as greater than 45 degrees of angulation or 1 cm of separation.



Other options

Option A: The Schatzker classification is a method of classifying tibial plateau fractures.

Option B: The Ideberg classification is a method of classification of intra-articular glenoid fossa fractures.

Option D: The Gartland classification is a method of classification of supracondylar fractures of the humerus.

Solution to Question 6:

The above X-ray shows fracture scaphoid.

Scaphoid fractures account for almost 75 percent of all carpal fractures although they are rare in the elderly and in children.

Mechanism of injury- The combination of forced carpal movement and compression, as in a fall on the dorsiflexed hand.

Diagnostic sign- precisely localized tenderness in the anatomical snuff box.

X-ray-

• Anteroposterior, lateral and oblique views are all essential

• often a recent fracture shows only in the oblique view.

• Usually, the fracture line is transverse, and through the narrowest part of the bone (waist), but it may be more proximally situated (proximal pole fracture).

Old, un-united fractures have hard borders, making it seem as if there is an extra carpal bone.

• Relative sclerosis of the proximal fragment is pathognomonic of avascular necrosis.

NOTE: The distal fragment, unrestrained by the scapholunate ligament, flexes and the proximal fragment tilt dorsally with the lunate (a DISI deformity). The hump-backed deformity of the scaphoid is permanent.

Solution to Question 7:

An open fracture with a wound with extensive soft tissue loss and periosteal stripping and bone damage belongs to classification IIIB.

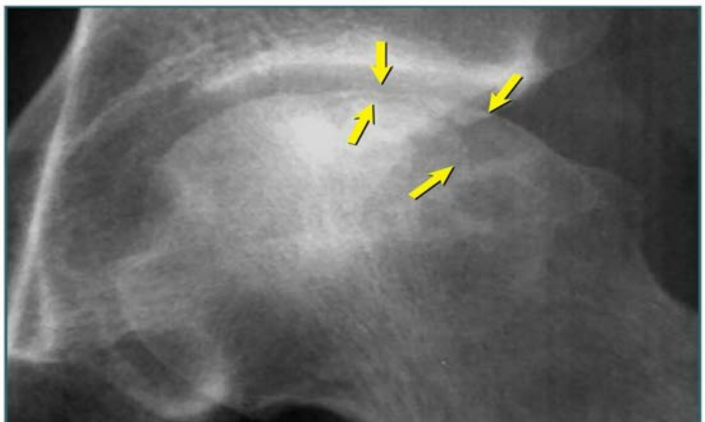
Solution to Question 8:

The complaints of hip pain in an overweight man with a history of long-term steroid use along with the radiological findings of subchondral cysts and positive crescent sign in the femoral head are characteristic of osteonecrosis of the femoral head.

Avascular necrosis or osteonecrosis is a painful bone condition that results from diminished blood supply mainly due to trauma or non-traumatic etiologies. Alcohol, corticosteroids, immunosuppressives, and cytotoxic drugs, either singly or in combination, are the most common causes of non-traumatic osteonecrosis. The typical features are pain with movements in or near the joint, click in the joint, joint stiffness, limping gait, and deformity. Restriction of movements is also a feature of this condition.

An x-ray shows reactive new bone formation, increased radiolucency in the subchondral bone, and a distorted articular surface. The crescent sign is a subchondral fracture overlying the necrotic segment of the femoral head seen on radiography. MRI is the investigation of choice in early disease. It allows the identification of AVN of the femoral head and also determines the exact stage and extent of the pathologic process.

Crescent sign



Avascular necrosis of femoral head

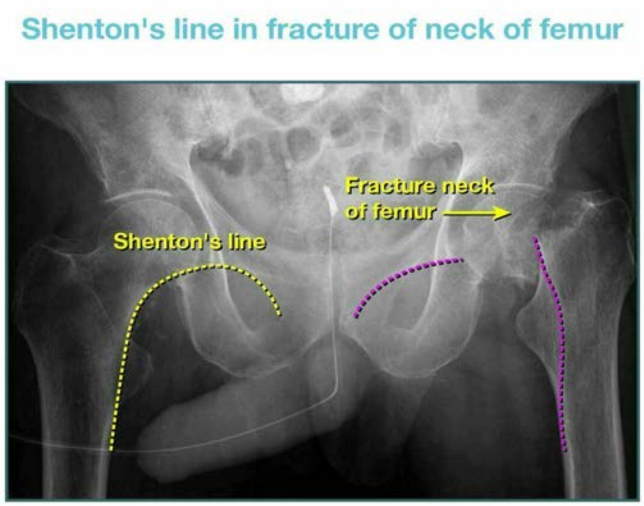
Management is based on Ficat and Arlet Classification:

Other options:

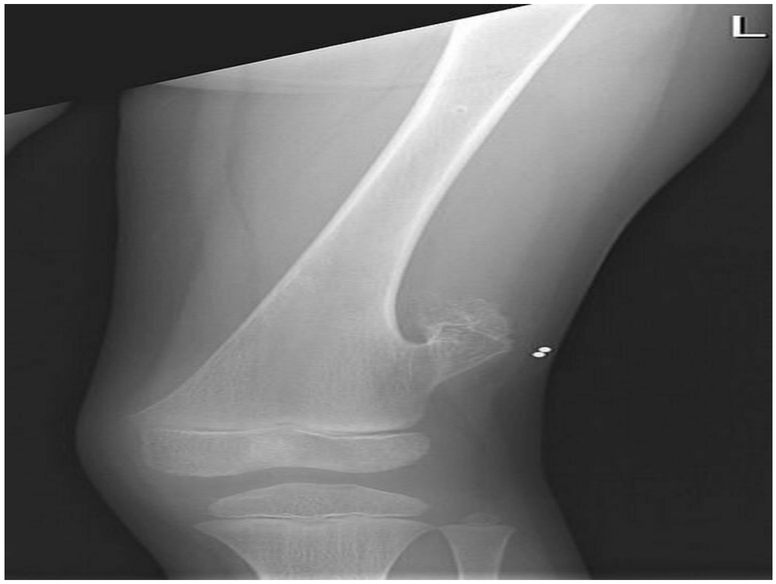
Option A: Tuberculosis of the hip is characterized by painful and restricted movements of the hip, especially at night. The radiographic findings are not appreciable in the early stages. Later stages show periarticular osteoporosis, hazy and irregular joint margins with a reduction in the joint space, and wandering acetabulum.

Stage	Findings	Management	Method
1	X-ray normal, MRI +ve	Conservative	Bisphosphonates
2a	X-ray: sclerosis, cysts, no collapse	Conservative	Bisphosphonates
2b	X-ray: crescent sign, no collapse	Surgical decompression	Core decompression
3	Collapse+loss of sphericity	Surgical	Osteotomy
4	Advanced arthritis	Reconstructive	Total hip replacement

Option B: Fracture neck of femur is common in elderly people after a trivial fall due to osteoporosis. Examination reveals tenderness in the groin along with shortening and external rotation of the leg. Xrays show a break in Shenton's line and trabecular stream, overriding of greater trochanter so that it lies at the level of the head of the femur, and external rotation of the femur is evident by prominent lesser trochanter.



Option C: Osteochondromas are common benign bone tumors commonly present in adolescents. The lesions consist of a bony mass, often in the form of a stalk, produced by endochondral ossification of a growing cartilaginous cap. They are usually painless. The common sites of involvement are the distal femur, proximal tibia, and proximal humerus.



Solution to Question 9:

Dextrocardia is not associated with Sprengel deformity.

Sprengel deformity is the congenital elevation of the scapula. It is the most common congenital shoulder abnormality.

The scapula is abnormally small and too high. Sometimes the clavicle is affected as well. Shoulder movements may be restricted and on abduction or elevation, the scapula moves very little or not at all.

Sprengel's deformity may be associated with other defects of the cervical spine (e.g. Klippel–Feil syndrome), and high thoracic kyphosis or scoliosis is quite common.

The high scapula may still be attached to the spine by a tough ■brous band or a cartilaginous bar (the omovertebral bar). Associated vertebral or rib anomalies are quite common.

An unappreciated association between Sprengel anomaly and diastematomyelia (congenital splitting of spinal cord) of the lumbar spine also raises questions about the embryologic origin of the scapula.

Associations of Sprengel deformity of Shoulder

• Klippel-Feil Syndrome

• Spina Bifida and Diastematomyelia

• Kyphosis or Scoliosis

• Torticollis

• Underdevelopment of clavicle or humerus

Solution to Question 10:

Decreased response to analgesia → suspect compartment syndrome.

Pain is the most common initial symptom in case of compartment syndrome. Red flags for compartment syndrome are pain out of proportion with the injury, and pain on passive stretch of the digits.

Later signs include pallor, paresthesia, paralysis, and pulselessness.

As the ischemia occurs at capillary level in compartment syndrome, pulses may still be felt and skin may not appear pale.

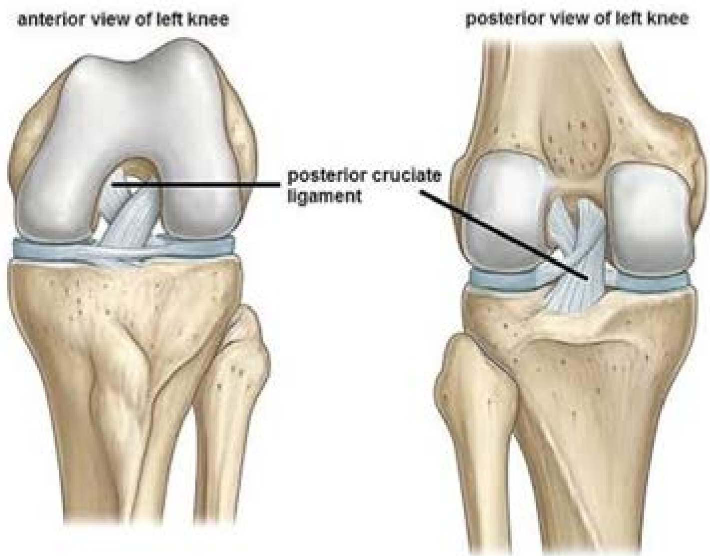
Solution to Question 11:

The lateral collateral ligament, not the posterior, is responsible for preventing the internal rotation of the knee joint

The posterior cruciate ligament (PCL) is an intra-articular but extra-synovial structure because of the presence of a synovial sheath that lines the ligament. It is the only primary restraint to posterior translation throughout the entire knee range of motion. It prevents the internal rotation of the tibia on the femur when there is varus stress on the knee.

It acts as a secondary restraint to valgus/varus, external rotation.

Anterior cruciate ligament	Posterior cruciate ligament
Attached to anterior part of the intercondylar area of tibia	Posterior part of intercondylar area of tibia
Direction – Upwards, backwards & laterally & attached to posterior aspect of medial surface of lateral condyle.	Upwards, forwards & medially & attached to the anterior aspect of the lateral surface of the medial condyle.
Prevents posterior dislocation of femur on tibia and forwards dislocation of tibia on femur.	Prevents anterior dislocation of femur on tibia and posterior dislocation of tibia on femur
Limits hyperextension of knee joint(taut during extension)	Limits hyperflexion of knee joint. (taut during flexion)
Weaker	Stronger



Solution to Question 12:

The test being performed by the examiner is Hoover's test. This test is useful in differentiating between the organic and functional causes of limb weakness.

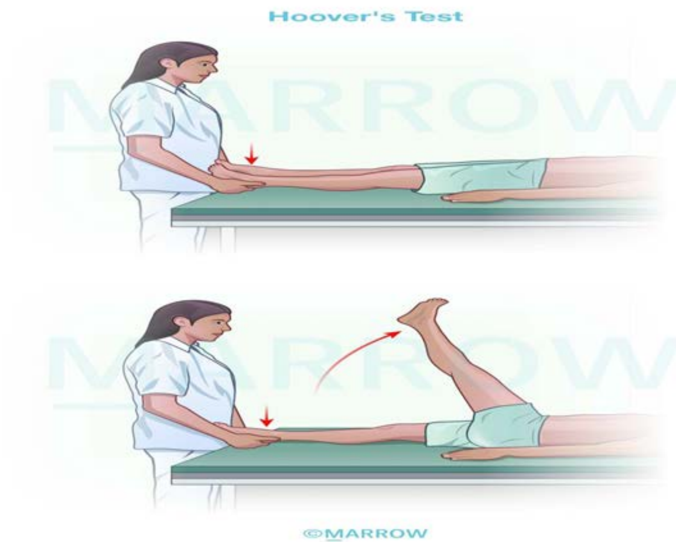
Hoover's test involves the following steps:

• The examiner places both their hands under the heels of the patient.

• The patient is asked to flex the affected hip while keeping the knee extended.

• Downward pressure in the healthy limb is felt by the examiner in an organic cause of limb weakness and in healthy individuals. This pressure is not felt in the functional cause of limb weakness and malingering.

In a normal individual, when raising one leg by flexing at the hip, the other limb exerts a downward pressure which can be felt at the ankle.



The McBride's test, Waddell’s test, and O’Donoghue’s test are other examination techniques that help in distinguishing between organic and non-organic causes of lower back pain.

Solution to Question 13:

March fracture/stress fracture is seen most commonly in the neck of the second metatarsal bone.

It results from long, repeated stresses like prolonged walking or running, usually seen in military men, athletes, and dancers. They may complain of gradual onset of pain over the 2nd metatarsal neck starting 2-4 weeks after the beginning of stressors. On examination, a tender lump may be noted. The x-ray may be normal initially but later show a hairline crack. After a period of 4-6 weeks, a mass of callus can be identified.

Management: Reduction and splinting are not necessary as it is an undisplaced fracture. Normal walking is permitted while using an elastic bandage on the forefoot.

The below image shows a March fracture of the second metatarsal bone.

Note: According to Campbell orthopedics (14th edition) in March fractures, patients often note the gradual onset of pain directly over the second metatarsal neck region. In Uptodate, it is mentioned that many of the stress fractures occur in the second metatarsal shaft, especially at the neck. Hence, neck is considered a more correct answer than shaft.

Solution to Question 14:

The image shows a bone holding forceps.

Solution to Question 15:

History of sudden onset breathlessness on the 3rd day of admission after femur fracture should raise suspicion of fat embolism.

Fat embolism is the presence of fat globules in vital organs and peripheral circulation after a fracture of the long bone or other major trauma. Its pathogenesis can be related to 2 factors: the release of free fatty acids by the action of lipases on the neutral fats and the actual obstruction of small pulmonary vessels by fat globules.

Clinical features:

- Early signs are fever and tachycardia
- Presenting features are of 2 types -
- Cerebral type - drowsiness, restlessness, disorientation
- Pulmonary type - tachypnoea, tachycardia
- Classical triad -
- Respiratory distress

- Neurological abnormalities
- Petechial hemorrhages
- Pathognomonic signs are petechiae on the trunk, axillae, conjunctival folds, and retina.

Treatment of fat embolism is mainly supportive as given below:

- Respiratory support with oxygen
- Heparinization increases lipoprotein lipase activity
- Intravenous low molecular weight dextran
- Corticosteroids to avoid pneumonitis
- Prompt stabilization of long-bone fractures

Solution to Question 16:

Polio commonly affects the quadriceps. The weakness of the quadriceps leads to the buckling (bending) of the knee while walking or bearing weight on the leg. So, in order to walk, the person has to prevent the buckling of the knee by supporting with the hand, producing a hand-knee gait.

Few important examination findings in paralytic polio:

- Asymmetric: involvement of affected muscles is haphazard.
- Most common muscle affected: Quadriceps, often partially paralyzed.
- Most common muscle to undergo complete paralysis: Tibialis anterior
- Most common muscle in hand: Opponens pollicis
- Motor paralysis is not associated with sensory loss.
- In the knee, triple deformity in severe cases: flexion, posterior subluxation, and external rotation

Solution to Question 17:

The most common sequela of traumatic shoulder dislocation in young adults is recurrent dislocations of the shoulder joint.

Traumatic shoulder dislocation is associated with tear of the glenoid labrum (Bankart lesion). This results in loss of attachment of the inferior glenohumeral ligament, which is the primary static stabilizer of the shoulder.

In addition, there is a loss of concavity due to damage to the labrum. Therefore, the shoulder joint becomes unstable. This predisposes the person to repeated shoulder dislocations. Shoulder instability/recurrent dislocation is an indication for surgical intervention.

Solution to Question 18:

Micro-movements at the fracture site stimulate callus formation.

Other options:

Option A: It is not necessary to immobilize all fractures (e.g., fracture ribs, scapula, etc) as they heal anyway. Some other fractures need strict immobilization (e.g., fracture of the neck of the femur) and may still not heal.

Options C, D: Kirschner wires (K-wire) and intra-medullary nails are used when a fracture is so unstable that it is difficult to maintain it in an acceptable position by nonoperative means. They are used in order to secure rigid immobilization. They do not affect the growth of the callus directly.

Solution to Question 19:

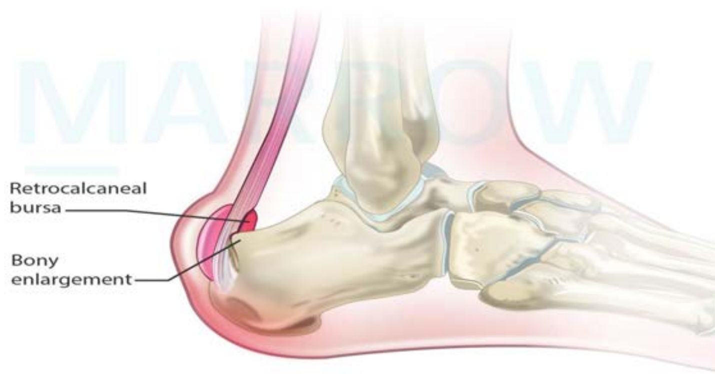
Haglund deformity, pump bump, or Mulholland deformity is seen around the ankle joint. It refers to retrocalcaneal exostosis formed at the posterosuperior aspect of the calcaneum.

retrocalcaneal bursa usually protects the Achilles tendon from the calcaneal tuberosity. But in Haglund deformity, there is chronic inflammation of this retrocalcaneal bursa due to increased or repetitive chafing between the tuberosity and Achilles tendon. Due to prolonged inflammation, degenerative changes occur and osteophytes form within the tendon leading to calcification at the site of the tendon insertion. This leads to deformity and pain symptoms.

Management is usually conservative, using a heel lift, silicone pads in closed shoes, ice application, anti-inflammatory drugs, and stretching exercises of the calf.

The image below shows the Haglund deformity.

Haglund deformity



©MARROW

Solution to Question 20:

The history of fall and hip injury in an elderly patient along with the radiological findings of a broken Shenton's line is suggestive of a neck of femur fracture. In this condition, the attitude of the leg will be shortened and externally rotated and movements of the hip joint will be painful.

A plain radiograph may show the following features:

• Fracture line in the neck of femur

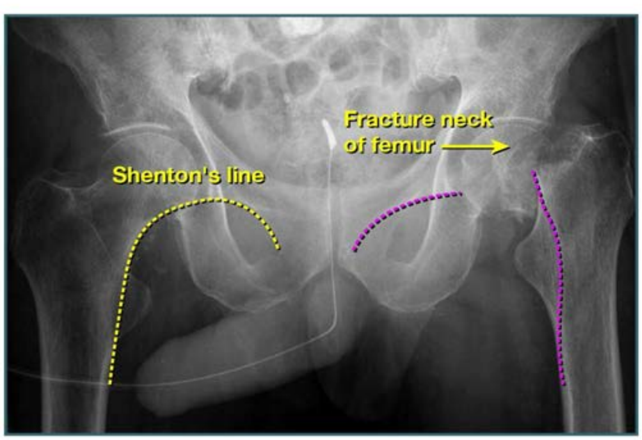
• External rotation of the femur with the lesser trochanter appearing more prominent

• Displacement of the greater trochanter

• Break in the trabecular stream

• Break in Shenton's line

Shenton's line in fracture of neck of femur



Orthopedics AIIMS 2018

Question 1:

The X-ray image of a patient that presented to the casualty is given below. How will you manage this case?

- **a)** Nailing
- **b)** Plating
- **c)** Tension band wiring
- **d)** Knee splint

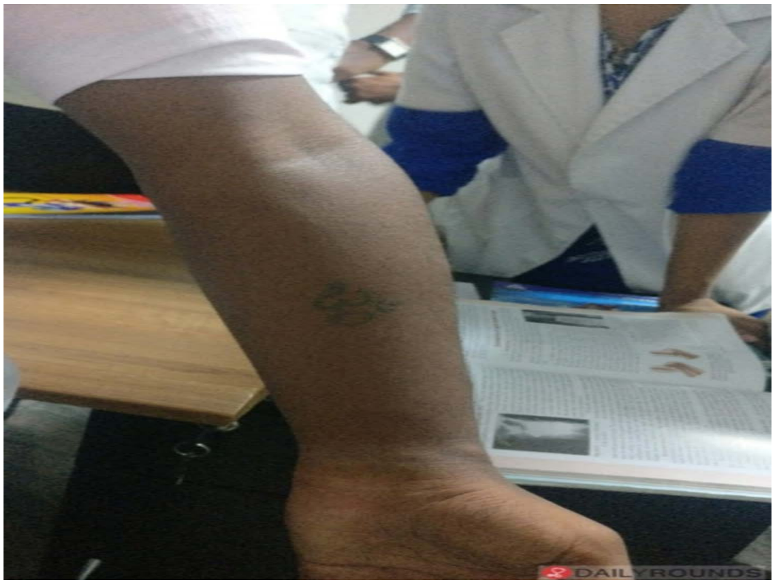
Question 2:

A 55-year-old female patient presented with a flexion contracture of the hip. Which of the following tests is useful in diagnosis?

- **a)** Allen's test
- **b)** Trendelenburg test
- **c)** Thomas test
- **d)** Ober test

Question 3:

A 10-year-old came to the OPD with the following deformity of his left upper limb. What is the most likely cause of the fracture?



a) Supracondylar Fracture

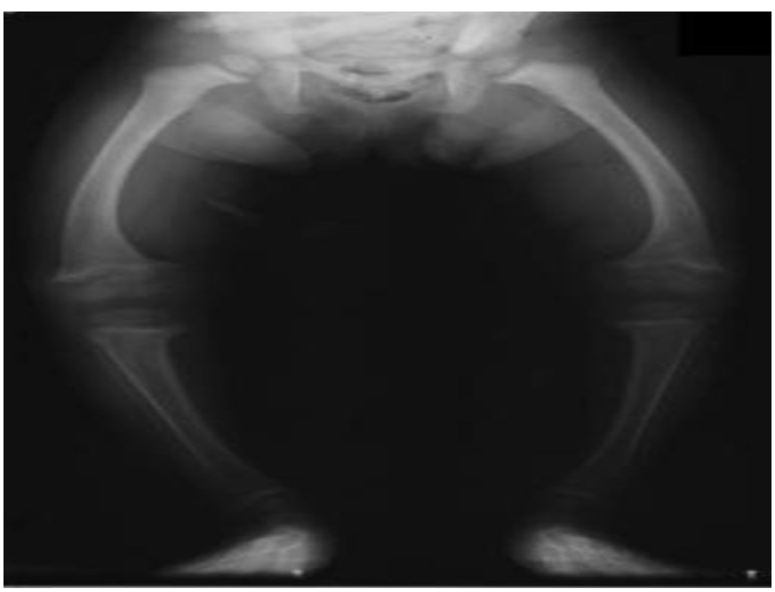
b) Olecranon Fracture

c) Radial Head fracture

d) Triad of elbow

Question 4:

A 4-year-old child was brought to the OPD with a painless deformity. A radiograph of the child's legs is shown below. The most common cause of this deformity is ______.



a) Skeletal dysplasia

b) Rickets

c) Scurvy

d) Blount disease

Question 5:

A 25 year old man involved in a fight, injured his finger and the deformity is shown in the following image. X-ray is normal. The most appropriate management at this stage is:

a) Splint the finger in flexion

b) Buddy strapping

c) Surgical repair of the flexor tendon

d) Splint the finger in hyperextension

Question 6:

What is your diagnosis from the below X-ray images?

a) Solitary bone cyst

b) Aneurysmal bone cyst

c) Giant cell tumor

d) Osteoblastoma

Question 7:

Which of the following is true regarding the scar tissue in patellar clunk syndrome?

a) Superior pole of patella, impinging during flexion

b) Superior pole of patella, impinging during extension

c) Inferior pole of patella, impinging during flexion

d) Inferior pole of patella, impinging during extension

Question 8:

The earliest symptom of acute compartment syndrome is _____________

a) Pain

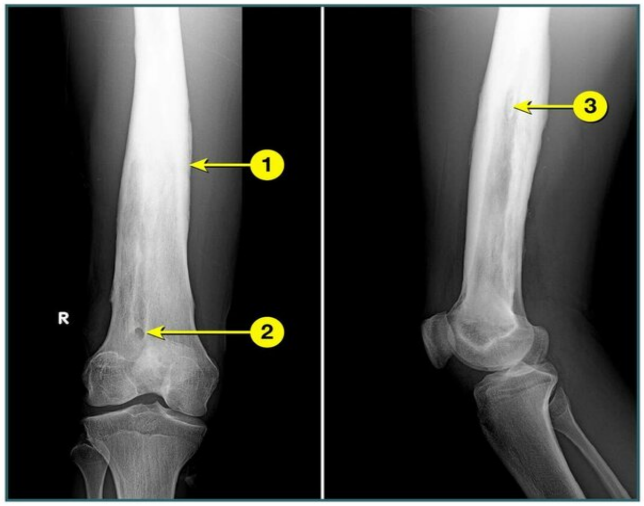
b) Pallor

c) Paresthesia

d) Paralysis

Question 9:

Identify the following marked structures.



a) 1 - Sequestrum, 2 - Involucrum, 3 - Cloacae

b) 1 - Sequestrum, 2 - Cloacae, 3 - Involucrum

c) 1 - Involucrum, 2 - Cloacae, 3 - Sequestrum

d) 1- Cloacae, 2 - Sequestrum, 3 - Involucrum

Question 10:

Which tumor does not cause osteomalacia and phosphaturia?

a) Osteosarcoma

b) Hemangiopericytoma

c) Fibrosarcoma

d) Breast carcinoma

Question 11:

A polytrauma patient with femur fracture and massive hemorrhage presents to the emergency department after a road traffic accident. Which of the following is not an immediate treatment consideration in this patient?

a) Administration of IV crystalloids

b) Administration of tranexamic acid

c) Check coagulation with thromboelastography

d) Internal fixation of femur fracture

Question 12:

What is the most common form of arthritis?

a) Rheumatoid arthritis

b) Psoriatic arthritis

c) Seronegative arthritis

d) Osteoarthritis

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	b
5	d
6	c
7	b
8	a
9	c
10	d
11	d
12	d

Detailed Explanations

Solution to Question 1:

The given radiograph shows a transverse fracture of the patella which should be managed by tension band wiring.

Patellar fractures can be undisplaced, transverse, comminuted, or longitudinal. The radiological examination includes AP and lateral views. A skyline view of the patella is required in some cases of undisplaced fractures.

Management of patellar fractures:

• Undisplaced fractures: Treated with a cylinder cast extending from the groin to just above the malleoli, with the knee in full extension, left for at least 3 weeks.

• Transverse fractures: Surgical reduction using tension-band wiring and K-wire fixation, followed by extensor retinaculum repair and physiotherapy.

• Comminuted fractures: Open reduction and fixation with patellectomy. Patella-saving operations are also being performed with improved fixation techniques.

• Longitudinal fractures: As the extensor apparatus is usually intact, they can be treated with a cylinder cast or extension brace initially, followed by range-of-motion exercises.

Solution to Question 2:

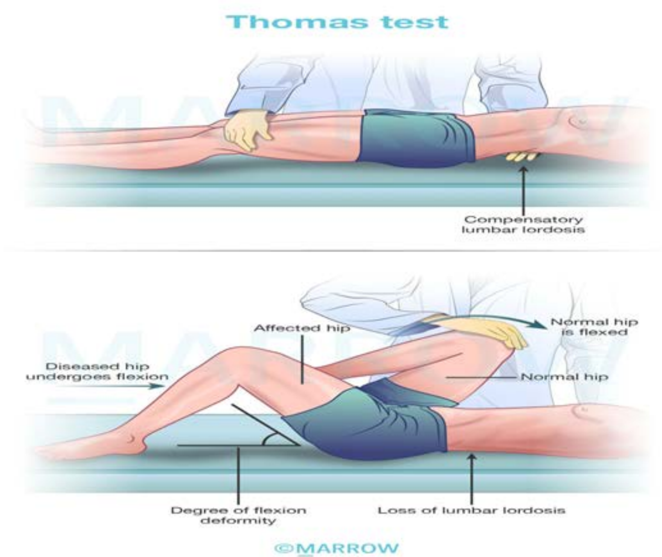
Thomas test is useful in diagnosing flexion contracture of the hip.

Flexion deformity is a common hip deformity, often caused by stronger flexor muscles overpowering the extensor muscles. When there is a spasm of the muscles, the stronger flexors pull the hip in flexion. Soft tissue contractures can convert this into a fixed flexion deformity, making locomotion impossible.

The Thomas test is used to evaluate the degree of flexion deformity. The aim is to eliminate compensatory lumbar lordosis and reveal the obvious flexion deformity for measurement.

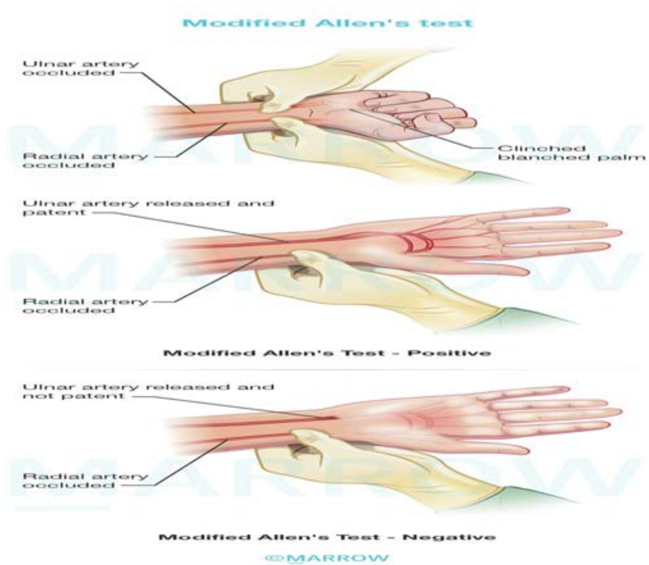
During the test, the patient lies supine on a hard surface with legs straight. The patient may compensate for the flexion deformity by arching the lower back, creating excessive lumbar lordosis. The easy passage of the examiner's hand between the bed and the back of the patient is considered a positive test as normally this is not possible.

By gradually flexing the unaffected hip, the pelvis tilts and eliminates lumbar lordosis, causing the affected hip to assume the flexed position. The angle between the affected thigh and the bed indicates the degree of flexion deformity.

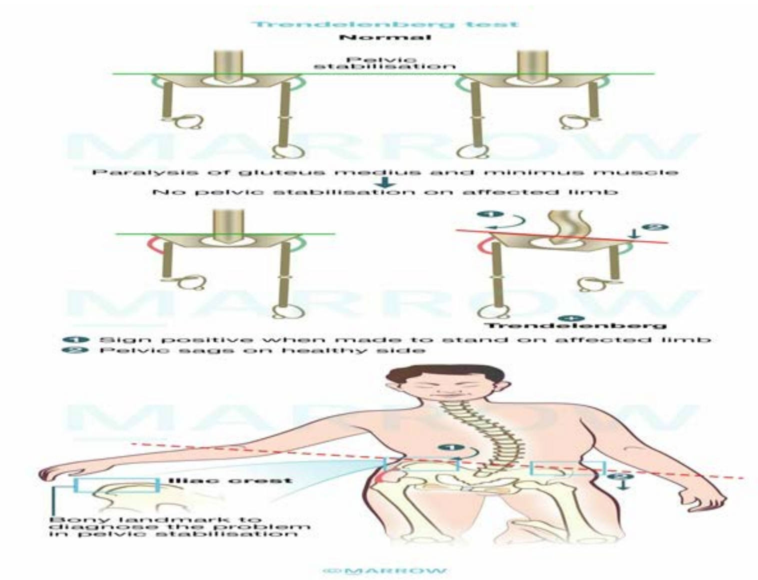


Other options:

Option A: Allen's Test is mainly used to test the adequacy of ulnar collateral blood flow to the hand, which is usually done prior to radial artery cannulation.



Option B: Trendelenberg test is used to assess the weakness or paralysis of the gluteus medius muscle.



Option D: Ober's test is used in physical examination to identify the tightness of the iliotibial band.

Solution to Question 3:

The given image shows cubitus varus deformity, where the forearm has deviated medially in relation to the upper arm, which is a common complication of supracondylar humeral fracture.

Malunion is the commonest complication of a supracondylar fracture and results in a cubitus varus deformity, also known as gunstock deformity. This occurs when the fracture unites with the distal fragment tilted medially and in internal rotation.

The image below shows gunstock deformity:

Cubitus varus is primarily a cosmetic concern with minimal functional impairment. Mild cases may not necessitate treatment, while severe deformities should be corrected through a lateral closed wedge osteotomy (French osteotomy).

Other options:

Option B: Olecranon fracture results from a direct injury as in a fall onto the point of the elbow. Complications include non-union, elbow stiffness, and osteoarthritis

Option C: Radial head fracture results from a fall on an outstretched hand and presents with pain over the lateral aspect of the forearm near the elbow along with restricted elbow movements, pronation, and supination. Complications include posterior interosseous nerve injury, myositis ossificans, and osteoarthritis.

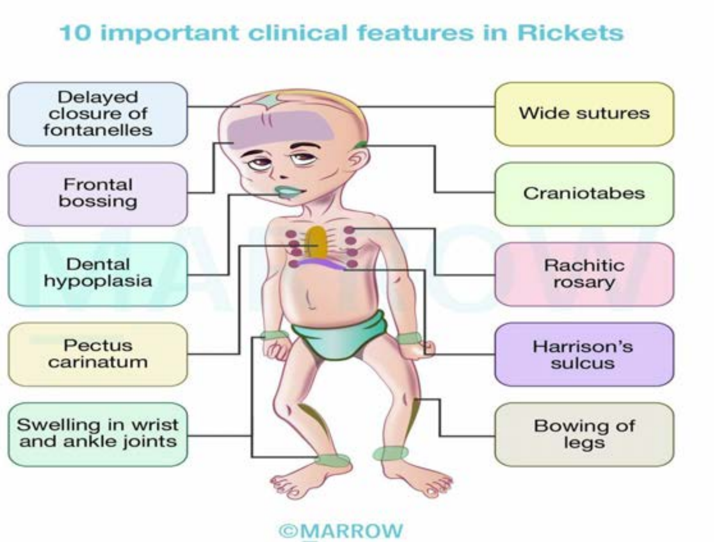
Option D: Terrible triad of elbow refers to elbow dislocation with fracture of the radial head, coronoid process, and medial collateral ligament rupture.

Solution to Question 4:

The given X-ray shows bow legs or genu varum which is most commonly seen in rickets.

Rickets is a bone mineralization disorder that primarily affects children with open growth plates, resulting from a deficiency of vitamin D, calcium, or phosphate.

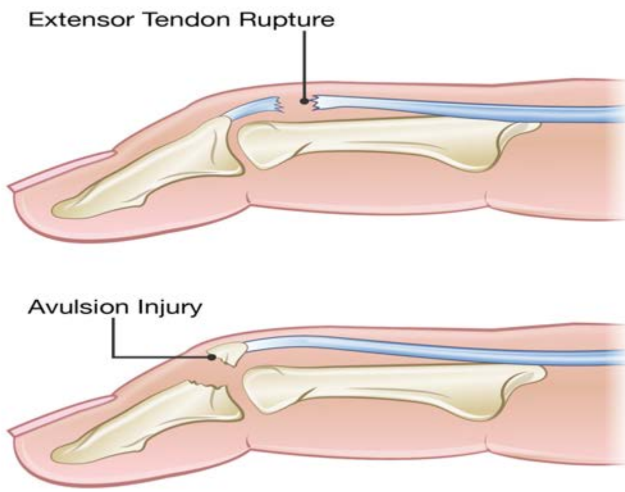
In toddlers, genu varum is one of the most common initial sign of rickets. In older children, genu valgum or coxa vara might be present.



Solution to Question 5:

Based on the mechanism of injury and the deformity characterized by flexion at the distal interphalangeal joint, along with a normal radiograph, the likely diagnosis is mallet finger, which is managed by splinting the finger in hyperextension.

Mallet finger, also known as baseball finger, occurs when there is a sudden passive flexion of the distal interphalangeal joint, causing avulsion of the extensor tendon from its insertion at the base of the distal phalanx.



Surgery is considered only if there is a large fragment (>50%) and subluxation of the DIP joint.

Solution to Question 6:

The given radiographs show the anteroposterior and lateral views of the wrist, demonstrating a solitary lytic lesion with cortical expansion and a soap-bubble appearance in the epiphysis of the distal radius, suggestive of a giant cell tumor (GCT).

GCT usually affects young adults aged 20 to 40 years, with slight female predominance. It is a benign but locally aggressive tumor. It is exclusively seen in adults and affects the ends of long bones, typically involving the epiphysis. The most common location of the tumor is the distal femur followed by the proximal tibia. Involvement of the distal end of the radius is also seen and the tumor is more aggressive in this location.

Important X-ray features of a GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis

• Eccentric location, most often sub-chondral

• Expansion of the overlying cortex

• Soap bubble appearance (the tumor is homogenously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance).

• No calcification within the tumor

• No or minimal reactive sclerosis around the tumor

• Thinned out cortex, may be perforated

The classical "soap bubble" appearance of a giant cell tumor (GCT) is shown below:

Other options:

Option A: A solitary bone cyst is most commonly occurs in the metaphysis of the bone. On an X-ray, a bone fracture fragment may be observed resting in a dependent portion of the cyst, referred to as the fallen fragment or fallen leaf sign. This sign is considered pathognomonic of a simple bone cyst.

The image below shows a bone cyst in the humerus along with a pathological fracture and fallen leaf sign:

Option B: On an X-ray, an aneurysmal bone cyst is seen as an eccentric, expansile, lytic lesion with well-defined margins in the metaphysis.

Option D: Osteoblastoma is a benign bone tumor usually occurring in the jaw and spine. On X-ray, a radiopaque central nidus with a surrounding radiolucent halo is seen. If occurring in the long bones, it involves the diaphysis or metaphysis, but never in the epiphysis.

The image below shows an osteoblastoma in C7 vertebrae:

Solution to Question 7:

In patellar clunk syndrome, the scar tissue which is formed at the superior pole of the patella becomes impinged during knee extension.

Patellar clunk syndrome is seen after posterior-stabilized knee arthroplasty. A fibrous nodule forms on the posterior surface of the quadriceps tendon just above the superior pole of the patella. This can become entrapped in the intercondylar notch of the femoral prosthesis and cause the knee to "clunk" at 30- 45 degrees to the femoral axis while the patient actively extends the knee.

The recommended treatment is arthroscopic debridement of the nodule.

Solution to Question 8:

The earliest symptom of acute compartment syndrome is pain.

Acute compartment syndrome is a surgical emergency caused by increased pressure within a fascial compartment, leading to tissue ischemia. It is characterized by rapid progression of pain and swelling in an extremity, commonly precipitated by traumatic injury.

The classic features of ischemia in acute compartment syndrome include the 5 P's:

• Pain - It is the first symptom of acute compartment syndrome. Ischemic muscle is highly sensitive to touch. On passive stretching of the muscles, severe pain can be elicited.

• Paresthesia - It is the first sign of nerve ischemia.

• Pallor - It may not be seen always. The skin outside the compartment may be normally perfused, as only the intra-compartmental perfusion may be affected.

• Paralysis - Paralysis of muscle groups affected in that compartment is a late sign.

• Pulselessness - Usually, the peripheral pulses and capillary return are intact in the early stages, unless there is a major arterial injury. If pulses are absent, arteriography is indicated.

The diagnosis can be confirmed by measuring the intra-compartmental pressure using a split catheter introduced into the compartment. The difference between diastolic pressure and compartment pressure is known as differential pressure (ΔP). A differential pressure (ΔP) of less than 30 mm Hg (4.00 kilopascals) is an indication for immediate compartment decompression by performing a fasciotomy.

Solution to Question 9:

In the given radiographs showing features of chronic osteomyelitis, the marked structures are as follows:

• Involucrum is the thick sheath of periosteal new bone that surrounds the sequestrum.

• Cloacae are openings in the involucrum that allow drainage of purulent and necrotic material out of the dead bone.

• Sequestrum is a piece of devitalized bone that has been separated from its surrounding bone during the process of necrosis.

Chronic Osteomyelitis

Solution to Question 10:

Breast carcinoma does not cause osteomalacia and phosphaturia.

Tumor-induced osteomalacia (TIO) is a rare paraneoplastic syndrome which presents with hypophosphatemia, osteomalacia and renal phosphate wasting. It is most commonly seen with mesenchymal tumors.

Tumors associated with osteomalacia are:

• Hemangiopericytoma (option B)

• Non-ossifying fibroma

• Osteosarcoma (option A)

• Osteoblastoma

• Fibrosarcoma (option C)

Solution to Question 11:

The internal fixation of a femur fracture is not an immediate concern in a patient with polytrauma and hemodynamic instability secondary to severe blood loss. Instead, splinting is preferred until the patient has been stabilized.

Damage Control Orthopaedics (DCO) in unstable/borderline polytrauma patients involves 3 components:

• Stage 1: Resuscitative procedures for rapid hemorrhage control

• Stage 2: Restoration of normal physiologic parameters

• Stage 3: Definitive surgical management

The first stage is early temporary stabilization of unstable fractures and the control of the bleeding. In femur fracture, the most suited approach is temporary stabilization using an external fixator.

The second stage involves resuscitation and stabilization of the patients in the ICU.

Finally, the third stage involves delayed definitive fracture management after the patient's condition becomes suitable. In femur fracture, the definitive procedure is intramedullary nailing which is performed after the patient becomes stable.

Solution to Question 12:

Osteoarthritis is the most common type of arthritis.

Osteoarthritis is strongly associated with age and is highly prevalent among older individuals. Studies suggest that over 80% of people aged 55 and above have osteoarthritis in at least one joint.

Orthopedics AIIMS 2019 (May & Nov)

Question 1:

Following a fall in the bathroom, an elderly lady is diagnosed with a fractured neck of the femur. Which of the following coexisting conditions is most likely to have predisposed to this injury?

a) Osteoporosis

b) Osteopetrosis

c) Osteoarthritis

d) Osteomalacia

Question 2:

At what age in women is a DEXA scan done routinely?

a) At 50 years

b) At 55 years

c) At 60 years

d) At 65 years

Question 3:

Most specific feature for ankylosing spondylitis:

a) HLA-B27 positivity

b) Bilateral sacroiliitis

c) Elevated ESR

d) Lumbar laxity

Question 4:

The treatment for the condition given below is:

a) Extended curettage with allograft

b) Bone biopsy

c) Curettage

d) Extended curettage with autograft

Question 5:

Identify the instrument given below.

a) Bone nibbler

b) Bone curette

c) Plate holding forceps

d) Bone holding forceps

Question 6:

A 30-year-old man presents with numbness of both hands. He was asked to perform the test demonstrated below. What is this test?

a) Adson's test

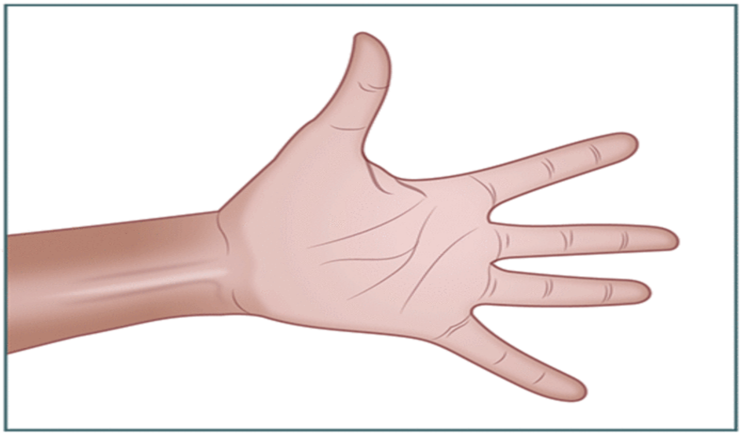
b) Pemberton test

c) Roos test

d) Wright test

Question 7:

The test performed in the following video is



- **a)** Allen's test
- **b)** Roos test
- **c)** Adson's test
- **d)** Wright test

Question 8:

A child presents with difficulty in walking. On biopsy of the hip, some cells with grooved nuclei and eosinophilic cytoplasm are present. Which investigation would you order next?

- **a)** MRI
- **b)** Serum calcium
- **c)** Serum PTH
- **d)** CD1a marker

Question 9:

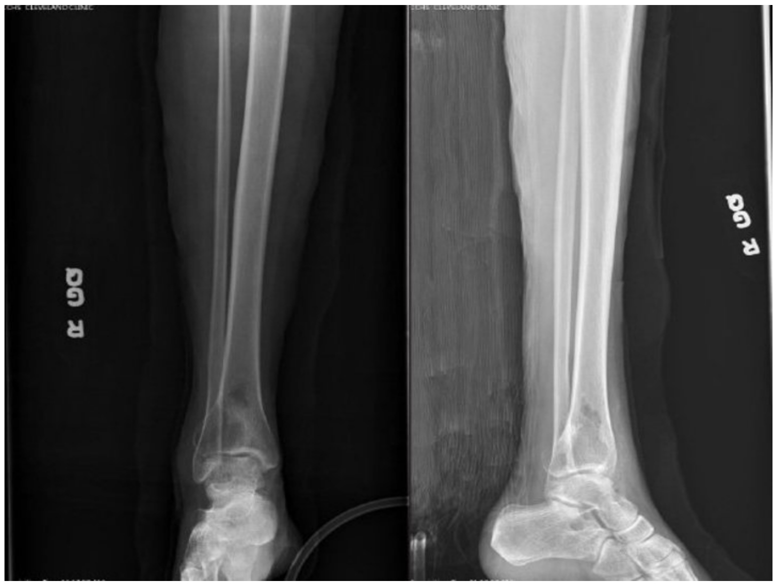
A 28-yr old man has recently finished a half- marathon. He complains of pain along the anterior and medial aspect of the tibia which increases on jogging & walking for a long time. Plain X-Ray radiograph of the leg was found to be normal. What is the probable diagnosis?

- **a)** Jones fracture
- **b)** Shin splint
- **c)** Lisfranc fracture

d) Nutcracker fracture

Question 10:

The following X-ray shows a lytic lesion of the lower limb. The biopsy shows fibroblastic proliferation, osteoclast and inflammatory cells. From the given options, the most likely diagnosis that can be inferred from the above clinical picture is______.



a) Aneurysmal bone cyst

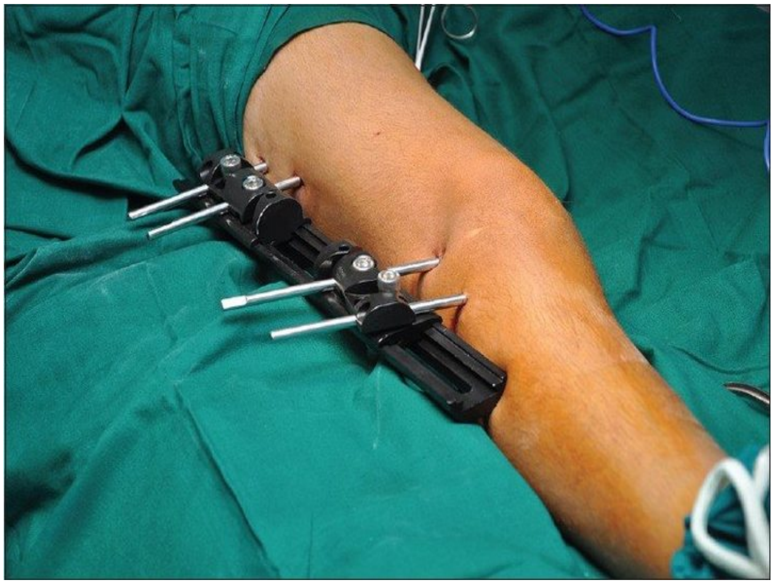
b) Giant cell tumor

c) Chondroblastoma

d) Langerhan cell histiocytosis

Question 11:

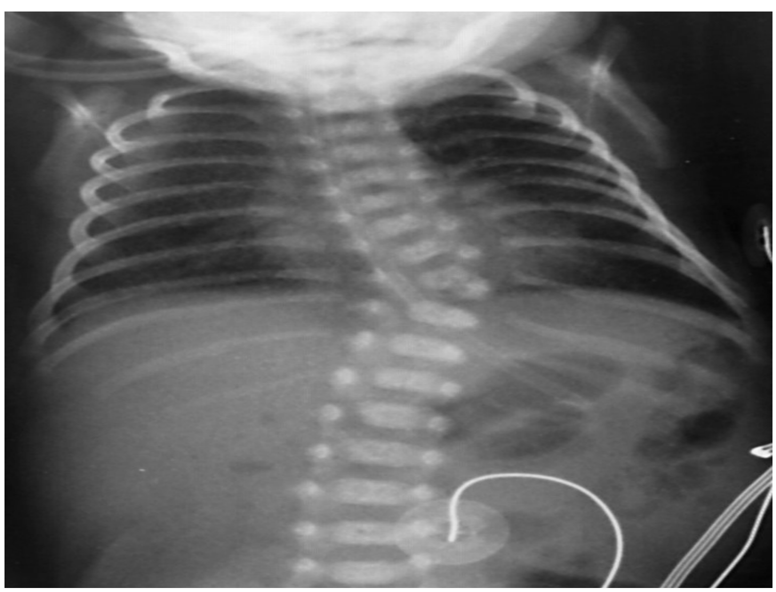
Which of the following is used to stabilise a fracture of the distal femur as shown in the image below?



- **a)** Rail fixator
- **b)** Ilizarov fixator
- **c)** K wire binding
- **d)** Interlocking nail

Question 12:

What is the type of scoliosis shown in the X-ray given below?



- **a)** Congenital scoliosis
- **b)** Postural scoliosis

c) Infantile scoliosis

d) Degenerative scoliosis

Question 13:

The following instrument is used for:

a) Nibbling the ends of the bones

b) Holding the bone and plate

c) Controlled traction of fracture segments

d) Cutting the K- wire

Question 14:

A patient presents to the emergency department after he fell on his outstretched hand. An X-ray was performed and is given below. What is the diagnosis?

a) Colle's fracture

b) Rolando's fracture

c) Galeazzi fracture

d) Montegia fracture

Question 15:

Identify the following device.

a) Cock-up splint

b) Colles' cast

c) Knuckle bender splint

d) Dynamic finger splint

Answer Key

Question No.	Correct Option
1	a
2	d
3	b
4	d
5	a
6	c
7	a
8	d
9	b
10	b
11	a
12	a
13	b
14	c
15	a

Detailed Explanations

Solution to Question 1:

Given the clinical history of an elderly woman experiencing a fracture following a trivial fall, which indicates a pathological fracture, the most probable predisposing condition would be osteoporosis.

Osteoporosis is the most common metabolic bone disorder characterized by abnormally low bone mass, which makes the bone unusually fragile and increases the risk of fractures.

A DEXA scan is performed in the following patients to assess bone mineral density and screen for osteoporosis, and the results are reported in terms of T-score or Z-score:

• Females aged 65 years or more

• Males aged 70 years or more

• Post-menopausal females under the age of 65 years with risk factors

• Fracture after the age of 50 years

• Males aged 50-69 with risk factors

Solution to Question 2:

Dual-energy x-ray absorptiometry (DEXA scan) is routinely done in women at 65 years of age to screen for osteoporosis.

Osteoporosis is the most common metabolic bone disorder characterized by abnormally low bone mass, which makes the bone unusually fragile and increases the risk of fractures.

The criteria to screen for osteoporosis according to the national osteoporosis foundation is as follows:

• Females aged 65 years or more

• Males aged 70 years or more

• Post-menopausal females under the age of 65 years with risk factors

• Fracture after the age of 50 years

• Males aged 50-69 with risk factors

A DEXA scan is performed in the above patients to assess bone mineral density, and the results are reported in terms of T-score or Z-score.

Solution to Question 3:

Bilateral sacroiliitis is the most specific sign of ankylosing spondylitis (AS).

While HLA-B27 is positive in up to 90% of patients with AS, it is not specific to AS (option A). Erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) are often, but not always, elevated and are not specific to AS (option C). Lumbar laxity is not a feature of AS (option D).

The Assessment of Spondyloarthritis International Society classification criteria for axial spondyloarthritis includes lower back pain for more than 3 months in patients less than 45 years of age and one of the following:

• Sacroiliitis confirmed on x-ray or MRI and at least one typical clinical or laboratory finding

• A positive HLA-B27 test and at least two typical clinical or laboratory findings

Clinical and laboratory findings used in the diagnostic criteria for AS include:

• Articular manifestations: inflammatory back pain, dactylitis, arthritis

• Extraarticular manifestations: enthesitis, uveitis, inflammatory bowel disease, psoriasis

• Patient history: positive family history, good response to NSAIDs

• Laboratory findings: positive for HLA-B27; elevated CRP and/or ESR

HLA-B27 is strongly associated with the following conditions:

• Ankylosing spondylitis

• Psoriatic arthritis

• Acute anterior uveitis

• IBD-associated ankylosing spondylitis

• Reactive arthritis

Solution to Question 4:

The provided radiograph of the knee joint shows an expansile, eccentric, subarticular, lytic lesion in the epiphysis of the distal femur with a soap bubble appearance, which suggests a giant cell tumor (GCT). The recommended approach for management is extended curettage and autograft.

GCT or osteoclastoma usually affects young adults in the age group of 20-40 years. The most common location of the tumor is the distal femur followed by the proximal tibia.

Important X-ray features of a GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis.

• Eccentric location, most often subchondral.

• Expansion of the overlying cortex.

• Soap bubble appearance - The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called soap bubble appearance.

• No calcification within the tumor.

• No or minimal reactive sclerosis around the tumor.

The image below shows an X-ray of a giant cell tumor showing the typical soap bubble appearance:

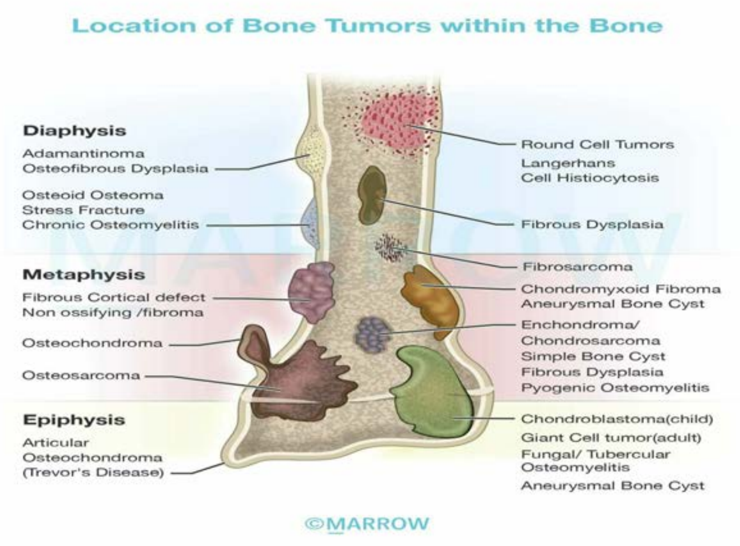
The treatment of choice for GCT is extended curettage and bone grafting.

• GCT in the relatively insignificant bone like the distal ulna, clavicle, and proximal fibula is treated by resection without reconstruction.

• Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease.

• Radiation/embolization can be used for inoperable spinal or sacral tumors.

• Denosumab is approved for use in unresectable tumors in skeletally mature patients.



Solution to Question 5:

The instrument given is a bone nibbler, which is designed to remove or shape small sections of bone by gently nibbling or biting away at the bone tissue.

It is available in various sizes and with different angles of the nose:

• Straight nibbler – for general use.

• Curved nibbler – for spinal surgery.

• Double-action nibbler – straight or curved. The double-action reduces the force required for cutting the bone and is mechanically superior.

Solution to Question 6:

The test performed in the image is Roos test. It is also referred to as the elevated arm stress test (EAST). It is a provocative test for the diagnosis of thoracic outlet syndrome.

For this test, the patient abducts the shoulders to 90 degrees and flexes the elbow to 90 degrees. The patient is made to open and close the hand repeatedly for three minutes. The test is positive if the patient is unable to complete the test or experiences cramping, pain, numbness, on the affected side.

Solution to Question 7:

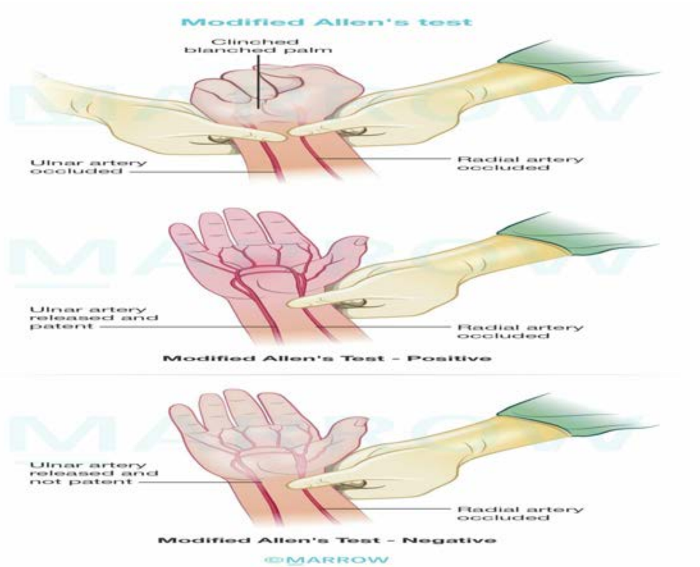
The test performed in the given video is Allen's test.

Allen's test is mainly used to test the adequacy of ulnar collateral blood flow to the hand, which is usually done prior to radial artery cannulation.

Procedure:

• The radial and ulnar arteries are occluded as the patient makes a tight fist so as to exsanguinate the palm.

• Then the patient is asked to open the hand and the occlusion over the ulnar artery is released.



Interpretation:

• As occlusion of the ulnar artery is released, the color of the open palm is observed. Normally, the color will return to the palm within 5 – 6 seconds.

• If the palm remains pale for more than 6-10 seconds, it indicates reduced ulnar collateral flow.

• The modified Allen’s test considers a threshold of 5 to 15 seconds.

Clinical significance:

• In recent years, studies have shown that, although the Allen test is often used to identify patients at increased risk for ischemic complications from radial artery cannulation, the predictive value of this test is poor.

• It is also used for identifying patients acceptable for radial artery harvest or use for coronary angiography.

Solution to Question 8:

Histopathologically, cells with grooved nuclei and eosinophilic cytoplasm can be observed in both Langerhans cell histiocytosis (LCH) and chondroblastoma. Therefore, the next step to differentiate between them would be to obtain a CD1a marker, which is typically positive in LCH.

The specific markers for LCH are CD1a and langerin (CD207)

It is worth noting that while an MRI can provide valuable information for evaluating the pathology, it is ideally performed before the biopsy.

Solution to Question 9:

The given clinical scenario, with a history of lower limb overuse (marathon running) and pain along the medial tibial border which increases with activity (jogging and long walks), along with a normal radiograph, suggests medial tibial stress syndrome (MTSS), also known as shin splints.

MTSS is a stress injury of the tibia which occurs due to the frequent overuse of lower limbs. It is commonly seen in athletes and military personnel.

Exercise-induced pain along the distal two-thirds of the medial tibial border is typical. Since it is a stress injury, the radiograph of the leg appears normal.

Management is mainly conservative with rest and activity modification with less repetitive and load-bearing exercises.

Solution to Question 10:

The provided X-ray reveals a lytic lesion in the epiphysis of the distal tibia, and the biopsy, which shows osteoclast cells, suggests a diagnosis of a giant cell tumor (GCT).

Important X-ray features of a GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis.

• Eccentric location, most often subchondral.

• Expansion of the overlying cortex.

• Soap bubble appearance - The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called soap bubble appearance.

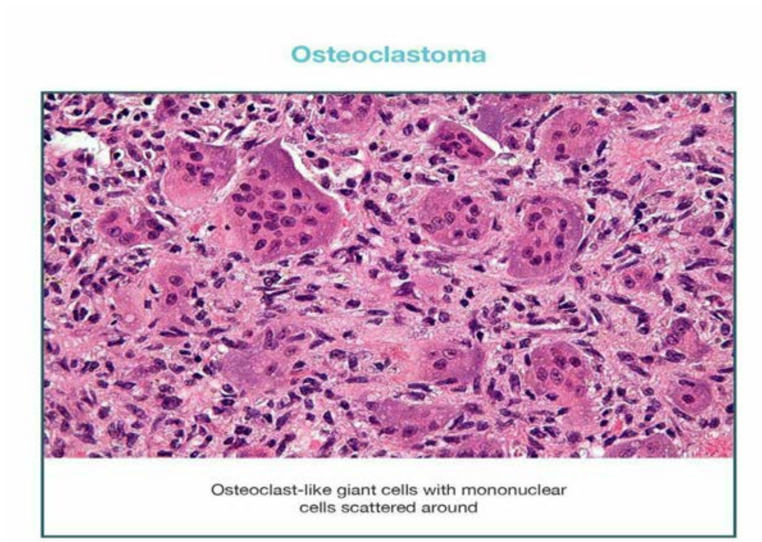
• No calcification within the tumor.

• No or minimal reactive sclerosis around the tumor.

The image below shows an X-ray of a giant cell tumor showing the typical soap bubble appearance:

Histologically, GCT is characterized by the presence of a large number of osteoclast-like giant cells among scattered round or spindle-shaped mononuclear cells. The tumor stroma is well-vascularized with bands of cellular or collagenous fibrous tissue.

The image below shows the histopathology of GCT:



Other options:

Option A: Aneurysmal bone cyst usually occurs in the metaphysis of long bones and the posterior elements of vertebral bodies. It is characterized by multiloculated blood-filled cystic spaces separated by septae, composed of multinucleated osteoclast-like giant cells and reactive woven bone. Radiographically, an aneurysmal bone cyst is seen as an eccentric, expansile, lesion with well-defined margins. Most lesions are completely lytic and often contain a thin shell of reactive bone at the periphery.

Option C: Chondroblastoma is a benign cartilaginous tumor that occurs mainly in bones around the knee joint. On x-ray, the tumor appears as a well-circumscribed, round, or oval lytic lesion in the center of the epiphysis of long bones, with areas of calcification within the tumor, giving the characteristic mottled appearance. On histology presence of "chicken wire calcification" is pathognomic.

Option D: Langerhans cell histiocytosis commonly involves the skull, pelvis, femur, and mandible. Radiographs show a lytic lesion with periosteal new bone formation. In the skull, lesions appear as a hole within a hole, and in the vertebra, vertebra plana (flattening of the vertebral body) can be seen. Histopathologically, Langerhans cells have a single large nucleus with a characteristic folded or grooved nucleus (“coffee-bean” appearance) with small inconspicuous nucleoli. On electron microscopy, the presence of Birbeck granules in the cytoplasm is characteristic.

Solution to Question 11:

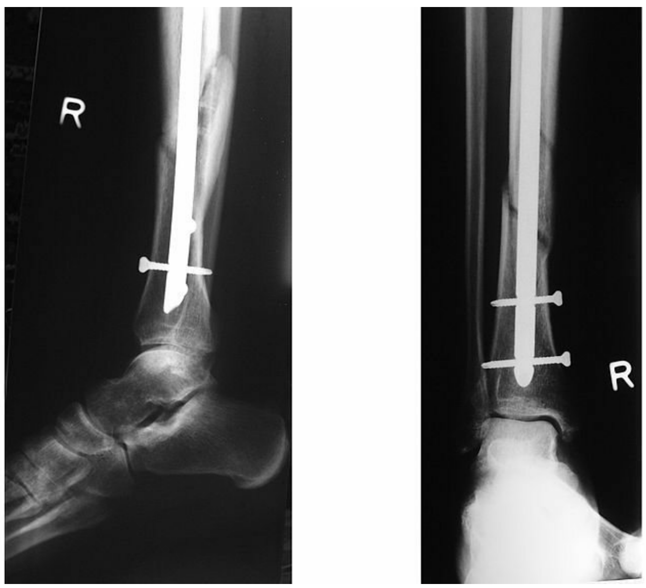
The above image shows a rail fixator.

In this type of external fixator, multiple pins are inserted into the bone. These pins are then attached to different blocks which can move over a track or rail.

Given below is an Ilizarov ring fixator. Here, pins are attached to an adjustable ring system.

Tension band wiring is used in the fixation of transverse patellar fractures. In this method, two K-wires are used to transfix the reduced patellar fragments and a flexible wire is looped around them tightly.

The following X-ray image shows tibial intramedullary interlocking nailing.



Solution to Question 12:

In the given radiograph, the presence of a single hemivertebra at T9 and concavity to the right in the lower thoracic region is suggestive of congenital scoliosis.

Congenital scoliosis is a lateral curvature of the spine which is always associated with a radiologically demonstrable anomaly of the vertebral bodies, which includes:

• Hemivertebrae (where only one-half of the vertebra grows)

• Block vertebrae (where two vertebral bodies fuse)

• An unsegmented bar (a bone bar joining two adjacent vertebrae on one side, which hinders growth on that side)

The image below shows a unilateral and unsegmented bar with contralateral hemivertebra:

In infantile scoliosis (option C), most curves are thoracic with convexity to the left and the anomalies described above are absent.

Note: The most progressive type of congenital scoliosis is a concave, unilateral unsegmented bar with a convex hemivertebra.

Solution to Question 13:

The image shown is of bone holding forceps (Verbrugge forceps). It is used to hold the bone and plate.

It consists of a ratcheted clamp with a curved distal end.

Solution to Question 14:

The provided history of falling on an outstretched arm and the X-ray revealing an oblique fracture in the lower third of the radius with dislocation of the distal radio-ulnar joint point to the diagnosis of Galeazzi's fracture.

Galeazzi's fracture is the fracture of the lower third of the radius and the subluxation or dislocation of the distal radioulnar joint. This injury usually happens when one falls on an outstretched hand.

Patients present with pain, swelling, and deformity at the lower end of the radius. On examination, the disruption of the radio-ulnar joint can be elicited by pressing the lower end of the ulna. The ulna moves down and comes back like the key of a piano (piano-key sign).

The aim of treatment for this fracture is to preserve supination and pronation of the forearm. In children, closed reduction is usually enough to achieve perfect reduction. However, in adults, open reduction using a k-wire fixation or compression plating is usually required. It is also important to look for ulnar nerve damage.

Galeazzi's fracture can cause complications such as malunion due to displacement of the fragment. This often results in deformity and limitation in supination and pronation.

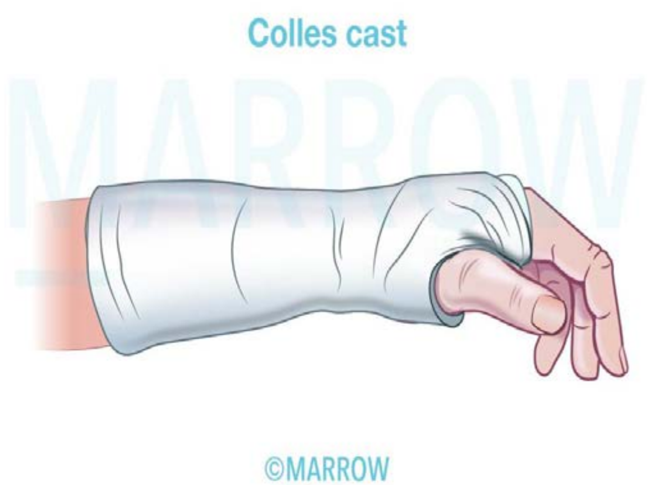
Solution to Question 15:

The device shown in the image is a cock-up splint. It is used to manage radial nerve palsy.

Radial nerve palsy manifests as an inability to dorsiflex the wrist and digits (wrist drop and finger drop). Numbness occurs on the dorsoradial aspect of the hand and the dorsal aspect of the radial 3 1/2 digits. The cock-up splint is designed to counteract the wrist drop position by holding the wrist in a slightly extended or neutral position, which helps prevent further contracture and allows for functional use of the hand.

Other options:

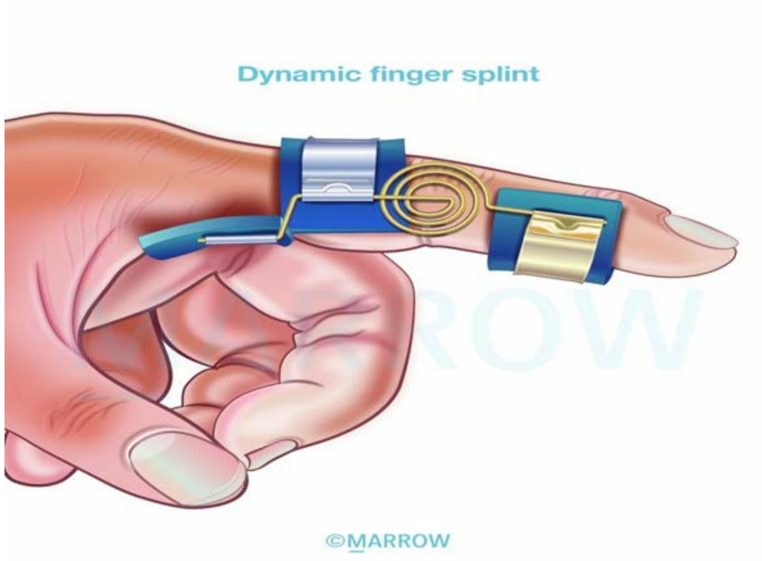
Option B: Undisplaced Colles' fracture is managed by immobilization in a below-elbow plaster cast for six weeks. Displaced fractures are managed by manipulative reduction followed by immobilization in Colles' cast in palmar flexion, pronation, and ulnar deviation.



Option C: Knuckle bender splint is used in cases of ulnar claw hand where ulnar nerve recovery is anticipated. The splint can be used to correct and prevent metacarpophalangeal hyperextension by maintaining flexion as shown below.

Option D: Dynamic finger splint can be used for:

• Immobilization in case of fracture of the proximal phalanx
• Extensor tendon injuries of the finger
• Prevention of deformities in rheumatoid arthritis.
• Post-operative immobilization.



Orthopedics AIIMS 2020 (May and Nov INI-CET)

Question 1:

All of the following are indications of surgery in a patient of Potts' spine except?

a) Cold abscess without neurological complication

b) Doubtful diagnosis

c) Drug resistant TB

d) Progressive cauda equina syndrome/conus medullaris syndrome

Question 2:

In case of a meniscal tear repair, which of the following zones heals best?

a) Red zone

b) White zone

c) Gray zone

d) Red+White zone

Question 3:

A 22-year-old male patient presents with a history of RTA. His leg is in abduction and external rotation. What could be the cause?

a) Anterior dislocation of the hip

b) Posterior dislocation of the hip

c) Intertrochanteric fracture

d) Fracture neck of femur

Question 4:

The most specific test for carpal tunnel syndrome is?

a) Durkan's test

b) Phalen test

c) Tinel test

d) Two point discrimination

Question 5:

A female patient with a history of fall few months back presents with the deformity shown in the image. Probing into history reveals it was managed with traditional treatment. This deformity is likely to be due to ___________

a) Intra-articular fracture of wrist

b) Extra-articular fracture of wrist

c) Boxer's fracture

d) Barton's fracture

Question 6:

Scoliosis screening in school children is done by?

a) Straight leg raising test

b) Forward bending test

c) Backward bending test

d) Single leg standing test

Question 7:

Fishtail deformity is seen in the fracture of ____________

a) Distal end of radius

b) Distal end of humerus

c) Distal end of tibia

d) Distal end of femur

Question 8:

Ring sequestrum is seen in?

a) Salmonella infection

b) TB osteomyelitis

c) Pin tract infection

d) Chronic osteomyelitis

Question 9:

A counterforce brace is used in?

a) Tennis elbow

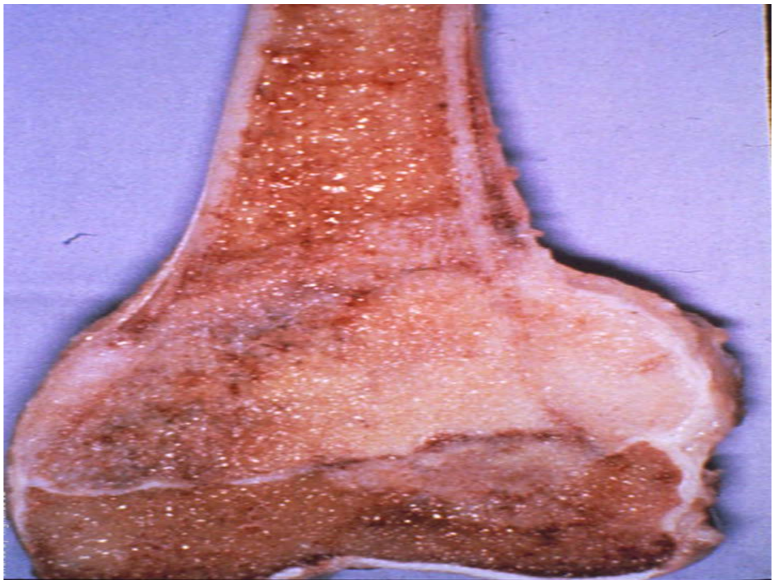
b) Ankle sprain

c) Ulnar claw hand

d) Foot drop

Question 10:

A 10-year-old boy presents with pain around the left knee joint. On X-ray, there is a lesion at the distal femur extending up to the terminal end. The gross image of the condition is given below. What can be the most probable diagnosis?



a) Giant cell tumour

b) Chondroblastoma

c) Osteoid osteoma

d) Osteosarcoma

Question 11:

Which of the following substances is used for bone formation in a patient with non-union?

a) PMMA

b) BMP

c) Ca PO4

d) CaSO4

Question 12:

The brace shown in the given image is _____.

a) Boston brace

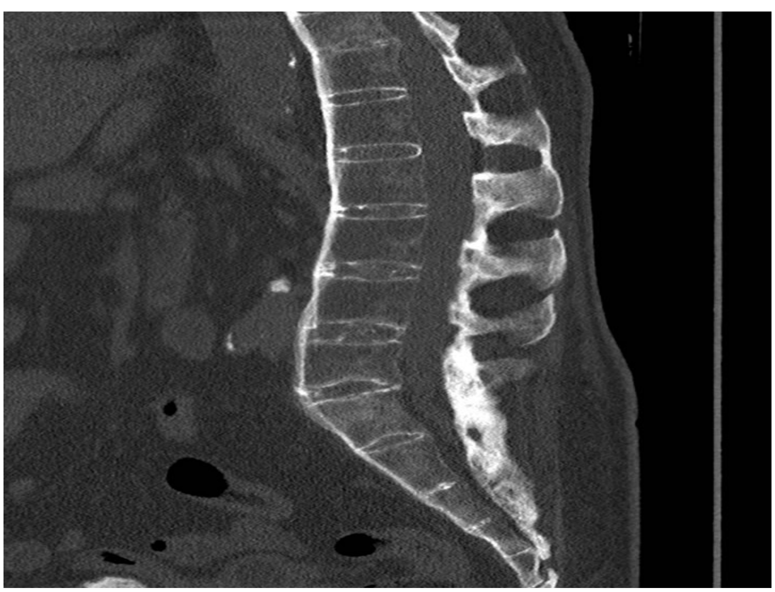
b) Milwaukee brace

c) Taylor brace

d) ASHE brace

Question 13:

A 40-year-old man is suffering from increasing difficulty in bending forward with increasing stiffness of the spine and reduced chest movements. A CT was performed and shown below, the likely diagnosis is?



a) Ankylosing spondylitis

b) Rheumatoid arthritis

c) Reiter's syndrome

d) Fluorosis

Question 14:

In spine fusion surgery, the screws are fixed to which part of the vertebra?

a) Spinous process

b) Lamina

c) Pedicle

d) Facet joint

Question 15:

Which of the following tests are not commonly used in osteoporosis:

a) 1, 2 & 3

b) 2 & 4

c) 2 & 3

d) 1, 3, 4 & 5

Answer Key

Question No.	Correct Option
1	a
2	a
3	a
4	a
5	b
6	b
7	b
8	c
9	a
10	d

11	b
12	b
13	a
14	c
15	c

Detailed Explanations

Solution to Question 1:

A cold abscess without neurological complications is not an indication for surgery in Pott's spine/TB spine.

Indications of surgery in TB spine are:

• Progressive neurologic deficit such as progressive cauda equina syndrome/conus medullaris syndrome (Option D)

• Progressive increase in spinal deformity

• Resistance to anti-tubercular therapy (Option C)

• Recurrence of disease

• Cord compression

• Severe pain due to abscess or spinal instability

• Progressive impairment of pulmonary function

• Uncertain diagnosis (Option B): Inability to obtain a microbiological diagnosis through microscopy, culture, or PCR techniques may occur. In cases where there is no accessible source for sampling (such as discharging sinus or superficial abscess) or insufficient specimen obtained from image-guided percutaneous biopsy, an open surgical biopsy is the only option to obtain a specific culture and sensitivity report.

Antero-lateral decompression is the surgery of choice for Pott's paraplegia as it helps in the removal of both solid and liquid debris.

Solution to Question 2:

The red zone heals best in case of a meniscal tear due to its rich blood supply.

The menisci are divided into three zones depending on the perfusion of blood. The menisci are supplied by perimeniscal blood vessels from superior and inferior branches of medial and lateral genicular vessels.

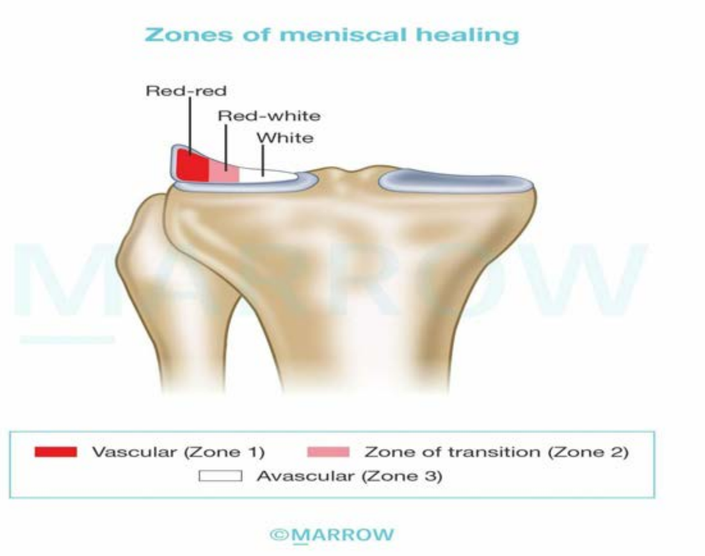
Following birth, the menisci become increasingly avascular. By adulthood, only the peripheral 10% to 25% of the tissue is perfused. This gives rise to 3 distinct regions of the menisci:

• The peripheral relatively vascularised region called the red-red zone (Zone 1)

• A zone of transition between the 2 zones, which is called the red-white zone (Zone 2)

• The completely avascular inner zone known as the white-white zone (Zone 3)

There is a direct relationship between the vascularization and capacity of the tissue to heal, predisposing the white-white zone to permanent post-traumatic and degenerative lesions.



Solution to Question 3:

The patient presents with abduction and external rotation of the lower limb following a road traffic accident, which is suggestive of anterior dislocation of the hip.

Anterior dislocation of the hip is usually sustained following a high-energy accident. On physical examination, the affected limb will show abduction, external rotation, and slight flexion. Lengthening of the limb is observed, along with a prominent anterior bulge in either the pubic or groin region.

In posterior dislocation of the hip, the lower limb is flexed, adducted, and internally rotated, and this condition is associated with limb shortening. The neck of femur fracture and intertrochanteric fracture present with external rotation and shortening of the lower limb.

The given image shows a patient with an externally rotated, abducted, and slightly flexed left leg due to anterior dislocation of the hip:

According to the position assumed by the femoral head, anterior dislocation of the hip can further be classified as:

• Type I - Superior or pubic

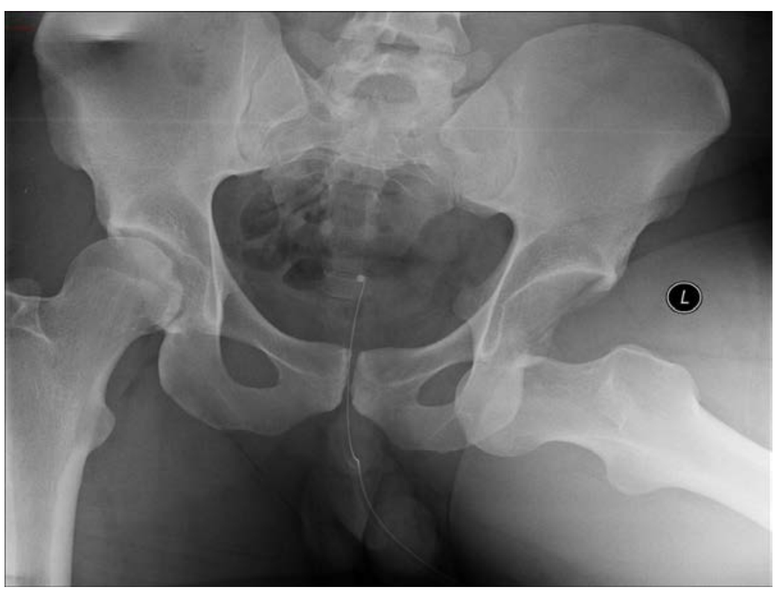
• Type II - Inferior or obturator

A lateral view X-ray will be confirmatory. Radiological findings include:

• The displaced femoral head will appear larger than the normal side

• As the thigh is externally rotated lesser trochanter is more prominent

X-RAY showing anterior dislocation of the hip:



Treatment of anterior dislocation includes a closed reduction under sedation or general anesthesia. If closed reduction fails, an open procedure may be planned.

Solution to Question 4:

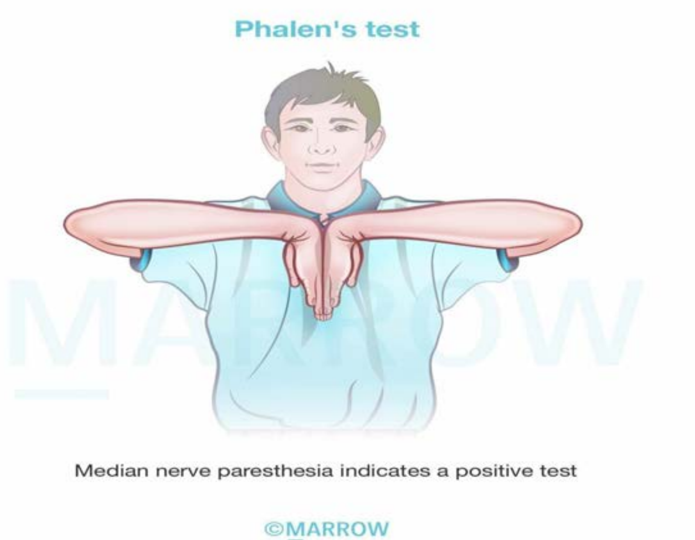
Among the given options, Durkan's test is the most specific test for the diagnosis of carpal tunnel syndrome (CTS).

CTS is a common entrapment mononeuropathy caused by median nerve compression as it passes through the fibro-osseous carpal tunnel beneath the flexor retinaculum. Patients present with pain and paresthesia over the radial 3 and a half digits. There may be a weakness of thumb opposition. In chronic cases, the thenar eminence is wasted. The diagnosis is usually made on clinical grounds but nerve conduction studies can confirm the diagnosis.

Durkan's test is the most specific test for the diagnosis of CTS. In Durkan's test, direct median nerve compression for about 30 seconds, elicits paresthesias and pain.

Other tests for CTS:

• Phalen's test - Elicited by holding the wrist flexed fully. The appearance of sensory symptoms within 60 seconds implies a positive test.



• Tinel's test - Sensory symptoms reproduced by percussing over the median nerve imply a positive test.

Two-point discrimination is done to assess sensory loss and is not specific to carpal tunnel syndrome.

Solution to Question 5:

The given image shows the dinner-fork deformity seen in Colle's fracture which is an extra-articular fracture of the wrist.

Colle's fracture is a fracture of corticocancellous junction of the distal end of the radius. The dinner fork deformity is the prominence on the back of the wrist and depression in the front. It is due to the dorsal angulation of the distal fracture fragment. It is treated by a Colles' cast, which immobilizes the wrist in palmar flexion, pronation, and ulnar deviation.

Displacements that may be seen in Colles' fracture:

• Dorsal tilt

• Dorsal shift

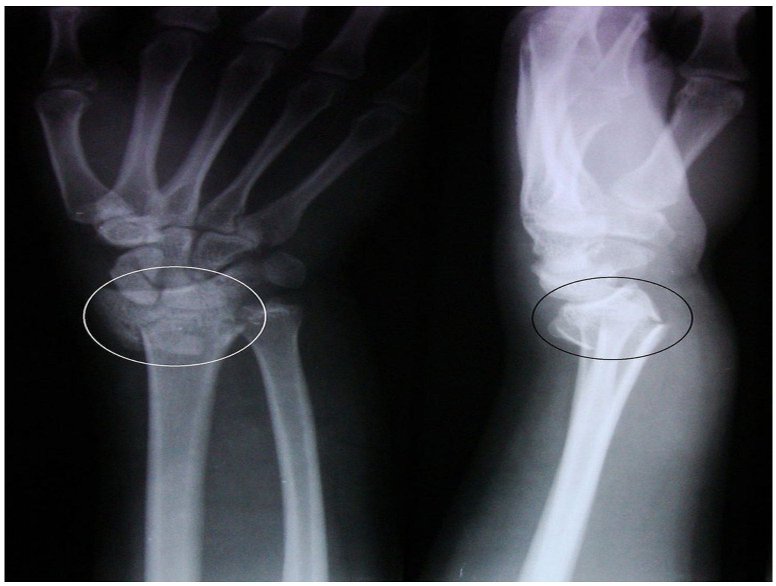
• Lateral tilt (Radial tilt)

• Lateral shift (Radial shift)

• Supination

• Impaction of fragments

The given radiograph shows the dorsal tilt of the distal fragment leading to dinner fork deformity.



Solution to Question 6:

The forward bending test is used as a screening tool in scoliosis.

Adam's forward bending test is the best clinical test for evaluating spinal curvature in scoliosis.

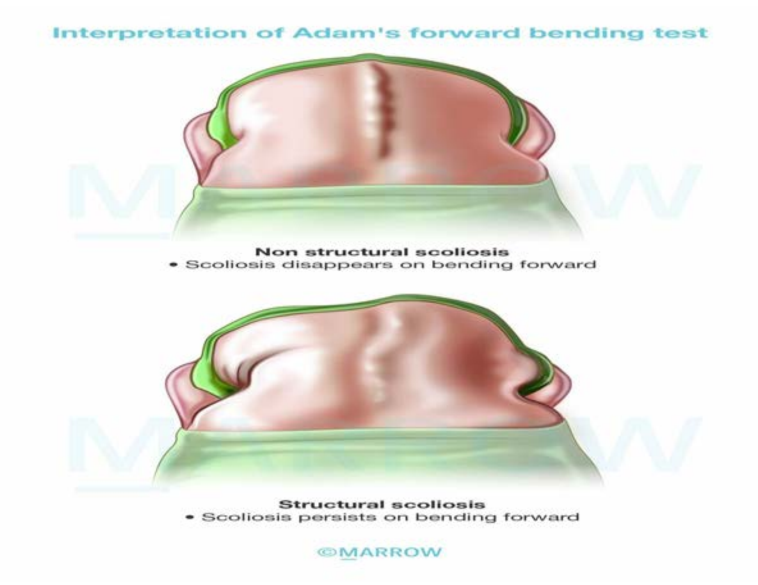
Steps of Adam's test :

• The patient is asked to bend forward at the waist until the spine is at the horizontal level

• Knees must be maintained straight, with the feet together, the arms dependent, and the palms in opposition

• The examiner observes the trunk from behind (to assess midthoracic and lumbar rotation) and from the front (to assess upper thoracic rotation)

It is used to differentiate between structural and non-structural curves in scoliosis. On bending forward, non-structural scoliosis disappears, while structural scoliosis persists.



A straight leg raising test is used to find nerve compression in the lumbosacral area.

Solution to Question 7:

Fishtail deformity is seen following a fracture of the distal end of the humerus.

Fishtail deformity of the elbow is a rare complication that can occur in childhood after a distal humeral fracture. It is characterized by a contour abnormality of the distal humerus due to the absence or resorption of the lateral trochlear ossification centers.

Fishtail deformity of the distal humerus can be of 2 types:

• The first type is a sharp-angled wedge, which commonly occurs after fractures of the lateral condyle. It is caused by the persistence of a gap between the lateral condylar physis ossification center and the medial ossification of the trochlea. Because of this gap, the lateral crista of the trochlea may be underdeveloped, which may represent a small “bony bar” in the distal humeral physis

• The second type is believed to be associated with osteonecrosis of the lateral part of the medial crista of the trochlea.

Solution to Question 8:

Ring sequestrum is seen in a pin tract infection.

Sequestrum is a fragment of dead bone lying within a cavity formed by necrosis. Infection at a pin site in an external fixation construct leads to a ring sequestrum.

Types of sequestra:

Type	Seen in
Tubular	Pyogenic infections like osteo myelitis
Ring	Around pin tracts in external fixators
Black	Actinomycosis
Coralliform	Perthes disease
Coke	Tuberculosis (cancellous bone)
Feathery	Syphilis, Tuberculosis
Sandy	Tuberculosis

Solution to Question 9:

A counterforce brace is used in tennis elbow.

Tennis elbow, also known as lateral epicondylitis, is a condition involving the lateral epicondyle caused by repetitive contraction of the extensor muscles during activities that require forceful wrist extension and supination, such as playing tennis or using a screwdriver.

A counterforce brace is a treatment option for tennis elbow (shown in the image below). It works by applying compression to the extensor muscles, specifically the extensor carpi radialis brevis (ECRB) region, which helps release the forces exerted on the lateral epicondyle. This compression prevents the transmission of tension up to the lateral epicondyle, providing relief and promoting healing in the affected area.

Management options in tennis elbow:

• Conservative management: Rest and physiotherapy

• Medical intervention: Injection of local anesthetic and steroids

• Surgical methods: Percutaneous release of epicondylar muscles

• Other management methods: Extracorporeal shock wave therapy, arthroscopic release, autologous blood injections, counterforce bracing, rehabilitative exercises, and ultrasound-guided percutaneous needle therapy.

Solution to Question 10:

The given clinical scenario, involving a child and a gross specimen revealing a metaphyseal tumor in the distal femur that has infiltrated through the cortex, lifted the periosteum, and extended through the epiphyseal plate, supports the diagnosis of osteosarcoma.

A giant cell tumor (option A) is primarily an epiphyseal tumor that usually affects adults between ages 20 and 40 years when skeletal maturity is complete. Chondroblastoma (option B) is a benign tumor of childhood most commonly seen in the epiphysis. Osteoid osteoma (option C) is a benign tumor most commonly seen in the diaphysis of the proximal femur.

Osteosarcoma is the most common non-hematologic primary malignancy of bone. It occurs most commonly in the second decade of life. It is typically metaphyseal. The distal femur, proximal tibia, and proximal humerus are common sites.

Morphologically, osteosarcomas are bulky, gritty, gray-white tumors, often with hemorrhage and cystic degeneration. They destroy cortices, form soft tissue masses, and spread extensively in the medullary canal, replacing hematopoietic marrow. Although infrequent, they may also penetrate the epiphyseal plate or enter the joint.

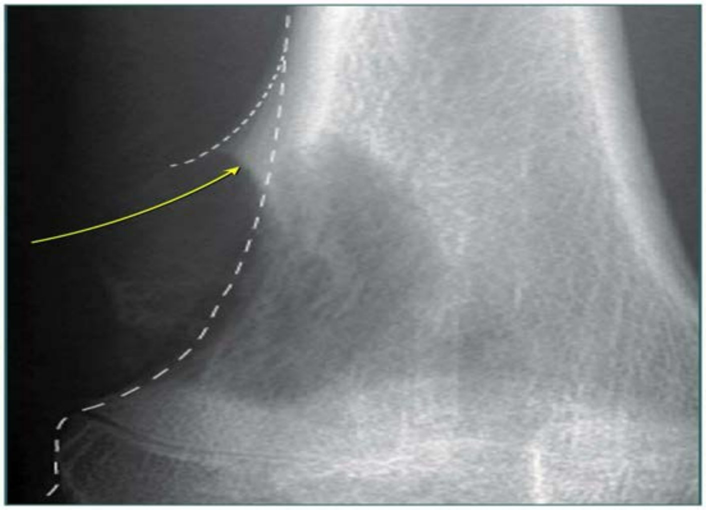
Radiological features:

• Irregular destruction in the metaphysis overshadowed by the formation of new bone

• Lifting of the periosteum (Codman triangle) due to the formation of new bone inciting a periosteal reaction

• Bone formation along the blood vessels within the tumor centrifugally, giving rise to a sunburst (sunray) appearance

Osteosarcoma - Codman Triangle



Solution to Question 11:

Among the given options, bone morphogenic protein (BMP) is the substance used for bone formation in a patient with non-union.

BMP is a bone-stimulating protein used as a bone graft in the treatment of non-union and open tibial fractures.

Bone grafts can be:

• Osteoinductive: They stimulate osteogenesis through the differentiation of mesenchymal cells into osteoprogenitor cells. BMPs are osteoinductive.

• Osteoconductive: They provide linkage across defects and a scaffold upon which new bone can form. Calcium phosphate (Option C), calcium hydroxyapatite, and calcium sulfate (Option D) are osteoconductive.

Polymethylmethacrylate (PMMA), also known as bone cement, holds bone and metal implants together by forming an interlocking network between the irregularities.

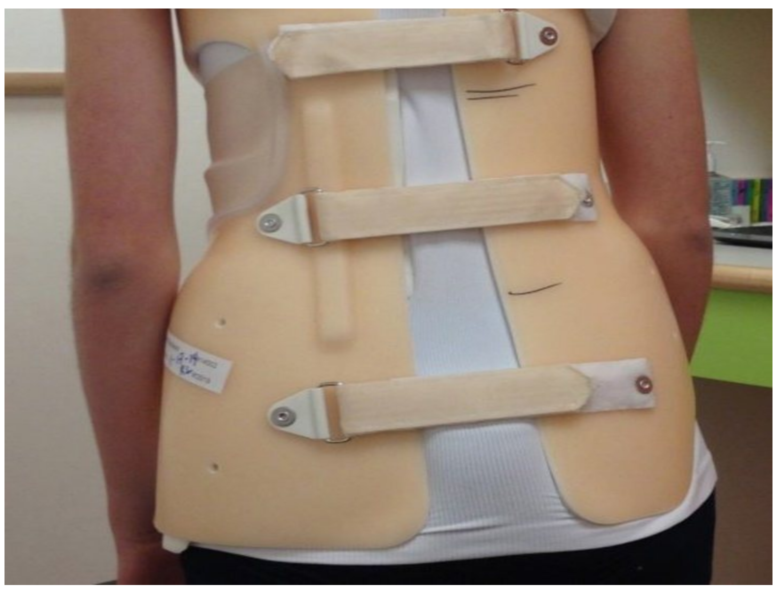
Solution to Question 12:

The brace shown in the image is the Milwaukee brace.

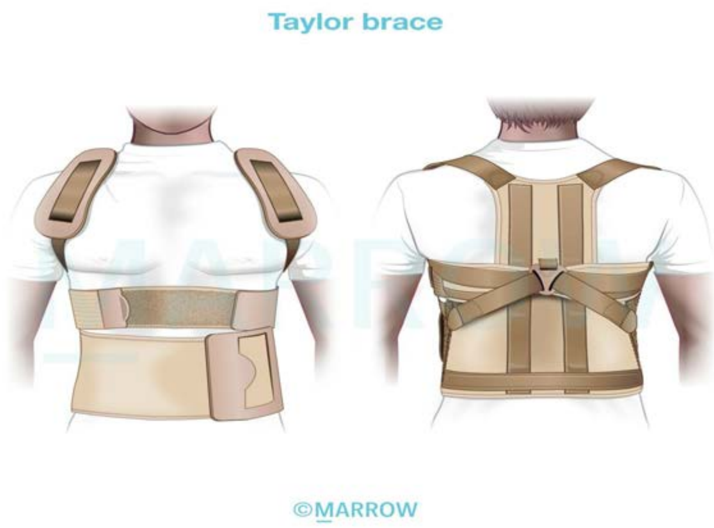
The Milwaukee brace is an active corrective spinal orthosis primarily used in the ambulatory treatment of structural scoliosis. Designed as a cervico-thoracic-lumbar-sacral orthosis (CTLSO), the main objective of the Milwaukee brace is to delay or eliminate the need for surgery. However, in recent years, the use of the Milwaukee brace has declined due to the availability and convenience of modern braces. Nonetheless, it is still employed for scoliosis cases characterized by high thoracic and cervical spine curves.

Milwaukee brace

Option A: Boston brace is a type of thoraco-lumbo-sacral-orthosis (TLSO brace). It has a hard polypropylene outer shell with a soft foam lining inside. It is used in patients whose scoliosis curve apex is at T7 or lower. The below image shows a Boston brace.



Option C: Taylor brace is designed to control the upper and lower regions of the spinal column and support your spine whilst standing, sitting, and walking after spinal surgery.



Option D: ASHE brace (Anterior spinal hyperextension brace) is used in dorso lumbar spinal injury.

Solution to Question 13:

The given clinical scenario of a 40-year-old man with increasing stiffness in the spine, along with CT spine showing ossification of the anterior longitudinal ligament and ankylosis involving the thoracolumbar vertebral bodies points to a diagnosis of ankylosing spondylitis (AS).

AS is a chronic inflammatory disease affecting the spine and sacroiliac joint. It is more common in young males and has a strong association with HLA-B27.

Features of lower back pain in AS:

• Age at onset < 40 years

• Insidious onset

• Morning stiffness > 30 minutes

• Improvement with exercise

• Pain persists with rest and is also present at night.

• Alternating gluteal pain

• Persistence for more than 3 months

Examination findings in AS:

• Loss of lumbar lordosis

• Increased thoracic kyphosis

• Diminished spinal movements

• Decreased chest expansion (<7 cm)

• Tenderness over sacroiliac joints

Important X-ray features of ankylosing spondylitis:

• Sacro-iliac joints:

• Cardinal sign/earliest sign is the erosion and fuzziness of the sacroiliac joints

• Later- periarticular sclerosis and bony ankylosis of the joint.

• Vertebral changes:

• Early spondylitis is characterized by small erosions at the corners of vertebral bodies with reactive sclerosis: Romanus lesions of the spine (Shiny corner sign)

• Squaring of vertebral bodies or flattening of the normal anterior concavity of the spine due to inflammatory resorption of the bone.

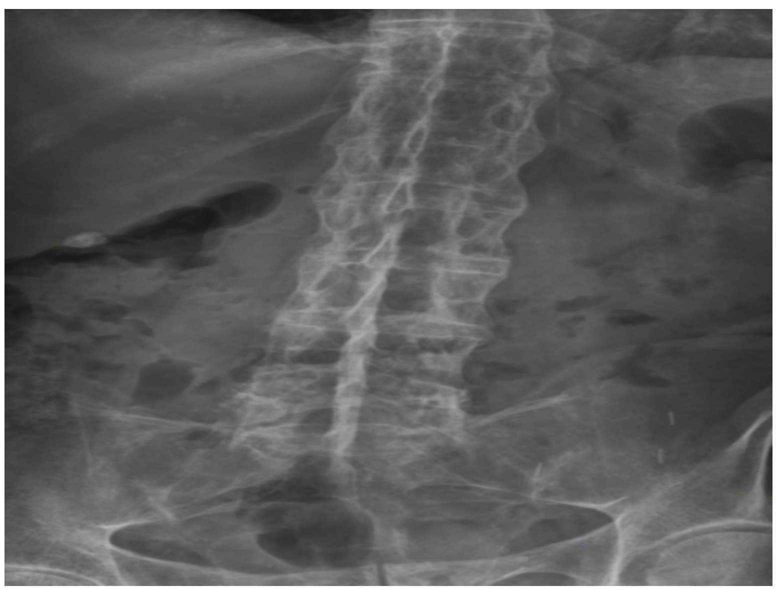
• Later signs are ossification of the ligaments around the intervertebral discs producing syndesmophytes (bridges) between adjacent vertebrae

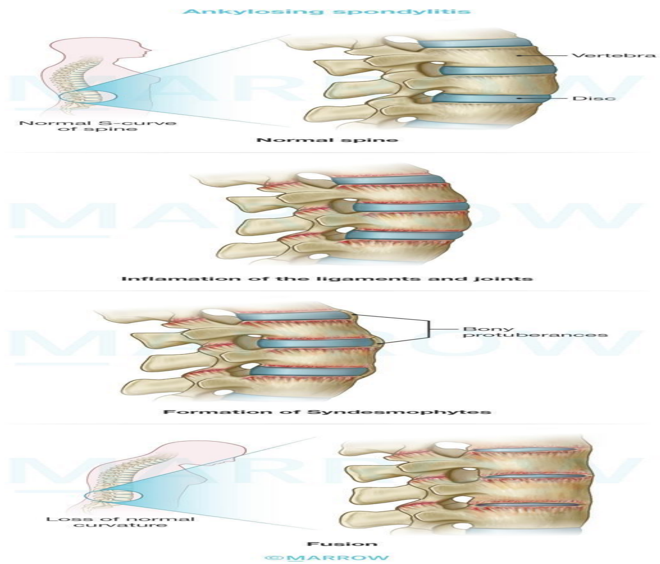
• Bamboo spine appearance is due to bridging at several vertebral levels

• Linear ossification along the central spine, representing interspinous ligament ossification can give a Dagger sign appearance on frontal radiographs.

CT imaging in AS helps to visualize bone erosions, osteoporosis/osteosclerosis, and new bone formation in the form of syndesmophytes, ligamentous ossification, and ankylosis.

The X-ray below shows the bamboo spine appearance in AS:





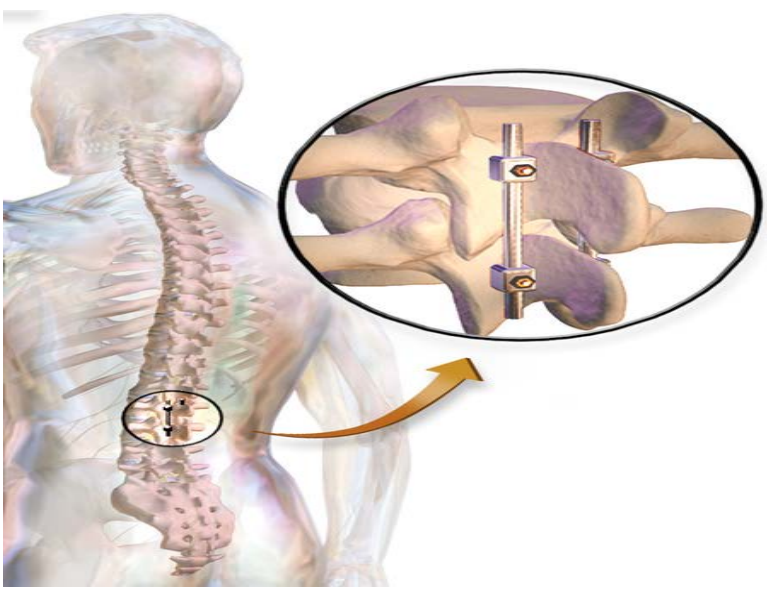
Solution to Question 14:

In spine fusion surgeries, screws are placed in the pedicle of the vertebrae.

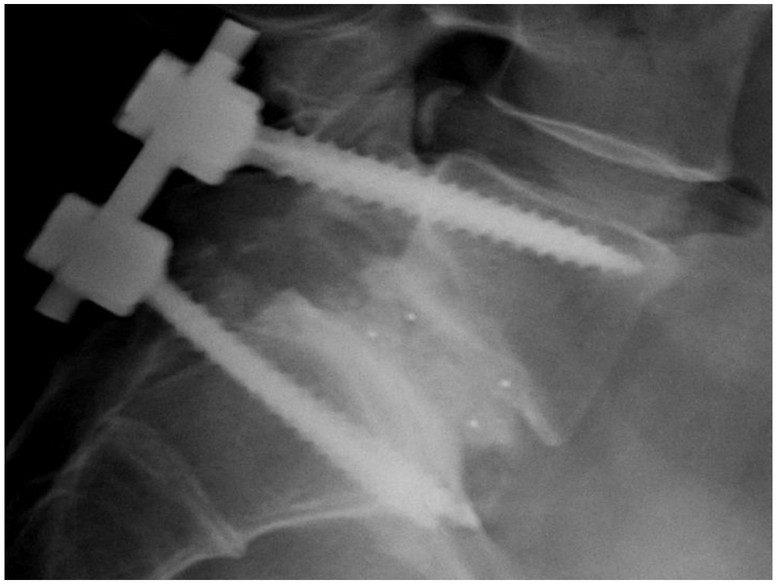
Spinal fusion or spondylodesis is a surgical technique that joins two or more vertebrae. This procedure prevents any movement between the fused vertebrae.

Pedicle screws are placed above and below the vertebrae that are to be fused. These are used in a spinal fusion to add extra support and strength to the fusion while it heals.

Rods are used to connect the screws as shown below. These prevent movement and allow the bone to heal.



The post-operative X-ray given below shows fusion of L5 and S1 vertebrae.



Solution to Question 15:

Among the given options, bone scan and quantitative CT are not commonly used in the diagnosis of osteoporosis.

Osteoporosis is a common metabolic bone disease characterised by a diffuse reduction in bone density due to a decrease in bone mass.

DEXA, chemical tests and X-ray are commonly used in the diagnosis of osteoporosis. While quantitative CT also helps to measure bone mass density at the spine and hip, it is rarely used clinically because the radiation exposure and cost are both much higher than with DXA. A bone scan uses a radioactive tracer injected into the bloodstream to detect altered metabolic activity in the bone with the aid of a gamma camera. It is not used in the diagnosis of osteoporosis.

DEXA - dual-energy X-ray absorptiometry, is the investigation of choice for the diagnosis of osteoporosis. T-score less than or equal to 2.5 in the lumbar spine, femoral neck, or total hip indicates osteoporosis.

Chemical tests like serum calcium, phosphorus and alkaline phosphatase levels are usually normal in primary osteoporosis and help to rule out other conditions before establishing a diagnosis of osteoporosis.

X-ray features of osteoporosis:

• Loss of vertical height of a vertebra occurs due to collapse

• Cod fish appearance characterized by the disc bulging into the adjacent vertebral bodies, resulting in a biconvex shape

• Ground glass appearance of the bones

Orthopedics NEET 2018

Question 1:

De Quervain's disease classically affects :

a) Extensor pollicis longus and abductor pollicis longus

b) Extensor pollicis longus and abductor pollicis brevis

c) Extensor pollicis brevis and abductor pollicis brevis

d) Extensor pollicis brevis and abductor pollicis longus

Question 2:

You are an intern in the casualty. For which of the following patients will you need to urgently call the orthopedic resident on call?

a) Patient with recurrent shoulder dislocation

b) Patient with a fractured arm with capillary refill time of less than 3 seconds in his fingers

c) Patient with a fractured arm with a 10 cm long incision over the arm

d) Patient with a fractured arm with capillary refill time of 5 seconds in his fingers

Question 3:

Which of the following is not associated with Sprengel deformity?

a) Diastematomyelia

b) Klippel-Feil syndrome

c) Dextrocardia

d) Congenital scoliosis

Question 4:

What is the most common bone to be fractured in children?

a) Distal forearm bones

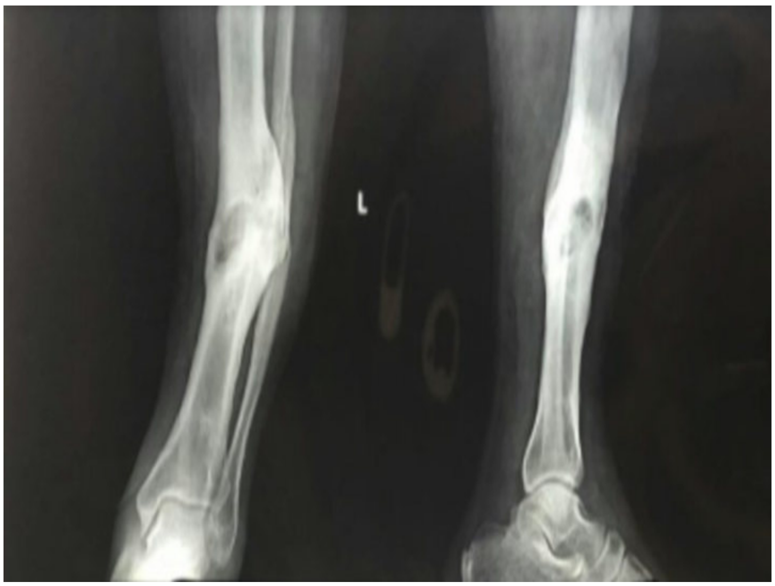
b) Humerus

c) Clavicle

d) Carpals

Question 5:

A patient has a history of a road traffic accident 2 years back, and has developed pain and swelling now at the same site. His X-ray image is shown in the image given below. What is the probable diagnosis?



a) Osteogenic sarcoma

b) Ewing's sarcoma

c) Chronic osteomyelitis

d) Multiple myeloma

Question 6:

A 35-year-old man comes with a history of musculoskeletal pain. Which of the following suggests that the pain is non-articular?

a) Swelling

b) Pain in both movement and at rest

c) Painless on passive movement

d) Presence of crepitation

Question 7:

The pathognomonic sign of a simple bone cyst is ______

a) Fallen fragment sign

b) Scalloping of cortex

c) Never breaches physis

d) Central radiolucent lesion

Question 8:

Which of the following is incorrect about perilunate dislocation?

a) Lunate is dislocated anteriorly but the rest of the carpals remains in position

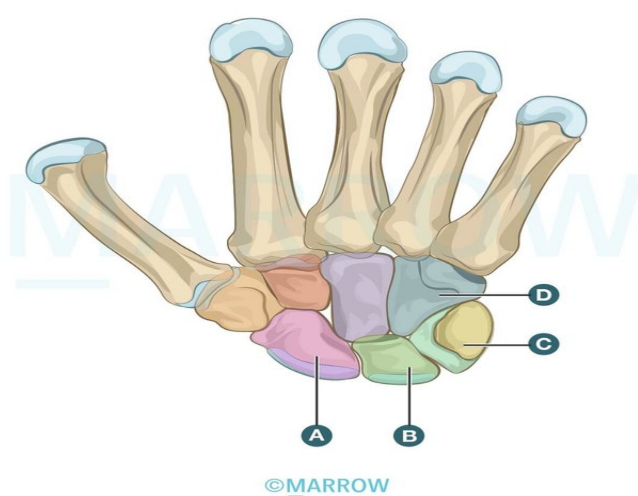
b) Lunate remains in position but rest other dislocate dorsally

c) Involves median nerve injury

d) Tavernier's manoeuvre is used for closed reduction

Question 9:

Identify the carpal bone that is most commonly fractured after fall on an outstretched hand.



a) A

b) C

c) B

d) D

Question 10:

For hypertrophic non-union following a fracture, the most appropriate treatment would be ___________

a) Stabilization

b) Bone grafting

c) Stabilization and bone grafting

d) None of the above

Answer Key

Question No.	Correct Option
1	d
2	d
3	c
4	a
5	c
6	c
7	a
8	a
9	a
10	a

Detailed Explanations

Solution to Question 1:

De Quervain's disease classically affects extensor pollicis brevis and abductor pollicis longus.

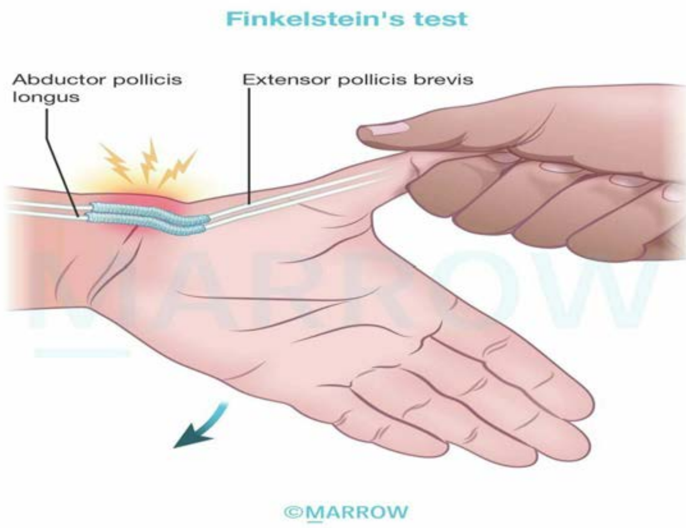
De Quervain's disease is a stenosing tenosynovitis of the extensor tendons of the first dorsal compartment, namely extensor pollicis brevis (EPB) and abductor pollicis longus (APL). It involves noninflammatory thickening of both the tendons and the tunnel or sheath through which they pass. This condition is prominent among females, especially women of childbearing age.

Key diagnostic factors:

• History of joint overuse at home or at work.

• Pain and tenderness at the tip of the radial styloid.

• Positive Finkelstein test: The examiner grasps the thumb of the patient and abducts the hand towards the ulnar side. This causes severe pain over the tip of the styloid process



Treatment in most patients includes conservative management with splinting and steroid injections. Only a few patients require surgical release of the 1st dorsal compartment (housing the APL and EPB tendons).

Solution to Question 2:

A patient with a fractured arm and a prolonged capillary refill time of 5 seconds in his fingers requires urgent evaluation and intervention by an orthopedic resident.

The capillary refill time is a simple clinical test used to evaluate peripheral circulation and assess blood flow to tissues. It can be assessed by applying pressure to the skin of the heel or toe pulp for approximately 5 seconds, causing blanching, and then releasing the pressure to allow the color to return. A capillary refill time of less than 3 seconds is considered normal, indicating good blood perfusion and adequate circulation.

In this patient with a fractured arm, a capillary refill time of 5 seconds suggests compromised blood flow to the fingers, which could be indicative of a potential vascular injury or compartment syndrome. Both of these conditions are surgical emergencies and require immediate attention to prevent further damage and ensure proper blood supply to the affected limb.

While the other patients should also be evaluated by an orthopedic resident, they do not present an immediate life-threatening situation.

Solution to Question 3:

Dextrocardia is not associated with Sprengel deformity.

Sprengel deformity is the congenital elevation of the scapula. It is characterized by an elevated shoulder on the affected side, with an abnormally high, smaller, and prominent scapula. The neck appears shorter, and there may be kyphosis or scoliosis of the upper thoracic spine. While shoulder movements are painless, abduction and elevation may be limited due to scapular fixation.

Abnormalities associated with Sprengel deformity include:

- Klippel-Feil Syndrome
- Kyphosis or Scoliosis
- Cervical rib
- Omovertebral bar (a bony bridge between the scapula and the cervical spine)
- Spina bifida
- Diastematomyelia (congenitally split spinal cord)
- Torticollis
- Renal abnormalities

The image below shows Sprengel deformity:

Solution to Question 4:

The most common bones to be fractured in children are the distal forearm bones.

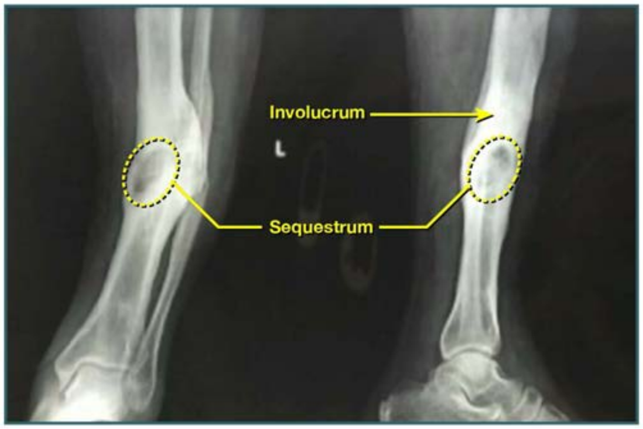
The distal radius is the most commonly fractured bone in childhood. It is more commonly fractured than the clavicle.

The clavicle is most commonly fractured during birth.

Solution to Question 5:

The given clinical scenario of a patient with a history of an accident 2 years ago, experiencing symptoms of pain and swelling at the site of injury, and an X-ray of the lower limb revealing sequestrum and involucrum in the tibia, strongly suggests a diagnosis of chronic osteomyelitis.

Chronic Osteomyelitis



Chronic osteomyelitis may occur by direct inoculation of the infective organism during trauma (like road traffic accidents). The infection continues to thrive within the bone leading to alternating quiescent and flare-up stages of chronic osteomyelitis.

Sequestrum is a piece of dead bone. It is formed when a fragment of bone gets separated from the healthy bone due to necrosis. It varies in size from a spicule to a large fragment and acts as a nidus for infection. It leads to abscess and persistent pus discharge in the bone. Sinus tracts are formed when the abscess from the infected bone drains out through the skin.

The involucrum is the dense new sclerotic bone surrounding the sequestrum. It is formed from deep layers of stripped periosteum. Cloacae are the perforations in the involucrum. Pus and tiny spicules of bone may be discharged through them.

The radiograph below shows sequestrum, involucrum and cloacae:

Option A: Osteogenic sarcoma characteristically shows the sun-burst appearance and Codman's triangle, as shown below.

Option B: Ewing's sarcoma shows an onion-skin appearance of the periosteum, as shown below.

Option D: Multiple myeloma shows multiple punched-out lesions of the bone, as shown below.

Solution to Question 6:

Non-articular disorders tend to be painless on passive movement.

Feature	Articular	Non-articular
Structures involved	Synovium, synovial fluid, articular cartilage, intraarticular ligaments, joint capsule, and juxta-articular bone	Extraarticular ligaments, tendons, bursae, muscles, fascia, bone, nerves, and overlying skin
Pain	Deep or diffuse	Localized

Feature	Articular	Non-articular
Pain with moveme nt	Both on active and passive m ovement	Only on active movement
Swelling	Present	Absent
Crepitation	Present	Absent
Instability	Present	Absent
Deformity	Present	Absent
Examples	Osteoarthritis, infectious arth ritis, ankylosing spondylitis	Fibromyalgia, myofascial pai n, acute muscle strain, myoto nic dystrophy

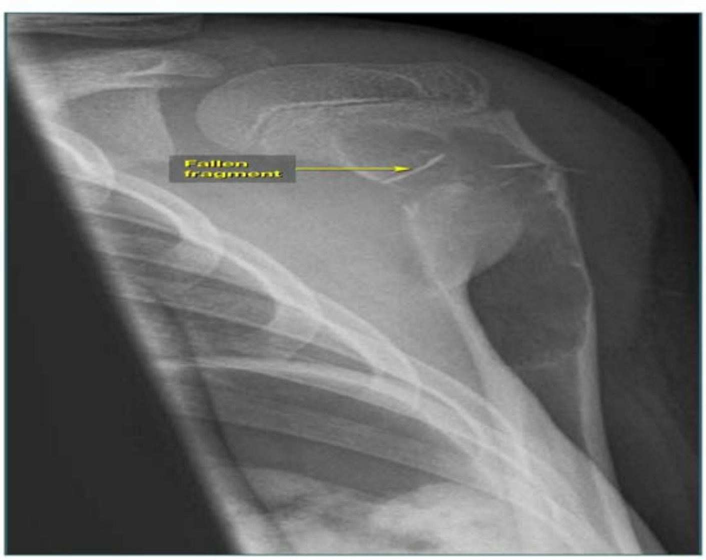
Solution to Question 7:

The fallen fragment sign is the pathognomonic radiographic sign of a simple bone cyst.

Simple bone cysts typically occur during the first two decades of life, predominantly between the ages of 4 and 10, with a 2:1 male preponderance. They are benign unilocular cysts filled with straw-colored serosanguinous fluid. The most common site for these cysts is the proximal humerus. Although they are usually asymptomatic, they can lead to pathological fractures through the cyst wall, resulting in pain and swelling. Additionally, due to their proximity to the growth plate, the cysts may cause growth disturbances.

On an X-ray, a bone fragment may be observed resting in a dependent portion of the cyst. This finding, known as the fallen fragment or fallen leaf sign, is considered pathognomonic of a simple bone cyst. Another variant of this sign is the trapdoor sign, in which a periosteal hinge prevents the fragment from falling down and enables it to change position with the patient's movement.

The radiograph below shows the fallen fragment sign:



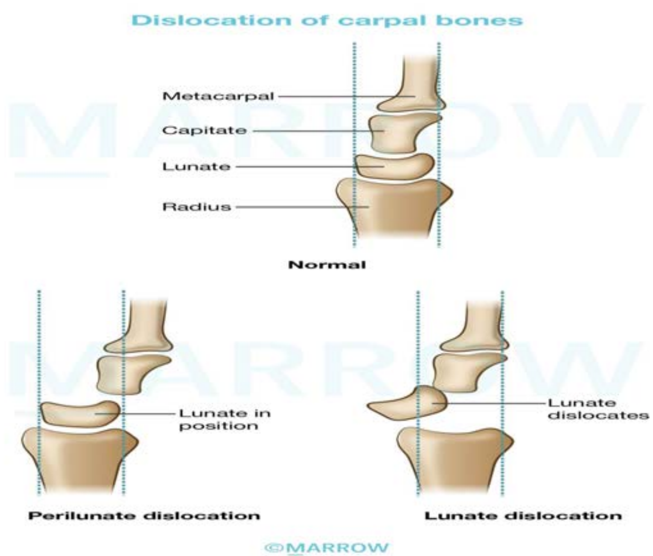
Solution to Question 8:

Among the given options, statement A is incorrect.

A fall on a dorsiflexed hand may tear the tough ligaments that normally bind the carpal bones, leading to dislocations. In peri-lunate dislocation, the lunate remains in position in the lunate fossa of the distal radius but the rest of the carpals dislocate dorsally. It is frequently associated with compression of the median nerve.

Prompt reduction reduces swelling and potential damage to the median nerve. The Tavernier maneuver is the most commonly used method of closed reduction. It involves locking the capitate into the distal concavity of the lunate. If the injury cannot be reduced using this method, urgent reduction in the theater should be performed.

In lunate dislocation, the lunate dislocates anteriorly while the rest of the carpal bones remain in place.

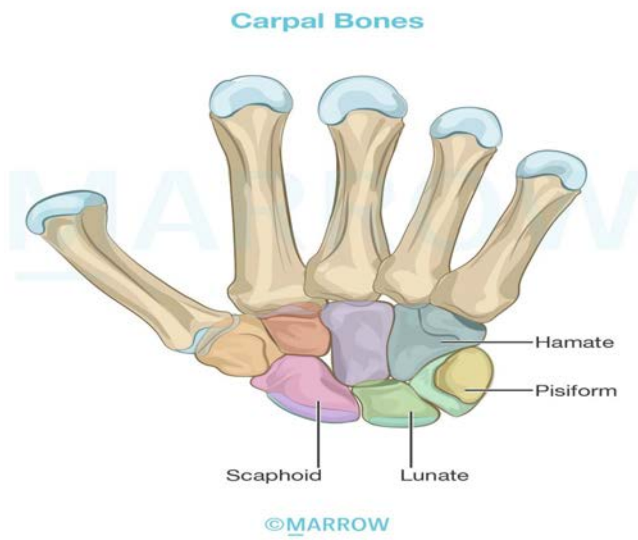


Solution to Question 9:

The scaphoid (labelled as A) is the most commonly fractured carpal bone after a fall on an outstretched hand.

It is more susceptible to fracture because it lies obliquely across the two rows of carpal bones. It also lies along the line of loading (i.e, in the line of weight transmission) between the forearm and thumb. A combination of forced carpal movement and compression, such as in a fall on the dorsiflexed (outstretched) hand, exerts severe stress on the scaphoid making it liable to fracture.

Due to its blood supply originating from the dorsal distal pole, the scaphoid's proximal pole possesses a poor blood flow, making it less prone to heal and resulting in non-union.



Solution to Question 10:

In cases of hypertrophic non-union following a fracture, stabilization alone has the potential to induce union.

Non-union following a fracture occurs when the healing process fails to progress, resulting in a failure to unite. There are two primary types of non-union:

• Hypertrophic - where callus formation is present but does not bridge the fracture site

• Atrophic - characterized by minimal or no attempt at callus formation

For hypertrophic non-union without deformity, conservative treatment using functional bracing or rigid fixation alone (internally or externally) can promote union. However, in cases of atrophic non-union, fixation alone is insufficient, requiring the excision of fibrous tissues in the fracture gap and sclerotic bone ends, followed by the application of bone grafts around the fracture.

Orthopedics NEET 2019

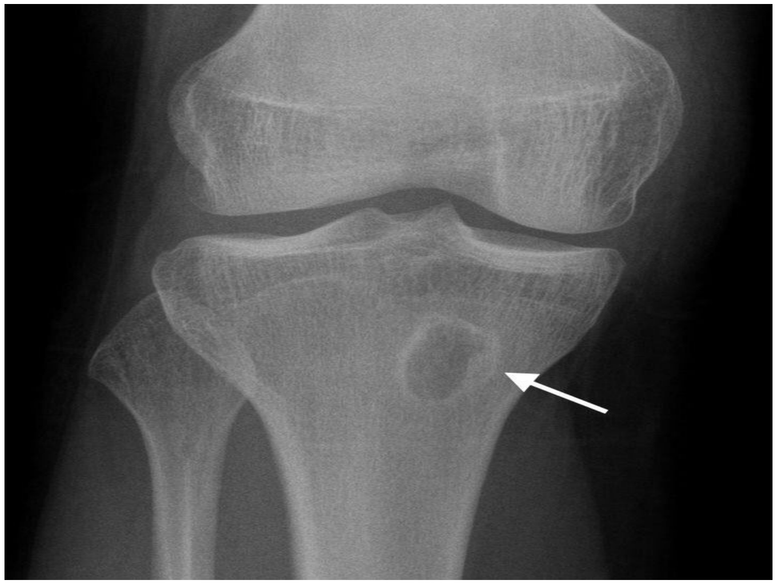
Question 1:

A 40-year-old patient presents to the emergency department following a road traffic accident. The radiograph of his pelvis is shown below. What is the most likely diagnosis?

- **a)** Central dislocation of the hip
- **b)** Fracture shaft of femur
- **c)** Posterior dislocation of the hip
- **d)** Anterior dislocation of the hip

Question 2:

A 12-year-old boy presents with swelling and pain for 6 months over the upper tibia. His radiographic image is shown below. The most likely diagnosis is:



a) Osteosarcoma

b) Osteoclastoma

c) Brodie's abscess

d) Ewing's sarcoma

Question 3:

Which of the following fractures is most prone to non-union?

a) Proximal scaphoid

b) Inter-trochanteric

c) Distal radius

d) Olecranon fracture

Question 4:

Painful arc syndrome is seen in all except:

a) Complete tear of supraspinatus

b) Fracture greater tuberosity

c) Subacromial bursitis

d) Supraspinatus tendinitis

Question 5:

De-gloving injury refers to ______

a) Skin and subcutaneous fat are stripped from the underlying fascia

b) Skin, subcutaneous fat and fascia are stripped from tendons

c) Skin, subcutaneous fat, fascia and tendons are stripped from bone

d) Only skin is stripped off

Question 6:

Rheumatoid arthritis management in a patient with deformity is:

a) Steroids

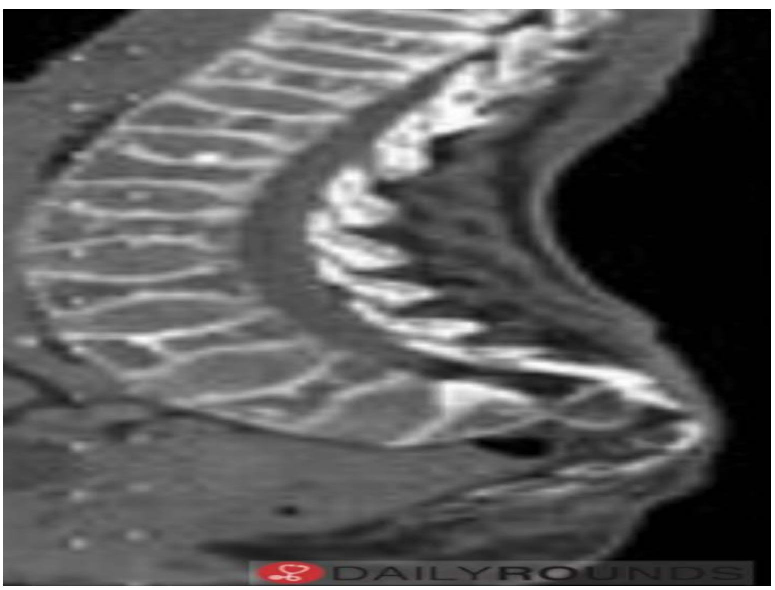
b) Methotrexate with steroids

c) Methotrexate with anti-TNF

d) Steroids only after NSAIDs fail

Question 7:

A 60 year old woman presents with low backache and on a lumbar CT the following is seen. What is your diagnosis?



a) Spondylolisthesis

b) Discitis

c) Osteoporosis

d) Pott's spine

Question 8:

An 18-year-old boy presents to the orthopedician with complaints of painful swelling below the knee joint. An X-Ray was obtained which is given below. Which of the following is the preferred treatment modality for this patient?

a) Extended curettage and bone grafting

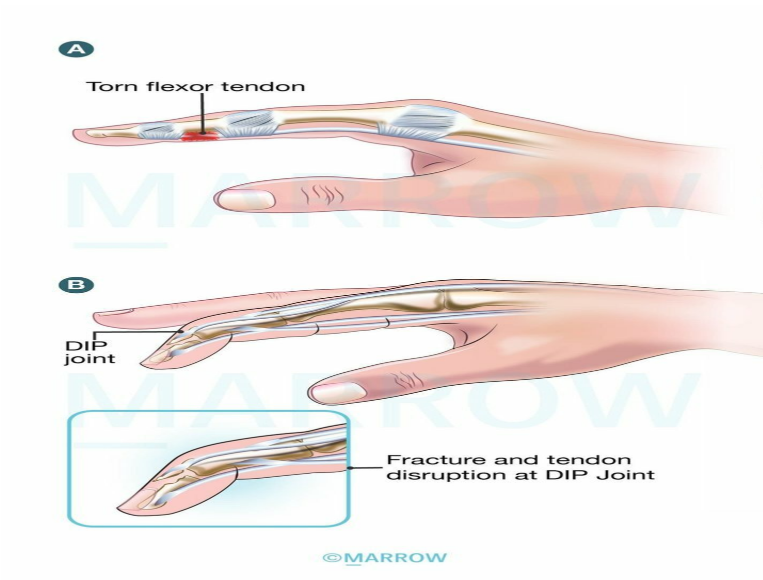
b) Simple curettage and bone grafting

c) Radiofrequency ablation

d) Radical excision

Question 9:

The following images A and B show:



a) Mallet finger & Jersey finger

b) Boxer fracture & march fracture

c) Mallet finger & boxer fracture

d) Jersey finger & baseball finger

Answer Key

Question No.	Correct Option
1	c
2	c
3	a
4	a
5	a
6	c
7	c
8	a
9	d

Detailed Explanations

Solution to Question 1:

The radiograph of the patient's pelvis following a road traffic accident reveals upward and lateral dislocation of the left femoral head, accompanied by an obscured lesser trochanter, indicating a posterior dislocation of the left hip.

Posterior hip dislocation is commonly seen following dashboard injuries and the patient presents with flexion, adduction, and internal rotation deformity of the lower limb.

Radiological findings:

• Loss of continuity of acetabulum with the femoral head (break in Shenton's line)

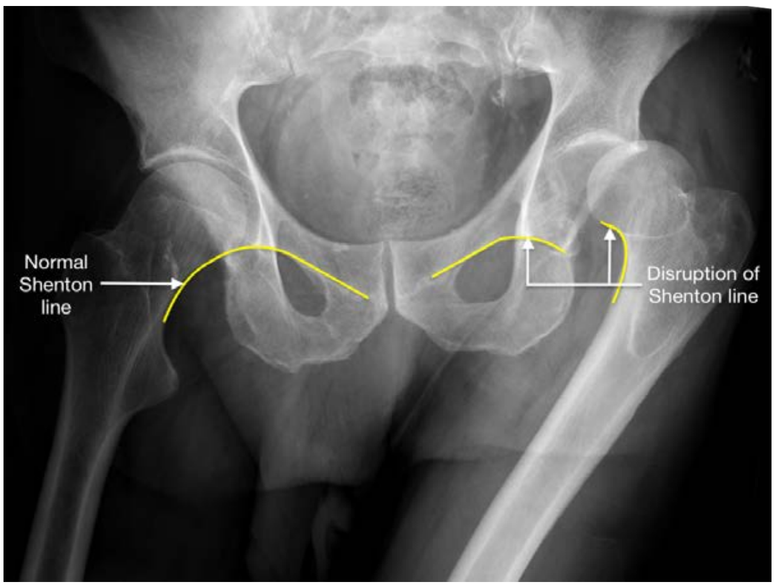
• Upward and lateral displacement of the femoral head

• The femoral head appears smaller than normal

• The lesser trochanter is not visualized as the thigh is internally rotated

• Femoral head and posterior acetabular wall fractures may also be seen

The image below shows posterior dislocation with disruption of Shenton's line:



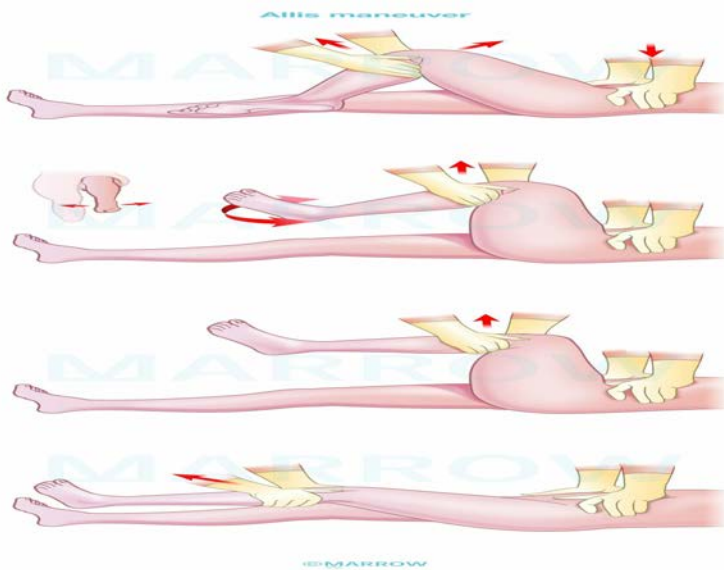
Treatment:

An early reduction should be done as the risk of osteonecrosis and subsequent osteoarthritis increases with time.

Closed reduction under sedation or general anaesthesia is done in most cases. If there is an associated femoral neck fracture, a closed reduction should not be attempted as it can lead to the displacement of the femoral neck and disrupt the blood supply to the femoral head.

In closed reduction the following steps are done:

The pelvis is kept in a steady position and traction is applied in line of the femur with the knee flexed at 90 degrees. The hip is then flexed maintaining the traction along with internal rotation and adduction. This is called Allis' maneuver. The below image describes this technique:



Other methods of closed reduction include - Stimson's maneuver, East Baltimore lift maneuver, and Bigelow's maneuver.

After reduction, the stability of the hip is tested by flexing the hip to 90 degrees, and a longitudinal and posteriorly directed force is applied. While performing this the joint is monitored in an image intensifier to look for any signs of subluxation. In case of evidence of subluxation or any significant fractures then surgical repair should be done. If there is a delay in surgical correction, skeletal traction with a distal femoral traction pin can be done.

If there is damage to the acetabulum, or intervening structures between the femoral head and the acetabular surface are present - open reduction can be considered.

Early complications:

- Sciatic nerve injury - It can be damaged in 10-20% of the cases
- Vascular injury - Superior gluteal artery tear can occur
- Concurrent femoral shaft fracture - When both occur together hip dislocation may be missed

Late complications:

- Osteonecrosis of the femoral head
- Myositis ossificans
- Secondary osteoarthritis
- Recurrent instability

Solution to Question 2:

The given clinical scenario of a child with prolonged swelling and pain over the upper tibia, along with a radiograph showing a lytic lesion with a dense sclerotic rim and minimal periosteal reaction, suggests the diagnosis of Brodie's abscess.

Brodie's abscess is a localized form of subacute osteomyelitis that occurs when organisms of low virulence infect an immunocompetent host. The abscess is walled off by the thickening of the surrounding bone trabeculae. It most commonly involves the metaphysis of the proximal tibia and distal femur.

The patient is typically a child or adolescent experiencing prolonged pain near a large joint for several weeks or months. Symptoms may include a limp, slight swelling, muscle wasting, and localized tenderness.

The typical radiographic finding is a circumscribed, round or oval radiolucent cavity measuring 1-2 cm in diameter, surrounded by a halo of sclerosis, with minimal or no periosteal reaction. The radioisotope scan shows markedly increased activity. MRI may be helpful in diagnosis. Open biopsy with curettage is indicated if there is any doubt in the diagnosis.

Differential diagnoses include:

• Cystic tuberculosis

• Eosinophilic granuloma

• Osteoid osteoma

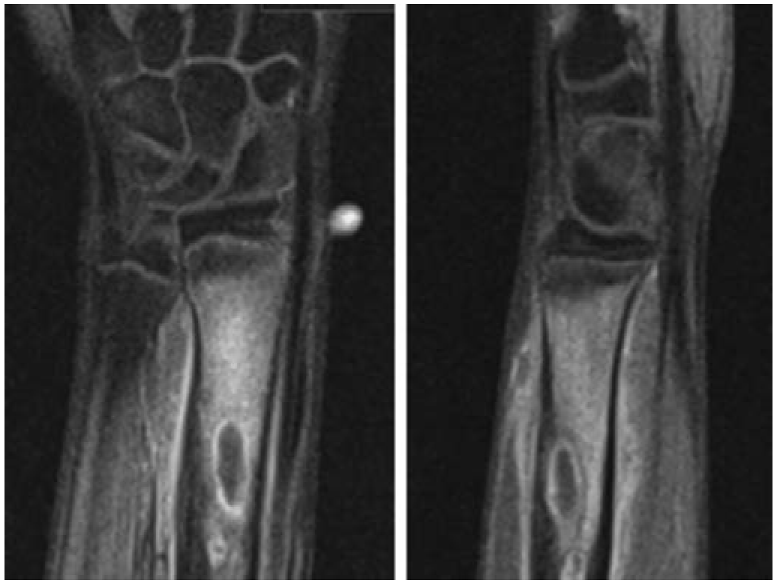
• Ewing's sarcoma

Treatment:

• Immobilization and antibiotics (flucloxacillin and fusidic acid) intravenous for 4-5 days, followed by oral administration for another 6 weeks; healing may take up to 12 months

• Curettage is needed if the x-rays show no healing after conservative management.

The following X-ray and MRI images show Brodie's abscess of the radius in a patient.



The differentiating features of each of the given conditions are as follows:

Condition	Age group	Radiographic findings
Brodie's abscess	Children, adolescents	Lytic lesion with dense sclero tic rim, minimal periosteal re action
Osteosarcoma (Option A)	Bimodal: Children and adoles cents, 70-80 yearsRapidly pr ogressivetumor	Periosteal reaction leading to sunray appearance and Cod man triangle
Osteoclastoma (Option B)	20-40 years (unlikely in child ren)	Lytic lesion withthin sclerotic rim, minimal periosteal reac tion
Ewing's sarcoma (Option D)	10-20 years	Diaphysealmass and perioste al reaction with onion-peel a ppearance

Solution to Question 3:

Among the given fractures, proximal scaphoid fractures are most prone to non-union.

Since the nutrient artery enters the scaphoid near its distal end, proximal fractures are associated with avascular necrosis and non-union. The more proximal the fracture in the scaphoid, the higher the chances of avascular necrosis and non-union.

Certain bones have a relatively limited or watershed blood supply that can potentially get interrupted by fracture. These include:

• Head of the femur: after fracture of the neck or posterior dislocation of the hip.

• Proximal scaphoid: due to fracture through the waist of the scaphoid.

• Lunate: following a dislocation.

• Body of talus: after fracture of its neck.

Fractures in these sites have a higher risk for non-union or development of osteonecrosis.

Other options:

Option B: Inter-trochanteric fractures involve either the greater or lesser trochanters or both, more common in the elderly following a history of a fall. Malunion is the most frequent complication. The vast metaphyseal region has an abundant blood supply, contributing to a higher union rate and fewer chances of avascular necrosis.

Option C: Distal radius fractures occur by fall on an outstretched hand and they include Colles' fracture (most common), Smith's fracture, Barton's fracture, and Chauffer's fracture. The complications include joint stiffness, malunion (more common), carpal tunnel syndrome, subluxation of the inferior radio-ulnar joint, Sudeck's dystrophy, and rupture of the extensor pollicis longus tendon. Avascular necrosis or non-union is rare.

Option D: Olecranon fracture results from a direct injury as in a fall onto the point of the elbow. Complications include non-union, elbow stiffness, and osteoarthritis.

Scaphoid Fracture

Solution to Question 4:

The partial tear of the supraspinatus tendon is responsible for painful arc syndrome rather than a complete tear.

Painful arc syndrome, also known as impingement syndrome is a condition where there is pain in the shoulder and upper arm during the mid-range of shoulder abduction (between 60-120 degrees), arising due to repetitive rubbing of the supraspinatus tendon under the coracoacromial arch.

Causes include:

• Minor tears of the supraspinatus tendon

• Supraspinatus tendinitis

• Calcification of supraspinatus tendon

• Fracture of greater tuberosity

• Subacromial bursitis

Important tests for impingement are:

• Painful arc test

• Hawkin-Kennedy's test

• Neer's impingement sign

• Jobe's test

Investigations:

• An x-ray of the shoulder helps to detect bony avulsions, spurs, calcific deposits, sclerotic areas etc.

• Ultrasound of the shoulder joint is highly reliable with a sensitivity of 98%.

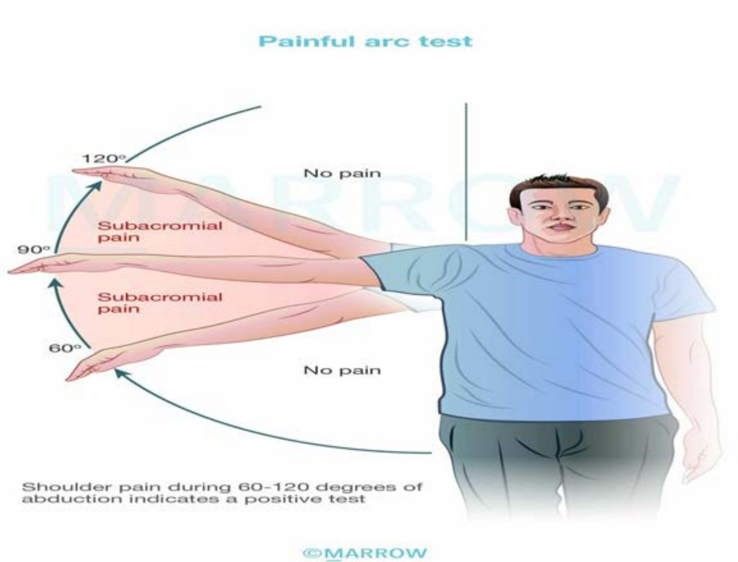
• Single contrast arthrogram is considered the gold standard in diagnosing rotator cuff tears

Management:

• Conservative: heat massage, exercises, NSAIDs, subacromial steroid injections, and temporary immobilization (90% of patients recover with these measures).

• Indications for surgery include failure of conservative measures for 3 months, increasing loss of shoulder function, and patients who are young and active.

• Surgical techniques include excision of adhesions and manipulation of the shoulder, excision of calcium deposits, repair of incomplete tear, acromioplasty, acromionectomy for more disabling pain with normal range of movements, direct suture for complete rupture of the rotator cuff, rotation and transposition of the flap, and free graft.



Hawkins–Kennedy test

The affected arm is elevated 90 degrees in the scapular plane with the elbow in 90 degrees flexion. The upper arm is then stabilized and rotated internally.

Neer's impingement sign

Keeping the scapula stable, the affected arm is raised fully in passive flexion, abduction, and internal rotation.

Jobe's test

Arms elevated in the scapular plane, elbows extended with the thumb pointing downwards, while the examiner applies downward pressure.

Solution to Question 5:

Degloving is the avulsion of skin and subcutaneous fat from the underlying fascia, muscle or bone.

Degloving injuries can be open or closed. An example of open degloving injury is finger avulsion injury with loss of skin. Closed degloving injuries result from shearing forces (e.g. motor vehicle collisions).

The extent of degloving injuries are often underappreciated and much of the skin may be non-viable. Disruption of perforating vascular and lymphatic vessels may result in a characteristic haemolymphatic collection between the fascial planes called a Morel-Lavallée lesion.

Patterns of degloving injury:

• Limited degloving with abrasion or avulsion

• Non-circumferential degloving

• Circumferential single plane degloving

• Circumferential multiplanar degloving

Assessing the viability of degloved tissue can be difficult and may therefore require more than one surgical exploration and debridement before definitive reconstruction. Non-viable skin may show fixed staining and thrombosis of subcutaneous veins. Most surgeons serially excise the degloved skin until punctate dermal bleeding is seen from viable tissue.

Intravenous fluorescein can help in delineating non-viable tissue, but it requires specialist equipment and there is a small risk of anaphylaxis. Indocyanine green has been found to have less risk of allergy.

Open degloving injuries are managed by reattaching the viable skin and grafting. Closed degloving injuries are managed by draining the accumulated fluid, removing the dead tissue and sclerotherapy. For severe, multiplanar degloving, amputation is done.

The given image below shows a degloving injury of the lower limb:

Solution to Question 6:

Methotrexate with anti-TNF is superior to any other treatment in limiting progression of joint destruction and disability.

In rheumatoid arthritis (RA) joint inflammation is the main cause of joint damage and is the most important factor of joint disability and deformities.

Methotrexate belongs to the category of drugs known as disease-modifying antirheumatic drugs (DMARDs). They slow down or prevent the structural progression of RA and hence are used in most combination therapies. Among the DMARDs, methotrexate is the preferred first-line drug.

Biologic DMARDs such as anti-TNF agents target cytokines and cell-surface molecules to slow the progression of joint inflammation. These drugs are typically used in combination with background methotrexate therapy.

Combination therapy with methotrexate and an anti-TNF drug is the most commonly used next step for the treatment of patients with deformities, having an inadequate response to methotrexate therapy.

Steroids are used when rapid control of the disease is required. They have a rapid onset of action, while the DMARDs take anywhere from 6-12 weeks to come into effect. They can be used as a cover during this period. Steroids can also be used in short courses for acute flare-up episodes of RA. Long-term use of steroids is not advised due to their adverse effects such as osteoporosis. They are usually not used alone (option A) but are used as an adjunct to DMARDs like methotrexate (option B).

Nonsteroidal anti-inflammatory drugs (NSAIDs) are used as adjunctive therapy for the management of uncontrollable symptoms because of their analgesic and anti-inflammatory properties. Chronic use is not advised due to their side effects namely gastritis and peptic ulcer disease as well as impairment of renal function (option D)

Hence, among the given options methotrexate with anti-TNF is the best therapy in a patient with deformity.

Solution to Question 7:

The given image depicts the ballooning of the intervertebral disc into the adjacent vertebrae leading to biconcave shape of vertebral body and biconvex shape of the disc which is known as the 'codfish appearance'. This is classically seen in osteoporosis.

Osteoporosis is a metabolic disease of the bone that is caused due to an equivalent decrease in both bone matrix and its mineralization, resulting in a diffuse decrease in both, bone density and bone mass. It is seen in various conditions namely in postmenopausal females, Cushing's disease, and long-term steroid therapy.

The various complications seen in osteoporosis are due to decreased bone mass which contributes to bone pains and increased frequency of pathological fractures such as Colle's fracture and neck of femur fractures. In the vertebrae, the dorso-lumbar spine is the most frequent site to be involved and can present as increased kyphosis due to compression of the anterior part of the vertebral bodies.

Other radiological features of osteoporosis:

• Loss of vertical height of vertebra due to collapse

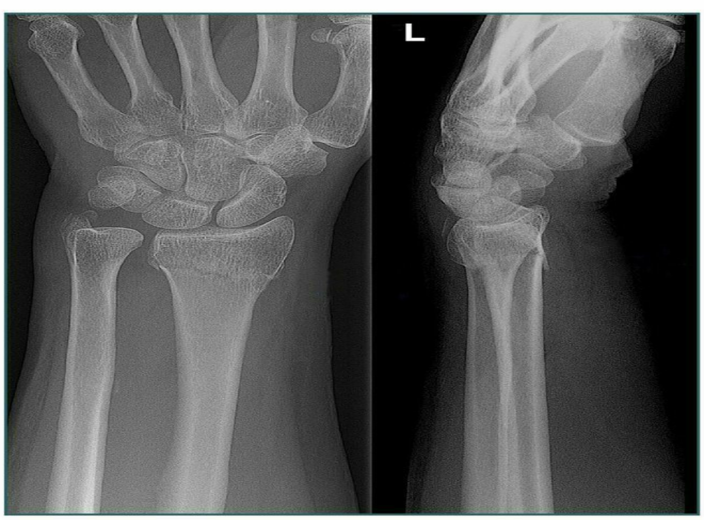
• Ground glass appearance of bones especially in the bones like pelvis

• Loss of cortical bone (picture frame vertebrae) and trabecular bone (ghost vertebrae)

• Loss of trabeculae in proximal femur area.

DEXA scan is the gold standard for quantifying bone mass in osteoporotic patients. A T-score of less than -2.5 is diagnostic of osteoporosis.

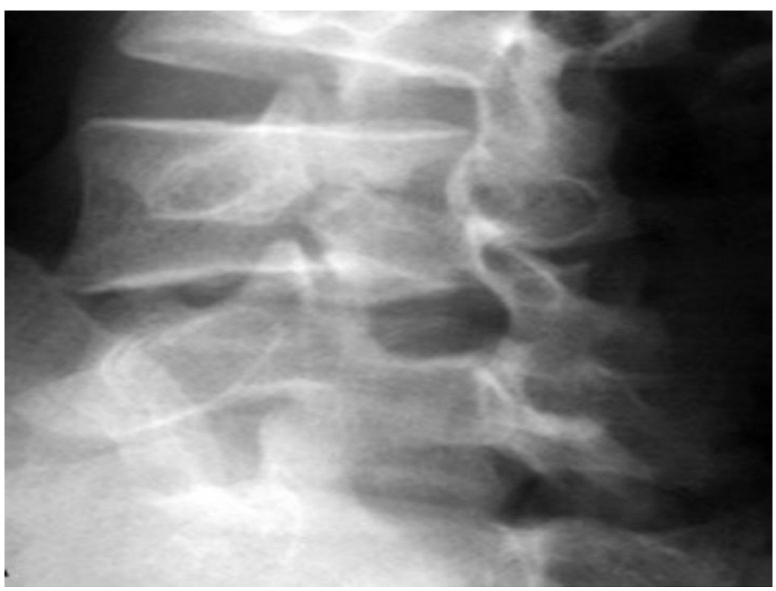
The X-ray given below is of a post-menopausal female with Colle's fracture after trivial trauma, caused secondary to osteoporosis.

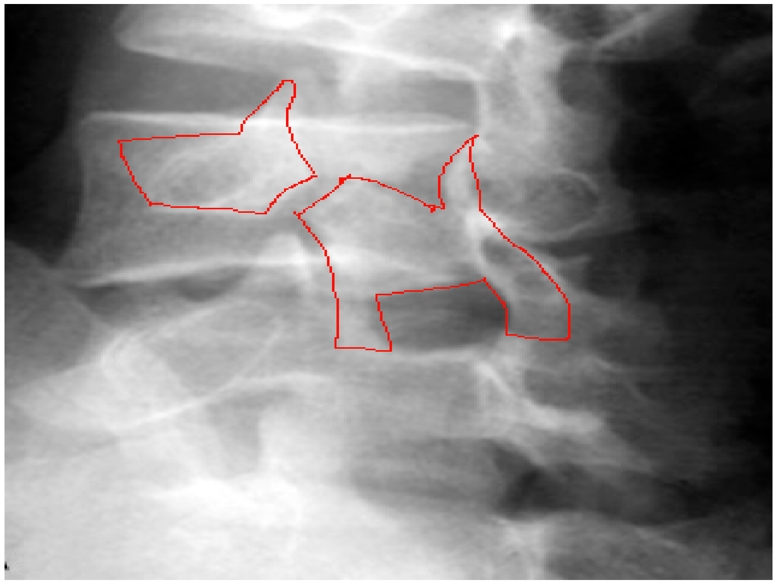


Option A: Spondylolisthesis is the anterior translation of one vertebra over another. It commonly occurs between L5-S1 followed by L4-L5. On examination, often there is a visible or palpable 'step'.

A lateral X-ray of the spine shows anterior displacement of one vertebra over another. The oblique view in severe spondylolisthesis shows the 'beheaded Scottish terrier sign' (due to a break in pars interarticularis).

The below-given images depict spondylolisthesis as seen on X-ray:





Option D: Pott's spine also known as TB spine occurs secondary to a pulmonary or gastrointestinal primary and spreads through the Batson's plexus of veins (paravertebral plexus). It most commonly involves the dorso-lumbar region.

On X-ray and CT imaging:

• A reduction in disc space is seen between the vertebrae and it is one of the earlieast signs of TB spine

• Vertebral body destruction can also be seen- based on the number of vertebrae involved they are called by different names. An involvement of 3-4 vertebrae is called as kyphosis

• Cold abscesses such as paravertebral abscess can be visible as a soft tissue shadow

• Advanced destruction and wedging of vertebrae

Solution to Question 8:

The given x-ray showing an expansile, eccentric, multiloculated lytic lesion, at the end of a long bone with a soap-bubble appearance suggests a giant cell tumor (GCT) or osteoclastoma. The treatment of choice for this condition is extended curettage and bone grafting.

Giant cell tumor is a benign but locally aggressive tumor seen involving the metaphysis and /or the epiphysis of long bones, as seen in the image, in young adults of age 20-45 years and rarely teenagers less than 20 years.

Although an aneurysmal bone cyst (ABC) is also a benign, locally destructive, eccentric multiloculated lytic tumor that can present with a similar radiographical picture in any age group (most commonly in adolescents), it is confined within the metaphysis of the long bones, unlike GCT, thus ruling it out in this case. On MRI a fluid sign can be seen.

The following is an x-ray image of the knee showing ABC confined to the metaphysis of the long bones:

An ABC is treated by extended curettage and grafting with a bone graft substitute (option B).

GCT is most commonly seen in the distal femur, proximal tibia, proximal humerus, and distal radius. It presents with complaints of swelling and vague pain and Sometimes, the patient can presents for the first time with a pathological fracture because of the lesion.

Important x-ray features of a GCT:

• Eccentric, solitary lytic lesion

• In epiphysis and metaphysis

• Expansile- expands overlying cortex

• Soap bubble appearance - due to the trabeculae of the remnants of bone traversing the lytic lesion, giving it a loculated appearance.

• No or minimal reactive sclerosis around the tumor

• No calcification

Histologically, GCTs consist of undifferentiated spindle cells profusely interspersed with multi-nucleated giant cells. The tumor stroma is well-vascularized with bands of cellular or collagenous fibrous tissue.

The treatment of choice for GCT is by

• Extended curettage and bone grafting (Option A).

• GCT in the relatively insignificant bone like the distal ulna, clavicle, and proximal fibula is treated by resection without reconstruction.

• Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease.

• Radiotherapy can be used for vertebral tumors

• Additional supportive therapy such as cryotherapy, when used reduce the reccurence rate.

Solution to Question 9:

The following image shows the Jersey finger and baseball finger respectively.

• Jersey finger - is an avulsion injury of the flexor digitorum profundus at the insertion of the base of the distal phalanx with the ring finger most commonly involved. It is caused due to sudden hyperextension of the distal joint.

• Baseball finger (also known as mallet finger)- is a disruption to the extensor tendon at the DIP joint due to collision with a hard object (like a ball) forcing it to flex past its normal range of motion. Clinically, the distal phalanx is in slight flexion in this condition.

Other options:

Boxer's fracture is the break in the metacarpal neck, most commonly of the fifth finger following a blow with the fist.

The x-ray given below shows Boxer's fracture:

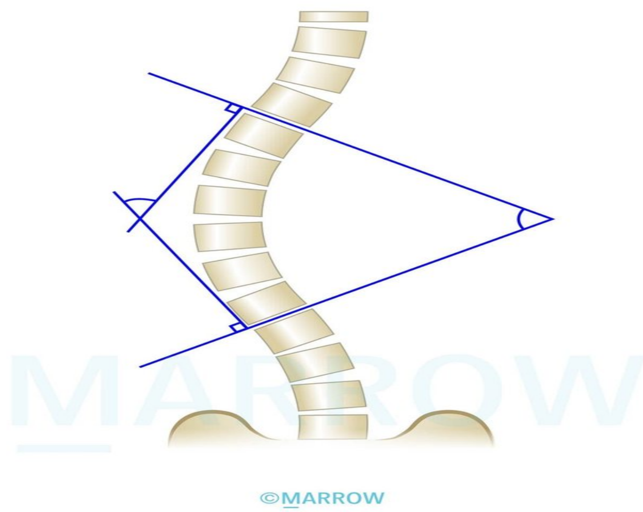
March fracture is a stress fracture of the neck of the 2nd or 3rd metatarsal

The X-ray given below shows March fracture:

Orthopedics NEET 2020

Question 1:

Which method of measurement of scoliosis has been depicted in the image given below?



a) Cobb's angle

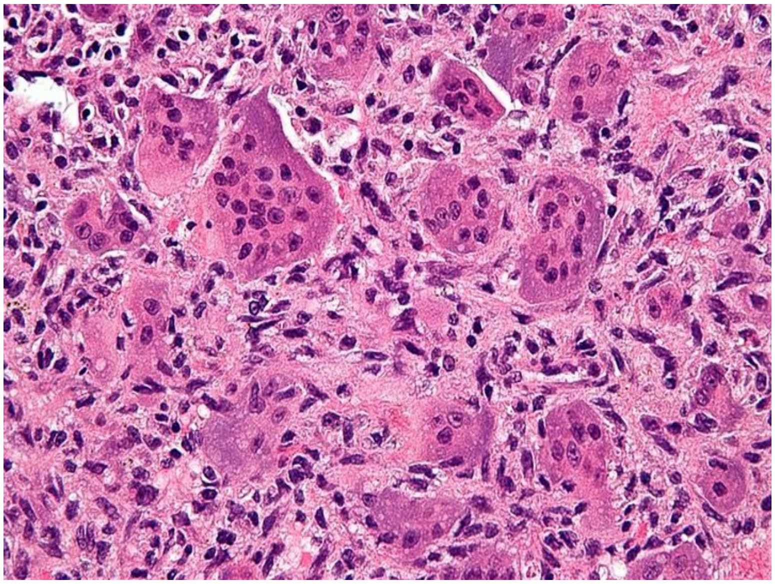
b) Baumann's angle

c) Ferguson angle

d) Bohler's angle

Question 2:

A 30-year-old woman developed a tumor at the knee joint. Biopsy-image is given below. What is the most likely condition?



a) Osteosarcoma

b) Ewing's sarcoma

c) Giant cell tumour

d) Osteoblastoma

Question 3:

A patient presented with pain in the hand. The joints involved were proximal interphalangeal joint, distal interphalangeal joint and first carpometacarpal joint. The wrist and metacarpophalangeal joints were spared. What is the likely diagnosis?

a) Osteoarthritis

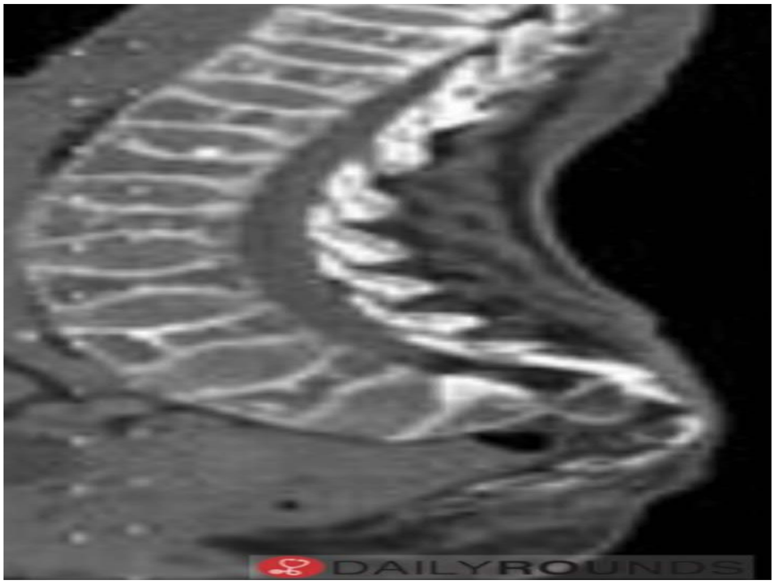
b) Rheumatoid arthritis

c) Psoriatic arthritis

d) Pseudogout

Question 4:

A 55 year old female complains of lower backache. The radiographic image of her lumbosacral spine is given below. What is the probable diagnosis?



a) Osteoporosis

b) Ankylosing spondylitis

c) Diffuse idiopathic skeletal hyperostosis

d) Renal osteodystrophy

Question 5:

A maid is playing with a child by spinning him while holding his hands. A few hours later, the child starts crying, does not use his arm, and does not let anybody touch him. What is the possible diagnosis?

a) Pulled elbow

b) Olecranon fracture

c) Fracture head of radius

d) Elbow dislocation

Question 6:

A patient was admitted to ICU 48 hours after the fracture of the femur. The saturation of oxygen in the rebreathing unit was 100% but his Spo2 remained 60%. The patient was in a state of confusion. Chest radiograph showed lung fields to be clear. What is the most likely diagnosis?

a) Pulmonary embolism

b) Fat embolism

c) ARDS

d) Occult pneumothorax

Answer Key

Question No.	Correct Option
1	a
2	c
3	a
4	a
5	a
6	b

Detailed Explanations

Solution to Question 1:

The given image depicts the measurement of the curvature of scoliosis using the Cobb's angle.

Scoliosis is characterized by the lateral curvature of the spine in the coronal plane when the patient is upright. Confirmation of the diagnosis is achieved through full-length lateral and posteroanterior X-rays of the spine. A Cobb's angle greater than 10% on the radiograph is diagnostic of scoliosis.

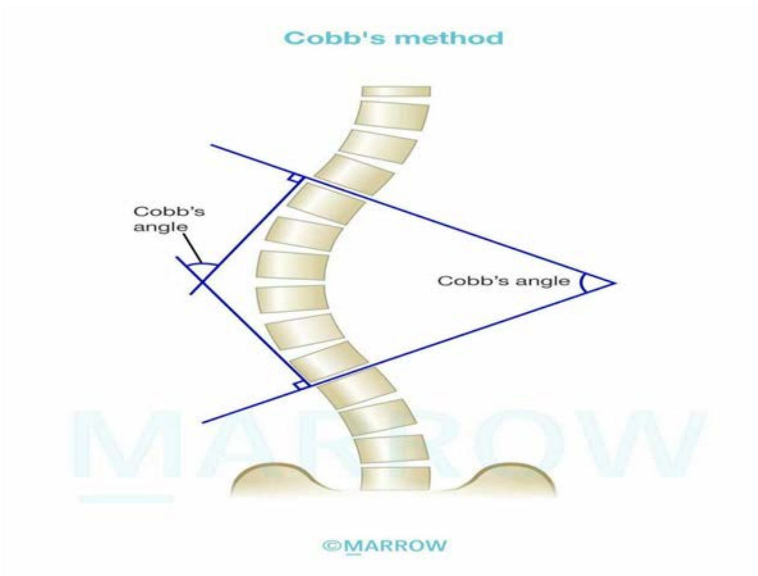
The measurement of Cobb's angle consists of three steps:

• Locating the superior-end vertebra of the curve.

• Locating the inferior-end vertebra of the curve.

• Drawing perpendicular lines from the superior surface of the superior-end vertebra and from the inferior surface of the inferior-end vertebra and making them intersect.

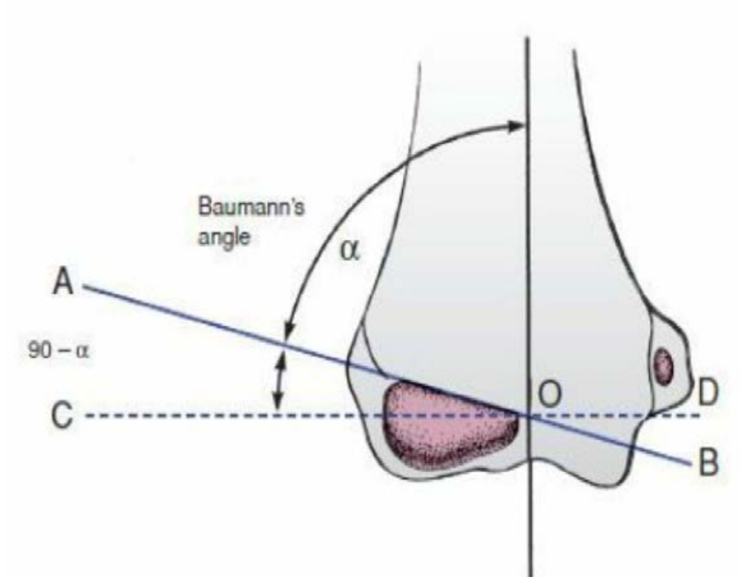
The angle formed by these perpendiculars represents the degree of spinal curvature, referred to as Cobb's angle.



Nash and Moe's method and Perdriolle and Vidal method are tools for assessing the vertebral rotation in scoliosis.

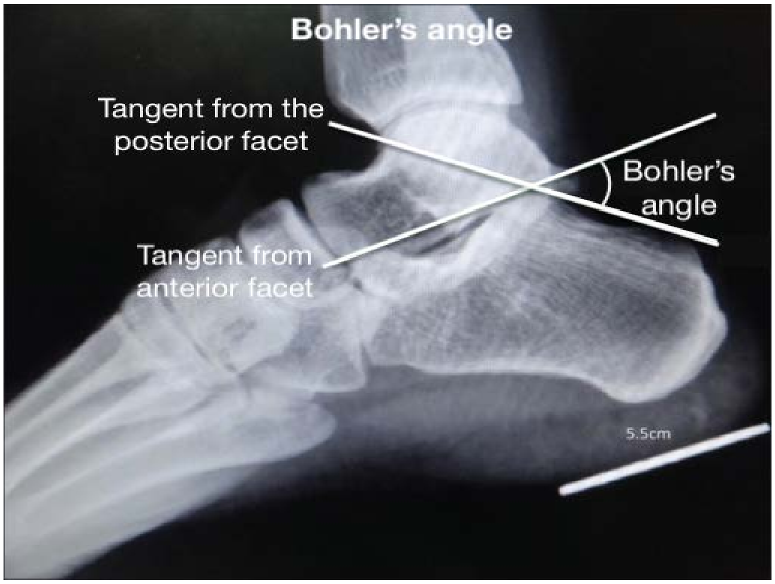
Other options:

Option B: Baumann's angle is the angle formed between a line drawn through the lateral epiphysis and the long axis of the humerus. Normally, it measures less than 80 degrees but increases in cases of supracondylar fractures.



Option C: Ferguson's angle is measured between the upper endplate of the S1 vertebra and a horizontal line. A normal Ferguson angle is typically around 34 degrees. A decreased Ferguson angle can be observed in advanced spondylolisthesis.

Option D: Bohler's angle is the angle formed by two lines tangent to the highest points of the anterior and posterior facets of the calcaneus. This angle is decreased in calcaneal fractures.



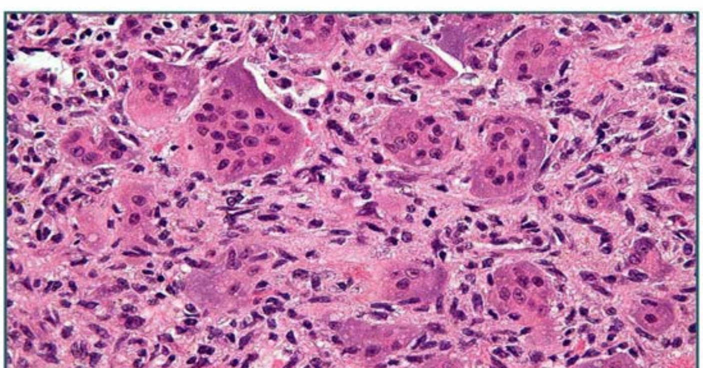
Solution to Question 2:

The biopsy image shows multinucleated osteoclast-like giant cells with scattered mononuclear cells, indicating a diagnosis of giant cell tumor (GCT)/osteoclastoma.

GCT usually affects young adults of age 20 to 40 years, with slight female predominance. It is a benign but locally aggressive tumor. It is exclusively seen in adults and affects the ends of long bones, typically involving the epiphysis. The most common location of the tumor is the distal femur followed by the proximal tibia.

Histologically, GCTs show uniform oval mononuclear tumor cells alongside numerous osteoclast-type giant cells containing 100 or more nuclei. Necrosis and mitotic activity may be prominent.

Osteoclastoma



Osteoclast-like giant cells with mononuclear cells scattered around

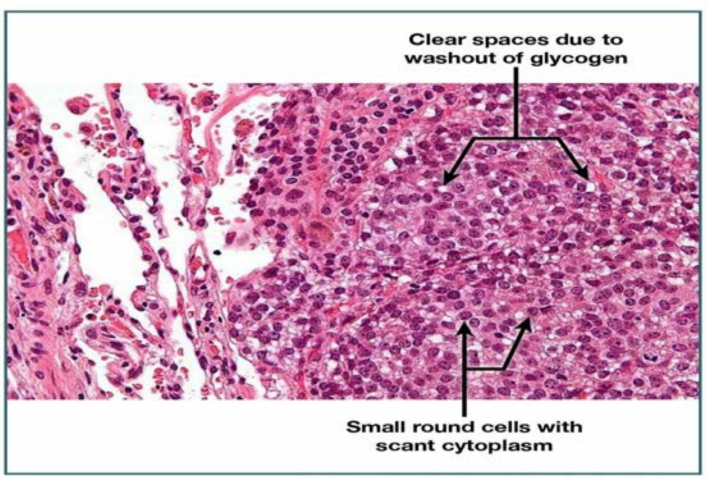
Other options:

Option A: Histologically, osteosarcoma shows pleomorphic tumor cells with hyperchromatic nuclei and abundant mitoses. A definitive diagnosis requires the presence of malignant tumor cells producing unmineralized osteoid or mineralized bone, which is typically fine and lacelike in appearance.

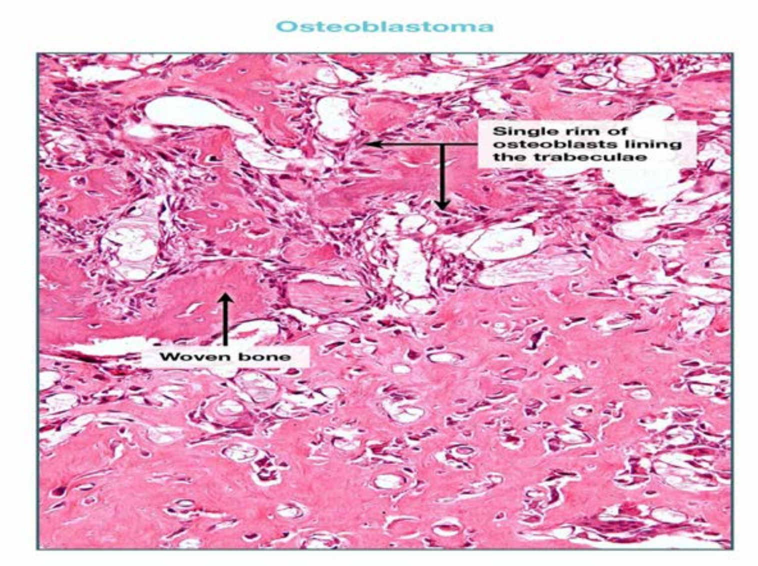
Osteosarcoma

Option B: Histologically, Ewing's sarcoma shows sheets of uniform, small, round cells with scant cytoplasm, which may appear clear because they are rich in glycogen. The presence of Homer–Wright rosettes indicates a greater degree of neuroectodermal differentiation.

Ewing Sarcoma



Option D: Histologically, osteoblastoma shows randomly interconnecting trabeculae of woven bone, which are prominently rimmed by a single layer of osteoblasts. The stroma surrounding the neoplastic bone is composed of loose connective tissue that contains numerous dilated and congested capillaries.

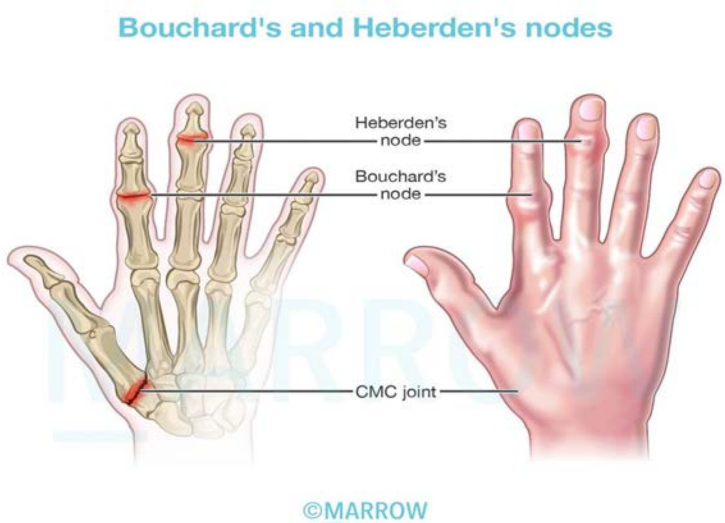


Solution to Question 3:

The most likely diagnosis is osteoarthritis (OA) of the hand.

In OA of the hand, the joints involved are the distal and proximal interphalangeal joints (DIP & PIP), and the first carpometacarpal joint (base of the thumb), while the metacarpophalangeal (MCP) joint, wrist, elbow, and ankle are usually spared.

The nodules palpable in the DIP are called Heberden's nodes, while those palpable in the PIP are called Bouchard's nodes.



The image below shows OA of the hand:

Other options:

Option B: There is relative sparing of the DIP joint in rheumatoid arthritis.

Option C: Psoriatic arthritis of the hands commonly involves the PIP, DIP, and metacarpophalangeal joints.

Option D: In pseudogout, the knee joint is most commonly affected, followed by the wrist, while the first metatarsophalangeal joint (podagra) is rarely affected.

Solution to Question 4:

The radiographic image of an elderly female's lumbosacral spine shows ballooning of disc spaces, resulting in the characteristic biconcave appearance of vertebral bodies known as the 'codfish' appearance, typically seen in osteoporosis.

Osteoporosis is a common metabolic bone disease characterised by a diffuse reduction in bone density due to a decrease in bone mass.

Risk factors for osteoporosis include:

• Increasing age

• Female sex

• Alcohol

• Smoking

• Previous history of fragility fractures

• Family history of hip fracture

• Low body mass index- less than 18.5 kg/m2

• Excessive usage of glucocorticoids

Osteoporosis presents with fractures that occur without any trauma or as a result of low-impact trauma. The commonly associated fractures are fractures of the distal part of the radius (Colle's fracture), fractures of the hip, and fractures of the vertebrae. In severe cases, the fracture of vertebrae can cause kyphosis.

Radiological findings:

• A decreased vertical height of the vertebra due to collapse

• Ground glass appearance - especially in bones like pelvis

• Codfish appearance - it occurs due to bulging of the disc into the adjacent vertebral bodies so that it becomes biconcave.

DEXA (dual-energy X-ray absorptiometry) is the investigation of choice for the diagnosis of osteoporosis. T-score less than -2.5 in the lumbar spine, femoral neck, or total hip indicates osteoporosis.

Solution to Question 5:

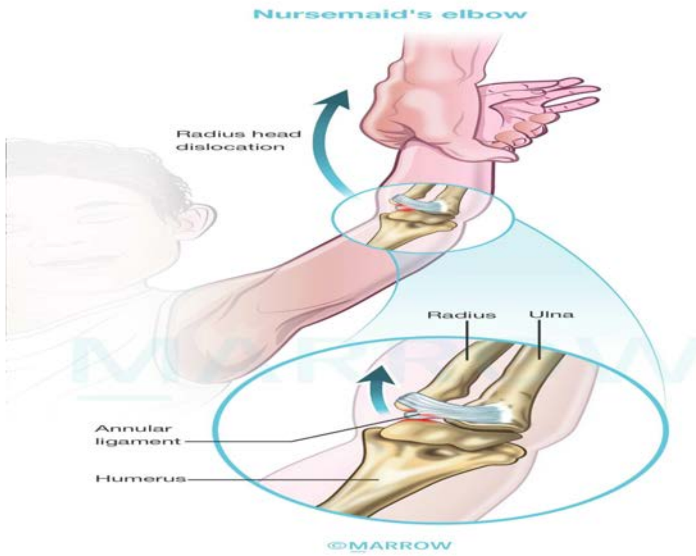
The given scenario of spinning the child by holding his hands, followed by features suggesting pain and inability to use the arm, points to a diagnosis of pulled elbow.

Pulled elbow or nursemaid's elbow refers to the dislocation of the radial head over the annular ligament, commonly seen in children under the age of 6 years. The mechanism of injury is through axial traction on the extended and pronated forearm. This happens when a child is pulled, lifted, or swung by the arm.

Following the dislocation, the child holds the arm relatively immobile, with the elbow extended and the forearm pronated, and resists supination.

The diagnosis is primarily clinical, though X-rays are usually obtained to exclude a fracture.

To treat the condition, simple analgesia is administered, and while the child's attention is diverted, the elbow is extended and pronated to relocate the radial head with a snap. This technique is preferred over the alternative method of supination and flexion, which tends to be more painful and less successful.



Solution to Question 6:

In this clinical scenario, the presence of reduced SpO2 and altered mental status in a patient presenting within 72 hours following a long bone fracture suggests the diagnosis of a fat embolism.

Fat embolism is caused by the fat globules originating from the bone marrow following the fracture of long bones. Clinical features of fat embolism are usually seen within 72 hours. They include tachycardia, fever, breathlessness, petechiae, confusion,and restlessness. Gurd's criteria is used for the diagnosis of fat embolism syndrome.

There is no specific diagnostic test for fat embolism. However, urine analysis to detect fat globules and monitoring of blood pO2 levels are conducted.

The management of fat embolism primarily involves supportive care, including oxygen therapy and prompt stabilization of long-bone fractures.

Option A: Pulmonary embolism most commonly occurs as a result of thrombi from deep vein thrombosis of the leg.

Option C: Acute respiratory distress syndrome is diagnosed using the Berlin's criteria. All four of the following conditions must be met:

• It develops within 1 week of clinical insult or worsening respiratory symptoms.

• Chest radiograph shows bilateral opacities consistent with pulmonary edema. These opacities are not fully explained by effusions, lobar or lung collapse, or nodules.

• The edema is not due to cardiac failure, or fluid overload.

• There is impaired oxygenation. The ratio of arterial oxygen partial pressure to the fraction of inspired oxygen (PaO_2/FiO_2) is ≤ 300 mmHg as assessed on 5 cm H_2O of positive end-expiratory pressure (PEEP).

Option D: Occult pneumothorax is a condition that is not suspected on the basis of the symptoms or plain radiography, but is detected with CT imaging. It usually follows a thoracic injury.

Orthopedics INI-CET 2021

Question 1:

Which of the following is seen in osteoporosis?

a) Normal calcium and increased ALP

b) Normal calcium and normal ALP

c) Decreased calcium and normal ALP

d) Increased calcium and increased ALP

Question 2:

A 15-year-old boy was brought to the ER following a motor vehicle collision complaining of pain over the hip. The x-ray is given below. What is the next step of management?

a) Closed reduction and assessment of hip stability

b) CT pelvis with 3D reconstruction

c) High weight skeletal traction

d) Open reduction and reconstruction of posterior pillar of acetabulum

Question 3:

Which of the following findings appear late in compartment syndrome?

a) Pallor

b) Paralysis

c) Pulselessness

d) Pain on passive strech

Question 4:

Which of the following statements are true about open fractures?

a) 1 and 2

b) 1 and 4

c) 2 and 4

d) 2 and 3

Question 5:

An old lady slipped and fell at home. She was diagnosed with a Colles' fracture and managed with a POP cast. What is the correct sequence of reduction?

a) Traction, POP, palmar flexion, ulnar deviation

b) Traction, palmar flexion, ulnar deviation, POP

c) Traction, ulnar deviation, palmar flexion, POP

d) Palmar flexion, ulnar deviation, traction, POP

Question 6:

Match the following fractures with their sites

a) A-1,B-3,C-2,D-4

b) A-2,B-3,C-1,D-4

c) A-4,B-2,C-4,D-3

d) A-2,B-4,C-1,D-3

Question 7:

Arrange the following in the correct sequence for the Thomas test:

a) 1,2,3,4,5

b) 3,2,1,4,6

c) 3,2,6,5,4

d) 3,6,2,5,4

Question 8:

Following a femoral shaft fracture, your consultant asks you to provide tibial traction. Which of the following will you request from the nurse?

a) 1, 2, 4

b) 3, 5, 6

c) 3, 4, 5

d) 1, 2, 3, 4, 5, 6

Question 9:

The deformity shown in the image below is noted when a patient tries to extend his fingers. Injury to which nerve can lead to this condition?

a) Ulnar nerve

b) Radial nerve

c) Median nerve

d) Posterior interosseous nerve

Question 10:

A 60-year-old woman with a history of Colles' fracture 8 months ago comes to the orthopedics outpatient department. Her T score is -2.5 and she attained menopause at the age of 52 years. She is currently asymptomatic. What treatment will you give this patient?

a) Hormone replacement therapy

b) Alendronate

c) Calcium and vitamin D

d) Repeat DEXA scan after 3 months

Question 11:

Following a road traffic accident, a patient develops type IIIa compound tibial fracture. Arrange the following external fixators in decreasing order of their stability (highest to lowest).

a) 1>2>3>4

b) 2>4>3>1

c) 1>4>3>2

d) 2>3>1>4

Question 12:

A 45-year-old patient with a tibial fracture in a cast presented to the emergency department with complaints of intense pain. The cast was removed and on examination, passive extension was painful. The dorsalis pedis and posterior tibial pulses were present and there was a loss of sensation over the first web space. What is the next step?

a) Duplex imaging

b) Re-apply cast

c) Measure anterior compartment pressure

d) Analgesics

Question 13:

Match the following with the best mode of management:

a) 1-d, 2-c, 3-b, 4-a

b) 1-c, 2-a, 3-d, 4-b

c) 1-a, 2-d, 3-b, 4-c

d) 1-d, 2-a, 3-b, 4-c

Question 14:

In nerve injuries, if the outer sheath and nerve fiber are intact while the inner axons are damaged, what is it known as?

a) Neurapraxia

b) Axonapraxia

c) Axonotmesis

d) Neurotmesis

Answer Key

Question No.	Correct Option
1	b
2	a
3	c
4	b
5	b
6	a
7	c
8	b
9	a
10	b
11	c
12	c
13	c
14	c

Detailed Explanations

Solution to Question 1:

Osteoporosis is associated with normal calcium, normal ALP levels and phosphorous.

Osteoporosis is a common metabolic bone disease characterised by a diffuse reduction in bone density due to a decrease in bone mass.

Risk factors for osteoporosis include:

• Increasing age

• Female sex

• Alcohol

• Smoking

• Previous history of fragility fractures

• Family history of hip fracture

• Low body mass index- less than 18.5 kg/m2

• Excessive usage of glucocorticoids

• They present with fractures that occur without any trauma or as a result of low-impact trauma. The commonly associated fractures are fractures of the distal part of the radius (Colle's fracture), fractures of the hip, and fractures of the vertebrae. In severe cases, the fracture of vertebrae can cause kyphosis.

Radiological findings:

• A decreased vertical height of the vertebra due to collapse

• Ground glass appearance - especially in bones like pelvis

• Codfish appearance - it occurs due to bulging of the disc into the adjacent vertebral bodies so that it becomes biconcave

DEXA (dual-energy X-ray absorptiometry) is the investigation of choice for the diagnosis of osteoporosis. T-score less than -2.5 in the lumbar spine, femoral neck, or total hip indicates osteoporosis.

Treatment - the mainstay is to reduce the risk of fractures. Following medications can be useful:

• Bisphosphonates. They are the first line of medication. They reduce osteoclastic bone resorption and prevent bone loss. Eg. zoledronate

• Denosumab- This is an antibody to the RANK ligand. It prevents the activation of RANK which is required for the proliferation of osteoclasts. Therefore, it decreases bone resorption

• Parathyroid hormone - Preotact and teriparatide are recombinant parathyroid hormone analogues. At low and intermittent doses, they can promote bone formation.

• Selective estrogen receptor modulators (SERMs)- They act on the estrogen receptors and prevent bone resorption. Eg. raloxifene.

• Strontium ranelate- It is thought to increase bone formation and reduce bone resorption.

• Cathepsin K inhibitor- inhibits the activity of collagenases (eg, odanacatib)

• Sclerostin inhibitor- Sclerostin inhibits the activity of osteoblasts. Romosozumab is a humanized antisclerostin monoclonal antibody that increases bone formation.

Operative treatment is required for femoral neck and other long bone fractures.

Solution to Question 2:

The radiograph shows upward and lateral dislocation of the left femoral head with an obscured lesser trochanter which is suggestive of posterior dislocation of the left hip and the next step in the management is closed reduction and assessment of hip stability.

Posterior hip dislocation:

It is commonly seen following dashboard injuries and in this condition, the patient presents with flexion, adduction, and internal rotation deformity of the lower limb.

Radiological findings:

• Upward dislocation of the femoral head

• The femoral head appears smaller than normal

• The lesser trochanter is not visualized as the thigh is internally rotated

• The femoral head and posterior acetabular wall fractures may also be seen

Treatment:

An early reduction should be done as the risk of osteonecrosis and subsequent osteoarthritis increases with time.

Closed reduction under sedation or general anaesthesia is done in most cases. If there is an associated femoral neck fracture, a closed reduction should not be attempted as it can lead to the displacement of the femoral neck and disrupt the blood supply to the femoral head.

In closed reduction the following steps are done:

The pelvis is kept in a steady position and traction is applied in line of the femur with the knee flexed at 90 degrees. The hip is then flexed maintaining the traction along with internal rotation and adduction.

After reduction, the stability of the hip is tested by flexing the hip to 90 degrees, and a longitudinal and posteriorly directed force is applied. While performing this the joint is monitored is an image intensifier to look for any signs of subluxation. If there is evidence of subluxation or any significant fractures then surgical repair should be done. If there is a delay in surgical correction, skeletal traction with a distal femoral traction pin can be done.

In case of damage to the acetabulum, or the presence of intervening structures between the femoral head and the acetabular surface open reduction can be considered.

Early complications:

• Sciatic nerve injury - It can be damaged in 10-20% of the cases

• Vascular injury - Superior gluteal artery tear can occur

• Concurrent femoral shaft fracture - When both occur together hip dislocation may be missed

Late complications:

• Osteonecrosis of the femoral head

• Myositis ossificans

• Secondary osteoarthritis

• Recurrent instability

Solution to Question 3:

Pulselessness is a late sign is compartment syndrome.

Acute compartment syndrome is a surgical emergency caused by increased pressure within a fascial compartment, leading to tissue ischemia. It is characterized by rapid progression of pain and swelling in an extremity, commonly precipitated by traumatic injury.

The classic features of ischemia in acute compartment syndrome include the 5 P's:

• Pain: Pain is the first symptom of acute compartment syndrome. Ischemic muscle groups have an increased sensitivity to touch. On passive stretching of the muscles, severe pain can be elicited.

• Paresthesia: Paresthesia and hypoesthesia are the first signs of ischemia and may be noted due to compression of nerves traversing the affected compartment.

• Pallor: Pallor of the skin may not always be seen as ischemia occurs at the capillary level (late sign).

• Paralysis: Paralysis of muscle groups affected in that compartment (late sign).

• Pulselessness: The absence of distal pulses (extremely late sign).

Treatment:

• Prompt decompression of the affected compartment is done.

• All the dressings, bandages, and cast on the affected limb are removed and the limb is nursed flat.

• Fasciotomy is done if the differential pressure is less than 30 mm Hg

• Fasciotomy wounds are left open and after 2 days if there is necrosis debridement is done. If the wounds are healthy it can be sutured without tension or skin grafting can be done.

The diagnosis of compartment syndrome can be confirmed by measuring the intra-compartmental pressure using a split catheter introduced into the compartment. The difference between diastolic pressure and compartment pressure is known as differential pressure (ΔP). If the differential pressure (ΔP) is less than 30 mm Hg (4.00 kilopascals), immediate compartment decompression is indicated.

Solution to Question 4:

Open fractures commonly involve the tibia and phalanges (statement 1) and necessitate early wound debridement (statement 4).

Open fractures are characterized by a break in the overlying skin and soft tissues, exposing the fracture fragment to the external environment. It is important to note that open fractures can be accompanied by multiple other injuries, and the presence of an open fracture does not exclude the possibility of developing compartment syndrome as a potential complication.

The most common sites involved in open fractures include:

• Phalanges (fingers and toes)

• Tibial diaphysis

• Distal radius

• Ankle

Major components of the management of open fractures are:

1) Antibiotic prophylaxis - It should be given as early as possible. Co-amoxiclav or cefuroxime are the preferred drugs.

2) Early debridement of the wound and fracture - This includes the following steps:

• Excision of the wound margins.

• Extension of the wound if required.

• Delivery of the fracture outside the wound for proper examination.

• All the devitalized tissue should be removed.

• Cleansing of the wound to remove all the foreign materials.

• Nerves and tendons are usually left alone, but if the wound is clean and expertise is available their repair can be considered.

3) Early wound cover:

• In Gustilo type 1 and 2 fractures wound cover can be done at the time of debridement.

• Delayed wound cover is usually practiced in type 3 fractures.

4) Fracture stabilization:

• The method of fracture stabilization depends on the degree of contamination, amount of soft tissue damage, and time from injury to operation.

• External fixation is used as a temporary measure if there is a delay in wound cover.

• If definitive wound cover can be achieved at the time of debridement and there is no significant contamination, then an open fracture can be managed as same as a closed injury.

Solution to Question 5:

The correct order of reduction of Colles' fracture is traction, palmar flexion, ulnar deviation, POP.

Colle's fracture refers to the extra-articular fracture of the distal radius with dorsal angulation of the distal bone fragment.

Displacements that may be seen in Colles' fracture are:

• Impaction of fragments

• Dorsal tilt

• Dorsal displacement

• Lateral tilt (radial tilt)

• Lateral displacement (radial displacement)

• Supination

The patient usually presents with pain, swelling, and deformity of the wrist. There may be a typical 'dinner-fork deformity' where the radial styloid process comes to lie at the same level or a little higher than the ulnar styloid process. A dorsal tilt is characteristic and is seen on a lateral x-ray.

Management of Colles' fracture:

• Undisplaced fractures: immobilization in a below-elbow plaster cast for 6 weeks

• Displaced fractures: closed manipulative reduction and immobilization in Colles' cast. The technique of reduction of Colles' fracture is as follows:

• Dis-impaction of the fracture fragments by longitudinal traction against counter traction provided by the assistant.

• The distal fragment is then pressed into palmar flexion followed by ulnar deviation.

• The patient's hand is drawn into pronation, palmar flexion, and ulnar deviation.

• A Colles' cast is then applied, extending from below the elbow to the metacarpal heads, with the wrist in palmar flexion and ulnar deviation.

• Surgical correction is advisable for comminuted fractures, especially in young adults where the dominant hand is involved. It can be with percutaneous K-wire fixation, external fixators, or locking compression plates.

Complications:

• Stiffness of joints - finger stiffness is commonly seen

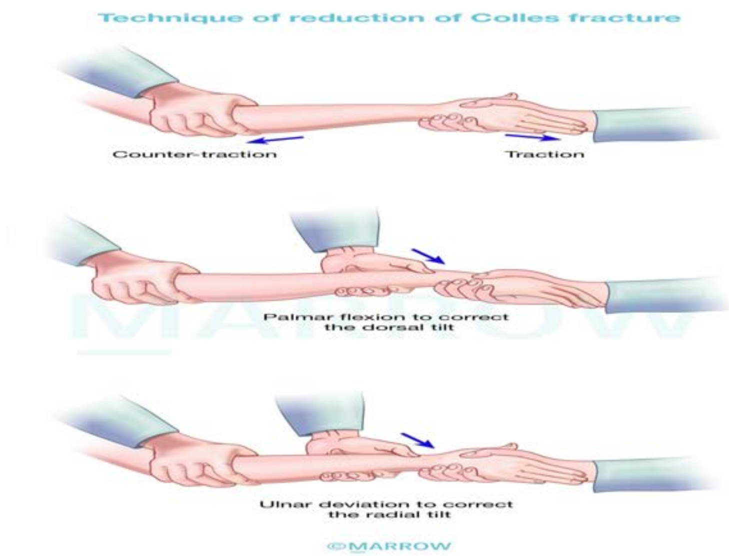
• Malunion - can produce dinner-fork deformity

• Subluxation of the inferior radio-ulnar joint

• Carpal tunnel syndrome - uncommon

• Sudek's osteodystrophy - Colles' fracture is the commonest cause of this condition in the upper limb.

• Rupture of extensor pollicis longus tendon - extremely rare



Solution to Question 6:

The correct match is A-1, B-3, C-2, D-4

• Jones fracture - 5th metatarsal

• Bennett's fracture - 1st metacarpal

• March fracture - 2nd metatarsal

• Boxer fracture - 5th metacarpal

Jones fracture - It refers to the fracture of the proximal part of the fifth metatarsal at the metaphyseal-diaphyseal junction. It is usually caused by an adduction injury of the forefoot. Conservative management is done with a short-leg non-weight bearing cast for a period of 6 weeks. Surgical correction by intramedullary screw fixation is advocated for faster healing in highly active individuals.

Bennett's fracture - It occurs at the base of the first metacarpal bone. It is sustained usually by a fall or during punching. The fracture line is oblique and extends into the carpometacarpal(CMC) joints. It is highly unstable due to the pull of abductor pollicis longus on the shaft of the metacarpal.

X-ray shows a triangular fragment in contact with the medial edge of the trapezium and proximal subluxation of the rest of the thumb.

Surgical correction with percutaneous K-wire or open reduction and internal fixation (ORIF) with a lag screw is the preferred treatment.

March fracture - These are stress fractures that usually involve the second metatarsal. It is typically seen in military recruits or runners. On examination, a tender lump may be noted in the midshaft region of the metatarsal. Similar lesions may be also seen in osteoporotic individuals.

The x-ray may be normal initially but later show a hairline crack. After a period of 4-6 weeks, a mass of callus can be identified.

Reduction and splintage are not necessary as it is undisplaced. Normal walking can be done and it's managed by elastic bandage of the forefoot.

Boxer's fracture - It refers to the fracture at the neck of the fifth metacarpal. It is usually produced by striking an object with the hand as seen with a boxer's punch.

It is managed using a gutter splint with flexed MCP joints and straight IP joints until the discomfort settles.

Solution to Question 7:

Among the given options, the correct sequence for the Thomas test is given by option C i.e. 3,2,6,5,4.

Thomas test is used to detect flexion deformity of the hip. Flexion deformity is a common hip deformity, often caused by stronger flexor muscles overpowering the extensor muscles. When there is a spasm of the muscles, the stronger flexors pull the hip in flexion. Soft tissue contractures can convert this into a fixed flexion deformity, making locomotion impossible.

Steps of Thomas test:

• Positioning: The patient is instructed to lie supine on a hard surface with their legs fully extended.

• Observing lumbar lordosis: As the patient assumes the supine position, the curvature of the lower back is visually assessed for any exaggerated lumbar lordosis. It is noted that excessive lumbar lordosis may be a compensatory response to a flexion deformity.

• Gradual flexion of the normal hip: The normal hip is gradually flexed, with care taken not to exceed its normal range of motion.

• Monitoring for overcorrection: During the flexion of the normal hip, signs of overcorrection on the normal side are observed, such as the pelvis starting to lift off the ground. To prevent this, both ischial tuberosities are ensured to remain in contact with the surface.

• Automatic positioning of the affected hip: As a result of the flexion of the normal hip, the affected hip automatically assumes the deformed position characterized by flexion.

• Measuring the degree of flexion: Once the affected hip is in the flexed position, the angle formed between the affected thigh and the bed is measured. This angle represents the actual degree of flexion deformity.

Solution to Question 8:

Among the given options, Steinmann pin, Bohler's stirrup, and Bohler Braun splint are used for tibial traction in femoral shaft fractures.

Steinmann pin: It is a stout, straight steel rod used to provide skeletal traction. Its diameter ranges from 3 to 6 mm. The common sites where it is used include the upper end of the tibia, the supracondylar region of the femur, and the calcaneum.

Bohler's stirrup: It is used to hold a Steinmann pin and apply traction. The screws on the sides of the stirrup hold the pin. The direction of traction can be changed without moving the pin inside the bone, which prevents the pin from loosening.

Bohler Braun splint: This refers to Bohler's modification of the Braun splint. It consists of a heavy metal frame with four pulleys. It is used for the treatment of fracture shaft femur and supracondylar fractures of the femur. It can also be used for the fractured shaft of the tibia and fibula.

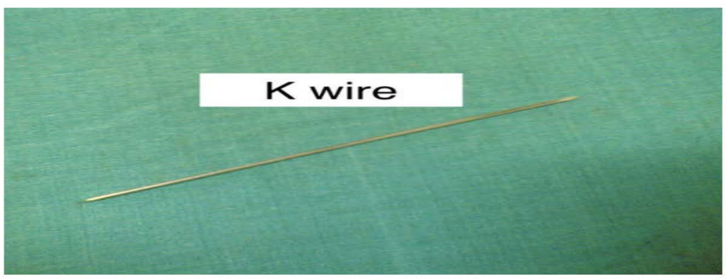
Other options:

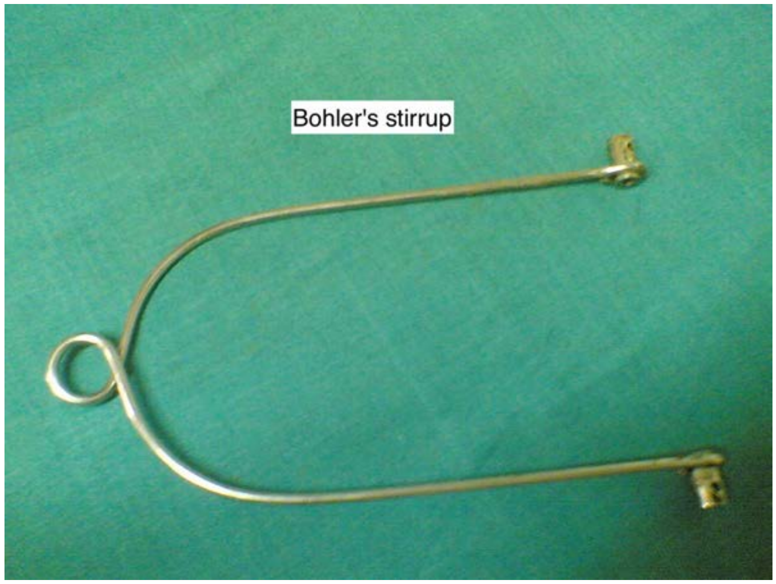
Denham pin: This pin is threaded in the center and engages the bony cortex, which reduces the risk of pin sliding. It is used when the bone is cancellous -like calcaneum and osteoporotic bones.

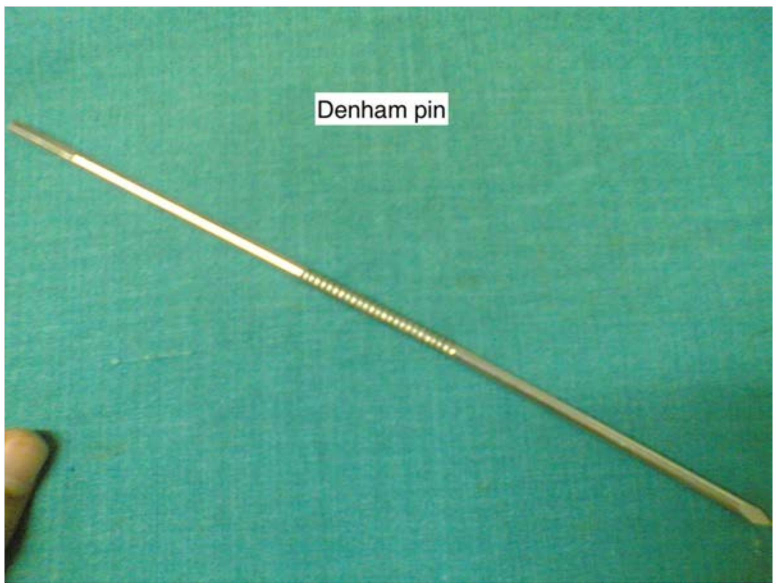
Kirschner wire or K-wire is 1 to 3 mm in diameter. The wire is pointed at both ends for both anterograde and retrograde insertion. The shape of the tips could be trocar, diamond or threaded. It is used in small bone fractures.

Thomas splint: It is used for immobilizing fractures of the femur.

Given below are the images of these instruments

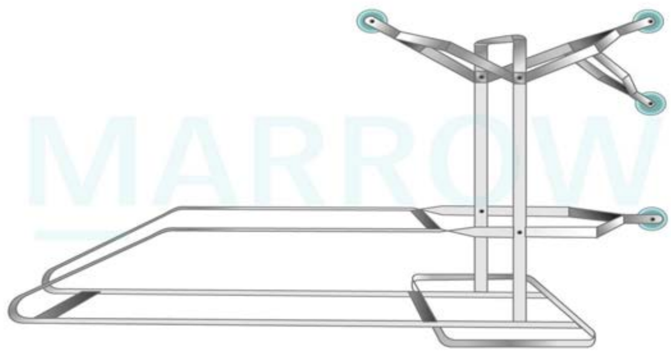



Bohler's stirrup


Denham pin


Thomas Splint
Inner diameter:
25 cm
Length: 102 cm
MARROW
©MARROW

Bohler Braun splint



Solution to Question 9:

The given image shows extension at the metacarpophalangeal joint and flexion at the interphalangeal joints of the ring and little fingers while the patient tries to extend his fingers, suggesting an ulnar nerve injury.

The ulnar nerve supplies the medial two lumbricals in the hand. The lumbricals have a combined action on metacarpophalangeal (MCP) joint flexion and interphalangeal (IP) joint extension (glass holding posture). In ulnar nerve injury, since the medial lumbricals are not functional, the forearm muscles are unopposed. Hence, there is hyperextension at MCP joints and flexion at IP joints. This causes claw hand deformity. There is also numbness of the ulnar one and a half fingers.

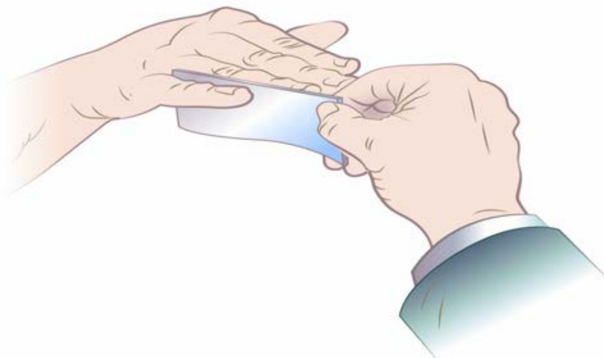
Elbow fractures or dislocations lead to high lesions of the ulnar nerve. Here, the hand is not markedly deformed because the ulnar half of the flexor digitorum profundus (which is responsible for flexing the fingers) is paralyzed and the fingers, therefore, appear less clawed. This is called the high ulnar paradox.

The following tests are done to test the integrity of the ulnar nerve:

• Egawa's test - In a positive test, due to the weakness of the dorsal interossei, side-to-side movements of the middle finger is weak.

• Book test or Froment's sign - The patient is asked to hold a book forcefully while the doctor tries to pull it. In a positive test, due to weakness of adductor pollicis muscle, adduction of thumb is lost. The patient tries to hold the book by using flexor pollicis longus muscle. This produces flexion at the inter-phalangeal joint while holding the book.

• Card test - In the case of weak palmar interossei, it is easy for the examiner to pull out the card when held between the fingers. This is due to the loss of adduction by palmar interossei muscles.



Positive Card Test
In the weakness of palmar interossei, there is a weakness of finger adduction. So the patient is unable to hold card firmly between fingers.

Other options

Option B - In radial nerve palsy, the patient presents with 'wrist drop' in case of paralyzed wrist extensors, and 'finger drop' when extensor digitorum is paralyzed. When there is compression in the axilla, radial nerve palsy is called Saturday night palsy. There is paralysis of the triceps, and weakness of the wrist and hand. In high lesions, there is sensory loss on the dorsum of the hand over the anatomical snuff box.

Option C - In median nerve injury, paralysis of opponens pollicis muscle leads to 'ape thumb' deformity. In forearm or elbow injuries, there is paralysis of long flexors of the thumb, index, and middle fingers, radial wrist flexors, and forearm pronators. This leads to the characteristic 'pointing index finger' when the patient tries to make a fist. There is sensory loss over three and a half digits on the radial side.

Option D -Posterior interosseous nerve injury, is seen in posterior interosseous syndrome. The wrist is in radial deviation and there is loss of metacarpophalangeal joint extension. However, wrist extension is preserved.

Solution to Question 10:

The clinical vignette of an elderly post-menopausal woman with a history of Colles' fracture and a T score of -2.5 suggests postmenopausal osteoporosis and the treatment of choice in this condition would be a bisphosphonate such as alendronate.

Options A and C: Hormone replacement therapy, calcium and vitamin D supplements, and calcitonin are not recommended treatments to reduce fracture risk in osteoporosis.

Option D: Repeating the DEXA scan after 3 months will not be required as the patient's T score (-2.5) already suggests osteoporosis. DEXA is done in the following patients to screen for osteoporosis:

• Females aged 65 years or more

• Males aged 70 years or more

• Postmenopausal females under the age of 65 years with risk factors

• Fracture after the age of 50 years

• Males aged 50-69 with risk factors

Osteoporosis is a common metabolic bone disease characterized by a diffuse reduction in bone density due to a decrease in bone mass. It is typically asymptomatic until fractures occur. These fractures are commonly low trauma in nature, defined as a fall from standing height or less. The initial fracture that often occurs is in the distal radius, known as Colles' fracture, followed by fractures in the vertebrae and hip unless treatment is initiated.

A DEXA scan is the investigation of choice in diagnosing osteoporosis and the results are reported in terms of T-score. According to WHO, a T score of

• -1 or higher is considered normal

• -1.01 to -2.49 is defined as osteopenia.

• -2.5 or lower is defined as osteoporosis

Bisphosphonates like alendronate are the drugs of choice for fracture prevention in postmenopausal women. They specifically inhibit bone resorption and thus are very effective in treating osteoporosis. They concentrate at sites of active remodeling by remaining in the matrix until the bone is remodeled. Then, they are released in the acidic environment of the resorption lacunae and induce apoptosis in osteoclasts.

Other treatment options for postmenopausal osteoporosis are:

• Denosumab - This is an antibody to RANKL that promotes osteoclastogenesis. Inhibiting RANKL thus decreases osteoclastic activity.

• Parathyroid hormone - Preotact and teriparatide are anabolic hormones. They promote bone formation when given at intermittent low doses. Aboloparatide is a new drug in this group that has been tried for this condition.

• Raloxifene - It is a selective estrogen receptor modulator (SERM) approved and shown to be effective in preventing vertebral fractures.

• Strontium ranelate - It is thought to promote osteogenesis but is strongly contraindicated in patients prone to thromboembolic events.

• Romosozumab - Sclerostin inhibits osteoblastic activity. Romosozumab shows anti-sclerostin activity, thereby increasing bone formation.

Solution to Question 11:

Among the given options, Ilizarov> Biplanar frame/Ring with cylindrical rod> Uniplanar with double rod> Uniplanar with a single rod (1>4>3>2) represents the correct order of stability of the given external fixators.

External fixation provides rigid fixation of the bones. This is commonly used in severe open types II and III fractures, as the application of a cast, or traction methods and internal fixation is not recommended

The principle of external fixation is the transfixion of bone above and below the fracture with Schanz screws or tensioned wires and these are then connected to each other by rigid bars. There are numerous types of external fixation devices:

• A unilateral frame with a support member and half-pins on one plane constitute a unilateral one-plane configuration. The addition of a second support member and the second plane of half-pins form a unilateral two-plane configuration.

• Transfixation pins connected to support members at either end are used to build the bilateral one-plane configuration. The addition of a second plane of half-pins forms a bilateral two-plane configuration.

• Ring fixators consist of complete or partial rings connected by rods. The rings are fixed to the bone via half pins or wires under high tension.

Circular external fixators confer greater stability and often negate the need to span joints. Unilateral external fixation is commonly used as a temporary method of fixation. Its disadvantages include the need to span joints for sufficient stability. Uniplanar fixation devices are quick and easy to apply but are less sturdy than multiplanar fixation. Bilateral frames are created when the pins are on both sides of the bone and can also add additional stability.

Given below is an Ilizarov ring fixator.

Ilizarov apparatus

Solution to Question 12:

The clinical presentation of this patient with a history of tibial fracture complaining of intense pain, with painful passive extension and loss of sensation of the first web space upon the removal of the cast, is suggestive of acute compartment syndrome. The next step would be to measure the anterior compartment pressure to confirm the diagnosis.

Acute compartment syndrome refers to the rise in intra-compartmental pressure leading to tissue ischemia and necrosis. The injuries at high risk of developing acute compartment syndrome are fractures of the proximal one-third of the tibia, soft tissue injury, distal radius fracture, and diaphyseal fracture of the forearm. Other high-risk injuries are crush injuries and circumferential burns.

The classic features of ischemia in acute compartment syndrome include the 5 Ps:

• Pain: is the first symptom of acute compartment syndrome. On passive stretching of the muscles, severe pain can be elicited.

• Paresthesia: paresthesia and hypoesthesia are the first signs of nerve ischemia.

• Paralysis: paralysis of muscle groups affected in that compartment is a late sign.

• Pulselessness: usually, the peripheral pulses and capillary return are intact in the early stages, unless there is a major arterial injury. If pulses are absent, arteriography is indicated.

• Pallor: may not always be seen as ischemia occurs at the capillary level.

The diagnosis can be confirmed by measuring the intra-compartmental pressure using a split catheter introduced into the compartment. The difference between diastolic pressure and compartment pressure is known as differential pressure (ΔP). If the differential pressure (ΔP) is less than 30 mm Hg, immediate compartment decompression is indicated.

Treatment:

• Prompt decompression of the affected compartment is done.

• All the dressings, bandages, and cast on the affected limb are removed and the limb is nursed flat.

• Fasciotomy is done if the differential pressure is less than 30 mm Hg

• Fasciotomy wounds are left open and observed after 2 days. If there is necrosis, debridement is done. If the wounds are healthy, they can be sutured without tension or skin grafting can be done.

Solution to Question 13:

Option C is the correct answer. The given lesions and the correctly matched best modes of management are as follows:

Osteosarcoma: is the most common primary malignant tumor of the bone. It is a high-grade medullary osteoid-producing tumor. The preferred method of treatment of osteosarcoma is wide resection and reconstruction. This is mainly owing to the possibility of limb salvage. Amputation is considered only when tumor-safe surgical margins are compromised.

Multi-agent neoadjuvant chemotherapy with agents like doxorubicin, cisplatin, ifosfamide, and methotrexate is started immediately after diagnosis for 8-12 weeks. The tumor is then re-staged followed by wide resection and limb reconstruction.

Radiographic findings include - a periosteal reaction is the typical finding and it may take the form of a Codman's triangle or it may have a sunburst (sunray) appearance on X-ray. It is shown in the image below:

Bone tumor	Management
1. Osteosarcoma	a. Wide excision

Bone tumor	Management
2. Simple bone cys t	d. Intralesional steroid
3. Giant cell tumo r	b. Curettage
4. Osteoid osteom a	c. Radiofrequency ablatio n

Simple bone cysts: these are cystic bone cavities that are often solitary and unilocular filled with serous or serosanguinous material. The lesions usually regress with bone maturity. In cases where the lesions are active, an injection of steroids is done. Only when there is a high risk of fracture, curettage or fixation and stabilization may be required.

It is seen as a well-outlined lytic centrally placed lesion with cortical thinning on X-ray. When there is a fracture, a small fragment may be visualized within the cavity, known as the fallen leaf sign. The image below shows a bone cyst in the humerus along with a pathological fracture and the fallen leaf sign:

Giant cell tumor: it is a benign, locally aggressive tumor of the bone due to the proliferation of mononuclear cells with scattered macrophages.

In stages 1 and 2, intralesional curettage with debridement of the lesional wall is effective. Stage 3 of the disease can be treated through curettage but some cases require en-bloc resection. Antibodies like denosumab (anti-RANKL) have been effective in controlling osteoclastic activity.

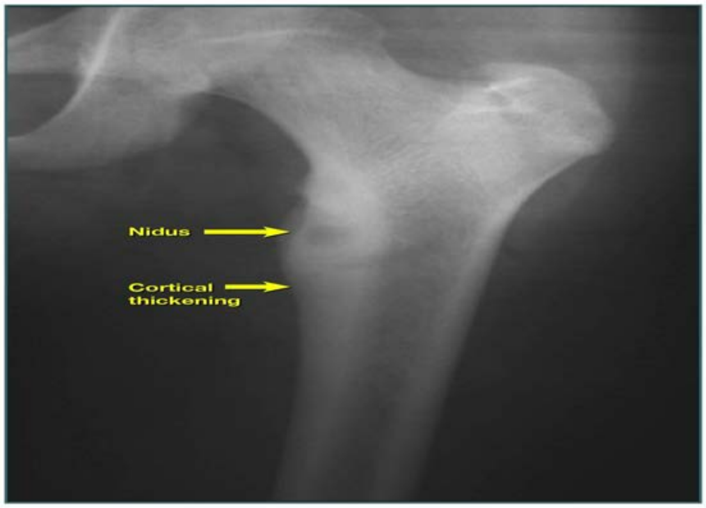
Radiographically, they are observed as an eccentrically located solitary lytic lesion in the metaphyses. They show a characteristic soap bubble appearance and periosteal new bone formation is absent.

Below image shows the soap bubble appearance seen in a giant cell tumor:

Osteoid osteoma: it is a benign tumor formed of osteoid and woven bone surrounded by a halo of reactive bone. The preferred method of treatment in this condition is CT-guided radiofrequency ablation to eradicate the lesion and improve symptoms.

Radiography shows a sclerotic peripheral rim with central radiolucent nidus:

Osteoid osteoma



Solution to Question 14:

In nerve injuries, an intact outer sheath and nerve fiber with damaged inner axons is referred to as axonotmesis.

According to Seddon's classification, peripheral nerve injuries can be classified as follows:

• Neurapraxia: This refers to the temporary failure of nerve function without physical axon disruption. There is no degeneration, and normal axonal function typically returns within hours to months, often within the 2-4 week range.

• Axonotmesis: This involves the disruption of axons and myelin while the surrounding connective tissue (including endoneurium and perineurium) remains intact. Axonal degeneration occurs both proximally and distally from the site of injury. Distal degeneration is known as Wallerian degeneration. Recovery may occur after 18 months.

• Neurotmesis: This refers to the disruption of axons and endoneurial tubes, with the degeneration of both proximal and distal axons.

Axonapraxia is a disruption to the axoplasmic flow of neurochemicals temporarily.

Orthopedics NEET 2021

Question 1:

A 20-year-old male patient presented with a history of lower backache and early morning stiffness for two years. He also gave a history of bilateral heel pain since 6 months. Which of the following is the most likely diagnosis?

a) Ankylosing spondylitis

b) Tuberculosis of spine

c) Disc prolapse

d) Mechanical pain

Question 2:

A 5-year-old child was brought to the pediatrician with complaints of bilateral knee joint pain. The x-ray image of the joints is given below. What is the most likely diagnosis?

a) Scurvy

b) Rickets

c) Metaphyseal dysplasia

d) Pycnodysostosis

Question 3:

The following test is done to assess

a) Posterior interosseous nerve

b) Median nerve

c) Ulnar nerve

d) Musculocutaneous nerve

Question 4:

A 30-year-old male patient presented with complaints of a gradually progressive swelling around his wrist joint for 3 months. Given below is the image of the swelling and the X-ray film. What is the most likely diagnosis?

a) Ewing's sarcoma

b) Osteosarcoma

c) Osteoclastoma

d) Osteochondroma

Question 5:

A patient was brought to the hospital with complaints of pain around the left hip joint following a road traffic accident. On examination, the affected limb was flexed, adducted, and medially rotated with obvious shortening. What is the most likely diagnosis?

a) Anterior hip dislocation

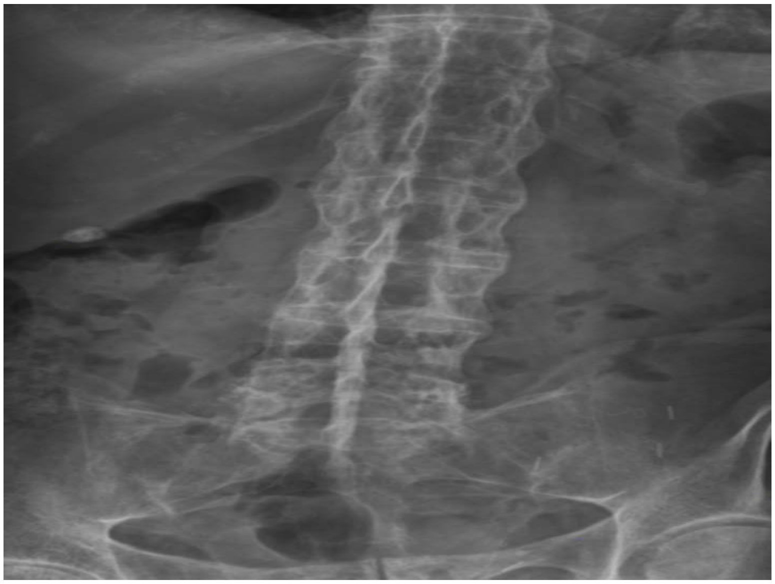
b) Posterior hip dislocation

c) Transcervical fracture

d) Intertrochanteric fracture

Question 6:

A 28-year-old man presented with complaints of backache, morning stiffness, and redness of the eyes. An X-ray image of the spine is given below. Which of the following is the most likely diagnosis?



a) Rheumatoid arthritis

b) Paget's disease

c) Osteopetrosis

d) Ankylosing spondylitis

Answer Key

Question No.	Correct Option
1	a
2	a
3	b
4	c
5	b
6	d

Detailed Explanations

Solution to Question 1:

The given clinical scenario of a young male presenting with lower backache, heel pain, and early morning stiffness is suggestive of ankylosing spondylitis.

Ankylosing spondylitis is commonly seen in young males and has a strong association with HLAB27. The patient usually presents with complaints of recurring backache and early morning stiffness. Later the patient can develop other symptoms like generalized fatigue, pain, swelling of the joints, foot strain, tenderness at insertion of the achilles tendon, and intercostal pain.

Extra-articular manifestations include iritis, pericarditis, aortic incompetence, apical lobe fibrosis, generalized osteoporosis, and atlantoaxial joint subluxation.

On examination, there is a loss of normal lumbar lordosis, increased thoracic kyphosis, diminished spinal movements, and decreased chest expansion.

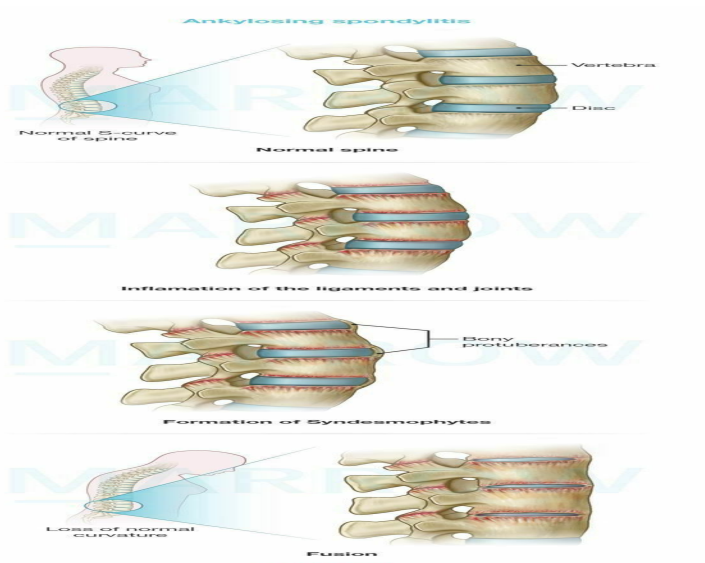
Other findings are:

• Positive Fleche's test - It is done to assess cervical spine involvement. Here the patient is asked to touch the wall with the back of the head without raising the chin.

• Positive tests for sacroiliac joint involvement such as the pump handle test, pelvic compression test, and Fabre’s test.

ASAS diagnostic criteria for ankylosing spondylitis:

Sacroiliitis on imaging plus ≥ 1 SpA feature:	or	HLA-B27 +ve plus ≥2 other SpA features
Sacroiliitis on imaging:Active (acute) inflammation on MRI highly suggestive of SpA-associated sacroiliitisand/orDefinite radiographic sacroiliitis according to modified New York criteria (grade >2 bilateral or grade 3 - 4 unilateral)		SpA features:Inflammatory back painArthritisEnthesitis (heel)Anterior uveitisDactylitisPsoriasisCrohn’s disease or ulcerativecolitisGood response to NSAIDsFamily history of SpAHLA-B27Elevated CRP



X-ray findings in ankylosing spondylitis:

• Sacro-iliac joints:

• Cardinal sign/earliest sign– erosion and fuzziness of the sacroiliac joints

• Later- peri-articular sclerosis and bony ankylosis of the joint

• Vertebral changes:

• Earliest- flattening of the normal anterior concavity of the vertebral body ('squaring')

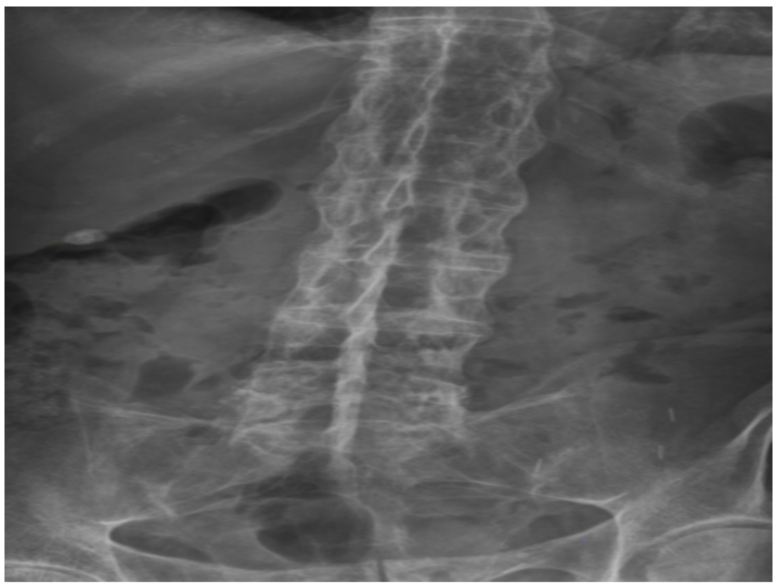
• Later- ossification of the ligaments around the intervertebral discs produces syndesmophytes (bridges) between adjacent vertebrae

• Bamboo spine appearance- due to bridging at several vertebral levels

• Long-standing cases are associated with osteoporosis. Wedging of the vertebral bodies produces hyperkyphosis of the spine.

• Progressive bony ankylosis or erosive arthritis may be seen in peripheral joints.

Bamboo-spine appearance is seen in the Xray of the lumbar spine below:



Other options:

Option B: TB spine (Pott's disease) is the commonest form of bone and joint tuberculosis. The clinical presentation can be variable with pain being the common symptom. The pain can be diffuse initially but later on, it gets localized to the affected segment. Stiffness is also seen as an early symptom. The patient can also present with swelling due to the formation of a cold abscess. Paraplegia and spinal deformities can occur at later stages.

Option C: Disc prolapse refers to protrusion or extrusion of the nucleus pulposus through a rent in the annulus fibrosis. The patient presents with low backache which can be acute or chronic. The patient can also present with pain radiating to the gluteal region, back of the thigh, and leg. This is referred to as sciatic pain. There can be paraesthesia in the dermatomes of the affected nerve roots.

Option D: Mechanical back pain is a broader term that includes all the causes that cause alternation in the spine mechanics. It includes conditions like osteoarthritis, lumbar spondylosis, osteoporosis, etc.

Solution to Question 2:

The clinical presentation and the x-ray showing Pelkan spurs, Trummerfeld zone, and Frenkel's line are suggestive of scurvy.

Rickets can also present with bilateral knee pain, but normal bone mineral density, lack of radiographical signs such as cupping and fraying of the metaphysis and widening of the growth plate makes it an unlikely diagnosis.

Radiological findings in scurvy:

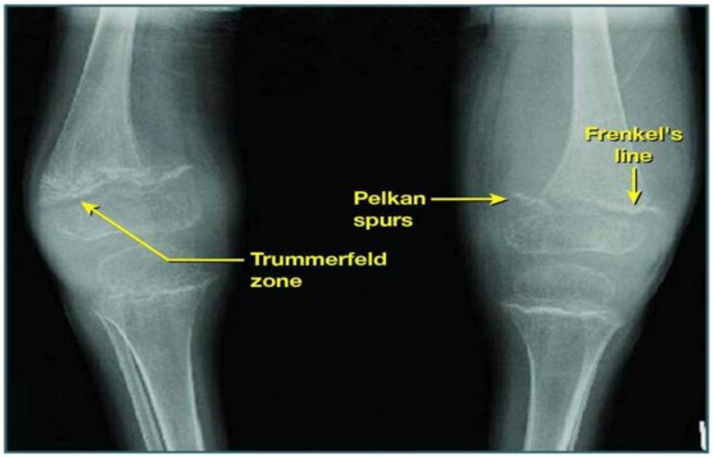
• Frenkel's line - The zone of provisional calcification at the growing metaphysis is seen as a dense, white line called the Frankel's line.

• Pelkan spurs - These are metaphyseal spurs formed due to fractures manifesting at the cortical margins.

• Wimberger's ring sign - The sclerotic rim around the epiphysis forms this appearance.

• Trummerfeld zone. - Lucent metaphyseal band underlying Frankel line.

Scurvy



Scurvy results from vitamin C deficiency, which is required for the hydroxylation of lysine and proline in procollagen. There is a defective formation of connective tissue and collagen in the skin, bone, cartilage, dentine, and blood vessels leading to their fragility.

Early manifestations include irritability, loss of appetite, musculoskeletal pain, and tenderness in the legs. Later on, the child can have swelling of the knees, ankles, and pseudo paralysis. Hemorrhagic manifestations of scurvy include petechiae, purpura, ecchymoses at pressure points, epistaxis, gum bleeding, and characteristic perifollicular hemorrhages.

Other manifestations are poor wound and fracture healing, hyperkeratosis of hair follicles, arthralgia, and muscle weakness. It also results in defective osteoid formation. Costo-chondral separation leads to the formation of tender nodules, which are sharply angulated also known as the scorbutic rosary.

Other options:

Option B: The findings in rickets are given in the pearl below.

Option C: Metaphyseal dysplasia (Pyle's disease) is an autosomal recessive condition caused due to failure of remodeling of the metaphysis. Erlenmeyer flask deformities of the distal femur and proximal tibia are seen in this condition. The patient can also develop genu valgum.

Option D: Pycnodysostosis is an autosomal recessive condition due to a mutation in the cathepsin K (CTSK) gene. It causes osteosclerosis and the affected individual has distinctive facial features and skeletal malformations. The patient has short stature and is prone to recurrent fractures due to their fragile bones.

Solution to Question 3:

The given image shows the pen test which is used to assess the median nerve.

It tests the abductor pollicis brevis muscle which is supplied by the median nerve. The hand must remain flat with the palm facing upwards; a pen is held above the palm and the patient is asked to touch the pen with his thumb - positive sign is unable to touch the pen.

Median nerve lesions can be high lesions or low lesions depending on the site of injury.

Low lesions - It occurs due to injuries at the distal third of the forearm such as carpal dislocation or cut wound of the wrist. There is paralysis of the hand muscles; the patient is unable to abduct the thumb and has a loss of sensation over the radial three-and-a-half digits. The forearm muscles supplied by the median nerve are spared in these lesions. Wasting of thenar eminence is seen in long-standing cases.

High lesions - It occurs due to injuries above the elbow such as forearm fractures or elbow dislocation. There is paralysis of the forearm muscles along with all the features of low lesions.

Muscles supplied by the median nerve are given in the table:

Other tests for median nerve:

Pointing index sign or Oschner's clasp test: The patient is asked to clasp both hands together. When this is done the index and the middle finger fail to flex due to the loss of action of long finger flexors which are supplied by the median nerve.

Benediction test: Due to loss of activity of the long finger flexors the patient is unable to flex the index and middle finger on lifting the hand. As this is the position a clergyman uses to bless the couple during the marriage, it is called the benediction test.

Level	Motor supply
In the arm	No muscles
Proximal one-third of the for earm	All the flexor muscles of the proximal forearm, except the flexor carpi ulnaris and the medial half of the flexor digit orum profundus

Level	Motor supply
Distal one-third of the forear m	No muscles
In the hand	Abductor pollicis brevisOppo nens pollicisFlexor pollicis br evisFirst two lumbricals

Solution to Question 4:

The clinical presentation and the x-ray showing an eccentric expansile lesion with a soap bubble appearance in a 30-year-old man are suggestive of osteoclastoma or giant cell tumor (GCT).

Osteoclastoma usually affects young adults in the age group of 20-40 years. The most common location of the tumor is the distal femur followed by the proximal tibia. Involvement of the distal end of the radius is also seen and the tumor is more aggressive in this location.

Radiographic findings in GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis

• Eccentric location, most often subchondral

• Expansion of the overlying cortex

• Soap bubble appearance - The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called soap bubble appearance

• No calcification within the tumor

• No or minimal reactive sclerosis around the tumor.

Treatment :

• Extended curettage and bone grafting.

• En-bloc wide resection and reconstruction are done for advanced stages, local recurrence, and tumors not responding to other methods.

• Resection without reconstruction is done for lesions in expandable bones like distal ulna, clavicle, etc.

• Irradiation and/or embolization are used for inoperable lesions in the spine and pelvis.

• Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease.

• Denosumab is approved for use in unresectable tumors in skeletally mature patients.

Other options:

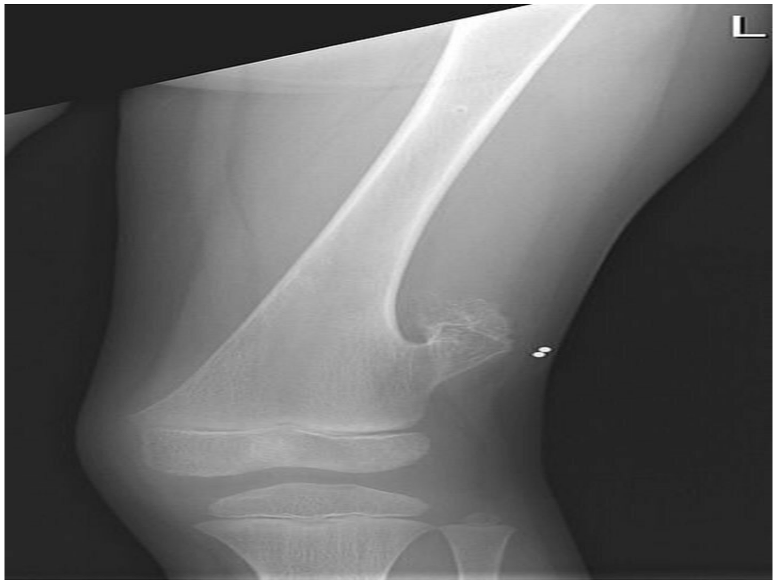
Option A: Ewing's sarcoma is a primary malignant tumor of the bone commonly involving the age group of 4-25 years involving the diaphysis of the long bones like the femur, tibia, fibula, and humerus.

X-ray shows onion-peel appearance due to deposition of periosteal reaction in layers is the typical finding.

Option B: Osteosarcoma is the most common non-hematologic primary malignancy of bone. It occurs most commonly in the second decade of life and is typically metaphyseal. The distal femur, proximal tibia, and proximal humerus are the common sites.

X-ray shows periosteal reaction and it may take the form of a Codman's triangle or have a sunburst (sunray) appearance.

Option D: Osteochondromas are common benign bone tumors. The lesions consist of a bony mass, often in the form of a stalk, produced by endochondral ossification of a growing cartilaginous cap. The common sites of involvement are the distal femur, proximal tibia, and proximal humerus.



Solution to Question 5:

The given clinical vignette showing a flexed, adducted, and medially rotated limb with apparent limb shortening following trauma is suggestive of posterior hip dislocation.

Posterior hip dislocation is the most common type of hip dislocation commonly seen following dashboard injuries.

X-ray findings:

• Upward dislocation of the femoral head

• Femoral head appears smaller than normal

• Lesser trochanter is not visualized as the thigh is internally rotated

• Femoral head and posterior acetabular wall fractures may also be seen.

Treatment: Early reduction should be done as the risk of osteonecrosis and subsequent osteoarthritis increases with time.

Closed reduction under sedation or general anesthesia is done in most cases. Closed reduction should not be attempted in associated femoral neck fractures as it can lead to the displacement of the femoral neck and disrupt the blood supply to the femoral head.

If there is a failure of closed reduction or any significant fractures then surgical repair should be done. If there is a delay in surgical correction skeletal traction with a distal femoral traction pin can be done.

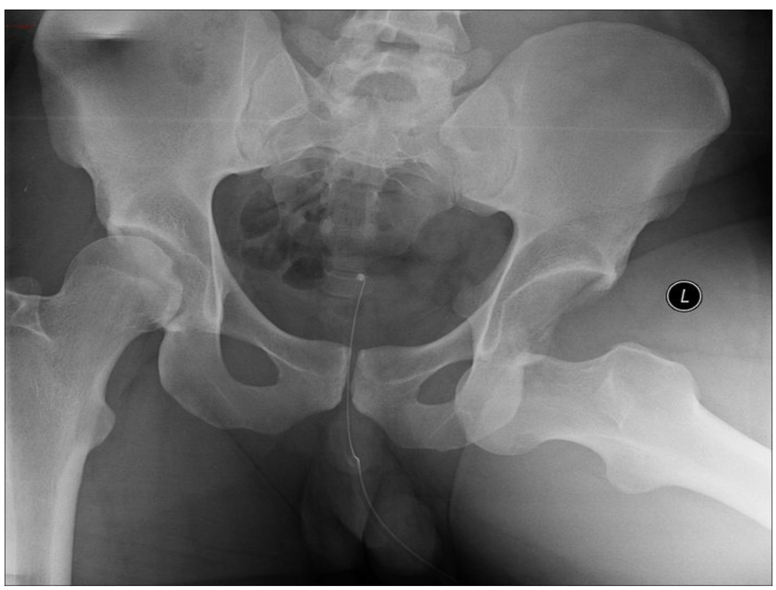
Early complications include sciatic nerve injury, vascular injury (superior gluteal artery tear can occur), and concurrent femoral shaft fracture (when both occur together hip dislocation may be missed).

Late complications include osteonecrosis of the femoral head, myositis ossificans, secondary osteoarthritis, and recurrent instability.

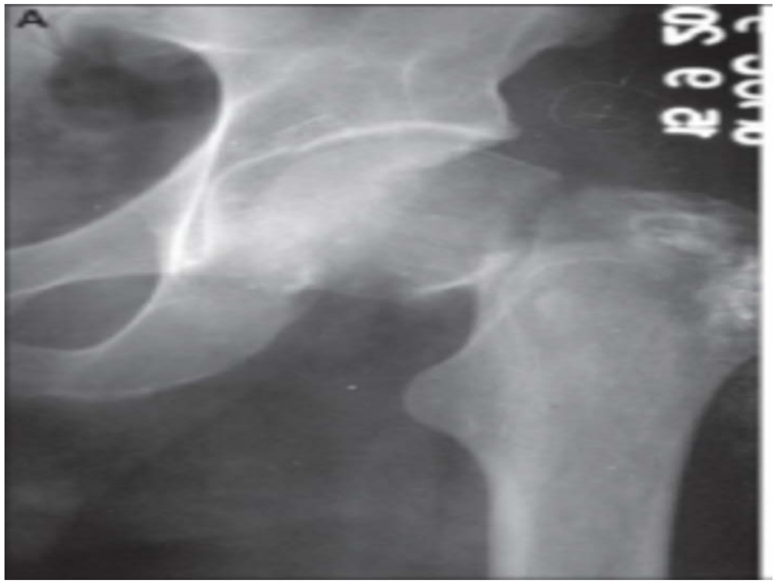
The image given below shows posterior hip dislocation on the left side.

Other options:

Option A: Anterior hip dislocations are less common than posterior hip dislocations. This injury is usually sustained following a high-energy accident. The affected limb will be abducted, externally rotated, and slightly flexed.



Option C: Transcervical fracture is a type of fracture neck of the femur where the fracture line is in the middle of the neck. In this case, there can be shortening and external rotation of the affected limb.



Option D: Intertrochanteric fracture is usually sustained following a fall or road traffic accident. The leg is short and externally rotated. Clinical findings are more marked than the neck of femur fractures.

Solution to Question 6:

The clinical presentation of a young male with backache, morning stiffness, redness of the eye, and the x-ray showing the classical bamboo-spine appearance are suggestive of ankylosing spondylitis.

Ankylosing spondylitis commonly involves young male adults. It has a strong association with HLAB27.

Radiographic findings in ankylosing spondylitis:

• Sacro-iliac joints:

• Cardinal sign/earliest sign– erosion and fuzziness of the sacroiliac joints

• Later- peri-articular sclerosis and bony ankylosis of the joint

• Vertebral changes:

• Earliest- flattening of the normal anterior concavity of the vertebral body ('squaring')

• Later- ossification of the ligaments around the intervertebral discs produces syndesmophytes (bridges) between adjacent vertebrae

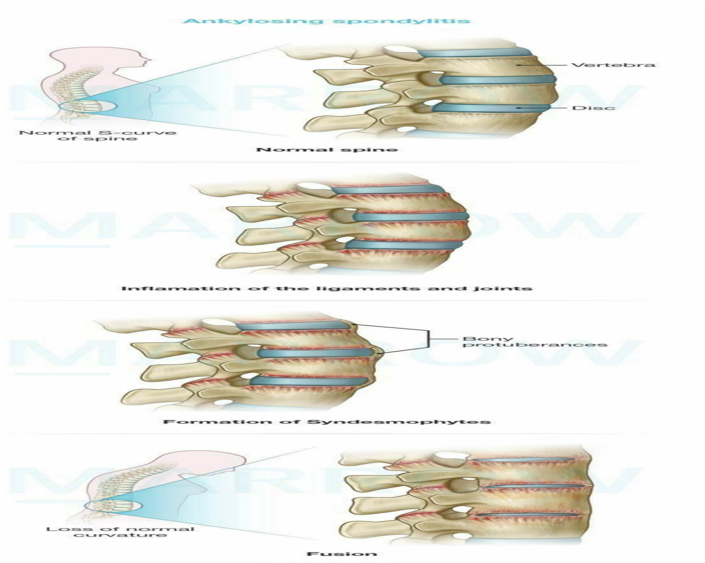
• Bamboo spine appearance- due to bridging at several vertebral levels

• Long-standing cases are associated with osteoporosis. Wedging of the vertebral bodies produces hyperkyphosis of the spine.

• Progressive bony ankylosis or erosive arthritis may be seen in peripheral joints.

ASAS diagnostic criteria for ankylosing spondylitis:

Sacroiliitis on imaging plus ≥ 1 SpA feature:	or	HLA-B27 +ve plus ≥2 other SpA features
Sacroiliitis on imaging:Active (acute) inflammation on MRI highly suggestive of SpA-associated sacroiliitisand/orDefinite radiographic sacroiliitis according to modified New York criteria (grade >2 bilateral or grade 3-4 unilateral).		SpA features:Inflammatory back painArthritisEnthesitis (heel)Anterior uveitisDactylitisPsoriasisCrohn's disease or ulcerativecolitisGood response to NSAIDsFamily history of SpAHLA-B27Elevated CRP



Other options:

Option A: Rheumatoid arthritis is a chronic inflammatory disease with an increased incidence between 25 and 55 years of age. Clinical features include swelling and pain in the small joints of the hands. Metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints are commonly involved. The patient also has morning stiffness in the affected joints.

Option B: Paget's disease, also known as osteitis deformans is characterized by an increased bone turnover due to a disorganized, alternating, and increased rate of osteoblastic and osteoclastic activity. It is seen more commonly in people over the age of 50 and most of the patients are asymptomatic. Symptomatic patients present with pain or deformity, with the pelvis and tibia being the commonly involved sites.

Option C: Osteopetrosis/Albers–Schönberg disease occurs due to a defect in the genes that control the osteoclastic activity. It can be of two types:

• Infantile form(autosomal recessive)- The infant presents with growth retardation, abnormal skull development, failure to thrive, and genu valgum. Pancytopenia and hepatosplenomegaly occur as a result of obliteration of the medullary canals and the subsequent extra-medullary hemopoiesis

• Adult form (autosomal dominant) - The patient can be asymptomatic or can present with recurrent fractures and compressive neuropathies. Hemopoiesis is unaffected.

Orthopedics INI-CET 2022

Question 1:

The most common pattern of involvement in Pott's spine is

- **a)** Paradiscal
- **b)** Central
- **c)** Anterior
- **d)** Posterior

Question 2:

Anterior dislocation of the shoulder is likely to result in the damage of which of the following nerves?

- **a)** Axillary nerve
- **b)** Median nerve
- **c)** Radial nerve
- **d)** Suprascapular nerve

Question 3:

A 60 year old patient presents with limping gait and left hip pain. What is the likely diagnosis?

a) Tuberculosis of the hip

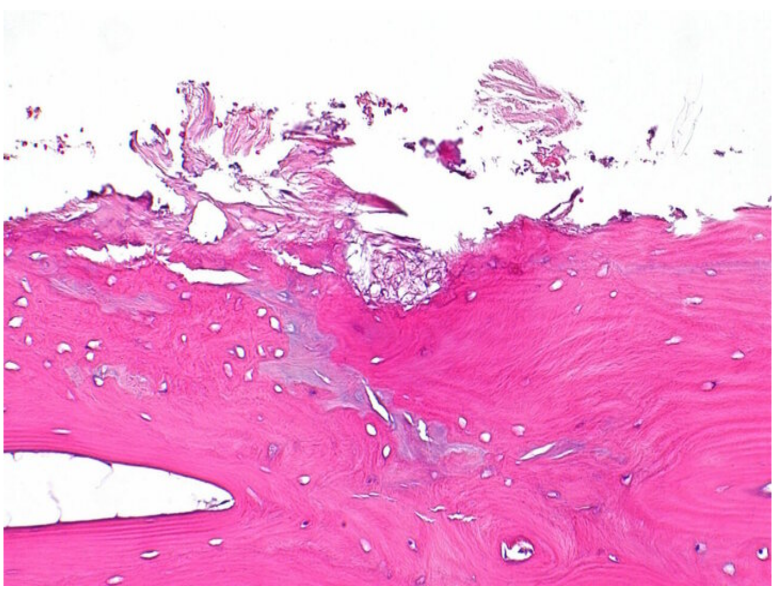
b) Rheumatoid arthritis

c) Ankylosing spondylitis

d) Avascular necrosis

Question 4:

A 72-year-old man came with complaints of unilateral knee pain. X-ray examination showed reduced joint space. Pathological findings shown below are of:



a) Osteoarthritis

b) Tuberculosis of the knee

c) Rheumatoid arthritis

d) Lyme disease

Question 5:

Which of the following is false regarding hip dislocation?

a) Posterior dislocation of hip has flexion and adduction

b) Vessel injury more common in anterior

c) Nerve injury more common in anterior

d) Posterior dislocation is more common than anterior

Question 6:

A 35-year-old man presented with knee pain. The x-ray of his knee joint is given below. What is the probable diagnosis?

a) Chondroblastoma

b) Osteochondroma

c) Osteoclastoma

d) Fibrous dysplasia

Question 7:

Book test is used to assess the function of adductor pollicis. Which of the following nerves supplies this muscle?

a) Posterior interosseous nerve

b) Ulnar nerve

c) Radial nerve

d) Median nerve

Question 8:

A 6-year-old girl sustained a fracture of the lateral condyle of the humerus which led to the development of elbow deformity. Which of the following findings can be seen in this patient?

a) Ape thumb deformity

b) Pointing index sign

c) Inability to adduct the fingers

d) Kiloh-Nevin sign

Question 9:

A 65-year-old postmenopausal female patient comes with persistent backache not responding to conservative treatment. She has a history of lifting heavy weights 4 months back and progressively increasing pain that worsens on walking. The pain starts radiating down her legs after walking for a distance of around 100 m. However, she feels no pain on walking uphill. What is the most probable diagnosis?

a) Osteoporotic compression fracture

b) Lumbar stenosis

c) Thromboangiitis obliterans

d) Atherosclerosis

Question 10:

Disruption of the axon with an intact endoneurium is seen in which of the following?

a) Neurapraxia

b) Axonapraxia

c) Axonotmesis

d) Neurotmesis

Question 11:

A patient presents to the OPD with Trendelenburg sign positive on one side. Which of the following muscles are involved?

a) 2 only

b) 1,3

c) 2,4

d) 1,4

Question 12:

You visit a newborn near your house and notice that the baby has CTEV, as shown below. When would you advise the baby's parents for a consultation to put on a cast?

a) In 1 month

b) In 6 months

c) In a few days

d) When the child starts walking

Question 13:

Combination of appearance in CTEV

a) Equinus, eversion, forefoot adduction, cavus

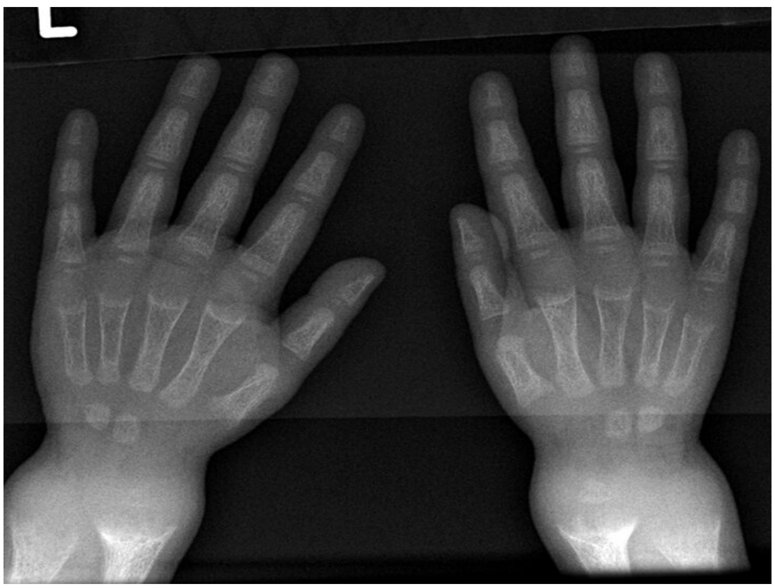
b) Equinus, inversion, forefoot adduction, cavus

c) Equinus, eversion, forefoot abduction, cavus

d) Equinus, inversion, forefoot adduction, planus

Question 14:

A child presents with poor growth and swelling at joints. A radiograph of his wrist is given below. Lab investigations reveal serum ALP levels of >1500. What is the possible diagnosis?



a) Scurvy

b) Rickets

c) Osteogenesis imperfecta

d) Paget's disease

Question 15:

What is not required on an emergency basis in an open fracture?

a) 1, 2, 3

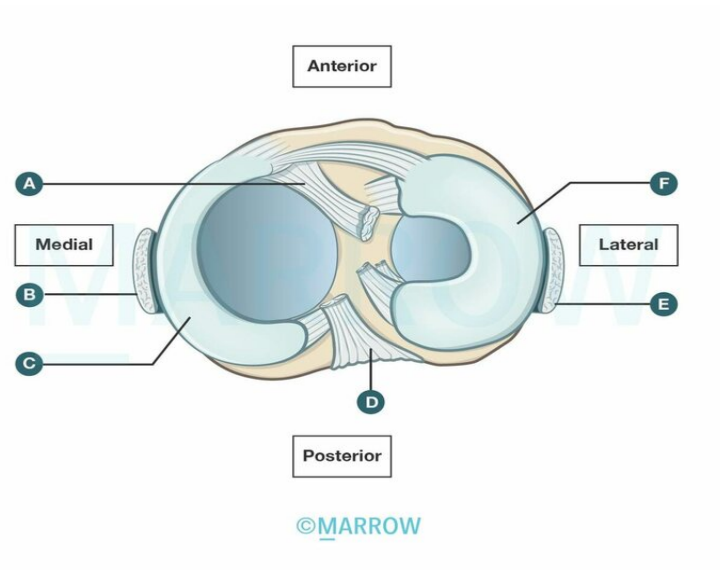
b) 2 only

c) 2 and 4

d) 2 and 3

Question 16:

Which of the following structures are involved in the unhappy triad?



a) A, D, E

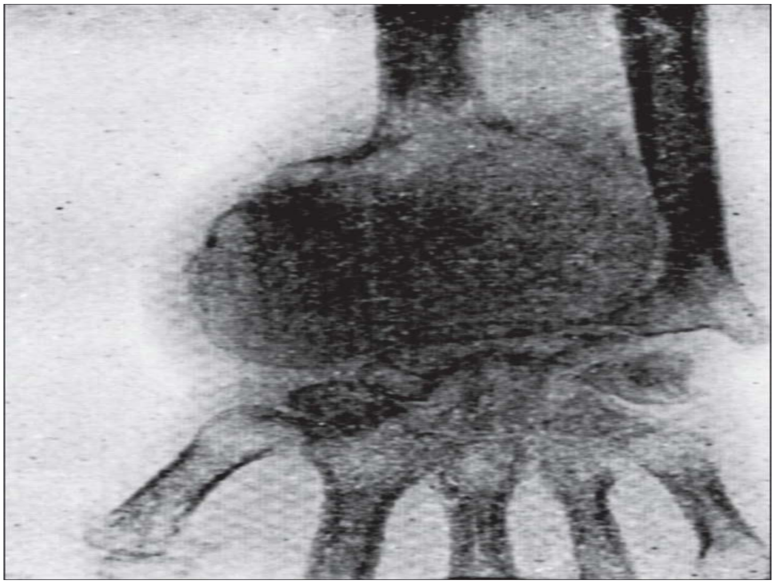
b) D, E, F

c) A, B, C

d) C, D, E

Question 17:

A radiograph of the wrist is given below. Identify the true statements regarding this condition.



a) 1, 3

b) 2, 3

c) 3 only

d) 2, 4

Answer Key

Question No.	Correct Option
1	a
2	a
3	d
4	a
5	c
6	c
7	b
8	c
9	b
10	c
11	a
12	c
13	b
14	b

15	b
16	c
17	c

Detailed Explanations

Solution to Question 1:

The most common pattern of involvement in Pott's spine is the paradiscal type.

Pott's spine occurs secondary to a pulmonary or gastrointestinal primary and spreads through the Batson's plexus of veins (paravertebral plexus) and most commonly involves the dorso-lumbar region.

Types of vertebral tuberculosis:

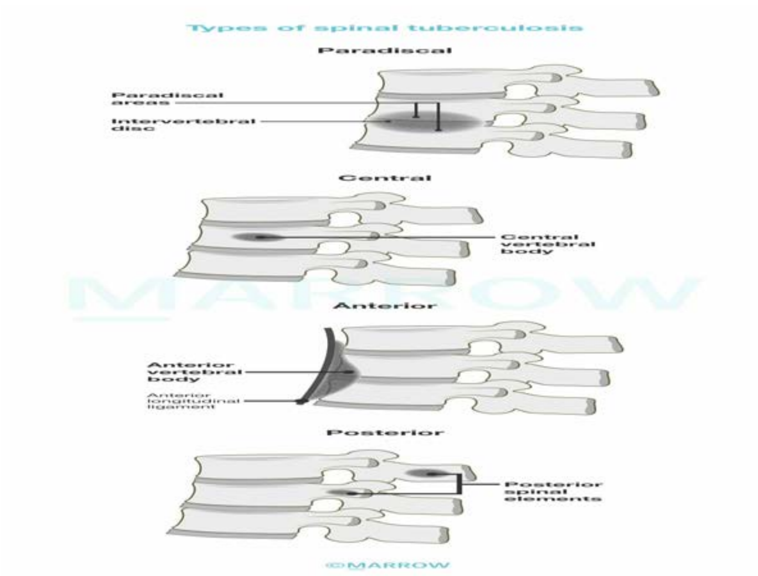
• Paradiscal - contiguous areas of two adjacent vertebrae and the middle intervening disc are affected. This is due to the high oxygen requirements of tubercular bacteria and the corresponding high blood supply in this region by the segmental vessels.

• Central - the body of the vertebra is involved leading to collapse (causing wedging or concertina collapse) of the affected vertebra. This leads to spinal deformities such as kyphosis and gibbus.

• Anterior - the anterior part of the body is affected and spreads through the anterior longitudinal ligament.

• Posterior - the posterior complex (pedicle, lamina, spinous process, and transverse process) is affected.

the image below shows various parts of the vertebra and patterns of involvement in Pott's spine:



Clinical features of spinal tuberculosis:

• Back pain, stiffness and reduced movements are the usual early complaints

• Cautious gait - a person with spinal tuberculosis will walk carefully with short steps to avoid pain resulting from movement

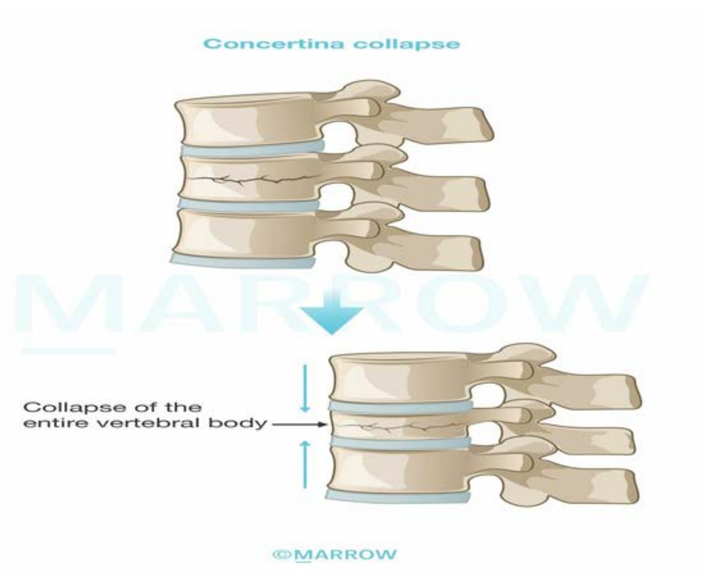
• Pott's paraplegia

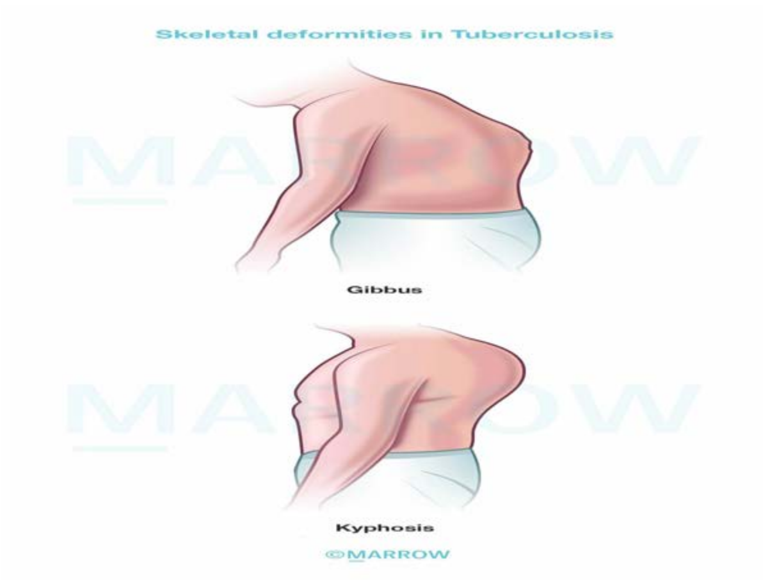
• Cold abscess

• Concertina collapse - a type of compression fracture without involvement of the interverterbral disc

• Deformity - kyphus and gibbus (gradually increasing prominence).

Images showing concertina collapse and the deformities seen TB spine.





Solution to Question 2:

Anterior shoulder dislocation is most likely to damage the axillary nerve.

Axillary nerve injury presents with an anaesthetic patch over the deltoid muscle, and deltoid muscle weakness (inability to abduct at the shoulder joint). It is usually a neuropraxia type of injury and resolves spontaneously. Other nerves that can be injured include the radial nerve, musculocutaneous nerve, median nerve or ulnar nerve.

Other complications of anterior dislocation of the shoulder joint:

• Early complications

• Rotator cuff injury is seen in older people and presents with difficult abduction (palpable contraction of deltoid muscle rules out nerve injury).

• Vascular injury: Axillary artery may be injured, especially in older individuals, either at the time of injury or during the reduction.

• Fracture-dislocation: There may be an associated fracture of the humerus (e.g. greater tuberosity) due to the traumatic injury. This may require open reduction and internal fixation.

• Late complications

• Recurrent dislocations: Most common sequelae of anterior dislocation. Usually seen when there is associated injury to the glenoid labrum or stripping of the capsule is present.

• Shoulder stiffness: Due to prolonged immobilization.

• Untreated dislocation: Sometimes it may go unnoticed, especially in older and unconscious individuals. The reduction can be tried up to 6 weeks after the dislocation, after which 'active neglect' (no treatment) is followed.

Radiograph showing anterior dislocation with humeral head lying below and medial to the glenoid socket.

Solution to Question 3:

The given scenario of limping gait and left hip pain with radiographic findings of the radiolucency and distorted articular surface is suggestive of avascular necrosis of the head of femur.

Avascular necrosis (AVN) or osteonecrosis is a painful bone condition as a result of diminished blood supply mainly due to trauma or non-traumatic etiologies.

The typical features are pain with movements in or near the joint, click in the joint, joint stiffness, limping gait, and deformity. Restriction of movements is also a feature of this condition.

An x-ray shows reactive new bone formation, increased radiolucency in the subchondral bone, and a distorted articular surface.

MRI is the investigation of choice. It allows the identification of AVN of the femoral head and also determines the exact stage and extent of the pathologic process.

Management is based on Ficat and Arlet Classification:

Other options:

Option A: Tuberculosis of the hip is characterized by painful and restricted movements of the hip. The radiographic findings are not appreciable in the early stages. But late stages show periarticular osteoporosis, hazy, and irregular joint margins with a reduction in the joint space, and wandering acetabulum.

Option B: Rheumatoid arthritis (RA) is usually seen in females in the 4th to 5th decade. Patients present with symmetrical polyarthritis of the proximal finger joints, tenosynovitis, early morning stiffness and pain. RA of the hip can be bilateral. Radiographic findings such as joint space narrowing due to hyaline cartilage destruction, cyst formation, and acetabular protrusion-type deformity can be seen.

Option C: Ankylosing spondylitis is a chronic inflammatory disease of the spine and sacroiliac joint. It typically occurs in the age group 15-25 years with a male preponderance. Features include backache, stiffness, sciatica, and fixed flexion contracture. Radiographic findings show erosion of sacroiliac joints, squaring, and bridging of the vertebral body leading to a bamboo spine appearance.

Stage	Findings	Management	Method
1	X-ray normal, MRI +ve	Conservative	Bisphosphonates
2a	X-ray: sclerosis, cysts, no collapse	Conservative	Bisphosphonates
2b	X-ray: crescent sign, no collapse	Surgical decompression	Core decompression
3	Collapse+loss of sphericity	Surgical	Osteotomy
4	Advanced arthritis	Reconstructive	Total hip replacement

Solution to Question 4:

The scenario of unilateral knee pain, the radiograph findings of reduced joint space, and the histopathology findings of loss of articular cartilage suggest osteoarthritis.

Osteoarthritis is a type of joint disease characterized by degeneration of cartilage leading to synovial joint destruction. Individuals with age > 50 years and overweight are usually affected.

The disease typically involves the knee and initially affects a single joint; eventually, both knees are affected. Pain is the predominant symptom and worsens with use. Other symptoms such as joint stiffness, swelling, giving away or locking can be present. Fixed flexion deformity can be seen in these patients.

A radiograph of the knee joint, the anteroposterior and lateral view is useful to establish a diagnosis. The characteristic findings are the diminution of joint space, subchondral sclerosis, subchondral cysts, osteophytes, and soft tissue calcifications.

Histopathology shows characteristic loss of the articular cartilage, congestion, fibrosis and inflammatory cells.

Management is as follows

• Conservative measures such as weight loss, exercise, physiotherapy for strengthening of the quadriceps muscle, analgesics for pain relief and usage of walking sticks for reducing the load of joints.

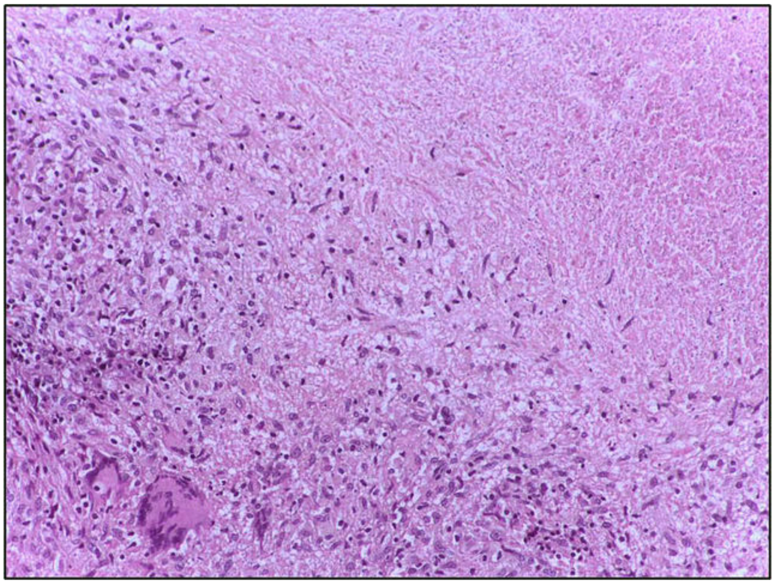
• Intraarticular injections of corticosteroids and hyaluronidase are useful.

• Operative measures such as osteotomy and total knee replacement surgery.

Other options:

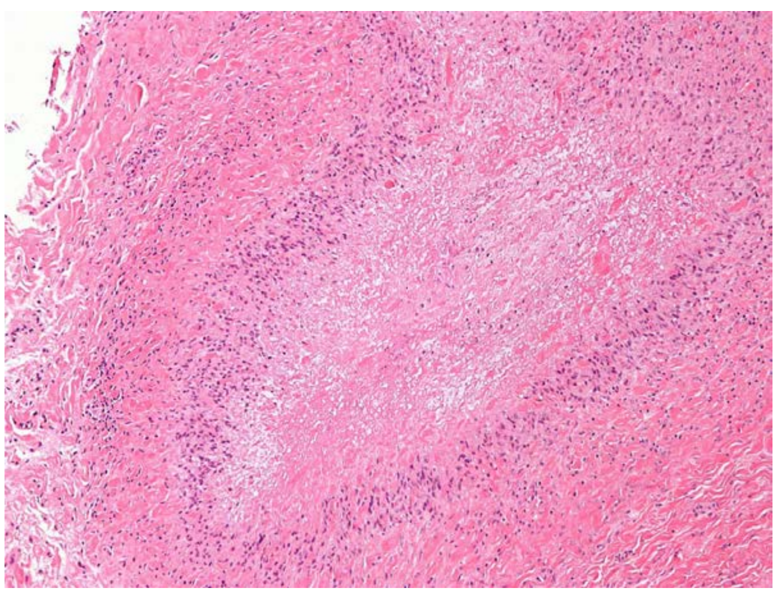
Option B: Tuberculosis of the knee is more common in childhood with pain and limping being the common symptoms. X-rays show an enlargement of the epiphysis. The characteristic histopathologic findings are caseating and noncaseating granulomas with surrounding epithelioid and multinucleate giant cells.

The below image shows caseating granuloma with Langhans giant cells:



Option C: Rheumatoid arthritis (RA) of the knee is less common. It is usually seen in females in the 4th to 5th decades. It presents as symmetrical polyarthritis of the proximal finger joints, tenosynovitis, early morning stiffness and pain. Histopathological features include synovial cell hyperplasia and proliferation with inflammatory infiltrates and rheumatoid nodules. These are necrotising granulomas surrounded by fibrinoid necrosis.

The below image shows rheumatoid nodules:



Option D: Lyme disease is a tick-borne infection with skin lesions and flu-like symptoms. It presents with asymmetric inflammatory polyarthritis of the large joints when left untreated.

Solution to Question 5:

Statement C is false. Nerve injury is more common in posterior hip dislocation. The sciatic nerve, lying behind the posterior wall of the acetabulum is most commonly involved here.

Sciatic nerve injury is seen in 10-20 % of cases with posterior hip dislocation. It is usually a neuropraxia type of injury and resolves spontaneously. But, if it is suspected after a reduction maneuver, immediate exploration must be carried out.

Posterior hip dislocation is the commonest type of hip dislocation (80%), commonly seen following dashboard injuries. The patient presents with flexion, adduction, and internal rotation deformities.

X-ray findings include:

• Femoral head lying above and behind the acetabulum and appears smaller than normal

• Lesser trochanter is not visualized as the thigh is internally rotated

• Femoral head and posterior acetabular wall fractures may also be seen

Treatment includes an early reduction to reduce the risk of osteonecrosis and osteoarthritis. Closed reduction under sedation or general anaesthesia is done in most cases. If there is an associated femoral neck fracture, the closed reduction should not be attempted as it can lead to the displacement of the femoral neck and disrupt the blood supply to the femoral head.

Anterior hip dislocation is less common than posterior hip dislocations and accounts for about 10-20% of all hip dislocations. This injury is usually sustained following a high-energy accident. The affected limb in this condition will be abducted, externally rotated, and slightly flexed.

Note: Although vascular injuries are seen in posterior dislocations as well, they are most likely with anterior dislocation. This is because of the femoral artery lying anterior to the head. (statement B)

Solution to Question 6:

The given clinical scenario and the x-ray showing a lytic lesion at the end of a long bone with a soap-bubble appearance suggest a giant cell tumor (GCT)/ osteoclastoma.

Osteoclastoma is a benign but locally aggressive tumor that usually affects young adults of age 20 to 40 years with slight female predominance. It affects the ends of long bones, involving metaphysis/epiphysis of long bones. GCT is most commonly seen in the distal femur, proximal tibia, proximal humerus, and distal radius. Common presenting complaints are swelling and vague pain. Sometimes, the patient presents for the first time with a pathological fracture through the lesion.

Important X-ray features of a GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis.

• Eccentric location, most often subchondral.

• Expansion of the overlying cortex.

• Soap bubble appearance - The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called soap bubble appearance.

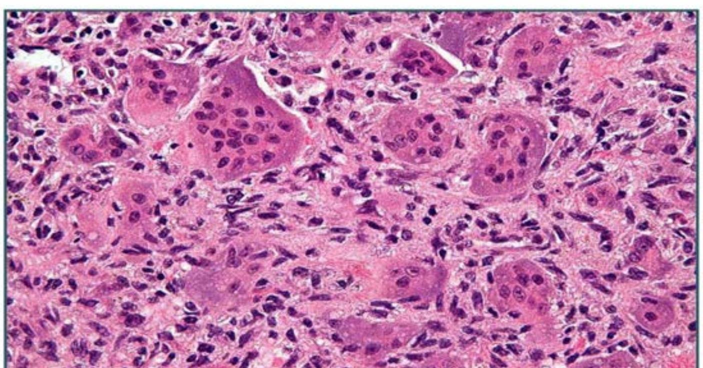
• No calcification within the tumor.

• No or minimal reactive sclerosis around the tumor.

Histologically, GCTs consist of undifferentiated spindle cells profusely interspersed with multi-nucleated giant cells. The tumor stroma is well-vascularized with bands of cellular or collagenous fibrous tissue.

The image below shows the histopathology of osteoclastoma:

Osteoclastoma



Osteoclast-like giant cells with mononuclear cells scattered around

The treatment of choice for GCT is extended curettage and bone grafting.

• GCT in the relatively insignificant bone like the distal ulna, clavicle, and proximal fibula is treated by resection without reconstruction.

• Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease.

• Radiation can be used for spinal or sacral tumors.

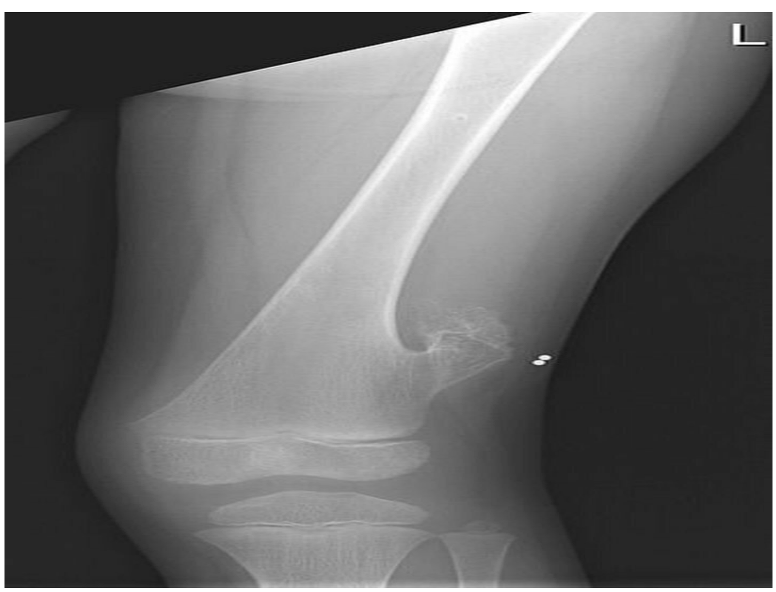
Other options:

Option A: Chondroblastoma is a benign tumor that usually appears as well-circumscribed round or oval lytic lesions in the center of the epiphysis of a long bone. It presents as a painful mass that causes effusion or stiffness. It is most commonly seen in ages 10-20 years.

The following is an x-ray image of the knee showing chondroblastoma:

Option B: Osteochondromas are common benign bone tumors commonly present in adolescents. The lesions consist of a bony mass, often in the form of a stalk, produced by endochondral ossification of a growing cartilaginous cap. They are usually painless. The common sites of involvement are the distal femur, proximal tibia, and proximal humerus.

The following is an x-ray image of the knee showing osteochondroma of the distal knee:



Option D: Fibrous dysplasia is a condition that affects bone development and is characterized by the replacement of the bone with fibrous tissue. The patient presents with a progressive limb deformity or a fracture through a previously asymptomatic lesion which often occurs in the second decade. Other symptoms include nonspecific bone pain, tenderness, and swelling in the affected area. The commonly involved sites are the proximal femur, tibia, pelvis, and foot.

Polyostotic fibrous dysplasia is associated with McCune-Albright syndrome.

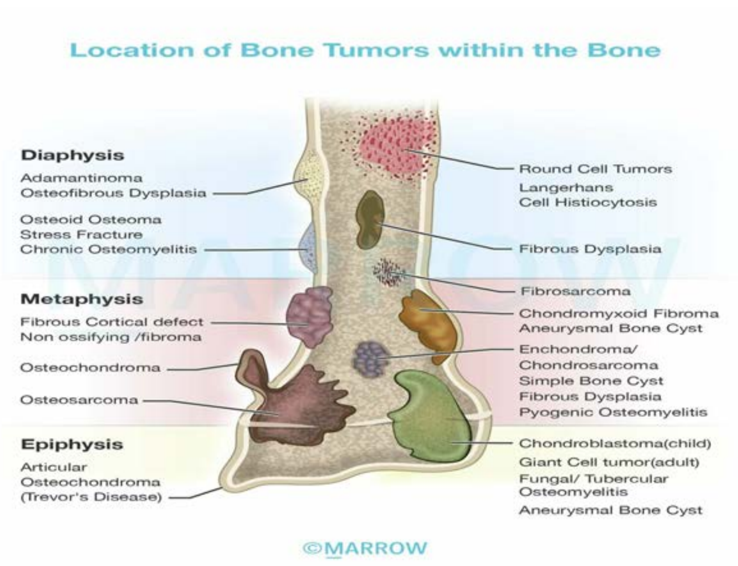
Radiological findings include:

• Well-defined, expansile intramedullary lesions

• Hazy ground glass appearance is seen in the majority of cases

• Shepherd's crook deformity - Expansion with marked varus in the proximal femur

The given image shows the common locations of various bone tumors:



Solution to Question 7:

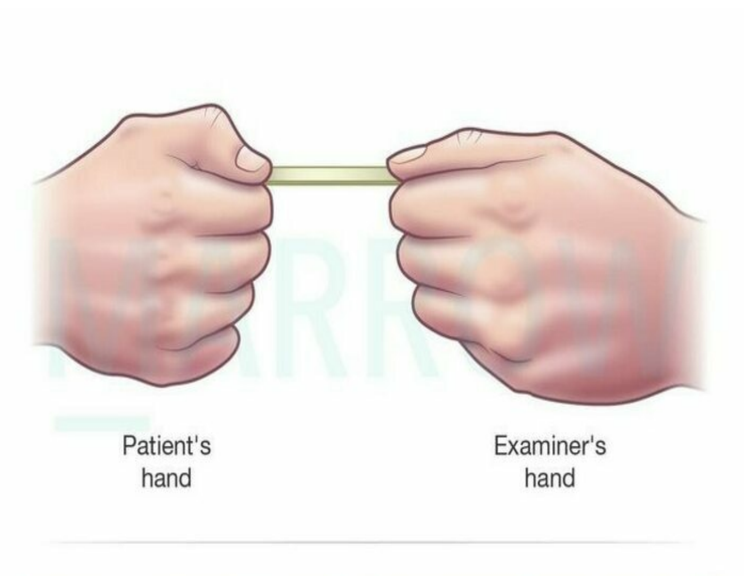
The deep terminal branch of the ulnar nerve supplies the adductor pollicis muscle.

A positive book test or Froment's sign indicates weakness of the adductor pollicis muscle, due to ulnar nerve weakness.

In this test, the patient is asked to hold a book between the thumb and index finger against resistance, and the sign is positive if there is flexion at the interphalangeal joint. This is because if the weakness of adductor pollicis is present, thumb adduction is weak; the flexor pollicis longus muscle compensates for this weakness by inducing flexion of the interphalangeal joint.

Adduction paralysis occurs as a result of low lesions of the ulnar nerve, which are commonly caused by cuts on shattered glass. The patient has numbness in the medial one-and-a-half fingers. The hand assumes the claw-hand deformity, with hyperextension of the metacarpophalangeal joints of the ring and little fingers, due to weakness of the intrinsic muscles. Hypothenar and interosseous wasting may be seen in the affected hand when compared to the normal hand.

The image below shows the test being demonstrated:



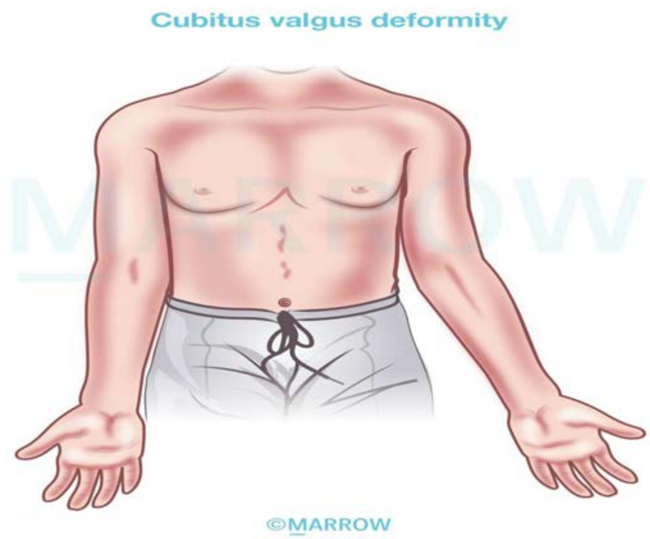
Solution to Question 8:

The finding that can be seen in this patient with a fracture of the lateral condyle of the humerus is the inability to adduct the fingers. This is due to tardy ulnar nerve palsy.

In this condition, there is stretching of the ulnar nerve due to cubitus valgus deformity that occurs as a result of long-standing non-union of a fractured lateral condyle of the humerus.

The normal carrying angle of the elbow is 13 degrees. A cubitus valgus deformity is an angle more than this or an asymmetry between the two sides.

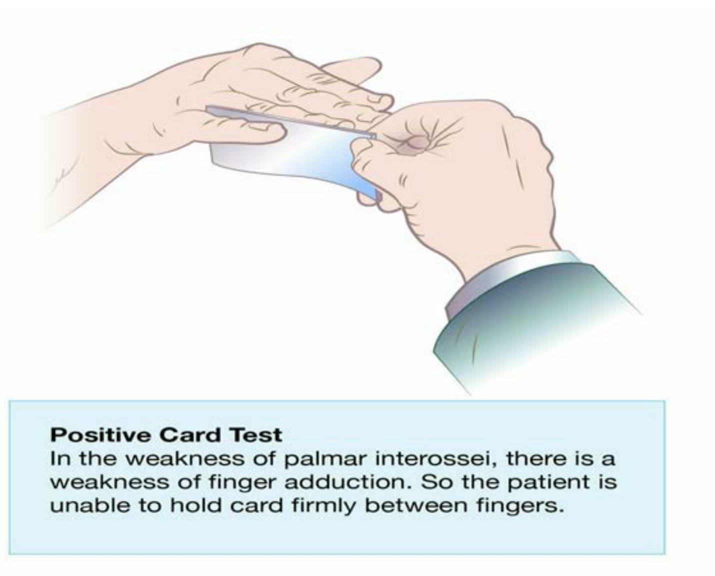
The image below shows cubitus valgus deformity.



The ulnar nerve is most commonly injured in flexion type of supracondylar fracture. In ulnar nerve injury at the wrist, there
is hyperextension at metacarpophalangeal joints and flexion at interphalangeal joints. This causes claw hand deformity. There is numbness in the medial one-third of the palm and the medial one-and-a-half fingers.

The patient is unable to abduct the fingers due to paralysis of the dorsal interossei. The loss of adduction is due to paralysis of the palmar interossei muscles.

A card test is done to test the function of palmar interossei muscles.



Other options:

Option A: Ape thumb deformity is characterized by the inability to abduct the thumb due to paralysis of the thenar muscles supplied by the median nerve. The thumb remains in the same plane as the wrist.

Option B: In median nerve injury of the forearm or elbow, there is paralysis of long flexors of the digits. This leads to the characteristic pointing index finger.

Option C: The inability to perform the 'OK' sign is called as Kiloh Nevin sign. It is seen in the involvement of the anterior interosseous nerve. The patient is unable to bend the terminal phalanx of the thumb and index finger joints due to weakness of the flexor pollicis longus and lateral half flexor digitorum profundus.

Solution to Question 9:

The given clinical scenario of an old woman with back pain that worsens on walking and relieves walking uphill points to the diagnosis of lumbar stenosis.

Walking uphill may be preferred as it flexes the spine and maximizes the spinal canal capacity, relieving the pain.

Spinal stenosis is the narrowing of the spinal canal, causing nerve compression. It can be congenital or acquired, with degeneration being the most common cause. Other factors include conditions like spondylolysis, trauma, infection, and metabolic disorders.

Degenerative spinal stenosis involves disc degeneration, facet joint hypertrophy, and ligamentum flavum calcification. These changes lead to reduced spinal canal dimensions and compression of the nerves resulting in neurogenic claudication.

They present with symptoms such as back pain (most common), sciatica, sensory disturbances in the legs, motor weakness, and sometimes urinary disturbances. Pain typically occurs after standing or walking for a few minutes and is relieved by sitting, squatting, or flexing the spine.

MRI is the imaging method of choice for the investigation of spinal stenosis. Conservative treatment is the first line for spinal stenosis, including reassurance, pain relief, exercise, and activity modification.

Surgery is considered if conservative measures fail and there is significant functional impairment and severe neurological symptoms. Posterior decompression is the surgical method. Laminectomy is performed for central stenosis.

Solution to Question 10:

Disruption of the axon with an intact endoneurium is seen in axonotmesis. There is an axonal disruption (inner axon is damaged) and loss of conductivity, with an intact outer sheath.

Axonotmesis (axonal interruption): There is a loss of conduction, but the continuity of the nerve is maintained and the neural tubes are intact. There is disruption of the axon i.e., the continuity of the axon is broken. But the endoneurium and perineurium are intact, maintaining the continuity of the nerve. It occurs following trauma such as closed fractures and dislocations.

Distal to the lesion, Wallerian degeneration of the disrupted axon occurs, where axons disintegrate and are resorbed by phagocytes. It is then followed by axonal regeneration where numerous fine unmyelinated fibres grow from the proximal stumps to the endoneurial tubes.

Other options:

Option A- Neurapraxia is the reversible physiological nerve conduction block causing some loss of sensation and muscle power. There is spontaneous recovery after a few days or weeks. It occurs following prolonged compression, as in Saturday night palsy, and crutch palsy. Here, there is segmental demyelination but the axon is intact. Nerve conduction is slowed because of the loss of myelin. It is followed by the regeneration of the myelin.

Option B- Neurotmesis is the complete transection of the nerve trunk. It is usually associated with open wounds. The endoneurial tubes are destroyed and the scar tissue prevents the regenerating axons from entering the distal segment. The regenerating fibers mingle with proliferating Schwann cells and fibroblasts to form a neuroma at the site of injury.

Option D- Axonapraxia is a disruption to the axoplasmic flow of neurochemicals temporarily.

Solution to Question 11:

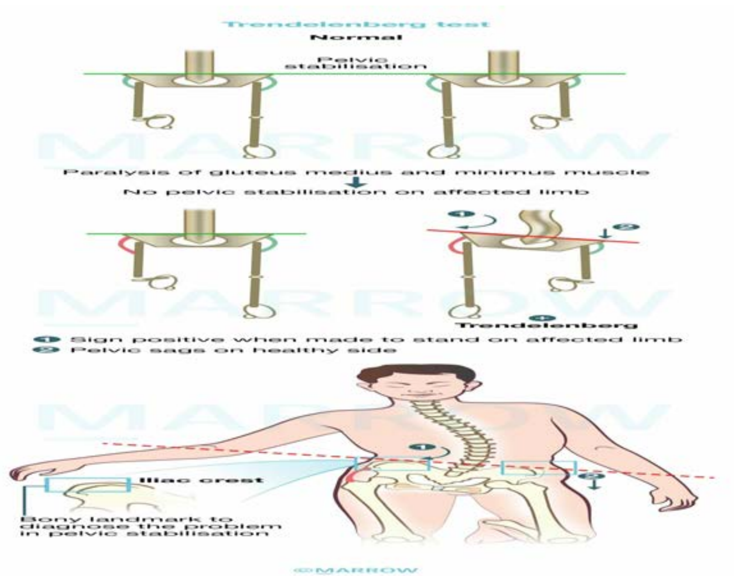
The muscles involved in a positive Trendelenburg sign are the hip abductors—the gluteus medius and gluteus minimus.

The gluteus medius and minimus are powerful abductors of the hip supplied by the superior gluteal nerve. Stabilizing the pelvis on the femur during walking is done by these muscles. It is demonstrated by asking the subject to stand on one limb. If the ipsilateral abductors are normal, the pelvis remains in a horizontal position. If the ipsilateral abductors are weak, then the pelvis tilts towards the opposite (unsupported) side. This is called positive Trendelenburg’s sign.

Other causes of positive Trendelenburg signs include shortening of the femoral neck, abductor tendon rupture, and dislocation or subluxation of the hip joint.

Patients are known to have a characteristic gait, which is called Trendelenburg or waddling gait. The trunk is thrown from side to side with each step, and the patient lurches toward the affected limb while walking.

The image below depicts Trendelenburg's gait.



Solution to Question 12:

Treatment of CTEV (congenital talipes equinovarus) should begin as early as the first 2 weeks after birth. In CTEV, or idiopathic clubfoot, the deformity is equinus, inversion, forefoot adduction, and cavus.

Ponseti's method is a successful method of manipulation in CTEV. The order of correction according to Ponseti's method is cavus-adduction-varus-equinus (CAVE). Weekly manipulation of the foot is done with the help of plaster casts.

The first metatarsal is elevated initially to correct the cavus. This creates a flat longitudinal medial border of the foot. Then, the forefoot is abducted in the plane of the metatarsals. This everts the hindfoot and corrects the forefoot out of supination. Finally, the equinus is corrected by dorsiflexing the foot at the ankle joint. In most cases, a percutaneous Achilles' tenotomy is required later for complete correction of the equinus deformity.

After correction, a foot abduction brace (FAB) is used to maintain the corrected position, initially full-time (for 3 months) and then night-time and nap-time for 4 years to prevent relapse of the deformity. CTEV shoes are worn during the day once the child starts walking for up to 5 years. They have a straight inner border (prevents adduction), outer shoe raise (prevents foot inversion), and no heel to prevent equinus.

The image below shows the foot abduction brace:

The image below shows the CTEV shoes:

Solution to Question 13:

In congenital talipes equinovarus (CTEV), or idiopathic clubfoot, the deformity is equinus, inversion, forefoot adduction, and cavus. CTEV is a congenital malformation of the foot. It is predominately seen in male children. Calf and medial foot tissues are underdeveloped.

Components of deformity:

• Cavus - This deformity occurs due to exaggerated arching of the foot at the mid-tarsal joint.

• Adduction - This deformity occurs at the talonavicular joint.

• Inversion - This deformity occurs at the subtalar joint. The calcaneum is inverted, and along with it, the whole foot is inverted. This leads to a medially facing sole.

• Equinus - This deformity occurs at the ankle joint and tarsal joints.

Equinovarus deformity causes inward twisting of the foot (shown below), with the sole facing posteromedially in severe cases. Associated disorders like congenital hip dislocation and spina bifida should be evaluated.

Ponseti's method is a successful method of manipulation in CTEV. It should begin as early as the first 2 weeks after birth. Weekly manipulation of the foot is done with the help of plaster casts. The order of correction according to Ponseti's method is cavus-adduction-varus-equinus (CAVE).

The first metatarsal is elevated initially to correct the cavus. This creates a flat longitudinal medial border of the foot. Then, the forefoot is abducted in the plane of the metatarsals. This everts the hindfoot and corrects the forefoot out of supination. Finally, the equinus is corrected by dorsiflexing the foot at the ankle joint. In most cases, a percutaneous Achilles' tenotomy is required later for complete correction of the equinus deformity.

Typically, five casts are required for correction, followed by a cast for 3 weeks. After correction, a foot abduction brace (FAB) is used to maintain the corrected position, initially full-time (for 3 months) and then night-time and nap-time for 4 years. This helps prevent relapse of the deformity. CTEV shoes are worn during the day once the child starts walking for up to 5 years.

Solution to Question 14:

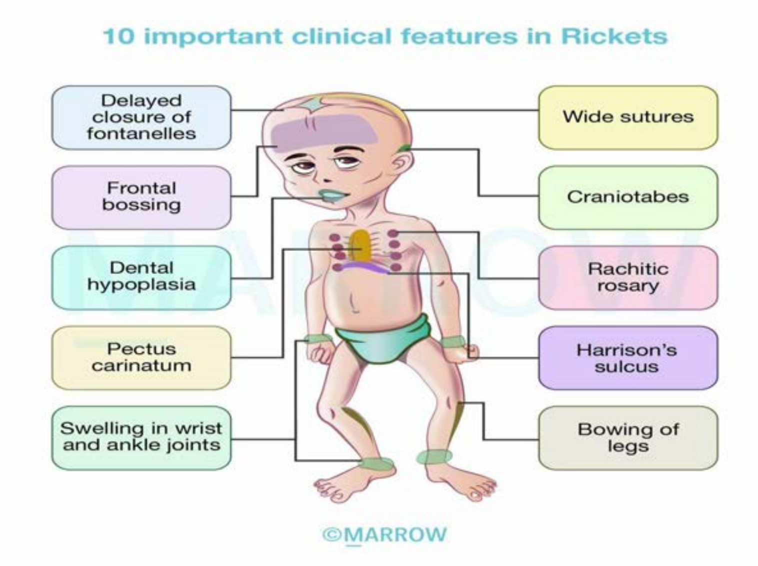
The clinical vignette describing a child with poor growth, the wrist radiograph findings showing cupping and fraying, and the raised ALP levels are suggestive of rickets.

Inadequate mineralization of bone in children is known as rickets. There is inadequate calcification of the growth plate, leading to irregular chondrocyte accumulation, widened growth plate, poor mineralization in the zone of calcification, and sparse bone formation.

Vitamin D disorders, such as vitamin D deficiency or resistance, calcium deficiency, phosphorus deficiency, and renal loss are among the common causes of rickets. Decreased vitamin D levels reduce calcium and phosphate absorption, leading to low blood calcium. This stimulates increased production of parathyroid hormone (PTH) and elevates alkaline phosphatase (ALP) levels involved in bone mineralization.

Infants with rickets may exhibit symptoms such as tetany, convulsions, failure to thrive, listlessness, and muscle weakness. Softening of the skull bones called craniotabes is seen. Delayed closure of fontanels, frontal bossing, and delayed dentition are also seen.

Widening of the costochondral junctions, forming a rachitic rosary along the ribs is seen in the chest. Harrison groove, a horizontal depression along the lower chest, can occur due to rib softening. Vertebral column abnormalities like scoliosis, kyphosis, and lordosis can also appear. In the distal extremities widening of the wrists and ankles, varus/ valgus deformities, windswept deformity (valgus on the side and varus on the contralateral side), and anterior tibial bowing are seen.



The laboratory findings on vitamin D deficiency rickets would be normal/decreased serum calcium, decreased phosphorous, elevated PTH, and increased ALP levels. Thickening and widening of the growth plate, cupping of the metaphysis, and potential bowing of the diaphysis are seen on X-ray.

Children with nutritional vitamin D deficiency should receive vitamin D supplementation and ensure adequate intake of calcium and phosphorus. Two strategies for vitamin D administration are stoss therapy (300,000-600,000 IU over 1 day) and daily supplementation (2,000 IU/day for at least 3 months).

Other options:

Option A: Scurvy is caused by vitamin C deficiency, resulting in bleeding tendencies and various manifestations such as petechiae, gum bleeding, and scorbutic rosary. Radiographically, it shows characteristic findings like the white line of Frankel, Pelkan spurs, and periosteal elevation.

Option C: Osteogenesis imperfecta (Brittle bone disease) is caused by deficiencies in the synthesis of type I collagen. It presents with the hallmark features of osteoporosis with increased fragility of bones along with blue sclera, dental fragility as well as hearing loss.

Option D: Paget's disease of the bone is a condition where there is an increased bone turnover followed by enlargement and thickening of the bone. In this condition, calcium levels are normal with raised ALP.

Solution to Question 15:

Emergency management of open fractures is mainly focused on the prevention of infection by providing antibiotic coverage and splinting (statements 3 and 4). Reduction of the fracture (statement 1) can be done in the emergency room. Internal fixation is not done on an emergency basis for an open fracture due to the risk of infection (statement 2). It is a part of definitive management and is done after the patient is stabilized.

Open fractures are characterized by a break in the overlying skin and soft tissues. Hence, the fracture fragment is exposed to the external environment. There are more prone to be infected.

The major components of the management of open fractures:

• Antibiotic prophylaxis - It should be given as early as possible (within 3 hours of injury). Amoxiclav is commonly used.

• Early debridement of the wound and fracture - This includes the following steps:

• Excision of the wound margins

• Extension of the wound if required

• Delivery of the fracture outside the wound for proper examination

• Removal of the devitalized tissue

• Cleansing of the wound to remove all the foreign materials

• Nerves and tendons are usually left alone, but if the wound is clean, a repair can be considered

• Wound cover:

• In Gustilo Type I and II fractures, wound cover can be done at the time of debridement (early wound cover)

• Delayed wound cover is usually practiced in Type III fractures

• Fracture stabilization: External fixators are used to stabilize the fractures in the emergency room. Definitive management is later done by open reduction and internal fixation (ORIF).

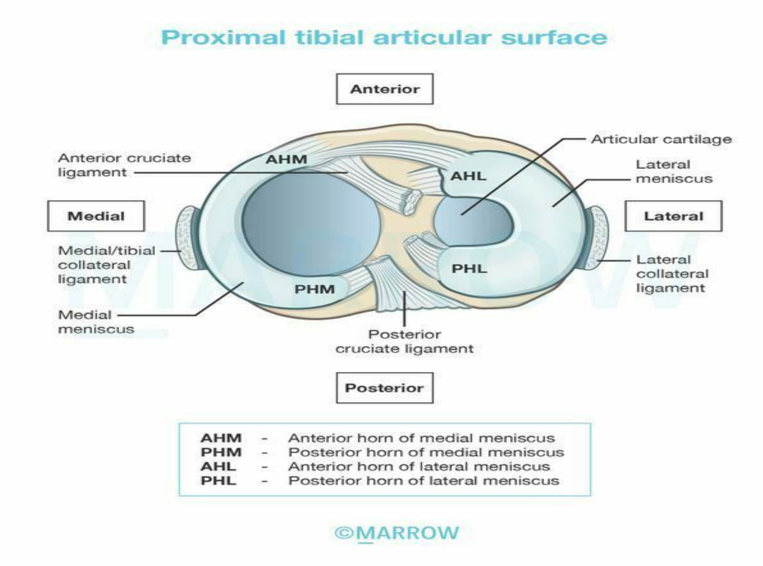
Solution to Question 16:

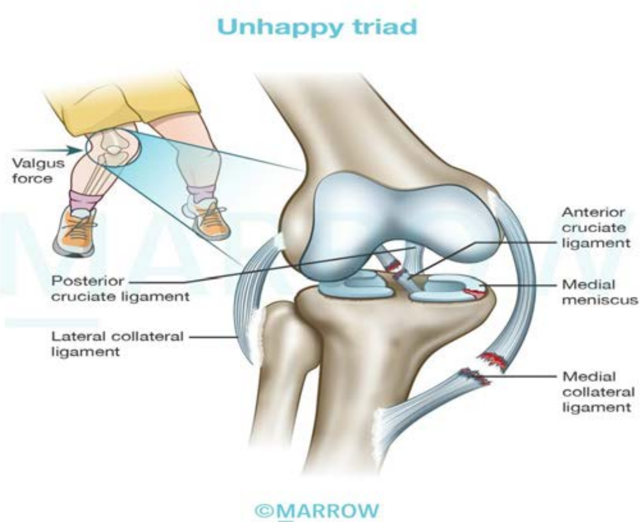
The unhappy triad of O'Donoghue consists of injury to

• Anterior cruciate ligament (A)

• Medial collateral ligament (B)

• Medial meniscus (C)





Solution to Question 17:

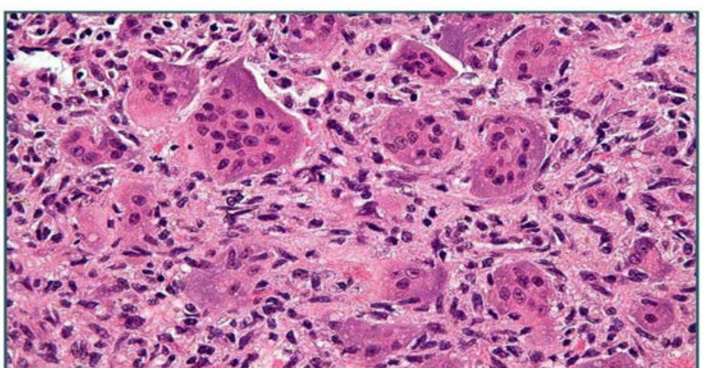
The given x-ray showing a lytic lesion at the end of a long bone is suggestive of a giant cell tumor (GCT)/osteoclastoma. It affects the ends of long bones, involving metaphysis/epiphysis (statement 3). GCT is most commonly seen in the distal femur, proximal tibia, proximal humerus, and distal radius.

GCTs usually affect young adults of age 20 to 40 years (statement 1), with slight female predominance. It is a benign but locally aggressive tumor. They present with swelling, vague pain, or pathological fracture through the lesion.

A solitary eccentric, expansile, lobulated lytic lesion in the epiphysis/metaphysis is seen on a plain X-ray, which is diagnostic (statement 2). The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called a soap bubble appearance (shown below). Soft-tissue extension with a mass can also be observed. They usually lack calcification, new bone formation, or periosteal reaction.

Histologically, GCTs display a mixture of meaty, reddish-purple tissue with soft yellow areas. They are characterized by numerous osteoclast-like giant cells dispersed among round or spindle-shaped mononuclear cells (shown below). The tumor stroma is well-vascularized with bands of cellular or collagenous fibrous tissue.

Osteoclastoma



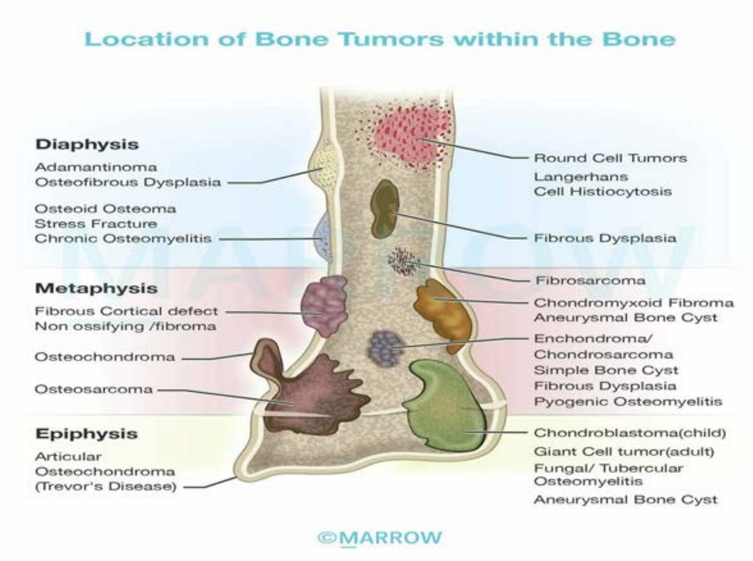
Osteoclast-like giant cells with mononuclear cells scattered around

Malignancy transformation is rare (<1%) in GCTs, though possible (statement 4).

The treatment of choice for GCT is extended curettage and bone grafting. GCT in the relatively insignificant bone like the distal ulna, clavicle, and proximal fibula is treated by resection without reconstruction. Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease. Radiation can be used for spinal or sacral tumors.

Denosumab is an anti-RANKL antibody that can be used to stop the osteolytic process. It can be used in advanced stage 3 lesions or lesions in inoperable locations.

The given image shows the common locations of various bone tumors:



Orthopedics NEET 2022

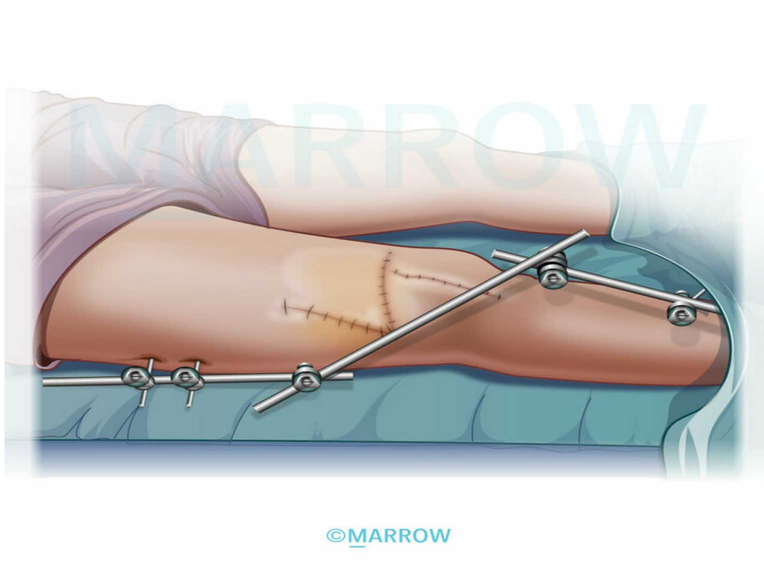
Question 1:

What could be the diagnosis of a woman with this appearance on x-ray?

- **a)** Multiple brown tumors
- **b)** Fibrous dysplasia
- **c)** Multiple enchondromas
- **d)** Multiple exostoses

Question 2:

Which of the following is true about the type of fixation shown in the image?



a) Fracture tibia, Illizarow fixator

b) Fracture tibia, spanning fixator

c) Fracture femur, spanning fixator

d) Periarticular fracture of knee, spanning fixator

Question 3:

A male patient presented with a bone fracture following a road traffic accident. After 2 days he developed dyspnea, petechiae involving the whole body, and a fall in oxygen saturation. What is the likely diagnosis?

a) Fat embolism

b) Air embolism

c) Venous thromboembolism

d) Pulmonary hypertension

Question 4:

An intrauterine scan at the 13th week of pregnancy showed a fetus with multiple long bone fractures. What is commonly associated with this finding?

a) Achondroplasia

b) Osteogenesis imperfecta

c) Cretinism

d) Marfan syndrome

Question 5:

What is the most common complication of this condition, if left untreated?

a) Malunion and stiffness

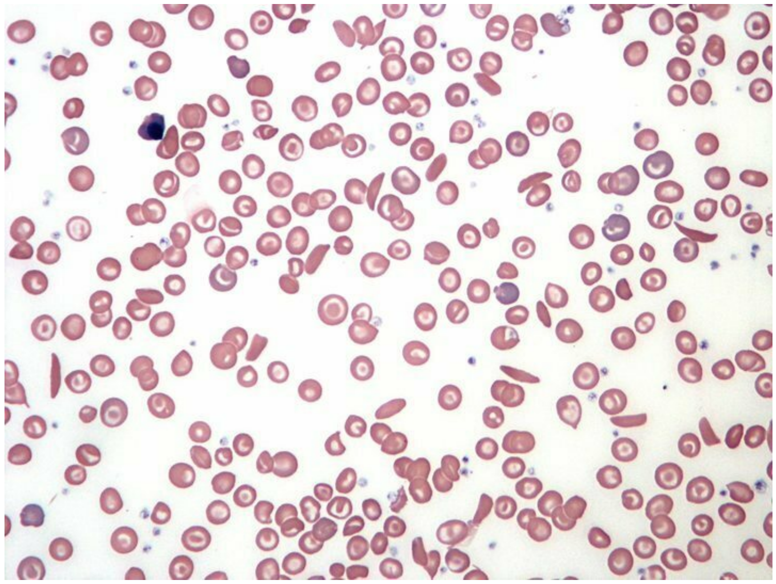
b) Non union and cubitus varus

c) Cubitus valgus

d) Myositis ossificans

Question 6:

A child with a history of recurrent respiratory infections presented with complaints of pain around the knee and a high fever. X-ray shows lytic and sclerotic bone. His peripheral blood smear is shown in the image below. Aspirate from the joint will most likely show:



a) Staphylococcus aureus

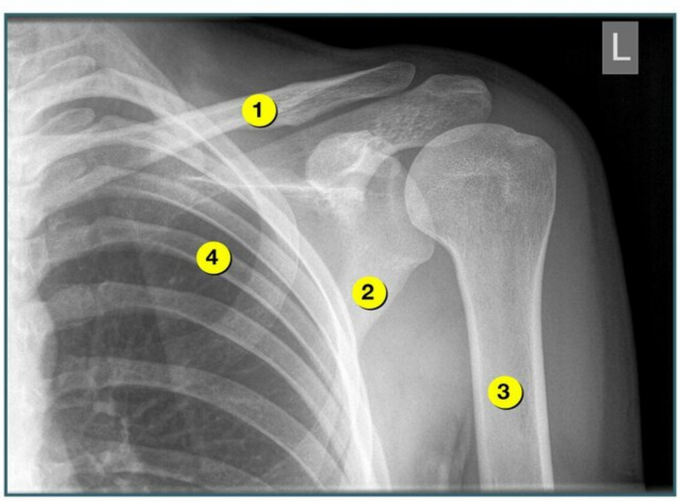
b) Escherichia coli

c) Salmonella

d) Streptococcus

Question 7:

A boy falls on the left shoulder joint and presents to the emergency department with shoulder pain. His left elbow is flexed and is supported by the right hand. Which bone might be most likely fractured?



a) 1

b) 2

c) 3

d) 4

Answer Key

Question No.	Correct Option
1	c
2	d
3	a
4	b
5	c
6	c
7	a

Detailed Explanations

Solution to Question 1:

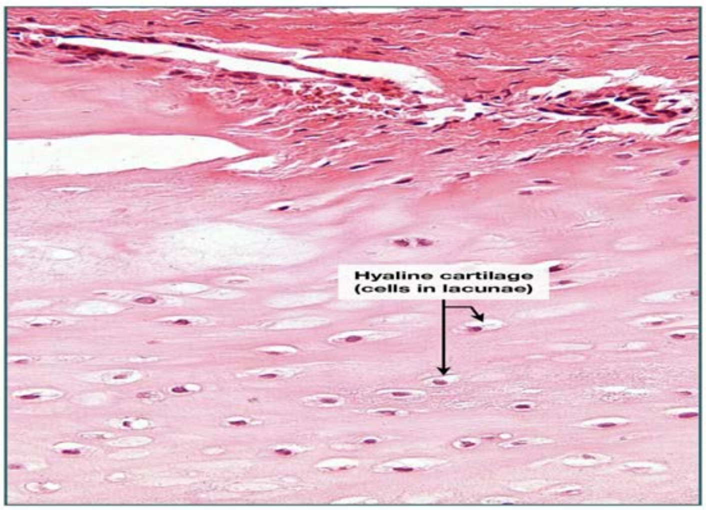
The diagnosis of the woman is multiple enchondromas. The X-ray shows multiple calcified lesions in the fingers.

Multiple enchondromatosis is also known as Ollier disease. It is a tumor in the metaphysis of long bones. There is a failure of bone formation and broadening of unossified cartilage within the bone. This results in shortening and angular deformity of the affected bones with an increased risk of pathological fractures. Enchondromas are also known as Maffucci syndrome if it is associated with overlying soft tissue hemangiomas. 25% of the patients with Ollier's disease will develop sarcomas by 40 years of age.

X-ray shows intralesional calcifications with a popcorn appearance. Histologically, enchondromas consist of hyaline cartilage with cytologically benign chondrocytes. Treatment of multiple enchondromas includes correction by osteotomy if the deformities are obvious.

Given below is an X-ray showing calcifications:

Enchondroma



Maffucci syndrome

Hemangiomas of the hand

Other options:

Option A: Multiple brown tumors are caused secondary to hyperparathyroidism and resemble a giant-cell tumor. It can be diagnosed by estimating the levels of calcium, alkaline phosphatase, and parathyroid hormone levels. The bone surrounding the lesion shows peritrabecular fibrosis.

The image below shows a brown tumor on the left hand:

Option B: Fibrous dysplasia is a condition characterized by the replacement of bone with fibrous tissue. The X-ray shows a ground-glass appearance with a well-defined sclerotic rim. It is seen in association with McCune-Albright syndrome. Shepherd’s crook deformity in the proximal femur is characteristic of fibrous dysplasia.

The image below shows a child with McCune-Albright syndrome:

Shepherd's crook deformity -
Fibrous Dysplasia

Option D: Multiple exostoses is an autosomal dominant type of osteochondroma that affects the phalanx, scaphoid, and metacarpals. The X-ray shows bony erosions and is excised.

The image below shows exostoses at the base of the thumb:

Solution to Question 2:

The image shows a periarticular fracture of the knee and a spanning external fixator used to stabilize the fracture.

External fixators are used if the fracture is associated with severe soft tissue damage. It is also used if internal fixation is risky due to wound contamination. The spanning external fixator will stabilize the fracture until the condition of soft tissue improves.

The technique of an external fixator is that the bone will be fixed above and below the fractures with screws and wires. These screws and wires are then connected to rigid bars. This method is used in fractures of the tibia, femur, pelvis, bones of the hand, etc.

In the image, the bones above and below the knee joint are fixed using a spanning external fixator and hence the fracture is likely to be a periarticular fracture of the knee.

The complications of external fixation include over-distraction, damage to soft tissue structures, and pin-track infections.

Another external fixator that is commonly used in an open type of fracture is the Ilizarov fixator. It is also used for the reconstruction of the limbs and for emergency stabilization in multiple long bone fractures.

Ilizarov apparatus

Solution to Question 3:

The likely diagnosis is fat embolism.

Fat embolism is caused by the fat globules originating from the bone marrow following fracture of long bones. Clinical features of fat embolism are usually seen within 72 hours. They include tachycardia, fever, breathlessness, petechiae, confusion, and restlessness.

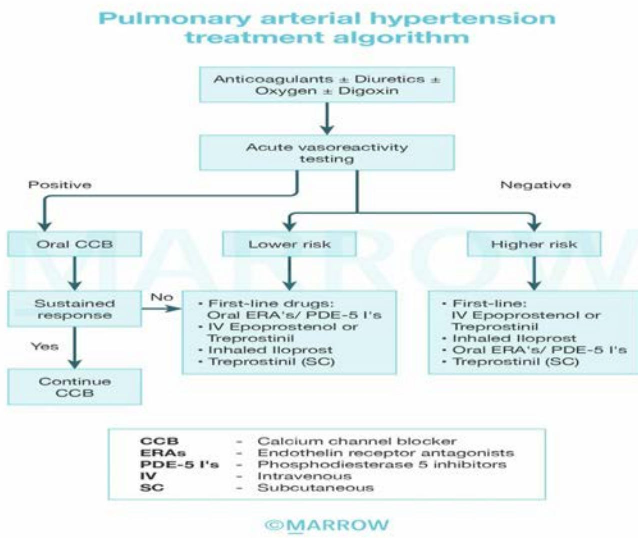
There is no specific test for diagnosis. However, urine analysis to look for fat globules and monitoring of blood pO2 are done. Management of fat embolism is mostly supportive, with oxygen therapy.

Other options:

Option B: Air embolism occurs when air bubbles entering the pulmonary circulation get absorbed in large quantities. Severe hypotension and hypoxemia are seen in this condition. The patient is placed in Trendelenburg's left lateral decubitus position.

Option C: Venous thromboembolism consists of pulmonary embolism (PE) and deep vein thrombosis (DVT). Bedridden patients and patients with chronic systemic diseases are more prone to thromboembolism. Clinical features of PE include dyspnea, syncope, cyanosis, hypotension etc. DVT is presented as leg cramps, erythema, and tenderness. Heparin, LMWH, or fondaparinux are used for the treatment.

Option D: Pulmonary hypertension is a disease that involves remodeling of the pulmonary vasculature. It results in increased pulmonary artery pressure and vascular resistance. It mainly occurs as a complication of left heart disease, other primary lung diseases, and pulmonary embolism. The clinical features include dyspnea, edema, chest pain, syncope, etc.



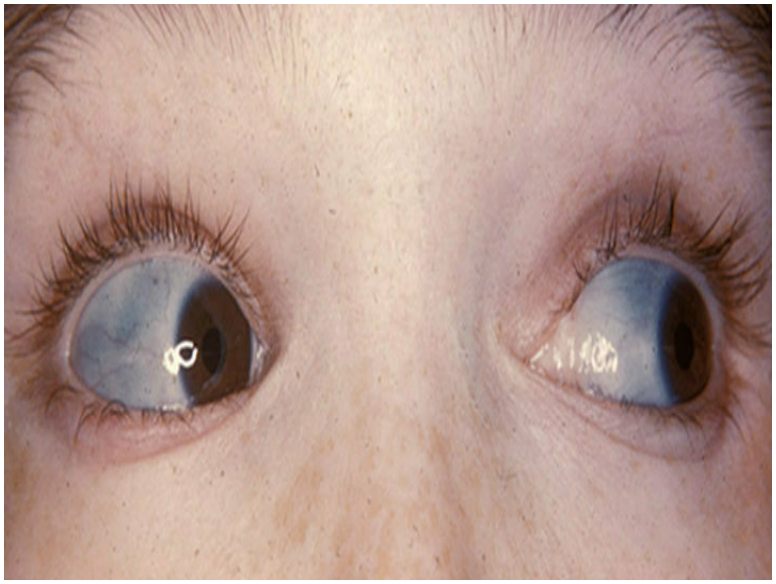
Solution to Question 4:

Osteogenesis imperfecta is commonly associated with this multiple long bone fractures.

Osteogenesis imperfecta is a connective tissue disorder that occurs due to abnormal type 1 collagen production. The main features are joints that are vulnerable to fracture, scleral alterations, dentinogenesis imperfecta, and deafness. Multiple bone fractures are a consequence of bone fragility. It is classified into four types based on scleral involvement, natural history, and mode of inheritance.

Protrusio acetabuli, wormian bones, and proximal femoral ‘shepherd’s crook’ deformities of the femurs are the common radiological findings that are observed. Management comprises the use of bisphosphonates, along with physiotherapy and walking aids.

The images below show blue sclera and wormian bones in osteogenesis imperfecta:



Other options:

Option A: Achondroplasia is an autosomal dominant skeletal dysplasia with disproportionately short limbs, sleep apnea, and kyphoscoliosis. Facial features include button nose, frontal bossing, macrocephaly, etc.

The images below show the features of achondroplasia:

(a) Prominent forehead and body asymmetry, (b) a low nasal bridge, (c) trident hands, (d) scoliosis

Option C: Cretinism occurs due to iodine deficiency. Its clinical features are deaf-mutism, squint, mental retardation, and spastic or rigid neuromotor disorder.

Option D: Marfan syndrome is an autosomal dominant disorder. The affected persons will be tall with long limbs and digits. Spontaneous pneumothorax, aortic dissection, and lens dislocation are also seen. An early diastolic murmur is heard on auscultation.

Solution to Question 5:

Cubitus valgus is the most common complication, as the given condition is untreated lateral condyle of humerus fracture.

5–15 degrees of valgus is the normal carrying angle of the elbow. An angle more than this or an asymmetry between the two sides is called a valgus deformity. Long-standing non-union of a fractured lateral condyle is considered the most common cause.

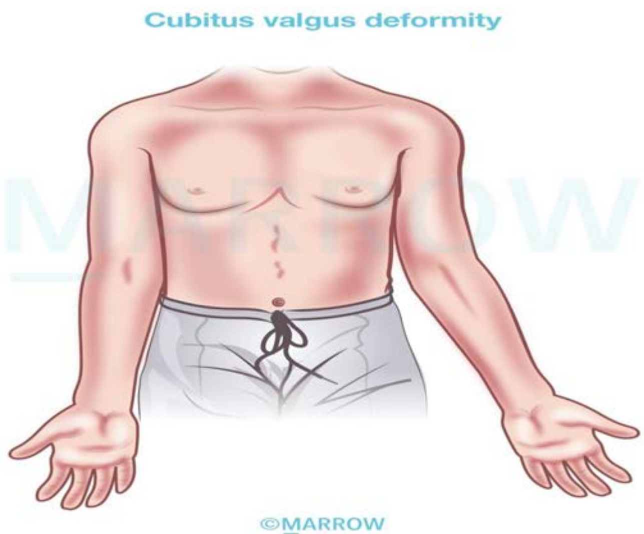
Features of ulnar nerve palsy, such as the weakness of the hand, numbness or tingling of the ulnar fingers are also seen with cubitus valgus

Lateral condyle of humerus fracture is seen mostly in children up to 16 years of age. Falling on the hand with an extended elbow is the mechanism of injury. It is classified into types 1 and 2: the fracture is lateral to the trochlea in type 1 and through the middle, in type 2.

A back slab with the elbow flexed 90 degrees is used if there is no displacement. Reduction and internal fixation are needed for a displaced fracture.

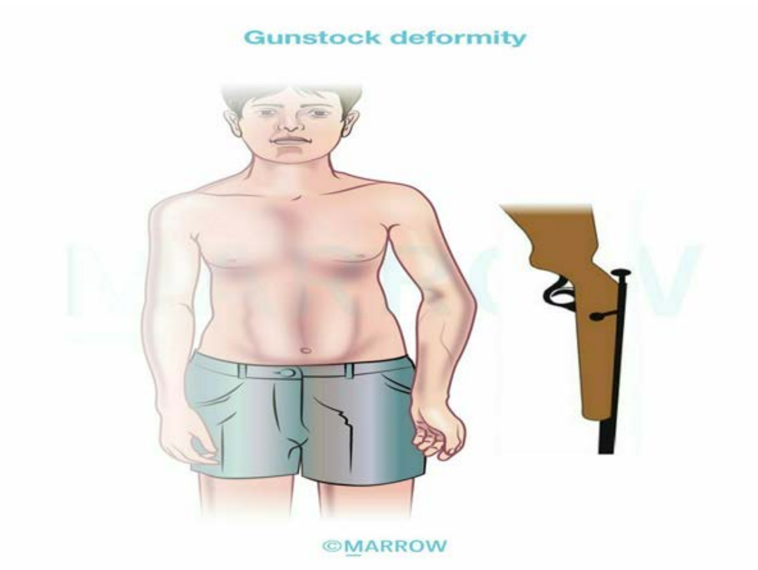
Recurrent dislocations and malunion are the other complications. Since it is a Salter-Harris type IV injury, an imperfect reduction can result in growth arrest.

The image below shows cubitus valgus deformity:



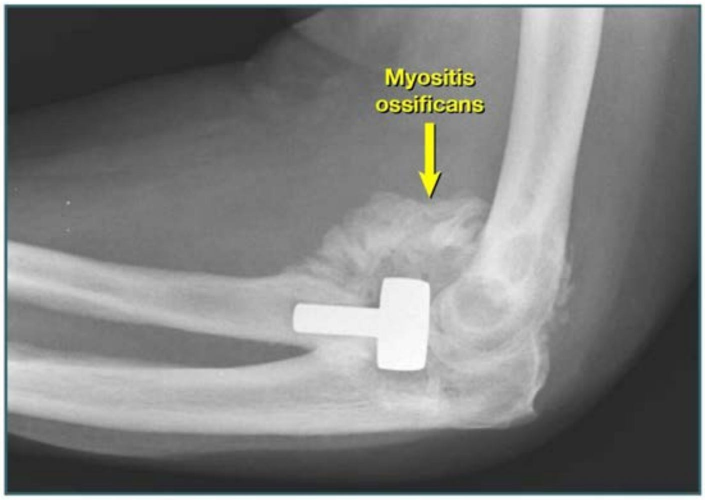
Other options:

Option B: Cubitus varus deformity is also called gunstock deformity. It happens due to the malunion of supracondylar fracture in childhood. Injury to the brachial artery is an important complication.



Option D: Myositis ossificans is a tender swelling seen in the area of injury. It is often confused as a malignant tumor. Calcified lesions are seen in radiological studies. However, the swelling soon becomes less painful.

The X-ray image below shows myositis ossificans:



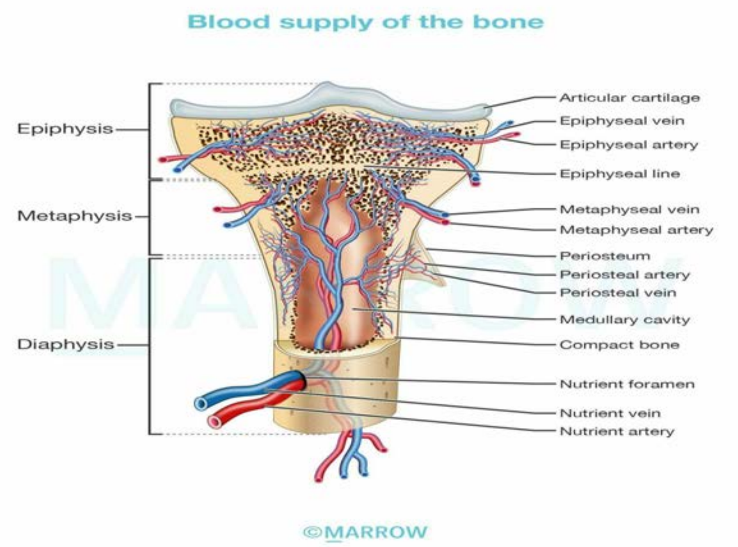
Solution to Question 6:

The given clinical scenario is suggestive of acute osteomyelitis. Aspirate from the joint will most likely show Salmonella.

The peripheral smear picture is indicative of sickle cell anemia. Patients with sickle cell anemia are prone to acute osteomyelitis caused by Salmonella.

Sickle cell anemia is a hematological disease caused by a mixture of genotypes. Its complication includes acute painful episodes, priapism, stroke, leg ulcers, acute chest syndrome (pneumonia-like illness with fever, cough, and hypoxia), etc.

Acute osteomyelitis refers to infection of the bone. It is mainly seen in children. Staphylococcus aureus is the most common organism causing osteomyelitis. However, in patients with sickle cell anemia, it is usually caused by Salmonella. Hemophilus influenza, E. coli, Streptococcus, and Pseudomonas aeruginosa can also cause the disease. The spread can be through hematogenous route or from exogenous routes such as open fractures, iatrogenic intervention, or spread from local infection.



Cardinal features of acute osteomyelitis in children include:

• Fever

• Pain

• Difficulty in weight-bearing

• Elevated WBC count

• Elevated CRP

• Elevated ESR

X-ray shows periosteal reaction and bony destruction after 10-12 days of infection. Management includes blood culture and sensitivity tests, followed by immediate empirical antibiotic therapy. Surgical drainage is done if there is an abscess or when the patient does not respond to antibiotics.

The complications of inadequately treated acute osteomyelitis include:

• Septicemia

• Epiphyseal damage and altered bone growth

• Chronic osteomyelitis

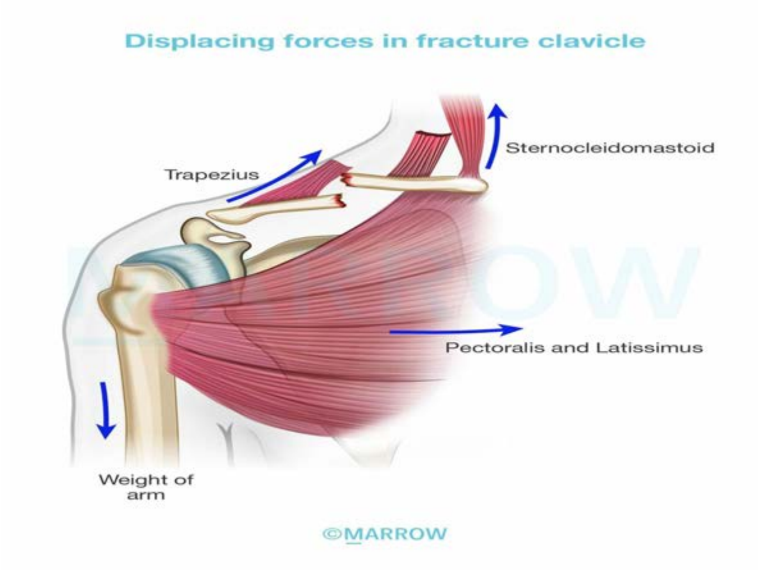
• Suppurative arthritis

• Pathological fracture

Solution to Question 7:

The clavicle marked as 1 might be most likely fractured.

One of the most frequently fractured bones in the body is the clavicle. Mostly, it occurs due to a direct blow or a fall on an outstretched arm. The midshaft of the clavicle is most commonly fractured. The lateral fragment is pulled down by the weight of the arm and the medial half is pulled up by the sternomastoid muscle.



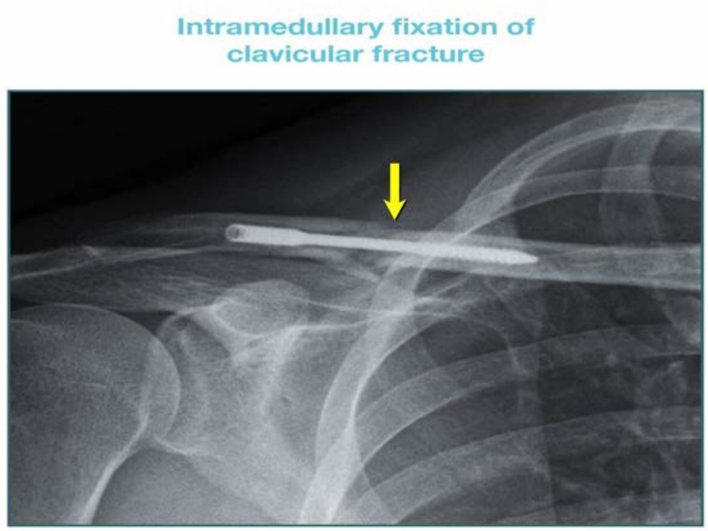
The patient will be presented with the flexed elbow as the arm will be held to the chest to prevent movement. It is divided into three groups, based on the location of the fractures.

• Group I – middle-third fractures

• Group II – lateral-third fractures

• Group III – medial-third fractures

Undisplaced fractures will be mostly treated non-operatively, with a simple sling for comfort. It can be used for 2-3 weeks and discarded once the pain subsides. Displaced fractures are treated with internal fixation either by contoured locking plates or intramedullary nails.



Complications include non-union, malunion, and stiffness of the shoulder.

Other options:

Option B: Scapula is marked as '2'. Fracture of the scapula occurs as a result of blunt trauma, falls, seizures, etc. The arm will be held immobile and bruising can be seen over the scapula and chest wall.

Option C: The shaft of the humerus is marked as '3'. Fracture of the shaft of the humerus occurs as a result of a fall on the hand, elbow, direct blow to the arm, or metastasis. The arm will be painful, bruised and swollen.

Option D: The 4th rib is marked as '4'. Rib fracture occurs due to the impact on the chest wall. Multiple rib fractures will result in the loss of structural integrity of the chest wall and cause flail chest. The lung cannot inflate as the flail segment is sucked in during inspiration.

Orthopedics NEET 2023

Question 1:

A child was brought with bony deformities as shown in the radiograph below. It could be due to the deficiency of which of the following?

- **a)** Vitamin A
- **b)** Vitamin D
- **c)** Vitamin K
- **d)** Vitamin C

Question 2:

In which of the following nerve injuries is the instrument shown below used?

a) Radial nerve

b) Median nerve

c) Ulnar nerve

d) Volkmann's ischemic contracture

Question 3:

A patient presented with chronic knee pain but has no history of trauma. Identify the condition shown in the radiograph below and the appropriate management.

a) Patellar avulsion fracture, TBW

b) Bipartite fracture patella, X-ray of other knee

c) Fracture of upper pole of patella, Cylindrical cast

d) Avulsion fracture , interfragmentary screw fixation

Question 4:

A patient at the orthopedics OPD complains of troubled sleep at night due to numbness and tingling sensation involving his lateral 3 digits. His symptoms are relieved as he lays his arms hanging from the bed. Which among the following options correctly describes his condition along with the test used to assess it?

a) Guyon's canal syndrome, Froment's test

b) Carpal tunnel syndrome, Froment's test

c) Guyon's canal syndrome, Durkan's test

d) Carpal tunnel syndrome, Durkan's test

Question 5:

A child is brought to the orthopedics OPD with a deformity in the lower limb and hyperpigmented skin lesions. The x-ray of her thigh is shown below. What is the most likely diagnosis?

a) Non-ossifying fibroma

b) Fibrous dysplasia

c) Paget's disease

d) Osteogenesis imperfecta

Question 6:

The given image is an x-ray of a 22-year-old female. What is the probable diagnosis?

a) Chondroblastoma

b) Osteochondroma

c) Giant cell tumor

d) Aneurysmal bone cyst

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	d
5	b
6	c

Detailed Explanations

Solution to Question 1:

The given pediatric radiograph depicts the bowing of legs, which occurs due to rickets that results from the deficiency of vitamin D. Deficiency of vitamin D and calcium results in poor mineralization of bone, leading to rickets in children and osteomalacia in adults.

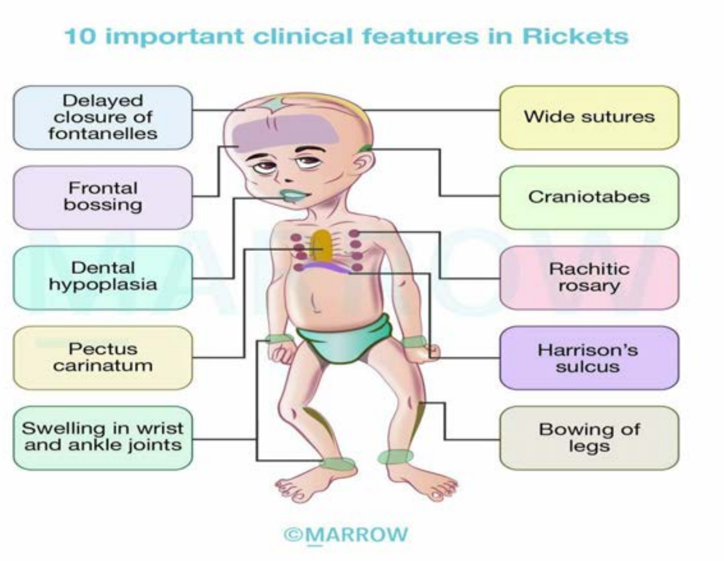
Rickets presents with skeletal changes like bowing of legs and bone pain once the child starts to walk. Radiological features show cupping at metaphyses, fraying, and widening of the growth plate. Prominent costochondral joints (rachitic rosary), craniotabes, frontal bossing, scoliosis, fractures, and dental anomalies may also be seen. The only cutaneous manifestation of vitamin D deficiency is alopecia.

Other options:

Option A: Vitamin A deficiency results in night blindness, xerophthalmia, and keratinization of the skin.

Option C: Vitamin K deficiency leads to hemorrhagic disease in newborns and elevated prothrombin time and bleeding. Vitamin K acts as a cofactor in the synthesis of coagulation proteins. Toxicity due to vitamin K results in hemolysis, hyperbilirubinemia, and kernicterus in the newborn.

Option D: Vitamin C deficiency leads to scurvy, which occurs due to defective collagen formation resulting in generalized bleeding tendencies and poor wound healing. Radiographic features of scurvy include the white line of Frankel, Trummerfeld zone, Pelkan spurs, and periosteal elevation. Tender nodules at the costochondral junctions, known as the scorbutic rosary, are characteristic.



Solution to Question 2:

The image shown above is that of a cock-up splint. It is used to manage radial nerve palsy.

Radial nerve palsy manifests as an inability to dorsiflex the wrist and digits (wrist drop and finger drop). Numbness occurs on the dorsoradial aspect of the hand and the dorsal aspect of the radial 3 1/2 digits.

Other options:

Option B: Median nerve injury happens in carpel tunnel syndrome where a nocturnal wrist splinting that helps to maintain the hand in a neutral position could be of help during the initial therapy to alleviate symptoms. It prevents prolonged flexion or extension of the wrist.

Option C: Ulnar nerve injury / ulnar claw hand could be managed with a knuckle bender splint if nerve recovery is anticipated. The splint can be used to correct and prevent metacarpophalangeal hyperextension.

Option D: Volkmann's ischaemic contracture (VIN) is most commonly seen following supracondylar humeral fractures. Mild cases of VIN are treated with a Volkmann's splint (Turnbuckle splint).

The first image below shows a knuckle-bender splint and the second image shows a Volkmann's splint

Solution to Question 3:

The given X-Ray shows a patella with a smooth discontinuity suggestive of an unfused accessory ossification center. Since this should arouse the suspicion of a bipartite patella, appropriate management would be to take an X-Ray of the other knee.

A bipartite patella (two-part patella) is a patella with an unfused accessory ossification centre, typically at the superolateral aspect. It usually is asymptomatic and is noted incidentally, however pain, when present, can be caused by overuse (such as in a cyclist or climber).

There are three types of bipartite patella

• Type I (5%) occurs at the inferior pole

• Type II (20%) occurs along the entire lateral border of the patella

• Type III (75%) the most common type, occurs as an elliptical area in the superolateral portion of the patella

Depending on the number of fragments they may be classified as bipartite, tripartite or multipartite. For diagnosis, a conventional skyline view is followed by a skyline view with the patient in a squatting weight-bearing position to assess if the pain is caused by a nonunion at the bipartite site. If the gap is higher in the squatting weight-bearing posture than in the standard skyline view, the test result is considered positive. MRI is not required to diagnose the condition.

If pain occur with a bipartite patella, limiting and restricting activity, correcting the activity that is causing an overuse syndrome, and prescribing nonsteroidal antiinflammatory agents can result in improvement in 3 weeks. Operative treatment with the Ogata technique is rarely necessary.

Solution to Question 4:

The given clinical vignette is suggestive of carpal tunnel syndrome and it can be assessed by Durkan's test.

Durkan's test is the most specific test for the diagnosis of carpal tunnel syndrome. In Durkan's test, direct compression of the median nerve is done for about 30 seconds, which elicits paresthesias and pain in cases of carpal tunnel syndrome.

Carpal tunnel syndrome is the most common entrapment mononeuropathy caused by compression of the median nerve as it passes through the fibro-osseous carpal tunnel beneath the flexor retinaculum. Patients present with pain and paresthesia over the radial 3 and a half digits. There may be a weakness of thumb opposition. In chronic cases, the thenar eminence is wasted. Carpal tunnel syndrome can be associated with rheumatoid arthritis, pregnancy, chronic renal failure, acromegaly, hypothyroidism, and gout. The diagnosis is usually made on clinical grounds but nerve conduction studies can confirm the diagnosis.

Other tests for CTS -

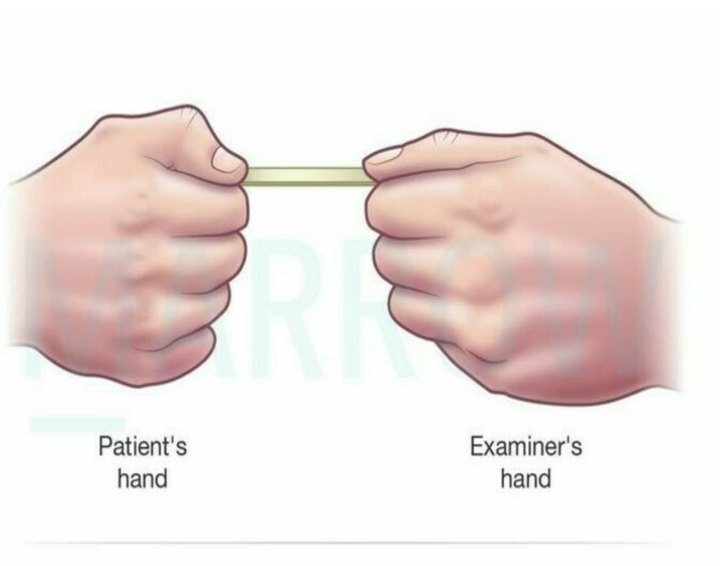
• Phalen's test - Elicited by holding the wrist flexed fully. The appearance of sensory symptoms within 60 seconds implies a positive test.

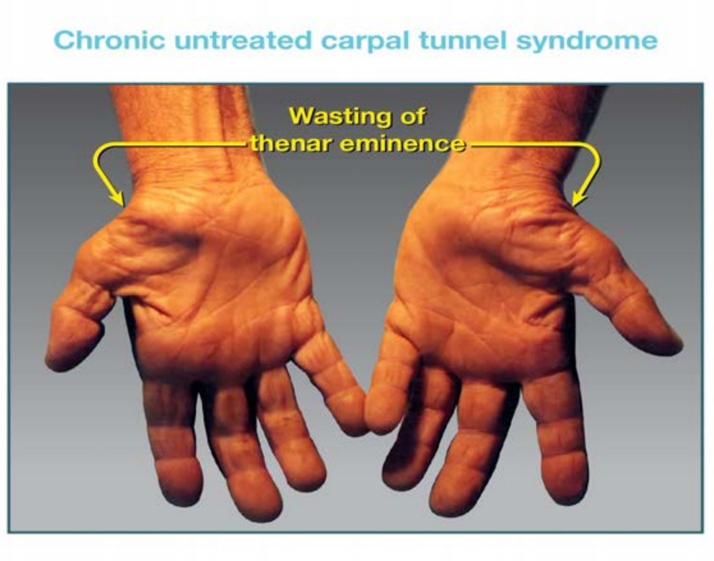
• Tinel's sign - Sensory symptoms reproduced by percussing over the median nerve imply a positive test.

Guyon’s canal is a fibro-osseous canal, approximately 4 cm long in the adult, on the anteromedial aspect of the wrist under the superficial part of the flexor retinaculum. The ulnar nerve passes in Guyon’s canal along with the ulnar artery and can be compressed here. Froment's test is used in a suspected ulnar nerve injury.

In Froment's test or Book Test, the patient is asked to grasp a book or a sheet of paper between the thumb and index finger. Due to weakness of the adductor pollicis muscle, adduction of the thumb is lost. When asked to hold a book, the patient tries to hold the book by using the flexor pollicis longus muscle instead of the adductor, this produces flexion at the interphalangeal joint.

The picture below shows Froment's test





Non-surgical Management of Carpal Tunnel Syndrome: In mild to moderate cases of carpal tunnel syndrome, non-surgical management options can be explored to alleviate symptoms and improve function. These conservative treatments include:

- Wrist splinting: Wearing a wrist splint at night can help maintain the wrist in a neutral position, reducing pressure on the median nerve and providing relief from nocturnal symptoms.

- Activity modification: Identifying and modifying activities that exacerbate symptoms can help in managing carpal tunnel syndrome. For example, taking frequent breaks during repetitive tasks or adjusting the workstation ergonomics can be beneficial.

- Nonsteroidal anti-inflammatory drugs (NSAIDs): Over-the-counter NSAIDs, such as ibuprofen, can help manage pain and inflammation associated with carpal tunnel syndrome.

- Corticosteroid injections: In some cases, corticosteroid injections into the carpal tunnel may provide temporary relief from symptoms by reducing inflammation and pressure on the median nerve.

- Physical and occupational therapy: Therapists can help patients learn exercises to stretch and strengthen the hand and wrist muscles, improving symptoms and function.

If conservative treatments do not provide sufficient relief or the condition worsens, surgical intervention, such as carpal tunnel release surgery, may be considered.

Solution to Question 5:

The above X-ray shows the bending of the femur known as Shepherd's crook deformity. It is seen in fibrous dysplasia of proximal femur.

Fibrous dysplasia is a condition that affects bone development and is characterized by the replacement of the bone with fibrous tissue. The patient presents with a progressive limb deformity or a fracture through a previously asymptomatic lesion. The presentation often occurs in the second decade. Other symptoms include nonspecific bone pain, tenderness, and swelling in the affected area. The commonly involved sites are the proximal femur, tibia, pelvis, and foot. Polyostotic fibrous dysplasia is associated with Albright-McCune syndrome.

Radiological findings include:

• Well-defined, expansile intramedullary lesions

• Hazy ground glass appearance is seen in the majority of cases

• Shepherd's crook deformity - Expansion with marked varus in the proximal femur

Other options:

Option A: Non-ossifying fibroma is the commonest variant of giant cell bone tumors. It is a benign, non-neoplastic lesion most commonly seen in the metaphyses of the long bones in children. Non-ossifying fibromas are usually asymptomatic, and often discovered incidentally on X-rays performed for other reasons. They commonly occur in the metaphyses of long bones and are characterized by well-defined, eccentric, cortically-based lytic lesions. The X-ray findings in the question do not match this description.

Option C: Paget's disease of the bone is a condition where there is an increased bone turnover followed by enlargement and thickening of the bone. In this condition, calcium levels are normal with raised ALP. Paget's disease typically presents in older adults and affects the axial skeleton, including the pelvis, spine, skull, and femur. It is characterized by increased bone turnover, leading to disorganized bone remodeling, resulting in enlarged and deformed bones. The X-ray findings and the patient's age do not match the characteristics of Paget's disease.

Option D: Osteogenesis imperfecta is a genetic disorder characterized by increased bone fragility and susceptibility to fractures. It is associated with blue sclera, dental fragility, and hearing loss. It presents with the hallmark features of osteoporosis with increased fragility of bones along with blue sclera, dental fragility as well as hearing loss. The X-ray findings in the question do not show the characteristic features of osteogenesis imperfecta, such as thin cortices, multiple fractures, or wormian bones.

Shepherd's crook deformity - Fibrous Dysplasia

Solution to Question 6:

The given x-ray showing a lytic lesion at the end of a long bone with a soap-bubble appearance suggests a giant cell tumor (GCT) or osteoclastoma.

Osteoclastoma usually affects young adults of age 20 to 40 years with a slight female predominance. It is a benign but locally aggressive tumor. It is exclusively seen in adults and affects the ends of long bones, involving metaphysis/ epiphysis of long bones. GCT is most commonly seen in the distal femur, proximal tibia, proximal humerus, and distal radius. Common presenting complaints are swelling and vague pain. Sometimes, the patient presents for the first time with a pathological fracture through the lesion.

Important x-ray features of a GCT:

• A solitary, possibly loculated, lytic lesion in the epiphysis

• Eccentric location, most often subchondral

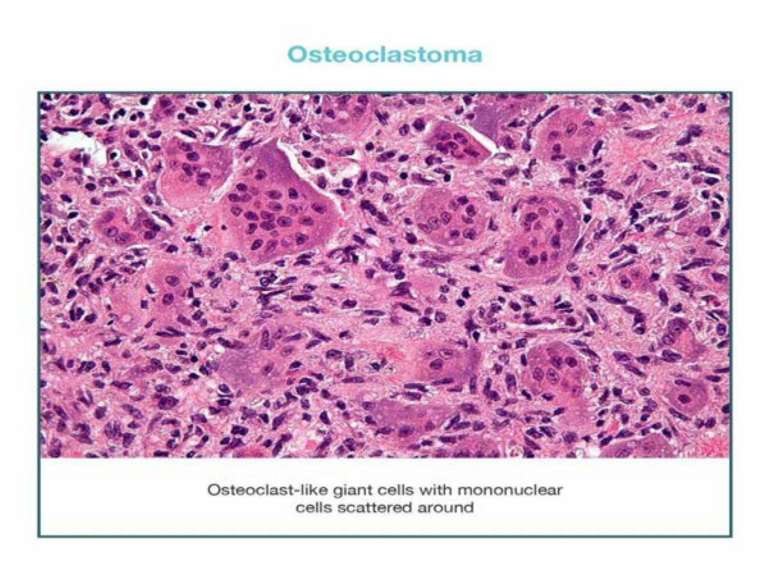
• Expansion of the overlying cortex

• Soap bubble appearance - The tumor is homogeneously lytic with trabeculae of the remnants of bone traversing it, giving it a loculated appearance called soap-bubble appearance.

• No calcification within the tumor

• No or minimal reactive sclerosis around the tumor

Histologically, GCTs consist of undifferentiated spindle cells profusely interspersed with multi-nucleated giant cells. The tumor stroma is well-vascularized with bands of cellular or collagenous fibrous tissue. The image below shows the histopathology of osteoclastoma.



Osteoclast-like giant cells with mononuclear cells scattered around

The treatment of choice for GCT is extended curettage and bone grafting.

• GCT in the relatively insignificant bone like the distal ulna, clavicle, and proximal fibula is treated by resection without reconstruction.

• Bisphosphonates have been shown to reduce the proliferation of osteoclasts and hence have been shown to stabilize local and metastatic disease.

• Radiation can be used for spinal or sacral tumors.

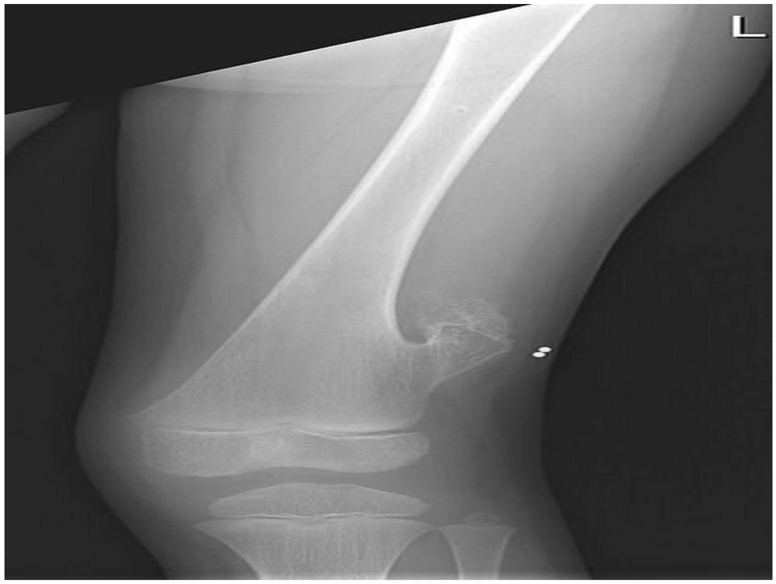
Other options:

Option A: Chondroblastoma is a benign tumor that usually appears as well-circumscribed round or oval lytic lesions in the center of the epiphysis of a long bone. It presents as a painful mass that causes effusion or stiffness. It is most commonly seen in ages 10-20 years.

The following is an x-ray image of the knee showing chondroblastoma:

Option B: Osteochondromas are common benign bone tumors commonly present in adolescents. The lesions consist of a bony mass, often in the form of a stalk, produced by endochondral ossification of a growing cartilaginous cap. They are usually painless. The common sites of involvement are the distal femur, proximal tibia, and proximal humerus.

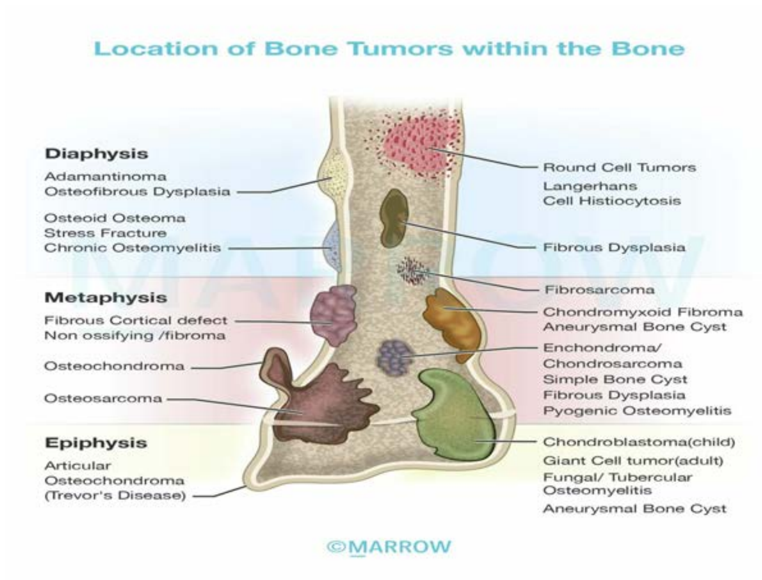
The following is an x-ray image of the knee showing osteochondroma of the distal knee:



Option D: Aneurysmal bone cyst is a tumor that affects all age groups but generally occurs more commonly in adolescents with an equal male: female distribution. It usually occurs in the metaphysis of long bones (like femur and tibia) and the posterior elements of vertebral bodies. Patients usually present with pain and swelling. Radiographically, ABC is seen as an eccentric, expansile, lytic lesion with well-defined margins.

The following is an x-ray image of the knee showing ABC:

The image below shows common locations of various bone tumors:


Location of Bone Tumors within the Bone
Diaphysis
Adamantinoma
Osteofibrous Dysplasia
Osteoid Osteoma
Stress Fracture
Chronic Osteomyelitis
Metaphysis
Fibrous Cortical defect
Non ossifying /fibroma
Osteochondroma
Osteosarcoma
Epiphysis
Articular
Osteochondroma
(Trevor's Disease)
Round Cell Tumors
Langerhans
Cell Histiocytosis
Fibrous Dysplasia
Fibrosarcoma
Chondromyxoid Fibroma
Aneurysmal Bone Cyst
Enchondroma/
Chondrosarcoma
Simple Bone Cyst
Fibrous Dysplasia
Pyogenic Osteomyelitis
Chondroblastoma(child)
Giant Cell tumor(adult)
Fungal/ Tubercular
Osteomyelitis
Aneurysmal Bone Cyst
©MARROW

Orthopedics INI-CET 2023

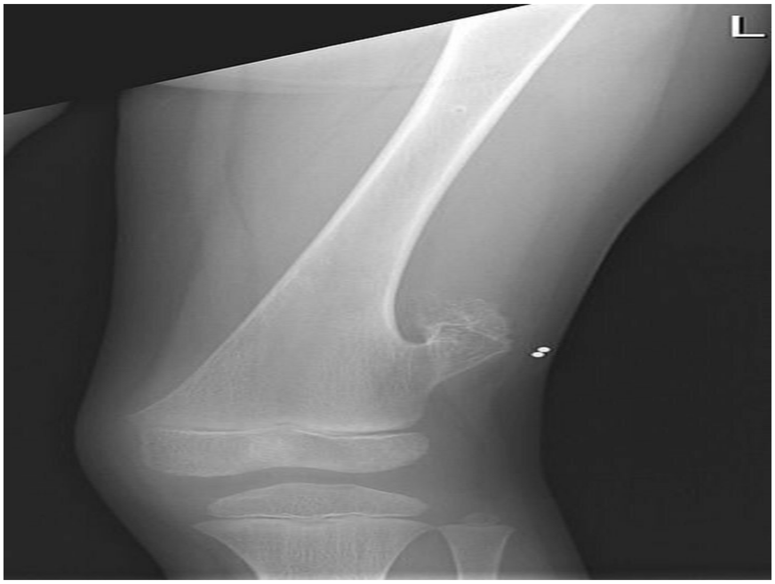
Question 1:

A 25-year-old male who sustained a shoulder injury presented with swelling around the shoulder joint. An X-ray of the shoulder joint is shown below. Which is/are the ligament(s) likely to have been injured?

- **a)** Both acromioclavicular and coracoclavicular ligament.
- **b)** Coracohumeral ligament
- **c)** Coracoacromial ligament
- **d)** Acromioclavicular ligament

Question 2:

A 14-year-old male presented with a mushroom-like tumor in the distal femur for the past 2 years. Which of the following features will help the clinician to identify a malignant transformation?



a) Initially growth of the tumor was away from the joint, now the growth of tumor is towards the joint.

b) Tumor does not have malignant transformation potential.

c) Tumor is in continuation with bone marrow cavity of the normal bone.

d) Hyaline cap thickness on MRI is 2cms.

Question 3:

A 16-year-old girl presents with multiple swellings and multiple brown rashes. She has increased uptake on bone scans over the femur, skull, and ribs. Biochemical parameters are abnormal. She also has a history of hypothyroidism. An X-ray of her femur is given below. Which of the following is the likely diagnosis?

a) Langerhans cell histiocytosis

b) McCune-Albright syndrome

c) Neuroma with bone involvement

d) Papillary carcinoma of thyroid

Question 4:

A patient came to the emergency department with a history of road traffic accident and a shaft of femur fracture. The patient was stabilized. After 12 hours, he developed sudden onset breathlessness, confusion, and petechial skin rashes over the chest and axilla. What is the probable diagnosis?

a) Fat embolism

b) Head injury

c) Blunt trauma to the chest

d) Hemorrhagic shock

Question 5:

Choose the true statements:

a) ii, iii and iv only

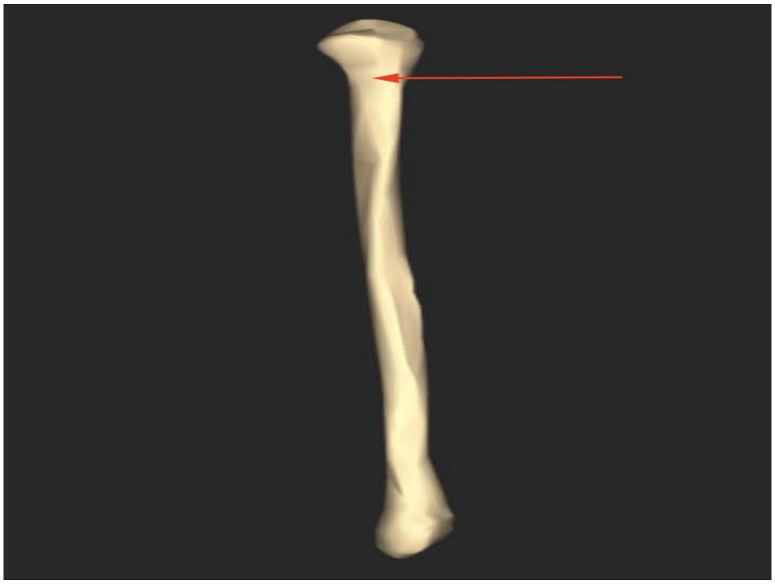
b) i, iii only

c) iii, iv only

d) i, ii only

Question 6:

What type of gait will be seen following a fracture in this region?



a) Waddling

b) High stepping

c) Wide-stepping gait

d) Stamping

Question 7:

True statements about bankart lesion?

a) Avulsion of the anteroinferior glenoid labrum

b) Avulsion of the anterosuperior glenoid labrum

c) Depression in posteroinferior part of the head of the humerus

d) Depression in posterosuperior part of the head of the humerus

Question 8:

Hawkin's classification is used for which fractures?

a) Talus

b) Navicular

c) Tibia

d) Calcaneum

Question 9:

A female patient presented with swollen and tender joints. There is symmetric involvement of multiple joints such as wrists, hands, and feet, which is bilateral, along with morning stiffness lasting more than 1 hour. Erythema is absent. What is the most probable diagnosis?

- **a)** Gout
- **b)** Rheumatoid arthritis
- **c)** Osteoarthritis
- **d)** Psoriatic arthritis

Question 10:

A 45-year-old patient with a tibial fracture in a cast presented to the emergency department with complaints of intense pain. The cast was removed and on examination, passive extension was painful. The dorsalis pedis and posterior tibial pulses were present and there was a loss of sensation over the first web space. What is the next step?

- **a)** Duplex imaging
- **b)** Re-apply cast
- **c)** Measure anterior compartment pressure
- **d)** Analgesics

Question 11:

Three-point bony relationship of the elbow will not be damaged in?

- **a)** Supracondlyar fracture humerus
- **b)** Intercondylar fracture humerus
- **c)** Elbow dislocation
- **d)** Lateral epicondyle fracture

Answer Key

Question No.	Correct Option
1	a

2	d
3	b
4	a
5	a
6	b
7	a
8	a
9	b
10	c
11	a

Detailed Explanations

Solution to Question 1:

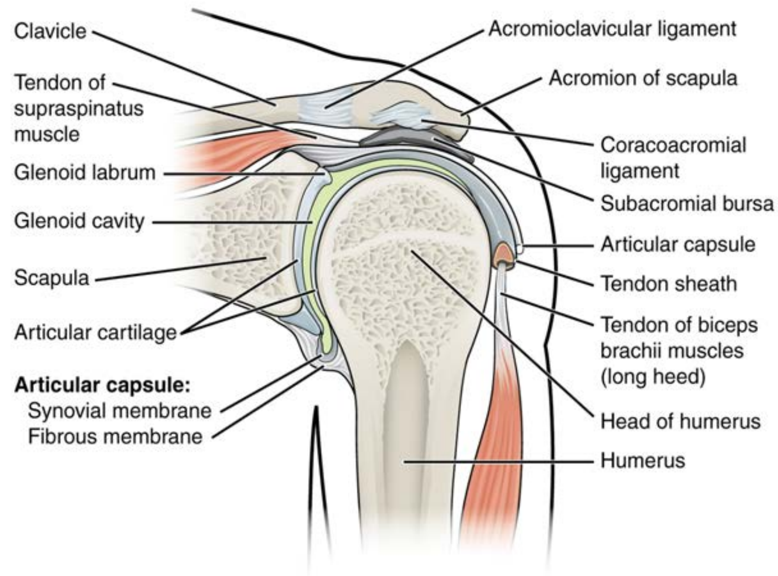
The given image shows that the clavicle has moved up and there is a gap between the clavicle and the acromion suggesting injury to both acromioclavicular and coracoclavicular ligament.

The acromioclavicular joint is a synovial plane type between the acromial end of the clavicle and the medial acromial margin. The two ligaments providing support are acromioclavicular(intrinsic) and coracoclavicular(extrinsic). The acromioclavicular ligament has a superior part that extends between the upper aspect of the clavicle's lateral end and the adjacent acromion. Its inferior part, extends between the lower surface of the clavicle's lateral end and the adjoining acromion. Additionally, it serves as an attachment site for the intra-articular disc.

The coracoclavicular ligament connects the clavicle with the coracoid process of the scapula. Although it is separate from the acromioclavicular joint, it plays a crucial role in maintaining the alignment of the acromion and the clavicle. Therefore, it contributes significantly to the suspension mechanism of the scapula. The ligament has two parts - trapezoid and conoid parts.

The primary cause of injury is direct impact to the lateral aspect of the shoulder or the acromion process while the arm is in an adducted position. It can also be caused by falling onto an outstretched hand or elbow.

Standard X-rays are sufficient for diagnosing acromioclavicular joint injuries and should be utilized to assess other potential sources of shoulder pain resulting from trauma. By comparing the coracoclavicular interspace distance with the contralateral view, the injury can be identified.



Solution to Question 2:

The given X-ray shows a well-defined exostosis pointing to the diagnosis of osteochondroma. 2cms thickness of hyaline cap thickness on MRI is suggestive of malignant transformation.

Osteochondroma can turn into low-grade chondrosarcoma but the malignant transformation is extremely rare(Option B). It should be suspected in the following cases:

• Growth after skeletal maturity

• Lucency in the bone

• Additional scintigraphic activity

• Cortical destruction

• Pain after puberty

• Soft tissue mass

• Thickened cartilage cap >2 cm

Osteochondromas are common benign bone tumors mostly arising from the metaphysis of long bones. The cortical and medullary continuity of the tumor with the underlying bone is a pathognomonic feature that establishes the diagnosis. (Option C). The lesions consist of a bony mass, often in the form of a stalk, produced by endochondral ossification of a growing cartilaginous cap. Most of these lesions are asymptomatic and are discovered incidentally.

These are of two types: Pedunculated and sessile, pedunculated types are more common. The definite stalk is directed away from the physis(Option A). The lesions are usually covered with a cartilaginous cap which cannot be seen radiologically and hence the lesions are larger clinically than what their X-Ray appearance suggests. The important feature is that the host bone flares from the cortex into the osteochondroma.

Most patients are asymptomatic and never seek medical attention. However, symptomatic people may undergo resection.

Solution to Question 3:

The above X-ray shows the bending of the femur known as Shepherd's crook deformity. It is seen in fibrous dysplasia of the proximal femur. The diagnosis based on the x-ray and clinical scenario points to McCune-Albright syndrome.

McCune Albright syndrome presents as a triad of cafe-au-lait spots (cutaneous pigmentation), polyostotic fibrous dysplasia, and precocious puberty. It is also associated with multiple endocrinopathies like hyperthyroidism, acromegaly, etc. The locus of involvement is the GNAS1 gene.

Cutaneous pigmentation is more prominent on the side with severe bone involvement. Fibrous dysplasia is a condition that affects bone development and is characterized by the replacement of the bone with fibrous tissue. The patient presents with a progressive limb deformity or a fracture through a previously asymptomatic lesion. The presentation often occurs in the second decade. Other symptoms include nonspecific bone pain, tenderness, and swelling in the affected area. The commonly involved sites are the proximal femur, tibia, pelvis, and foot.

Radiological findings include:

• Well-defined, expansile intramedullary lesions

• Hazy ground glass appearance is seen in the majority of cases

• Shepherd's crook deformity - expansion with marked varus in the proximal femur.

Shepherd's crook deformity - Fibrous Dysplasia

Other options:

Option A: Langerhans cell histiocytosis (LCH) occurs due to the proliferation of a special type of immature dendritic cells, called Langerhans cells. Their normal function is to present antigens. The most common mutation is the BRAFV600E gain-of-function mutation.

It is characterized by single or multiple osteolytic bone lesions and can also infiltrate nearly every organ, like the skin (of the scalp), lungs, liver, spleen, bone marrow, or the central nervous system. It presents with pain and swelling in the affected area. Spinal involvement results in compression fractures and neurological involvement. Vertebral involvement results in a rapid flattening of the vertebral body (vertebra plana).

LCH can present as the following clinical entities:

• Multifocal multisystem Langerhans cell histiocytosis (Letterer-Siwe disease) which is mostly seen in infants

• Unifocal and multifocal system Langerhans cell histiocytosis (eosinophilic granuloma) - The combination of bone defects, diabetes insipidus, and exophthalmos is called the Hand-Schüller-Christian triad that is mostly seen in younger children

• Pulmonary LCH is mostly seen in adults.

Option D: Papillary carcinoma of the thyroid (PTC) presents as a painless nodule or mass in the neck or enlargement of cervical lymph nodes. It usually occurs in women between 25 to 50 years of age. PTC is the most common thyroid cancer associated with previous exposure to ionizing radiation. Conventional PTCs have translocations involving RET or NTRK, and point mutations involving the BRAF gene.

Patients present with painless swelling. Initial presentation may be with cervical lymphadenopathy due to lymph node metastases.

Solution to Question 4:

The most likely diagnosis based on the given clinical scenario is fat embolism.

Fat embolism is caused by the fat globules originating from the bone marrow following the fracture of long bones. Clinical features of fat embolism are usually seen within 72 hours. They include tachycardia, fever, breathlessness, petechiae, confusion, and restlessness. Gurd's criteria is used for the diagnosis of fat embolism syndrome:

There is no specific test for diagnosis. However, urine analysis to look for fat globules and monitoring of blood pO2 are done. Management of fat embolism is mostly supportive, with oxygen therapy.

Other options:

Option C: A majority of chest trauma is caused by blunt injuries. Six life-threatening conditions such as airway obstruction, tension pneumothorax, open pneumothorax, massive hemothorax, flail chest, and pericardial tamponade should be immediately investigated and treated. Only 10% or less of blunt chest trauma patients require surgical treatment, and the remaining patients can be treated conservatively.

Option D: Hemorrhagic shock is the most common cause of shock in surgical or trauma patients. Here the patient presents with agitation, cold clammy extremities, hypotension, tachycardia, and weak peripheral pulses. Acidosis and elevated lactate levels may be seen due to hypoperfusion of tissues.

Solution to Question 5:

Statements ii, iii, and iv are true, and statement i is incorrect.

Statement i: The most commonly fractured carpal bone is the lunate is a false statement. The most common fractured bone is the scaphoid.

Statement ii: The scaphoid is most commonly fractured at the waist is a true statement. A scaphoid fracture is more common in young adults. Commonly, the fracture occurs through the waist of the scaphoid.

Statement iii: The cast for scaphoid fracture is set in a glass holding position is a true statement. The affected hand is immobilized in a scaphoid cast for 3-4 months. The scaphoid cast extends from below the elbow to the metacarpal heads including the thumb up to the inter-phalangeal joint. The wrist is maintained in a little dorsiflexion and radial deviation (glass holding position).

Statement iv: Kienböck's disease is due to osteonecrosis of the lunate is a true statement. Kienböck's disease is usually seen in young adults, who present with pain and stiffness in the wrist. A history of trauma may be present. On examination, localized tenderness can be appreciated over the lunate, and a reduction in grip strength can be noted. Limited and painful wrist movements are noted in the later stages of the disease.

Solution to Question 6:

In the above scenario, the structure marked in the given image is the neck of the fibula. Injury at this site causes failure of dorsiflexion (foot drop) due to injury to the common fibular nerve (also known as common peroneal nerve). Foot drop leads to high stepping gait.

The common peroneal nerve supplies the extensors and evertors of the foot. Injury to the common peroneal nerve results in foot drop. So the patient attempts to lift the leg high enough during walking so that the foot does not drag on the floor. This is known as a high-stepping gait.

The common fibular is derived from the dorsal branches of the L4 - L5 and S1 - S2 ventral rami. It innervates the short head of the biceps femoris in the posterior compartment of the thigh and then continues into the lateral and anterior compartments of the leg and onto the foot.

It gives the following branches:

• Sural communicating nerve (cutaneous branch)

• Lateral sural cutaneous nerve (cutaneous branch)

• Superficial fibular nerve

• Deep fibular nerve.

Injury to the common peroneal nerve will lead to:

• Weakness of dorsiflexion of foot or foot drop

• Loss of dorsiflexion

• Loss of sensation over the lateral aspect of the leg and dorsum of the foot.

• Wasting of fibular and anterior tibial muscles.

Other Options:

Option A: Waddling gait is seen in developmental dysplasia of the hip and in proximal muscle weakness.

Option C: Wide-stepping gait is a broad-based unstable gait and is indicative of cerebellar dysfunction.

Option D: Stamping of the foot is seen in tarsal tunnel syndrome. Tarsal tunnel syndrome is a condition with pain and sensory disturbance occurring in the medial part of the forefoot, unrelated to weight-bearing. This is due to compression of the posterior tibial nerve behind and below the medial malleolus. Typically the pain worsens at night and gets relieved on walking or stamping the foot.

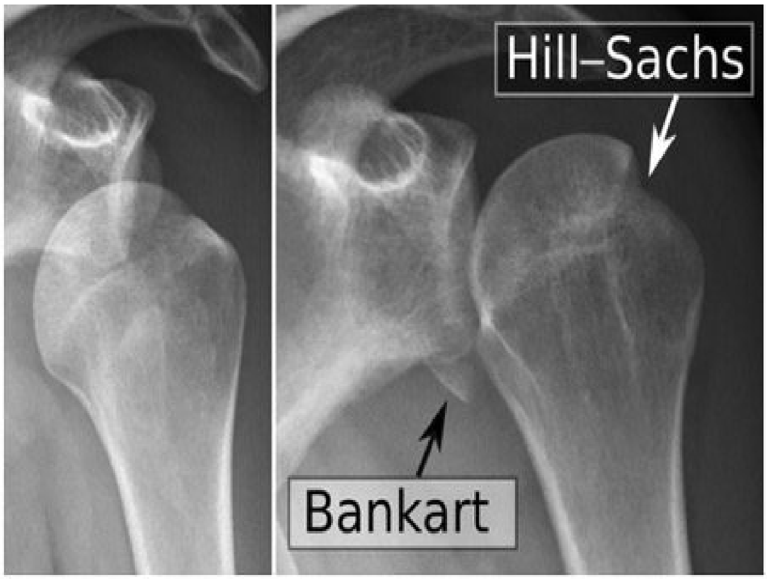
Solution to Question 7:

Bankart lesion is an avulsion in the anteroinferior part of the glenoid labrum caused due to the recurrent anterior dislocation of the humeral head.

Hill-Sachs lesion is a compression fracture on the posterolateral aspect of the humeral head. It is caused by the pressure of the anterior glenoid rim on the humeral head in recurrent dislocations.

Anterior dislocation is the commonest shoulder dislocation and can be caused due to trauma from behind the shoulder pushing the head out anteriorly. It is also caused due to falling on an outstretched hand in an abducted and externally rotated shoulder. It can be diagnosed on clinical examination with flattening of the shoulder contour and ability to place a rule (Hamilton ruler test) and the inability to touch his/her opposite shoulder (Duga's test). X-ray confirms the diagnosis and may show associated Hill Sachs or bony bankart lesion. The most common complication is recurrent shoulder dislocation.

The following X-ray shows the Bankart lesion and Hill-Sachs lesion:



Solution to Question 8:

Hawkin's classification is used for the fracture of the neck of the talus.

The fracture of the neck of the talus most commonly occur following a high-energy injury causing forced dorsiflexion with axial load. They can be further classified using Hawkin's classification:

• Type I - Undisplaced

• Type II - Subtalar dislocation

• Type III - Subtalar and tibiotalar dislocation

• Type IV - Subtalar, tibiotalar, and talonavicular dislocation

Undisplaced fractures are managed conservatively in a short leg cast for 8-12 weeks and a non-weight bearing cast for 6 weeks. Hawkin's type II-IV are managed by open reduction with internal fixation (ORIF).

The below image shows the Hawkins classification of talar neck fractures.

Solution to Question 9:

In the above clinical scenario, a female patient presenting with swollen and tender joints, symmetric involvement of multiple joints such as wrists, hands, and feet, bilateral in nature, along with morning stiffness lasting for more than 1 hour, and the absence of erythema is rheumatoid arthritis.

Rheumatoid arthritis (RA) is a chronic inflammatory disease of symmetric polyarthritis, mainly affecting the small joints of the hands and feet. Its incidence increases between 25 and 55 years of age. Clinically, patients present with early morning joint pain and stiffness lasting >1 hour that relieves with activity.

Signs and symptoms -

• Swelling and pain of the small joints of the hands initially- metacarpophalangeal (MCP) and proximal interphalangeal (PIP) joints are commonly involved. Distal interphalangeal (DIP) joint involvement is usually due to coexistent osteoarthritis.

• Morning stiffness

• Swan-neck deformity - Hyperextension of the PIP joint with flexion of the DIP joint

• Boutonnière deformity - Flexion of the PIP joint with hyperextension of the DIP joint,

• Z-line deformity - Subluxation of the first MCP joint with hyperextension of the first interphalangeal (IP) joint

• Ulnar deviation of fingers

• Constitutional findings such as fever, weight loss, etc

• Monoarthritis of large joints can occur later

• Tenosynovitis of palm(trigger finger)

• Atlantoaxial joint involvement

Joints commonly affected in Rheumatoid Arthritis: (bilateral and symmetrically involved)

- MCP joints
- Wrists
- PIP joints
- MTP joints
- Atlantoaxial joint

Laboratory investigations - Detection of Rheumatoid factor and anti-CCP antibodies help in differentiating this condition from other forms of arthritis.

Diagnosis is made with clinical signs and symptoms with corroboration from lab and imaging findings.

Other options-

Option (A): Gout is characterized by episodic acute arthritis or chronic arthritis due to the deposition of monosodium urate(MSU) crystals in the joints and connective tissue. It commonly affects postmenopausal women and middle-aged to elderly men. The metatarsophalangeal joint (MTP) of the first toe is commonly involved. Other joints that are commonly affected include the tarsal joints, ankle, and knee.

Option (C): Osteoarthritis is the most common form of arthritis with a high prevalence in the elderly population. Commonly involved joints are the knee, hip, first MTP, and the cervical and lumbosacral spine. In the hands, the DIP and PIP joints and the base of the thumb are commonly involved. The wrist, elbow, and ankle are usually spared.

Option (D): Psoriatic arthritis- In most cases psoriasis precedes arthritis but in some cases, both of them present in a span of 1 year, or arthritis precedes psoriasis. There can be involvement of DIP and PIP joints and the joint involvement tends to follow "ray "distribution. Ray distribution refers to the involvement of all the joints of one finger while the adjacent finger is totally spared. Nail changes in the affected fingers are a differentiating feature of this condition.

Solution to Question 10:

The clinical presentation of this patient with a history of tibial fracture complaining of intense pain, with painful passive extension and loss of sensation of the first web space upon the removal of the cast, is suggestive of acute compartment syndrome. The next step would be to measure the anterior compartment pressure to confirm the diagnosis.

Acute compartment syndrome refers to the rise in intra-compartmental pressure leading to tissue ischemia and necrosis. The injuries at high risk of developing acute compartment syndrome are fractures of the proximal one-third of the tibia, soft tissue injury, distal radius fracture, and diaphyseal fracture of the forearm. Other high-risk injuries are crush injuries and circumferential burns.

The classic features of ischemia in acute compartment syndrome include the 5 Ps:

- Pain: is the first symptom of acute compartment syndrome. On passive stretching of the muscles, severe pain can be elicited.

• Paresthesia: paresthesia and hypoesthesia are the first signs of nerve ischemia.

• Paralysis: paralysis of muscle groups affected in that compartment is a late sign.

• Pulselessness: usually, the peripheral pulses and capillary return are intact in the early stages, unless there is a major arterial injury. If pulses are absent, arteriography is indicated.

• Pallor: may not always be seen as ischemia occurs at the capillary level.

The diagnosis can be confirmed by measuring the intra-compartmental pressure using a split catheter introduced into the compartment. The difference between diastolic pressure and compartment pressure is known as differential pressure (ΔP). If the differential pressure (ΔP) is less than 30 mm Hg, immediate compartment decompression is indicated.

Treatment:

• Prompt decompression of the affected compartment is done.

• All the dressings, bandages, and cast on the affected limb are removed and the limb is nursed flat.

• Fasciotomy is done if the differential pressure is less than 30 mm Hg

• Fasciotomy wounds are left open and observed after 2 days. If there is necrosis, debridement is done. If the wounds are healthy, they can be sutured without tension or skin grafting can be done.

Solution to Question 11:

The three-point bony relationship will not be damaged in supracondylar fracture.

Supracondylar fracture of the humerus is most commonly seen in children of 5-7 years after a fall on an outstretched hand with the elbow in full extension. It is most commonly of extension type followed by flexion type. Gartland classification is used for supracondylar fractures. The three-point bony relationship is maintained in supracondylar fracture (option B). Three-point bony landmarks in the elbow are the tips of medial and lateral epicondyles and the olecranon.

Nerve and vessel injury:

• Most common nerve to be injured in extension type - anterior interosseous nerve > Median nerve > Radial nerve > Ulnar nerve. (AMRU)

• Most common nerve to be injured in flexion type - Ulnar nerve.

• Most common vessel to be injured - brachial artery.

Radiological findings:

• An undisplaced fracture may show a triangular lucency like a sail of a yacht (sail sign), also called a fat pad sign. This is due to the fat pad around the elbow being pushed forward by the hematoma.

• Baumann’s angle (humeral-capitellar angle) can help in assessing the degree of medial angulation before and after reduction.

The most common complication of supracondylar fracture is malunion. This leads to cubitus varus or gunstock deformity. Malunion occurs due to improper reduction and not due to physeal damage, as the fracture is extraphyseal (option D). Other complications include myositis ossificans, compartment syndrome, and Volkmann ischemic contracture.

Undisplaced supracondylar humerus fractures are treated by an above-elbow plaster slab with the elbow in 90° flexion. Displaced fractures are treated by closed reduction and K-wire fixation since they tend to slip after reduction. Open fractures or those associated with brachial artery injury are treated by open reduction and internal fixation.

Cubitus varus deformity is corrected after skeletal maturation and lateral closed-wedge osteotomy (French osteotomy) is the method employed.

Orthopedics INI-CET 2024

Question 1:

Which of the following layers is affected in Salter Harris type 1 fracture?

a) Through the whole of epiphysis

b) Resting zone of the growth plate

c) Calcific layer of the growth plate

d) Mineralisation zone

Question 2:

A 75 year old man presents with fracture of intracapsular neck of femur. What is the "standard of care" management done for this patient?

a) Boot and bar cast

b) Russell's traction

c) Intramedullary nailing

d) Hemiarthroplasty

Question 3:

What is the diagnosis of the following X Ray?

a) Tibial condylar fracture

b) Rheumatoid arthritis

c) Patellar fracture

d) Hemarthrosis

Question 4:

A 22 year old came with thigh pain. X ray was done as shown below. What is the diagnosis?

a) Osteosarcoma

b) Osteomalacia

c) Chronic osteomyelitis

d) Ewings sarcoma

Question 5:

Out of the following, which is the most common site of osteoporotic fracture?

a) Vertebral fracture

b) Glenoid fracture

c) Subtrochanteric fracture

d) Mid shaft of humerus

Question 6:

A 70 yr old female presents with extracapsular neck of femur fracture. What is the best step in management?

- **a)** Russell's traction
- **b)** Interlocking plate
- **c)** Dynamic hip screw
- **d)** Boot and bar cast

Question 7:

A 26 year old female presented with proximal one-third fracture of shaft of femur. What is the treatment of choice ?

- **a)** Intramedullary nailing
- **b)** Above knee slab
- **c)** Hip spica
- **d)** Above knee cast

Question 8:

All of the following statements regarding ankylosing spondylitis are true except?

- **a)** Most commonly seen between the age of 20 to 40.
- **b)** Affects males more than females
- **c)** In 85-95% patients, HLA-B27 is positive
- **d)** Non Erosive arthritis of the joint with sclerosis is seen

Question 9:

A 60-year-old female comes with complaint of unable to bear weight following trivial trauma. X ray of hip and leg is given. What is the diagnosis?

a) Brown tumor

b) Disseminated TB

c) Eosinophilic granulomas

d) Metastasis

Question 10:

A 40-year-old male comes with an intracapsular fracture of the femur. What is the surgery of choice?

a) Open reduction internal fixation

b) Closed reduction internal fixation

c) Hemiarthroplasty

d) Total arthroplasty

Question 11:

FRAX is related to?

a) Fracture risk assessment

b) Fracture rehabilitation assessment

c) Fracture repair assessment

d) Fracture reduction assessment

Answer Key

Question No.	Correct Option
1	c
2	d
3	c
4	c
5	a
6	c
7	a
8	d
9	d
10	b
11	a

Detailed Explanations

Solution to Question 1:

Salter Harris type 1 fracture is a transverse fracture through the hypertrophic or calci■ed zone of the growth plate. The growing zone of the physis is usually not injured and growth disturbance is uncommon.

Solution to Question 2:

In patients over 65 years old with an intracapsular neck of femur fracture and a normal hip otherwise, hemiarthroplasty is the standard management approach.

Reduction and ■xation of fractures in the elderly patient is not recommended due to the associated high rates of subsequent reoperation , non-union and osteonecrosis of the femoral head. In young patients presenting with fracture within 3 weeks, closed reduction internal fixation with cannulated cancellous screws is performed.

Bipolar hip prosthesis is preferred as it has 2 articulating sites. Austin Moore and Thompson prosthesis are also used. If the patient has pre exixting hip conditions like osteoarthritis, total hip replacement is preferred.

Solution to Question 3:

The X-ray shows a transverse fracture of the patella.

The patient presents with pain and swelling over the affected knee. Patellar tenderness and bruise over the front of the joint can be present. Crepitus may be felt in comminuted fractures. The patient cannot lift his leg in full extension(extension lag) due to the disruption of the extensor apparatus. Hemarthrosis can be associated.

Patellar fractures can be transverse, longitudinal, polar, or comminuted.

The radiological examination includes AP and lateral views. A skyline view of the patella is required in some cases of undisplaced fractures.

Management of patellar fractures:

• Undisplaced fractures: It is treated with a plaster cast extending from the groin to just above the malleoli, with the knee in full extension. This is called a cylinder cast and is usually left in place for a period of at least 3 weeks

• A transverse fracture is treated by a surgical procedure consisting of the reduction of the fragments known as tension-band wiring and K- wire fixation. This is followed by the repair of the extensor retinacula and physiotherapy.

• Comminuted fractures are treated by open reduction and fixation. Patellectomy is the preferred option in this case as restoration of a smooth articular surface is difficult in this condition. Patella-saving operations are recently more performed with advancements in fixation techniques.

• Longitudinal fractures usually have intact extensor apparatus. They can be treated with a cylinder cast or extension brace initially, followed by a range of motion.

Solution to Question 4:

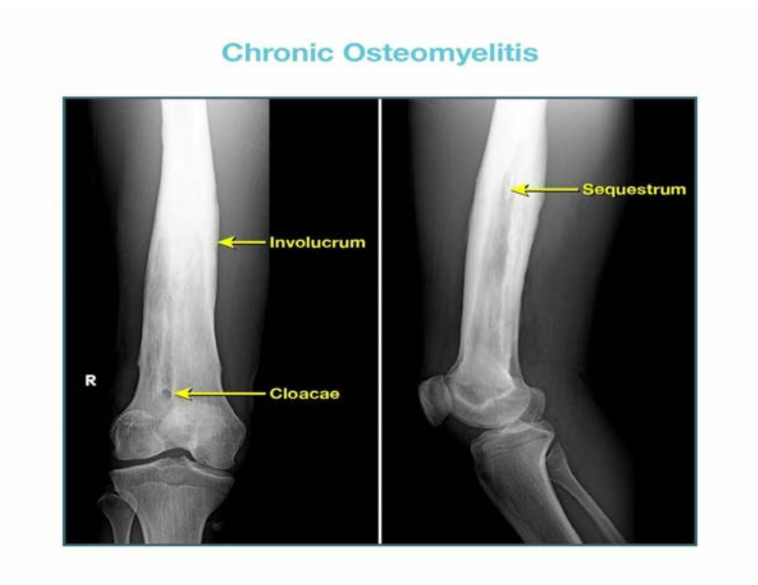
The given clinical scenario of a patient with thigh pain and an X-ray of the lower limb revealing involucrum and cloaca in the femur, strongly suggests a diagnosis of chronic osteomyelitis.

Chronic osteomyelitis may occur by direct inoculation of the infective organism during trauma (like road traffic accidents). The infection continues to thrive within the bone leading to alternating quiescent and flare-up stages of chronic osteomyelitis.

Sequestrum is a piece of dead bone. It is formed when a fragment of bone gets separated from the healthy bone due to necrosis. It varies in size from a spicule to a large fragment and acts as a nidus for infection. It leads to abscess and persistent pus discharge in the bone. Sinus tracts are formed when the abscess from the infected bone drains out through the skin.

The involucrum is the dense new sclerotic bone surrounding the sequestrum. It is formed from deep layers of stripped periosteum. Cloacae are the perforations in the involucrum. Pus and tiny spicules of bone may be discharged through them.

The radiograph below shows sequestrum, involucrum and cloacae:



Solution to Question 5:

Out of the following options, vertebral fractures are most commonly seen in osteoporosis

The fractures are classically low trauma de■ned as a fall from standing height or less. Fracture ofthe distal radius (Colles' fracture) is usually the ■rst fracture to occur, followed by vertebrae and hip unless treatment is initiated. Osteoporotic vertebral fractures are particularly dif■cult to diagnose as they may be clinically silent. In severe cases, signi■cant height loss (often exceeding 4 cm)and thoracic kyphosis can occur due to multiple vertebral fractures.

Osteoporosis is a common metabolic bone disease characterized by a diffuse reduction in bone density due to a decrease in bone mass. A DEXA (dual-energy X-ray absorptiometry) scan is the investigation of choice for the diagnosis and assessment of severity.

The mainstay of treatment is to reduce the risk of fractures. The following medications can be useful in the treatment:

• Bisphosphonates - They are the first line of medication. They reduce osteoclastic bone resorption and prevent bone loss. Eg. Zoledronate.

• Denosumab - This is an antibody to the RANK ligand. It prevents the activation of RANK which is required for the proliferation of osteoclasts. Therefore, it decreases bone resorption.

• Parathyroid hormone - Teriparatide is a recombinant parathyroid hormone analog. At low and intermittent doses, they can promote bone formation.

• Selective estrogen receptor modulators (SERMs) - They act on the estrogen receptors and prevent bone resorption. Eg. raloxifene.

• Strontium ranelate - It is thought to increase bone formation and reduce bone resorption.

Solution to Question 6:

Intertrochanteric fractures are the most common type of extracapsular fracture. Dynamic hip screw is the managament for undisplaced intertrochanteric fractures.

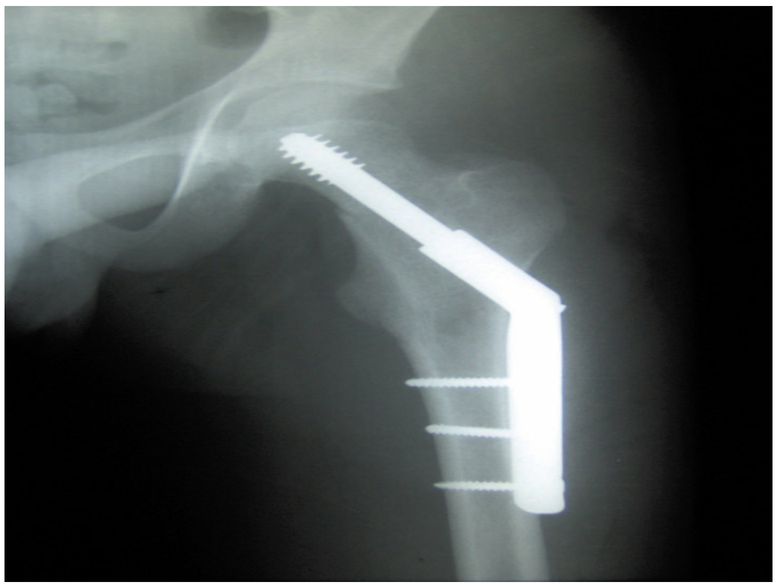
In elderly patients, internal fixation is preferred because inthem prolonged bed rest in traction may cause complications related to recumbency i.e., bed sores, pneumonia etc.

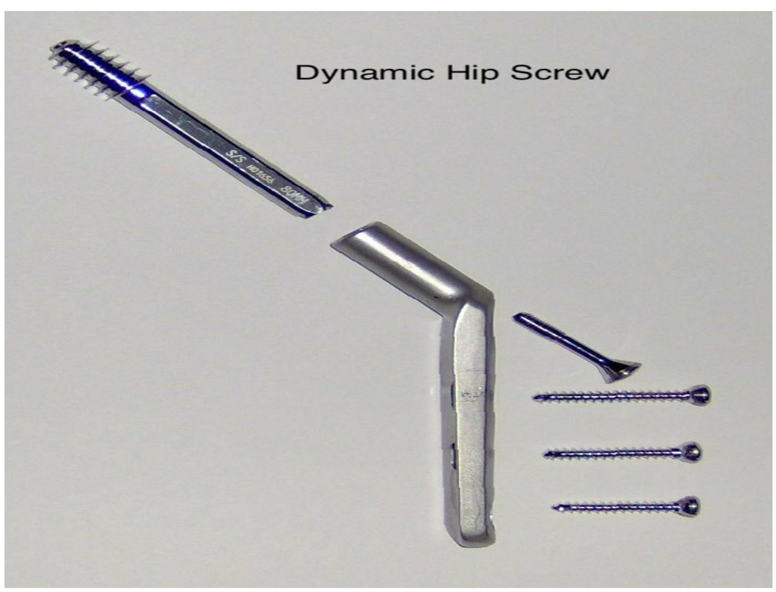
Intertrochanteric fractures of the femur can be treated by:

• Extramedullary implants - dynamic hip screw

• Intramedullary implants - proximal femoral nail and gamma nail.

Dynamic hip screws are used in undisplaced intertrochanteric fractures, whereas proximal femoral nails are used in displaced intertrochanteric fractures.





Conservative methods like Russels traction or skeleton traction in Thomas Splint are nowadays not preferred due to the success rate of operative methods.

Solution to Question 7:

Closed intramedullary nail with interlocking screws is the preferred treatment of most femoral shaft fractures.

Newer technique uses Gross-Kempf nails which have perforations at the proximal and distal ends to be locked in place, using transverse screws. Thus, it ensures stability, prevents rotation of the fractured fragments and telescoping of the implant.

In modern practice, conservative treatments like traction, bracing, and spica casts are rarely used. Traction can reduce and maintain alignment in most fractures, except for those in the upper third of the femur. Joint mobility can be preserved through active exercises. However, the major drawback is the extended bed rest required (10–14 weeks for adults), which makes it difficult to keep the femur aligned long enough for callus formation and can lead to increased patient discomfort and frustration.

After surgery, full weight-bearing should be commenced as soon as the ■xation allows. The fracture usually heals within 20 weeks and the complication rate is low; sometimes malunion or delayed union occur.

The image below shows an intramedullary nail in place for a fracture of the shaft of the femur.

Solution to Question 8:

The cardinal and the earliest sign of ankylosing spondylitis is erosion of the sacroiliac joints. Later there may be periarticular sclerosis, especially on the iliac side of the joint and ■nally bony ankylosis

Ankylosing spondylitis is a chronic inflammatory disease affecting the spine and sacroiliac joint. It is more common in young males and has a strong association with HLAB27. Males are affected more frequently than females and the usual age at onset is between 15 and 25 years. Clinically, patients present with recurring backache which worsens in the early morning or with inactivity and improves with exercise. Later they develop fatigue, pain, joint swellings, and heel pain.

Extra-articular manifestations include anterior uveitis, aortic valve insufficiency, heart blocks, and pulmonary fibrosis. Examination findings include loss of lumbar lordosis, increased thoracic kyphosis, diminished spinal movements, and decreased chest expansion.

The ESR and CRP are usually elevated during active phases of the disease. HLA-B27 is present in 95% of cases. Serological tests for rheumatoid factor are usually negative

X-ray findings:

• Erosion of sacroiliac joints- earliest sign

• Periarticular sclerosis in later stages

• Squaring of vertebrae due to the flattening of normal anterior concavity of the vertebral body is the earliest vertebral change

• Bamboo spine appearance due to bridging of adjacent vertebrae

• Wedging of the vertebra leading to hyperkyphosis

Treatment consists of maintaining posture and preserving mobility. Pain and stiffness can be controlled with anti-inflammatory drugs. In severe cases, TNF inhibitors are used. Surgery can be considered to correct the deformity.

Option D: Non erosive arthritis is a feature of systemic lupus erythematosus (SLE).

Solution to Question 9:

The given scenario of a 60-year-old female with trivial trauma unable to bear weight and X ray findings suggestive of destruction of the hip joint and periarticular bone and shaft of the tibia fracture is most likely due to metastasis.

Its more likely this patient has metastasis because:

• Age: 60-year (old age favors cancer)

• Overall bone denisty is normal

Pathological fractures are fractures that occur in a weakened bone that has structurally changed due to some condition. They occur either spontaneously or due to minor trauma. The features suggestive of a pathological fracture include:

• Pain at the site before the fracture

• Multiple recent fractures

• Elderly patient

• History of primary malignancy

• Unusual fracture pattern (Banana fracture - transverse fracture through an abnormal area of bone, similar to breaking a segment from a banana)

Pathological fracture
(Banana fracture pattern)

Other Options:

Option A: Brown tumor: Seen in hyperparathyroidism. X-rays show lytic lesions, but this is rare and typically involves long-standing disease in younger patients. Focal osteoblastic activity. Overall bone denisty is low.

Option B: Disseminated TB: Pott's disease or monoarticular destruction (e.g., hip or spine). Hip joint is usually destroyed and articular surface erosion is present. Disseminated TB rarely causes widespread fractures. Here there is no joint involvement.

Option C: Eosinophilic granulomas: Typically seen in younger individuals (children/young adults). X-rays reveal punched-out lesions, not diffuse destruction.

Solution to Question 10:

Closed reduction internal fixation is the surgery of choice for a 40-year-old patient with an intracapsular fracture of the femur.

A displaced fracture neck of the femur in an adult who is physiologically less than 65 years of age is best treated by closed reduction and internal fixation with multiple cannulated cancellous screws. Open reduction is only done if closed reduction fails, as the complication rate is higher.

Proximal femur fractures are classified based on their location near the hip joint. The main types include:

• Intracapsular Fractures:: It occurs within the joint capsule and involves the femoral neck and head.

Subtypes:

• Femoral head fracture

• Femoral neck fracture (common site for avascular necrosis and non-union).

2. Extracapsular Fractures: Occur outside the joint capsule.

Subtypes:

• Intertrochanteric fracture: Between the greater and lesser trochanters.

• Subtrochanteric fracture: Below the lesser trochanter.

• Trochanteric fracture: Isolated fracture of the greater or lesser trochanter.

Management of Intracapsular Fractures depends on the age of the patient:

1. > 65 years: Replacement - Hemi/Total Replacement (Option C and D)

2. < 65 years: is based on duration of fracture

• <3 weeks- CRIF with cannulated cancellous screws (avoid hematoma)

• >3 weeks- Check the viability of head- further management depends on the viability of head

Solution to Question 11:

FRAX is related to Fracture risk assessment.

It is used to predict the risk of fracture in a person with osteoporosis within the next 10 years.

It is used to recommend treatment or suggest ways of preventing osteoporosis.

The FRAX tool can be used to guide treatment decisions in people who meet the following three conditions:

• Postmenopausal women or men age 50 and older.

• People with low bone density (osteopenia).

• People who have not taken an osteoporosis medicine.

Orthopedics NEET 2024

Question 1:

A patient met with road traffic accident and complained of severe back ache. The X-ray is shown below. What fracture does the patient have?

a) Chance fracture

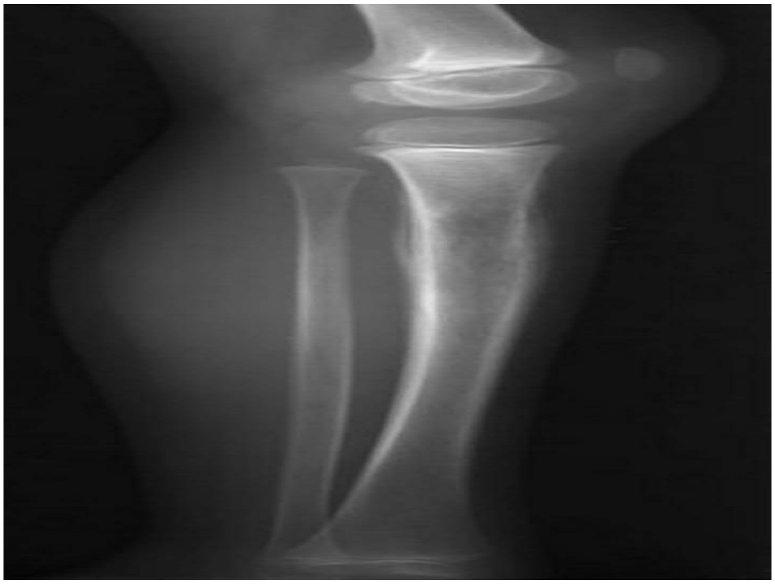
b) Burst fracture

c) Compression fracture

d) Fracture of spinous process

Question 2:

An 18-year-old boy complains of fever and pain in the lower limb. There is a localised rise of temperature and tenderness in the region shown in the radiograph. ESR is raised. What is the most probable diagnosis?



a) Osteosarcoma

b) Ewing's sarcoma

c) Giant cell tumour

d) Osteomyelitis

Question 3:

A child sustained an injury as shown in the below radiograph. What is the classification used and the grade used for this type of injury?

a) Salter harris classification-type III

b) Gartland's classification-type IV

c) Gartland's classification-type III

d) Salter harris classification-type IV

Question 4:

A patient sustained a fracture of the base of the fifth metatarsal in the right foot. He was advised below knee pop cast. How long should the cast be put?

a) 6-8 weeks

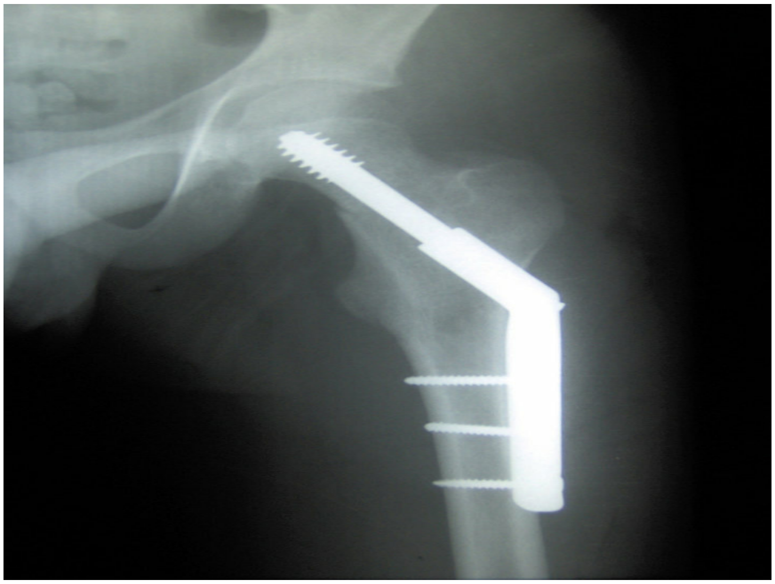
b) 2-3 weeks

c) 3-5 weeks

d) 16-20 weeks

Question 5:

Following a road traffic accident, a patient with an intertrochanteric fracture of the femur was treated with the implant shown below. Identify the implant.



a) Proximal femoral nail

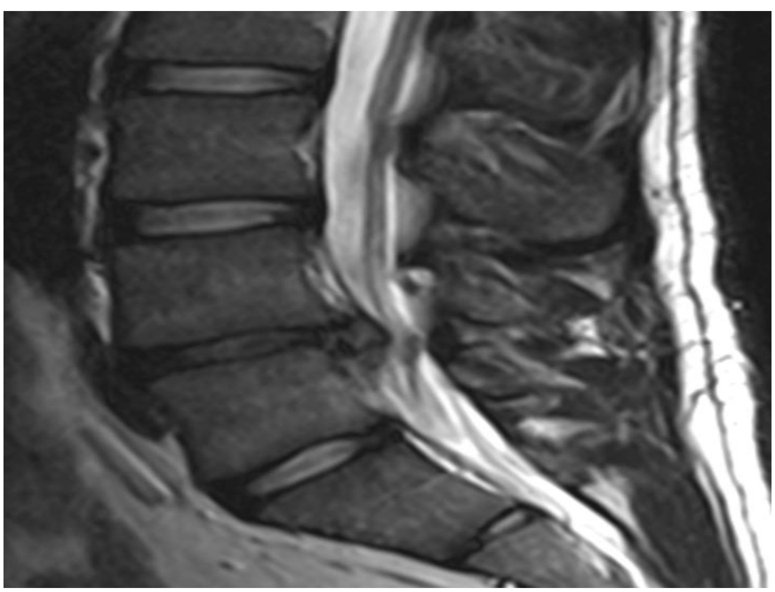
b) Dynamic hip screw

c) Austin Moore prosthesis

d) Herbert screw

Question 6:

A young woman, following an intense workout involving heavy weight lifting, presents with lower back pain radiating to her right hip and extending down to her right ankle. MRI shows the following finding. Based on the location and radiation of her pain, which of the following nerve roots is most likely involved?



a) L4

b) L5

c) S1

d) L3

Question 7:

A child with rickets not responding to treatment is being evaluated. Investigations show a serum calcium level of 9 mg/dL, low phosphate level of 2 mg/dL, elevated alkaline phosphatase, and normal parathyroid hormone levels with normal arterial blood gas levels. What is the diagnosis?

a) Vitamin D resistant rickets type 2

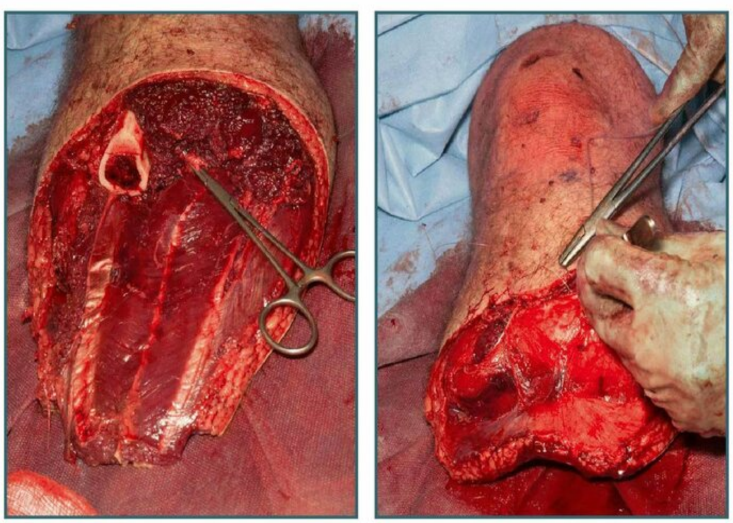
b) Hypophosphatemic rickets

c) Vitamin D resistant rickets type 1

d) Nutritional rickets

Question 8:

What is this type of amputation called?



a) Above knee amputation

b) Below knee amputation

c) Knee disarticulation

d) Syme's amputation

Question 9:

A patient complains of decreased sensation in the lateral three-and-a-half fingers on the palmar side of the right hand, along with difficulty moving the upper phalanx of the thumb. Examination reveals thenar atrophy. Which of the following structures, when affected, lead to this presentation?

a) High radial nerve

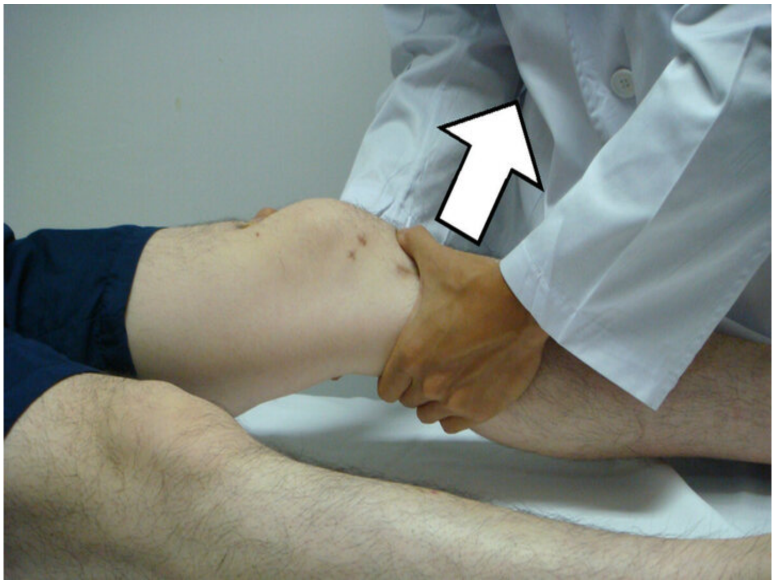
b) High median nerve

c) High ulnar nerve

d) Carpal tunnel

Question 10:

Patient presented with a twisting injury of the knee. Identify the test being performed.



a) McMurray's test

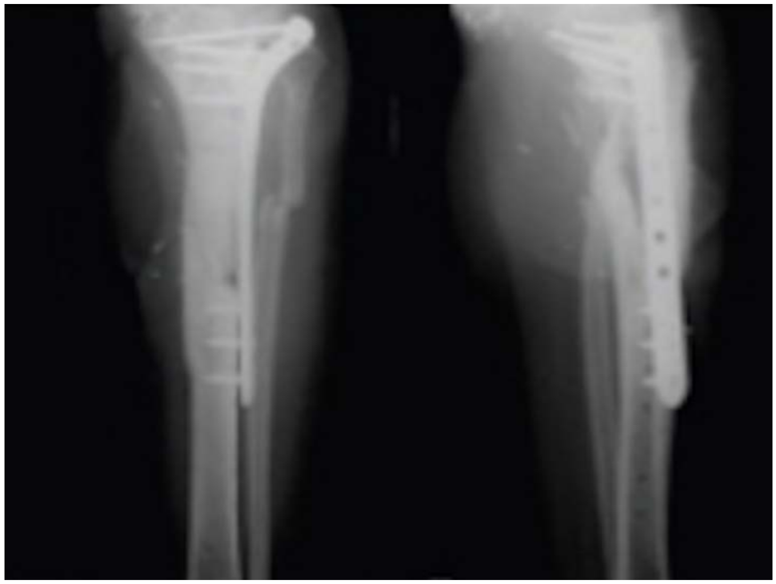
b) Lachman test

c) Anterior drawer test

d) Posterior drawer test

Question 11:

A child sustained a fracture after a fall. An X-ray taken after fracture management is shown below. By which of the following mechanisms will fracture healing occur?



a) Secondary healing

b) Primary healing

c) Creeping osteogenesis

d) Intermembranous ossification

Question 12:

A 58-year-old female patient presents with a pathological fracture of the hip and femur. Her serum calcium and phosphate are increased. DEXA scan confirms decreased bone density. Which of the following parameters will increase in this condition?

a) Lactate dehydrogenase

b) Aspartate aminotransferase

c) Alanine transaminase

d) Tartrate resistant acid phosphatase

Question 13:

A fracture line passing through which of these will impair longitudinal growth of the bone?

a) Epiphyseal plate

b) Epiphysis

c) Metaphysis

d) Diaphysis

Question 14:

A 56-year-old woman presented with chronic lower back pain. MRI of the lower back shows the following findings. Identify the diagnosis.

a) Renal osteodystrophy

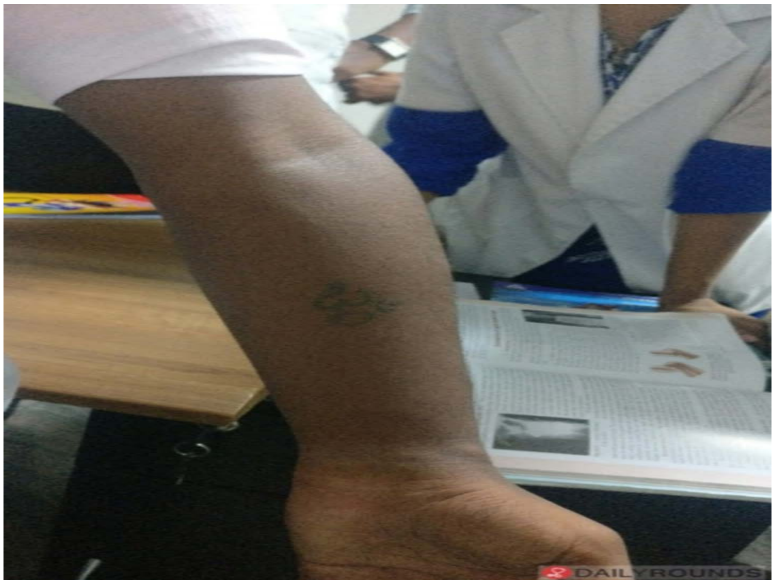
b) Spinal Tuberculosis

c) Spinal metastasis

d) Spondylolisthesis

Question 15:

A 21-year-old patient with a history of a fracture had a cast applied. On removal of the cast, the following deformity was seen. What is the most likely cause of this deformity?



a) Suprachondylar humerus fracture

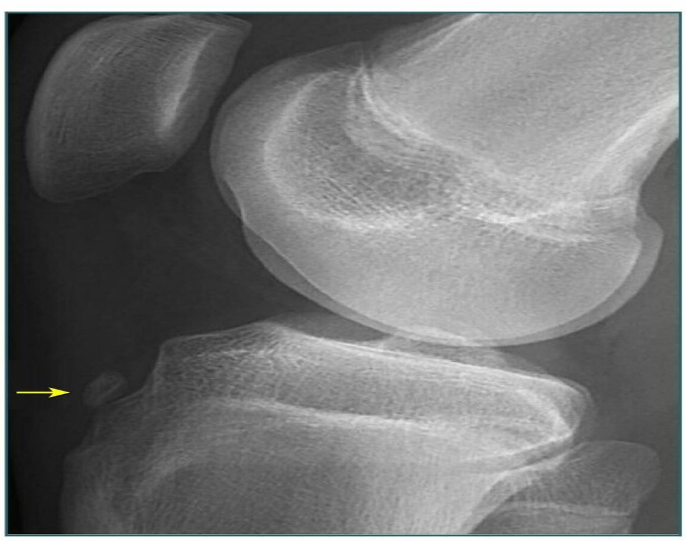
b) Radial fracure

c) Mid shaft fracture of humerus

d) Lateral condylar fracture

Question 16:

A 10-year-old child fell down the stairs. An X-ray was done. Identify the structure injured as marked in the X-ray.



a) Medial tubercle

b) Tibial tuberosity

c) Gerdy's tubercle

d) Anterior intercondylar area

Answer Key

Question No.	Correct Option
1	a
2	b
3	c
4	a
5	b
6	b
7	b
8	b
9	d
10	b
11	b
12	d
13	a
14	d
15	a
16	b

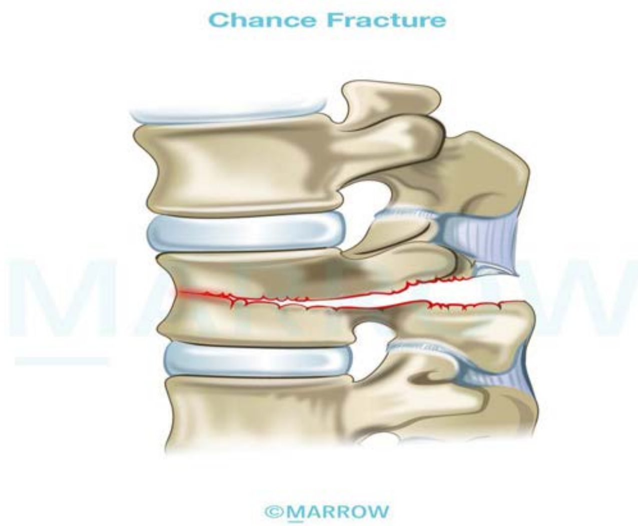
Detailed Explanations

Solution to Question 1:

The X-ray shows a vertebral fracture with a horizontal fracture line passing through the body of the vertebra. This indicates Chance fracture.

Chance fracture is also known as seat-belt injury or the jack-knife injury. This injury is typically located in the upper lumbar spine in people involved in motor vehicle collisions while wearing a lap seat belt without a shoulder harness. The sudden deceleration causes flexion and posterior distraction of the spine. This causes the mid-lumbar spine to bend like a jack-knife around an axis anterior to the vertebral column.

There is a tear in the vertebra that passes transversely through the bones (bony Chance fracture), or the ligamentous structures (soft Chance fracture), or both. There is little or no crushing of the vertebral body. However, it is an unstable injury as the posterior and middle columns are distracted. Neurological damage is uncommon. Associated intra-abdominal injuries are seen in 50% of these injuries.



X-rays may show a horizontal fracture with the split running through the vertebra. It may involve the spinous process, the transverse processes, pedicles and the vertebral body. There may be opening up of the disc space posteriorly.

Most Chance fractures can be managed conservatively with bed rest and bracing. If posterior fixation is done, it will allow early mobilization. Ligamentous Chance fractures require posterior instrumented fusion.

Other options:

Option B: Burst fracture results from severe axial compression that explodes the vertebral body. The lateral view CT image showing the L4 vertebra burst fracture is given below.

Option C: Spinal compression fractures occur as a result of injury, common fall onto the buttock or pressure from normal activities, to the weakened vertebrae due to osteoporosis. Compression causes a short oblique fracture. The image showing the L4 vertebra compression fracture is given below:

Option D: Fracture of the spinous process of C7(Clay Shoveller's Fracture) is an example of an avulsion injury which may occur with severe voluntary contraction of the muscles at the back of the neck.

Solution to Question 2:

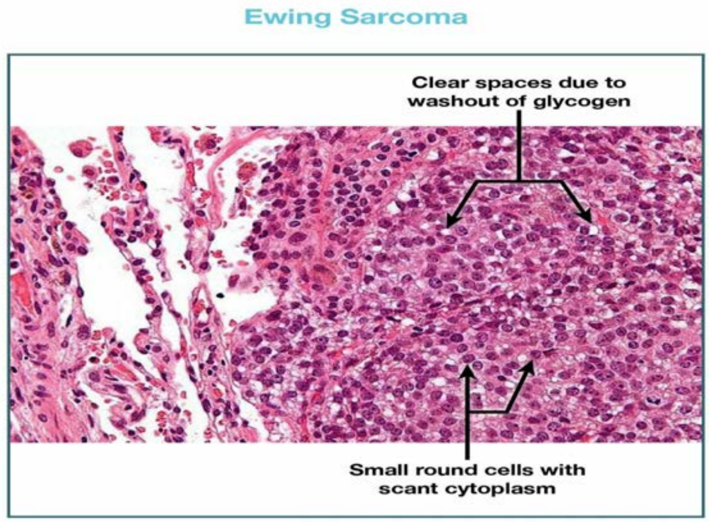
The clinical features and radiograph showing onion peel type of periosteal reaction point towards a diagnosis of Ewing's sarcoma.

Ewing's sarcoma is a malignant tumour that occurs more frequently in males, typically between 10 and 20 years of age and has a diaphyseal long-bone location. The femur is the most common long-bone site, followed by the tibia, humerus and fibula. Ewing's sarcoma is believed to arise from mesenchymal stem cells in the bone marrow.

Pain is the earliest symptom, followed by swelling and a low-grade fever. Radiographically, an aggressive, permeative, poorly defined osteolytic lesion with cortical destruction, periosteal reaction and large, radiolucent soft-tissue mass may be found. Periosteal reaction is common in young patients with the lamellar 'onion-skin' appearance causing fusiform bone enlargement. Ewing's sarcoma is composed of primitive, undifferentiated, small, round blue cells with large nuclei and scant cytoplasm. The characteristic translocation between chromosomes 11 and 22 results in a fusion gene, EWSR1-FL11, which is present in 90% of cases.

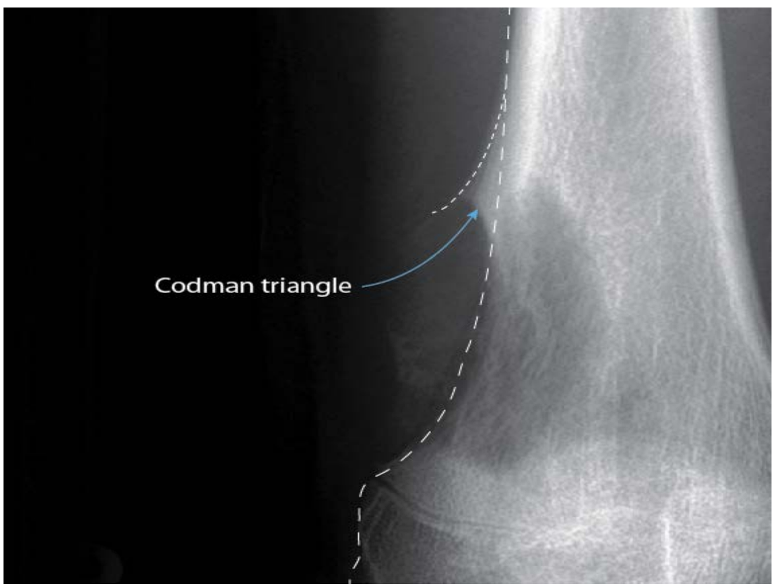
Preoperative chemotherapy frequently results in extensive tumour necrosis and shrinkage. Agents used in Ewing's sarcoma include vincristine, ifosfamide, doxorubicin and etoposide. Wide excision may be combined with adjuvant radiotherapy if surgical margins are poor. A more favourable outcome is expected with younger age, distal appendicular sites, small tumour volume, normal

ESR and greater chemotherapy necrosis response.



Other options:

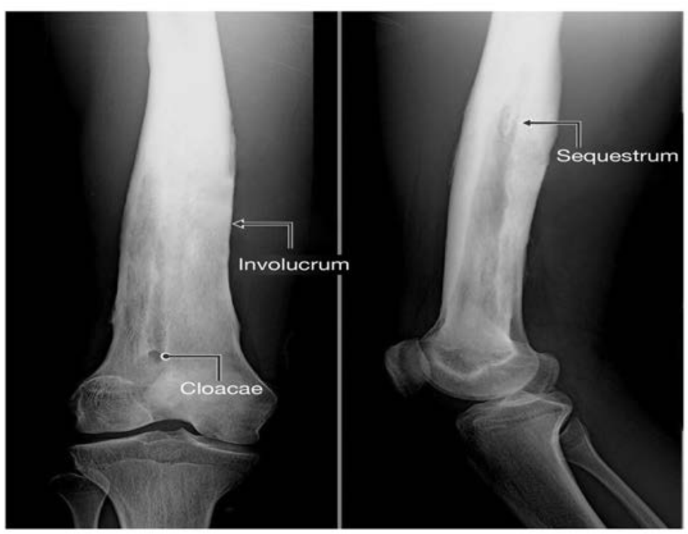
Option A: Osteosarcoma has a periosteal reaction characterised as the Codman triangle. It can also be seen in Ewing's sarcoma.



Option C: Giant cell tumor has a characteristic soap bubble appearance.

Option D: Acute osteomyelitis in children presents with fever, pain, refusal to bear weight, elevated WBC count, elevated CRP, and elevated ESR. During the first week, the X-ray appears to be normal and by the second week, a faint extracortical line may be visible due to periosteal new bone formation. This is a classic sign of early pyogenic osteomyelitis. A combination of regional osteoporosis with a localized segment of apparently increased density is an important late finding in this condition.

Chronic osteomyelitis presents with erythema, chronic pain, swelling, tenderness, and discharging sinus tract. X-ray shows sequestrum is a piece of devitalized bone that has been separated from its surrounding bone during the process of necrosis, involucrum is the thick sheath of periosteal new bone that surrounds the sequestrum and cloacae are openings in the involucrum that allow drainage of purulent and necrotic material out of the dead bone.



Solution to Question 3:

The given clinical features and X-ray image are suggestive of a supracondylar fracture of the humerus (extension type) and Gartland classification is used in this case. The extension type fracture which is displaced with no intact posterior cortex is type 3 as per this classification.

Gartland classification (in children):

• Type I: Undisplaced

• Type II: Displaced but posterior cortex is intact

• Type III: Displaced but no intact posterior cortex and distal fragment could be either displaced:

• Posteromedial

• Posterolateral

Supracondylar fracture of the humerus is most commonly seen in children of 5-7 years after a fall on an outstretched hand with the elbow in full extension. It is most commonly of extension type followed by flexion type. Gartland classification is used for supracondylar fractures. The three-point bony relationship is maintained in supracondylar fracture (option B). Three-point bony landmarks in the elbow are the tips of medial and lateral epicondyles and the olecranon.

Nerve and vessel injury:

• The most common nerve to be injured in extension type - anterior interosseous nerve > Median nerve > Radial nerve > Ulnar nerve. (AMRU)

• The most common nerve to be injured in the flexion type the ulnar nerve.

• The most common vessel to be injured - the brachial artery.

Radiological findings: An undisplaced fracture may show a triangular lucency like a sail of a yacht (sail sign), also called a fat pad sign. This is due to the fat pad around the elbow being pushed forward by the hematoma. Baumann's angle (humeral-capitellar angle) can help in assessing the degree of medial angulation before and after reduction.

The most common complication of supracondylar fracture is malunion. This leads to cubitus varus or gunstock deformity. Malunion occurs due to improper reduction and not due to physeal damage, as the fracture is extraphyseal Other complications include myositis ossificans, compartment syndrome, and Volkmann ischemic contracture.

Undisplaced supracondylar humerus fractures are treated by an above-elbow plaster slab with the elbow in 90° flexion. Displaced fractures are treated by closed reduction and K-wire fixation since they tend to slip after reduction. Open fractures or those associated with brachial artery injury are treated by open reduction and internal fixation.

Other options:

Options A and D: Salter and Harris created a classification of physeal injuries for risk stratification.

• Type 3: Here, the fracture splits the epiphysis and then goes transversely in either direction into the hypertrophic layer of the physis. This leads to damage of the growth plate leading to growth disturbance.

• Type 4: The fracture splits the epiphysis and extends into the metaphysis. Displacement can occur in these types leading to asymmetrical growth.

Solution to Question 4:

The given X-ray indicates a Jones fracture. Conservative management involves a short-leg non-weight bearing cast for 6-8 weeks.

A Jones fracture is typically caused by forced inversion of the foot, such as during a pothole injury, leading to the avulsion of the base of the fifth metatarsal. This injury is due to the pull of the peroneus brevis tendon or the lateral band of the plantar fascia, which inserts onto the base of the fifth metatarsal.

Surgical correction, such as intramedullary screw fixation, is recommended for faster healing in highly active individuals. Proximal avulsion fractures can generally be managed symptomatically with rest and support, followed by early mobilization and return to function.

Solution to Question 5:

The above radiograph shows an extramedullary implant called a dynamic hip screw.

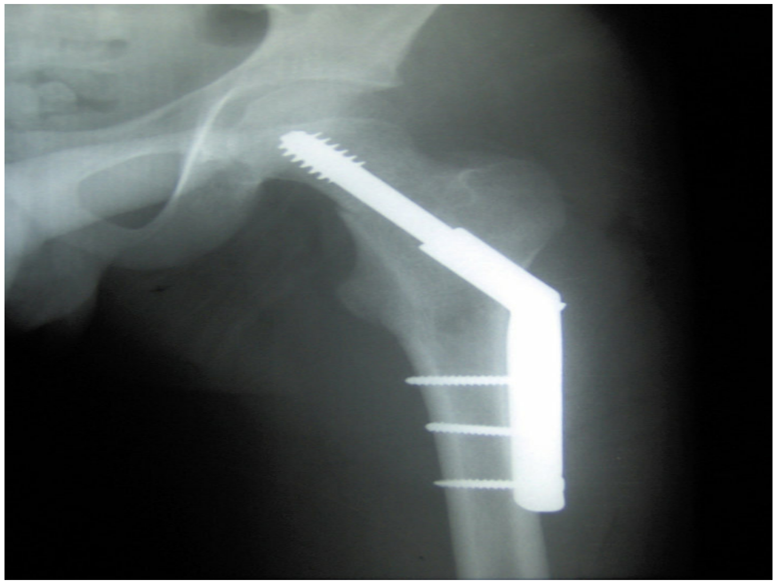
Intertrochanteric fractures of the femur can be treated by:

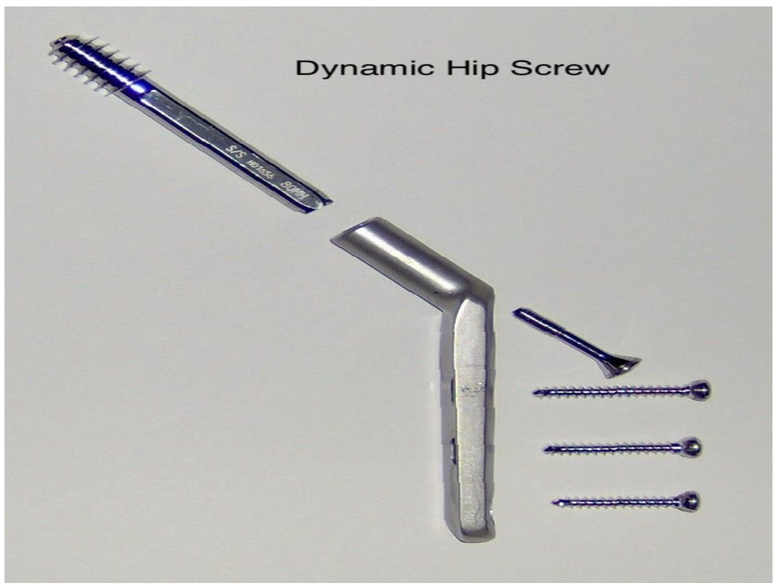
• Extramedullary implants - dynamic hip screw

• Intramedullary implants - proximal femoral nail and gamma nail.

Dynamic hip screws are used in undisplaced intertrochanteric fractures, whereas proximal femoral nails are used in displaced intertrochanteric fractures.

The below image shows a dynamic hip screw.





Advantages of using a proximal femoral nail over a dynamic hip screw include:

• Smaller incisions

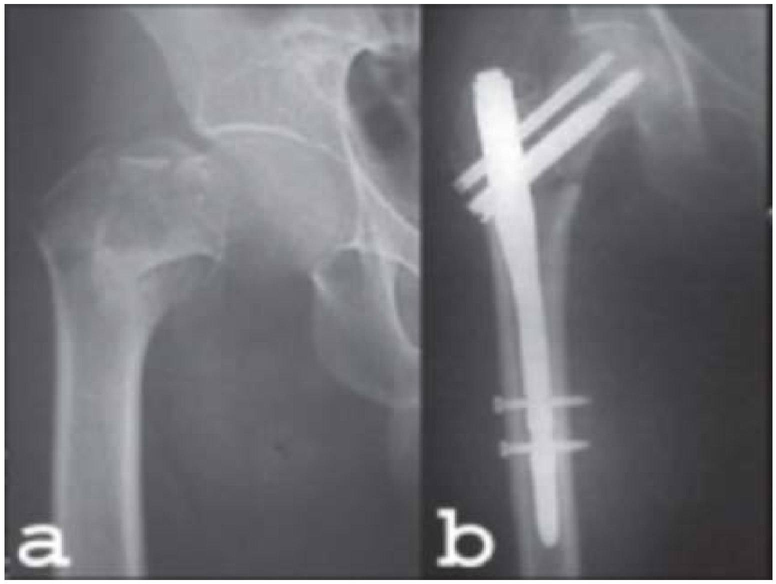
• Reduced blood loss

• Decreased femoral neck shortening

• Decreased risk of infection

Other options:

Option A: The below image shows a proximal femoral nail with locking and stabilization screws.



Option C: Austin Moore's implant is an older implant used in unipolar hemiarthroplasty. It can be identified by the fenestrations on the shaft of the prosthesis, as shown below.

Option D: Herbert screw is cannulated and threaded at both ends. It is designed for use in fractures of small articular bones such as carpals and scaphoid.

Solution to Question 6:

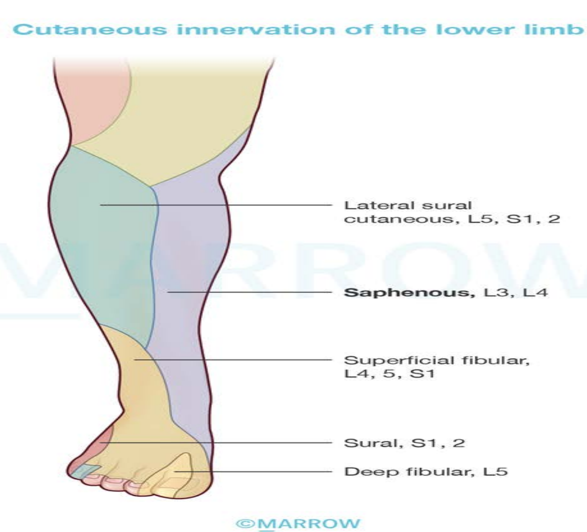
Based on the clinical presentation of lower back pain radiating to the right hip and ankle, along with an MRI showing L4-L5 disc prolapse, the likely nerve root involved is L5. In a typical L4-L5 disc herniation, the L5 nerve root is most commonly compressed.

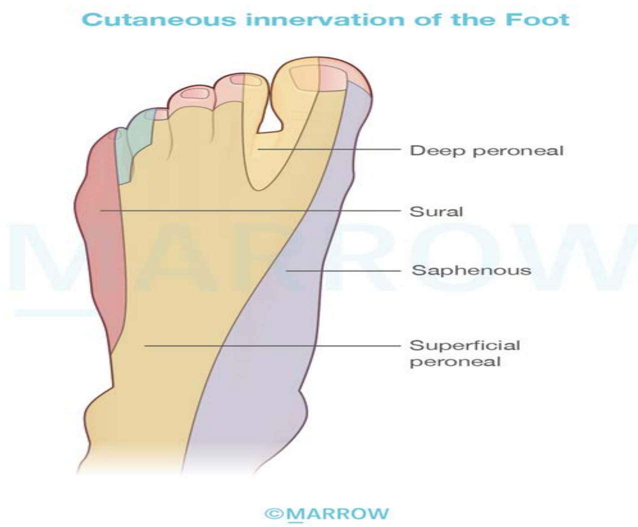
Other options:

Option A: L4: Involvement typically causes symptoms around the knee and anterior lower leg, but not the ankle.

Option C: S1: Compression here causes symptoms in the posterior leg, calf, and heel, differing from this case's distribution of pain.

Option D: L3: Affects the anterior thigh, with no radiating pain below the knee.





Solution to Question 7:

The given scenario indicates the diagnosis of hypophosphatemic rickets because of the following features:

• normal calcium levels (Normal: 9-11 mg/dL)

• normal parathyroid hormone levels

• decreased phosphate (Normal: 2.5-4.3 mg/dL)

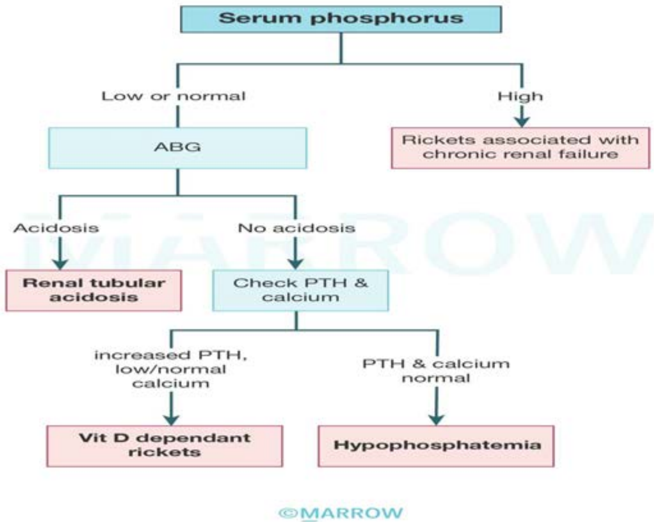
• elevated levels of alkaline phosphatase (Normal: 33-96 U/L)

• normal arterial blood gas levels

Hypophosphatemic rickets:

X-linked dominant hypophosphatemic rickets are the most common genetic cause of rickets while the most common, overall is nutritional vitamin D deficiency. PHEX gene (phosphate-regulating endopeptidase on x chromosome) products are responsible for the inactivation of phosphatonin. Mutation of this PHEX leads to increased phosphatonin (FGF 23) levels, and hence, the excretion of phosphorous is increased. This is also associated with impaired renal tubular reabsorption of phosphate leading to its excess excretion.

The chart attached below summarizes the evaluation of hypophosphatemic rickets.



Other options:

Options A and C: In vitamin D-resistant rickets, the renal tubule's inability to retain phosphate results in the loss of excessive phosphate through urine. Therefore, even normal or higher supplementations of vitamin D are not sufficient for bone mineralization.

• Vitamin D-dependent Type 1 rickets is due to decreased activation of vitamin D to its active form in the kidney.

• Vitamin D-dependent Type 2 rickets is due to resistance to vitamin D in peripheral tissues. This resistance to vitamin D can be overcome with high levels of this vitamin by supplementation.

Option D: Nutritional rickets occur due to a deficiency of vitamin D. This leads to low calcium levels in the blood. To raise the serum calcium, the secretion of the parathyroid hormone rises. The parathyroid hormone tries to raise the calcium levels by reabsorbing calcium in the kidneys and secreting phosphate in exchange leading to low phosphate levels and increasing bone resorption.

Solution to Question 8:

The given image shows below-knee amputation.

By retaining the knee joint, a below-knee amputation allows for better mobility and function compared to an above-knee amputation. After surgery, patients can usually be fitted with a prosthetic limb, enabling them to walk with a relatively normal gait.

Other options:

Option A: Above-knee amputation (AKA): Surgical removal of the lower limb above the knee joint, typically at the femur, resulting in the loss of the knee and requiring a more complex prosthetic for ambulation.

Option C: Knee disarticulation: Amputation through the knee joint, preserving the femur while removing the tibia and fibula, allowing for better weight-bearing but with a longer residual limb than AKA.

Option D: Syme's amputation: Removal of the foot at the level of the ankle joint, preserving the heel pad for weight-bearing and creating a durable limb for ambulation without complex prosthetics.

Solution to Question 9:

The most likely nerve injured here is the median nerve, which occurs in Carpal tunnel syndrome.

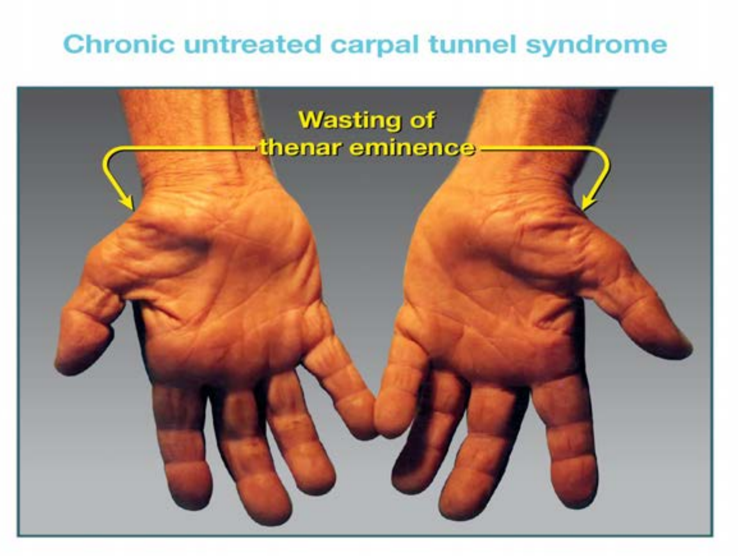
Carpal tunnel syndrome is the most common entrapment mononeuropathy caused by compression of the median nerve as it passes through the fibro-osseous carpal tunnel beneath the flexor retinaculum. Patients present with pain and paresthesia over the lateral 3 and a half digits. There may be a weakness of thumb opposition. In chronic cases, the thenar eminence is lost due to thenar atrophy. Carpal tunnel syndrome can be associated with rheumatoid arthritis, pregnancy, chronic renal failure, acromegaly, hypothyroidism, and gout. The diagnosis is usually made on clinical grounds but nerve conduction studies can confirm the diagnosis.

Other tests for CTS -

• Phalen's test - Elicited by holding the wrist flexed fully. The appearance of sensory symptoms within 60 seconds implies a positive test.

• Tinel's sign - Sensory symptoms reproduced by percussing over the median nerve imply a positive test.

The given image shows carpal tunnel syndrome:



Non-surgical Management of Carpal Tunnel Syndrome: In mild to moderate cases of carpal tunnel syndrome, non-surgical management options can be explored to alleviate symptoms and improve function. These conservative treatments include:

•

Wrist splinting: Wearing a wrist splint at night can help maintain the wrist in a neutral position, reducing pressure on the median nerve and providing relief from nocturnal symptoms.

•

Activity modification: Identifying and modifying activities that exacerbate symptoms can help in managing carpal tunnel syndrome. For example, taking frequent breaks during repetitive tasks or adjusting the workstation ergonomics can be beneficial.

•

Nonsteroidal anti-inflammatory drugs (NSAIDs): Over-the-counter NSAIDs, such as ibuprofen, can help manage pain and inflammation associated with carpal tunnel syndrome.

•

Corticosteroid injections: In some cases, corticosteroid injections into the carpal tunnel may provide temporary relief from symptoms by reducing inflammation and pressure on the median nerve.

•

Physical and occupational therapy: Therapists can help patients learn exercises to stretch and strengthen the hand and wrist muscles, improving symptoms and function.

If conservative treatments do not provide sufficient relief or the condition worsens, surgical intervention, such as carpal tunnel release surgery, may be considered.

Other options:

Option A: High radial nerve injuries typically cause wrist drop due to weakness in wrist and finger extensors, with sensory loss in the dorsal aspect of the hand, not the lateral three and a half fingers on the palmar side.

Option B: High median nerve injuries can affect sensation in the lateral three-and-a-half fingers on the palmar side and impair thumb flexion and opposition, particularly at the proximal phalanx. There is a loss of pronation due to the paralysis of the pronator teres and pronator quadratus. Paralysis of flexor digitorum superficialis and lateral half flexor digitorum profundus results in loss of wrist flexion. Ape-like hand and pointing index are due to this injury.

Option C: High ulnar nerve injuries result in sensory loss in the medial one-and-a-half fingers (pinky and ring finger) and hand intrinsic muscle weakness, unrelated to thenar muscle involvement.

Solution to Question 10:

The image shows an examination being performed with the knee flexed to about 20 degrees which indicates that it is the Lachman test. The presence of free gliding indicates that it is a positive Lachman test.

Lachman test: The patient’s knee is flexed to 20 degrees. The lower part of the thigh is grasped with one hand and the upper part of the leg with the other hand. The joint surfaces are shifted backward and forward upon each other. If the knee is stable, there should be no gliding. The presence of gliding indicates the laxity of the anterior cruciate ligament.

Other options:

Anterior and posterior drawer test: The test is performed with the patient's knees flexed to 90 degrees, and the feet resting on the examining bed. The examiner then grasps the upper part of the leg with both hands and rocks it forward and backward. Excessive anterior movement denotes anterior cruciate laxity (anterior drawer test) Excessive posterior movement denotes posterior cruciate laxity (posterior drawer test).

McMurray's test: The test is used to identify torn meniscus. The knee is flexed as far as possible and the joint is stabilized with one hand. With the other hand, the examiner rotates the leg medially and laterally. If there is a clicking sound while rotating the leg, the test is positive. The sound is produced due to the trapping of the loose tag of the torn meniscus between the articular surfaces.

Solution to Question 11:

The clinical vignette describes a child sustaining a fracture, and the X-rays show compression plating. In this case, the fracture heals through primary healing without callus formation.

The stages of fracture healing are as follows:

• Hematoma Stage (up to 7 days): The periosteum and soft tissues detach from the fracture ends, leading to ischemic necrosis. Some osteocytes die from lack of blood flow, while others transform into daughter cells that aid in healing.

• Granulation Tissue Stage (2-3 weeks): Sensitized daughter cells differentiate into blood vessels, fibroblasts, and osteoblasts, forming soft granulation tissue that fills the fracture gap.

• Callus Stage (4-12 weeks): Granulation tissue differentiates into osteoblasts that produce an intercellular matrix infused with calcium salts, forming a callus (woven bone), which is visible on X-rays around 3 weeks after the fracture.

• Remodeling Stage (1-2 years): The woven bone is replaced by mature lamellar bone, known as the secondary callus, through a multicellular process of substitution.

• Modeling Stage: The bone undergoes gradual strengthening and reshaping of its periosteal and endosteal surfaces, driven by local strains from weight-bearing stresses and muscle forces during normal activities.

Other options:

Option A: Secondary healing is more typical for fractures that have not been perfectly aligned or stabilized, where a callus is formed (soft callus forms first, followed by a hard callus as healing progresses).

Option C: Creeping osteogenesis refers to the process where new bone forms along the surface of existing bone, typically seen in the context of distraction osteogenesis or in cases where there is a gap that needs to be filled.

Option D: Intramembranous ossification is a type of bone formation that occurs primarily in flat bones and does not typically apply to the healing of fractures.

Solution to Question 12:

In this patient, the pathological fracture and elevated serum calcium and phosphate suggest a possible metabolic bone disease such as Paget's disease of bone. The parameter likely to be increased is TRAP (tartrate-resistant acid phosphatase), which is a marker of osteoclast activity.

TRAP is an enzyme produced by osteoclasts, which are cells responsible for bone resorption. Elevated levels of TRAP are often seen in conditions with increased bone turnover, such as Paget's disease, hyperparathyroidism, and certain bone malignancies.

Solution to Question 13:

A fracture line passing through the epiphyseal plate (also known as the growth plate) will impair the longitudinal growth of the bone.

The epiphyseal plate is the area of growing tissue near the ends of the long bones in children and adolescents. It is responsible for bone lengthening (longitudinal growth). If the fracture involves the epiphyseal plate, it can disrupt this process and lead to growth disturbances or deformities.

Other options:

Option B: The epiphysis is the end of a long bone. While fractures involving the epiphysis can affect joint function, they don't directly impair longitudinal growth unless the epiphyseal plate is involved.

Option C: The metaphysis is the region of bone between the diaphysis and the epiphysis, where bone growth in width occurs. A fracture here doesn't directly impact longitudinal growth.

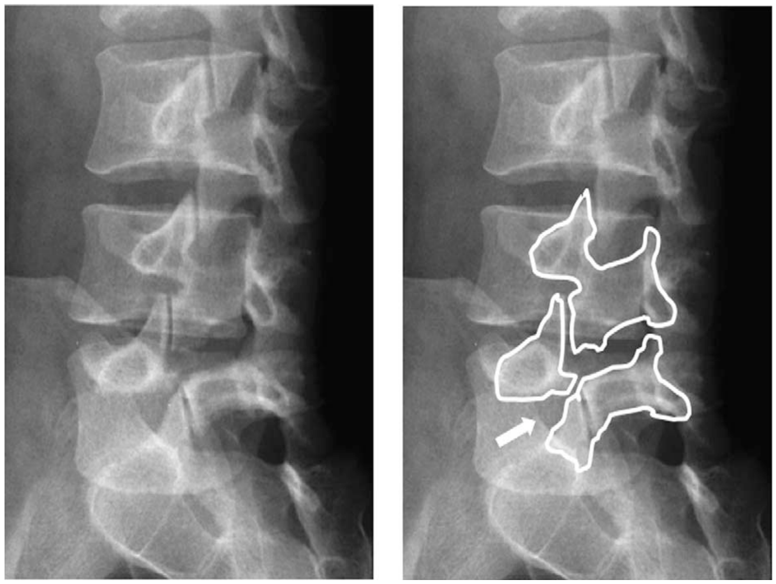
Option D: Diaphysis describes the shaft of the bone. A diaphyseal fracture typically affects the strength and structural integrity of the bone but doesn't impair longitudinal growth.

Solution to Question 14:

The above clinical scenario, the history and MRI scan are suggestive of spondylolisthesis.

Spondylolisthesis is the anterior displacement of a vertebra over another, usually at the lower lumbar spine. Occasionally, the displacement is backward (retrolisthesis). It is characterized by chronic back pain, lumbar lordosis, and step sign. Normally, spinous processes of adjacent vertebrae can be palpated on the back smoothly. In spondylolisthesis, a step-off or step sign can be felt over the spinous process at the affected level.

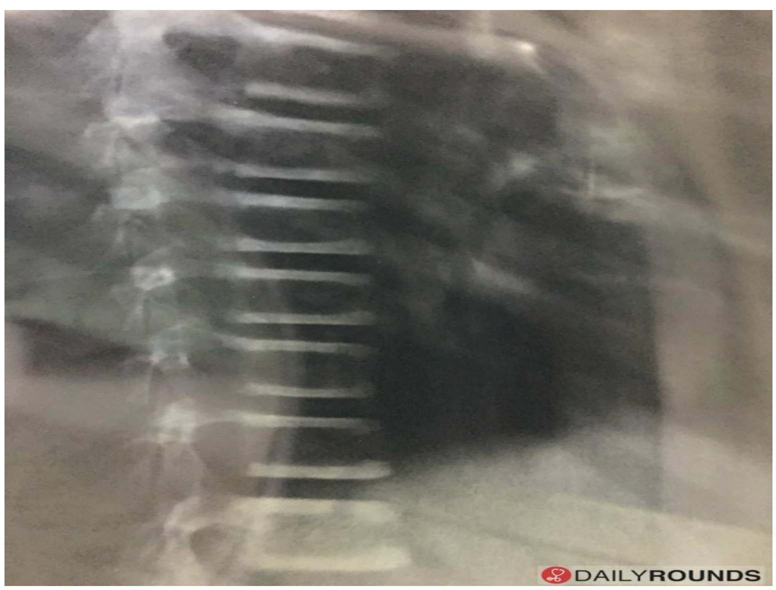
The Beheaded Scottish terrier sign (shown in the image below) on the x-ray is diagnostic. It is due to a break in the pars interarticularis of the vertebra.



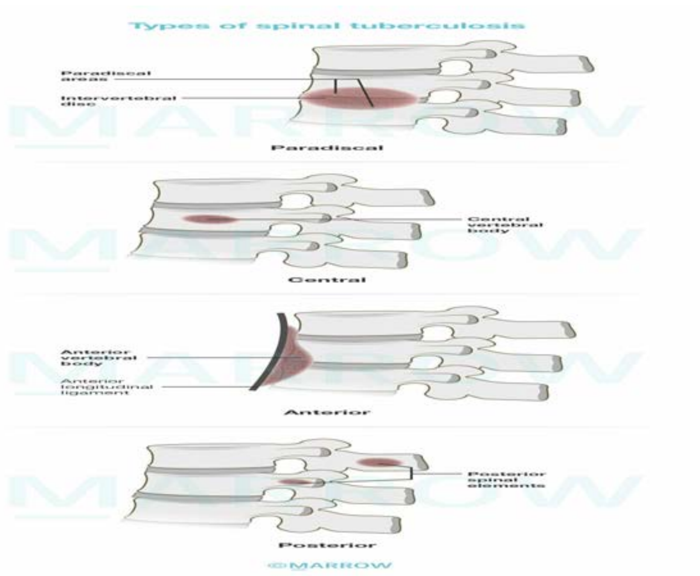
It is usually managed conservatively based on symptoms with analgesics, temporary activity restriction, muscle relaxants, physical therapy, and strengthening exercises.

Other options:

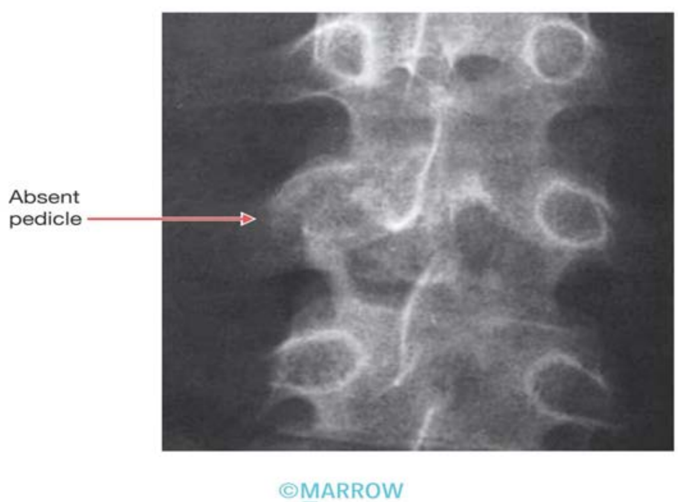
Option A: Renal osteodystrophy is a complex condition that results from the kidneys' inability to maintain the proper levels of calcium and phosphate in the blood, leading to secondary hyperparathyroidism. The below image represents Rugger Jersey spine which is characteristic of hyperparathyroidism seen in renal osteodystrophy.



Option B: TB spine occurs secondary to a pulmonary or gastrointestinal primary and spreads through the Batson's plexus of veins (paravertebral plexus). It most commonly involves the dorso-lumbar region.



Option C: Spinal metastasis: Represents the spread of malignancy to the spine, causing bone destruction. MRI would demonstrate vertebral lesions, bone marrow involvement, or spinal cord compression, without evidence of vertebral slippage. The following image shows a radiograph of spinal metastasis.

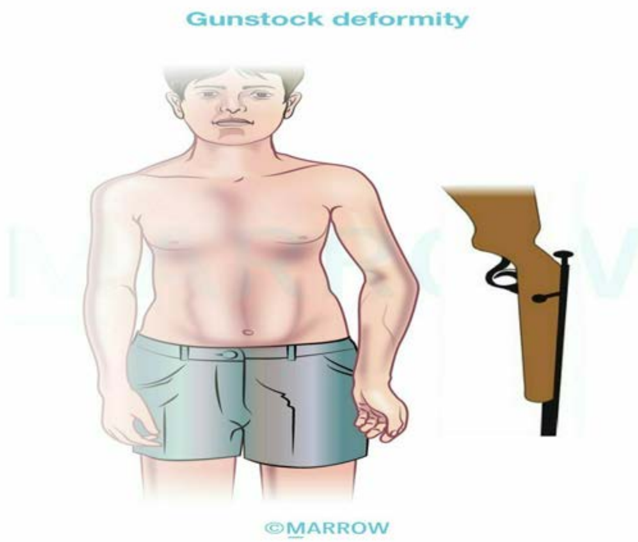


Solution to Question 15:

The given image shows cubitus varus deformity, where the forearm has deviated medially in relation to the upper arm, which is a common complication of supracondylar humeral fracture.

Malunion is the commonest complication of a supracondylar fracture and results in a cubitus varus deformity, also known as gunstock deformity. This occurs when the fracture unites with the distal fragment tilted medially and in internal rotation.

The image below shows gunstock deformity:



Cubitus varus is primarily a cosmetic concern with minimal functional impairment. Mild cases may not necessitate treatment, while severe deformities should be corrected through a lateral closed wedge osteotomy (French osteotomy).

Solution to Question 16:

The given X-ray points towards the tibial tuberosity. This image shows an avulsion fracture of the tibial tuberosity.

The tibial tuberosity is a bony prominence located on the anterior part of the tibia, just below the knee. It is the attachment point for the patellar tendon. Injury to the tibial tuberosity is common in adolescents, especially those involved in sports or after trauma like a fall.

Other options:

Option A: The medial tubercle refers to part of the medial condyle of the tibia.

Option C: Gerdy's tubercle is located on the lateral aspect of the tibia, where the iliotibial band (IT band) inserts. It is less commonly injured in falls but can be affected in conditions related to the IT band, like IT band syndrome.

Option D: Anterior intercondylar area is an area is located between the medial and lateral condyles of the tibia and serves as an attachment point for the anterior cruciate ligament (ACL). Injuries in this region are more commonly associated with ACL tears.